

CUDA Kernel 面试背题笔记

本书整理自 LeetCUDA 项目，系统汇编了 CUDA 面试中最高频的 GPU Kernel 实现，涵盖从 Warp/Block 原语、Elementwise、Softmax、GEMM、FlashAttention 等面试核心知识体系。

每个 Kernel 配备详细的中文面试要点注释 (WHY + HOW)，并以递进式优化路线 (naive → tiling → vectorize → tensor core → warp specialized) 展示 GPU 性能调优思维。

30 个 Kernel · 9 个 Phase · 全中文注释

项目主页: <https://github.com/xlite-dev/LeetCUDA>

更新日期: 2026 年 7 月 22 日

```

1 // =====
2 // notes-v2.cu — CUDA Kernel 面试刷题笔记
3 // =====
4 //
5 // 整理自 LeetCUDA 项目 (https://github.com/xlite-dev/LeetCUDA) , 涵盖:
6 //   - 面试高频 CUDA kernel 的完整实现 (~30 个 kernel)
7 //   - 每类 kernel 附带详细的面试要点注释 (WHY + HOW)
8 //   - 优化技术的递进式讲解 (naive → tiling → vectorize → tensor core → ws)
9 //   - BLAS 语义: N=col-major(Normal), T=row-major(Transposed)
10 //
11 // 10 个 Phase 覆盖:
12 //   Phase 0 — 面试框架速查 (GPU 架构 / Memory Hierarchy / Roofline / 优化清单)
13 //   Phase 1 — 基础原语: Warp Reduce / Block Reduce / Dot Product (含 broadcast 增强版)
14 //   Phase 2 — Elementwise: ReLU / Elementwise Add / Histogram (基础 + float4 向量化 + atomic)
15 //   Phase 3 — Softmax: naive → safe → online + RMS/Layer Norm
16 //   Phase 4 — RoPE: 旋转位置编码 (Llama 风格 theta=10000)
17 //   Phase 5 — Mat Transpose: 基础版 + BCF merge_write 最佳版 (Bank Conflict专题)
18 //   Phase 6 — GEMV: SGEMV K32/K128/K16 (warp-per-row)
19 //   Phase 7 — GEMM *: SGEMM → HGEMM → MMA m16n8k16(TN布局) → WGMMMA m64n128k16
20 //   Phase 8 — FlashAttention-2split_q (FA-2, 含 online softmax + P@V 寄存器复用)
21 //
22 // =====
23 // Phase 0: 面试框架速查 (纯注释, 面试开场必备的基础知识)
24 // =====
25
26 // ---- GPU 架构速查 ----
27 //
28 // SM (Streaming Multiprocessor) 内部结构:
29 //   - Warp Scheduler x4: 每 SM 4 个 warp scheduler, 每个每周期可发射 1 条指令
30 //   - Register File: 每 SM 65536 × 32-bit (4 bytes) = 256KB
31 //   - Shared Memory / L1: 可配置, 最大 shared memory ~228KB (Hopper)
32 //   - Tensor Cores: Hopper 每 SM 4 个; Blackwell 数量随型号/定义不同, 建议以官方 ISV guide 为准
33 //   - Warp = 32 threads: 最小调度单元, SIMT 执行模型
34 //
35 // Memory Hierarchy 带宽数量级 (H100 参考):
36 //   HBM3: ~3.35 TB/s (理论), 实际 ~2.5-3.0 TB/s
37 //   L2 Cache: ~12 TB/s (50MB, 跨 SM 共享)
38 //   L1/SMEM: ~19 TB/s (每 SM ~228KB)
39 //   Register: ~0 延迟, ~100+ TB/s 等效带宽
40 //
41 // 关键瓶颈判断:
42 //   Memory-bound: AI (Arithmetic Intensity) < 机器 FLOPS/带宽 比值
43 //   Compute-bound: AI 足够大, 受限于计算吞吐
44 //   Latency-bound: 线程不够多, 无法隐藏内存延迟
45 //
46 // Occupancy 公式:
47 //   occupancy = active_warps / max_warps_per_SM
48 //   受三类资源分别取下限: 每线程寄存器数 → threads/SM; 每 block shared memory
49 //   → blocks/SM; block 大小 → blocks/SM
50
51 // ---- 常见优化手段速查清单 ----
52 //
53 // 1. Coalesced Memory Access (合并访问)
54 //   - 同一 warp 的线程访问连续的 128B 对齐地址 → 1 次内存事务
55 //   - 否则产生多次事务 (最坏 32 次)
56 //
57 // 2. Tiling (分块)
58 //   - 将数据从 HBM 分块加载到 shared memory 复用, 减少 HBM 访问
59 //   - GEMM: Block Tile (BM×BN) + K Tile (BK)
60 //
61 // 3. Vectorized Memory Access (向量化)
62 //   - 使用 float4/half2 等向量类型, 减少 load/store 指令数

```

```

63 // - float4 = 128-bit, 单条指令加载 16 bytes
64 //
65 // 4. Thread Tile (寄存器分块)
66 // - 每个线程计算多个输出元素 (TM×TN), 提高计算密度
67 // - 减少线程总数, 降低同步开销
68 //
69 // 5. Bank Conflict Avoidance
70 // - Shared memory 有 32 banks × 4 bytes
71 // - 同 warp 多线程访问同一 bank 的不同地址 → bank conflict → 串行化
72 // - 解决方案: PAD (在每行末尾加 1 个元素打破对齐)
73 //
74 // 6. Pipeline / Double Buffering (流水线)
75 // - cp.async 异步拷贝下一批数据时同时做当前批的计算
76 // - Stage 数 = 2/3/4, 权衡 shared memory 占用和延迟隐藏
77 //
78 // 7. Tensor Core (MMA / WGMMa)
79 // - MMA m16n8k16 (Ampere): warp 级指令, 单 warp 完成 16×8×16 的矩阵乘
80 // - WGMMa m64n128k16 (Hopper): warpgroup 级指令 (128 threads), 异步执行
81 //
82 // 8. Warp Specialization (Hopper+)
83 // - Producer warpgroup 做 TMA 数据搬运, Consumer warpgroup 做计算
84 // - 通过 cuda::barrier 同步, 完全解耦数据搬运和计算
85 //
86 // 9. TMA (Tensor Memory Accelerator, Hopper+)
87 // - 硬件 DMA 引擎, 支持 1D~5D 寻址, 低寄存器开销
88 // - 配合 cp.async.bulk 实现异步数据搬运
89
90 // ---- Roofline 分析公式 ----
91 //
92 // AI (Arithmetic Intensity) = FLOPs / Bytes_transferred
93 //
94 // GEMM (M=N=K=4096):
95 // FLOPs = 2 × M × N × K = 2 × 40963 ≈ 137 GFLOPs
96 // Bytes = (M×K + K×N + M×N) × sizeof(float) ≈ 200 MB
97 // AI ≈ 137G / 200M ≈ 685 FLOPs/Byte → compute-bound (远超 H100 ridge point:
98 // FP16 TC ≈ 295:1, FP32 ≈ 20:1)
99 //
100 // GEMV (M=4096, K=4096):
101 // Bytes = (M×K + K + M) × sizeof(float) ≈ 67 MB
102 // AI ≈ 33M / 67M ≈ 0.5 FLOPs/Byte → severely memory-bound
103 //
104 // Softmax (N=4096): AI ≈ (5×N) / (2×N×4) = 5/8 ≈ 0.625 FLOPs/Byte → memory-bound
105
106 // =====
107 // Phase 1: 头文件 + 宏定义 + 基础原语 (Warp Reduce / Block Reduce)
108 // =====
109 // 面试要点:
110 // - warp_reduce: 用 __shfl_xor_sync 做蝶形归约, O(logN) 步, 无需 shared
111 //   memory
112 // - block_reduce: 两级归约 (warp → shared memory → warp0 broadcast),
113 //   注意最后必须 broadcast 回所有线程 (__shfl_sync), 否则只有 warp0 知道结果
114 // - 为什么不用 __shfl_down_sync? xor 模式所有线程做相同工作量, 更均衡
115
116 #include <algorithm>
117 #include <cstring>
118 #include <cuda_fp16.h>
119 #include <cuda_runtime.h>
120 #include <float.h>
121 #include <stdio.h>
122 #include <stdlib.h>
123 #include <math.h>
124 #include <cublas_v2.h>
125 #include <cuda.h>

```

```

126 #include <cuda/barrier>
127 #include <cuda/ptx>
128
129 #if defined(NOTES_V2_ENABLE_CUDNN)
130 #pragma nv_diag_suppress 128
131 #include <cuda/cudnn_frontend.h>
132 #pragma nv_diag_default 128
133 namespace fe = cudnn_frontend;
134 #endif
135
136 #if defined(NOTES_V2_ENABLE_TMA_MMA_WS) && CUDART_VERSION < 13000
137 #error "NOTES_V2_ENABLE_TMA_MMA_WS requires CUDA Toolkit 13.0 or newer"
138 #endif
139
140 #define INT4(value) (reinterpret_cast<int4 *>(&(value)))[0])
141 #define FLOAT4(value) (reinterpret_cast<float4 *>(&(value)))[0])
142 #define HALF2(value) (reinterpret_cast<half2 *>(&(value)))[0])
143
144 static constexpr int kWarpSize = 32;
145
146 // =====
147 // Phase 1a: Warp Reduce (warp 内归约, 纯寄存器操作, 无需 shared memory)
148 // =====
149
150 // Warp Reduce Sum — generic (used by both FP32 and FP16 contexts)
151 // 使用 __shfl_xor_sync 做蝶形归约 (butterfly reduction)
152 // 复杂度 O(logN), N=32 时只需 5 步
153 // 模板参数: T=数据类型, kWarpWidth=segment width (默认 32)
154 // kWarpWidth 会作为 __shfl_xor_sync 的第 4 个实参 width, 限制 shuffle 在同一 segment 内
155 // 当 kWarpWidth < 32 (如 FA 中 kWarpWidth=4) 时, 只有同 segment 的 lane 参与通信
156 // 蝶形归约示意 (以 warpSize=8 为例, 实际 warpSize=32 有 5 次迭代):
157 //
158 // 初始: 每个 lane 持有自己的值 v0..v7
159 // lane: 0 1 2 3 4 5 6 7
160 // val: v0 v1 v2 v3 v4 v5 v6 v7
161 //
162 // mask=4 (第1次迭代, lane i 与 lane i^4 交换并累加):
163 //
164 //      ┌──────────┐
165 // 对: (0,4) (1,5) (2,6) (3,7)
166 //
167 // lane: 0 1 2 3 4 5 6 7
168 // val: v0+v4 v1+v5 v2+v6 v3+v7 v4+v0 v5+v1 v6+v2 v7+v3
169 //
170 // mask=2 (第2次迭代, lane i 与 lane i^2 交换并累加):
171 //
172 //      ┌───┐ ┌───┐ ┌───┐ ┌───┐
173 // 对: (0,2) (1,3) (4,6) (5,7)
174 //      ─── 前4个一组 ───      ─── 后4个一组 ───
175 //
176 // lane: 0 1 2 3 4 5 6 7 (每 lane 持有前一轮2个值的和再加本轮配对)
177 // val: Σ{0,2,4,6} Σ{1,3,5,7} Σ{0,2,4,6} Σ{1,3,5,7} Σ{0,2,4,6} Σ{1,3,5,7} Σ{0,2,4,6} Σ{1,3,5,7}
178 //      = v0+v2+v4+v6 ... (逐步归约)
179 //
180 // mask=1 (第3次迭代, lane i 与 lane i^1 交换并累加):
181 //
182 //      ┌─┐ ┌─┐ ┌─┐ ┌─┐ ┌─┐ ┌─┐ ┌─┐ ┌─┐
183 // 对: (0,1) (2,3) (4,5) (6,7)
184 //
185 // lane: 0 1 2 3 4 5 6 7
186 // val: Σall Σall Σall Σall Σall Σall Σall Σall ← 所有 lane 拥有全归约结果!
187 //
188 // mask=0: 循环终止, 归约完成。
189 //
190 // 关键性质:
191 // - XOR 配对是对称的: lane i 的配对对象是 lane i^mask, 而 (i^mask)^mask = i

```

```

189 // - 每轮每个 lane 只和恰好 1 个其他 lane 通信 (一对一, 无冲突)
190 // - 每轮信息传递距离减半: 16→8→4→2→1 (距离减半, 信息翻倍)
191 // - 0(log2 N) 步完成, 无需 shared memory, 纯寄存器操作
192 // - __shfl_xor_sync 第四个参数 kWarpWidth 限制 segment width:
193 // 当 kWarpWidth=4 时, 只有同 segment(4个一组)内的 lane 参与 shuffle
194 template <const int kWarpWidth = kWarpSize, typename T = float>
195 __device__ __forceinline__ T warp_reduce_sum(T val) {
196 #pragma unroll
197     for (int mask = kWarpWidth >> 1; mask >= 1; mask >>= 1) {
198         val += __shfl_xor_sync(0xffffffff, val, mask, kWarpWidth);
199     }
200     return val;
201 }
202
203 // Warp Reduce Max — generic
204 template <const int kWarpWidth = kWarpSize, typename T = float>
205 __device__ __forceinline__ T warp_reduce_max(T val) {
206 #pragma unroll
207     for (int mask = kWarpWidth >> 1; mask >= 1; mask >>= 1) {
208         val = max(val, __shfl_xor_sync(0xffffffff, val, mask, kWarpWidth));
209     }
210     return val;
211 }
212
213 // =====
214 // Phase 1b: Block Reduce (block 内归约, 两级: warp → shared memory → warp reduce)
215 // =====
216
217 // Block Reduce Sum — FP32 (增强版, 带 broadcast)
218 // 两级归约流程:
219 // 1. 每个 warp 内做 warp_reduce_sum → 得到每 warp 的一个值
220 // 2. warp leader (lane=0) 写入 shared memory
221 // 3. syncthreads 后, lane 0~kNumWarps-1 读取 shared memory
222 // 4. 所有 warp 内再做一次 warp_reduce<kNumWarps> → 得到最终结果
223 // 5. __shfl_sync broadcast 到所有线程 (关键! 否则每个warp只有lane<kNumWarps 知道结果)
224 template <const int kNumThreads = 256>
225 __device__ float block_reduce_sum(float val) {
226     constexpr int kNumWarps = (kNumThreads + kWarpSize - 1) / kWarpSize;
227     int warp = threadIdx.x / kWarpSize;
228     int lane = threadIdx.x % kWarpSize;
229     __shared__ float shared[kNumWarps];
230
231     float value = warp_reduce_sum<kWarpSize>(val);
232     if (lane == 0)
233         shared[warp] = value;
234     __syncthreads();
235     value = (lane < kNumWarps) ? shared[lane] : 0.0f;
236     value = warp_reduce_sum<kNumWarps>(value);
237     // 关键: broadcast 结果到所有线程, 后续用 result
238     // 做除法等操作时所有线程都能拿到
239     value = __shfl_sync(0xffffffff, value, 0, 32);
240     return value;
241 }
242
243 // Block Reduce Max — FP32 (增强版, 带 broadcast)
244 template <const int kNumThreads = 256>
245 __device__ float block_reduce_max(float val) {
246     constexpr int kNumWarps = (kNumThreads + kWarpSize - 1) / kWarpSize;
247     int warp = threadIdx.x / kWarpSize;
248     int lane = threadIdx.x % kWarpSize;
249     __shared__ float shared[kNumWarps];
250
251     float value = warp_reduce_max<kWarpSize>(val);

```

```

252     if (lane == 0)
253         shared[warp] = value;
254     __syncthreads();
255     value = (lane < kNumWarps) ? shared[lane] : -FLT_MAX;
256     value = warp_reduce_max<kNumWarps>(value);
257     value = __shfl_sync(0xffffffff, value, 0, 32);
258     return value;
259 }
260
261 // ---- Block Reduce Sum All: y = sum(a[0..N-1]) ----
262 // 多 block 各自做 warp->smem->warp0 reduce, 然后 atomicAdd 到全局 y
263 // 跨 block 求和的常见模式, 适合 N 较大时使用;
264 // Grid: ((N + 255) / 256, 1, 1)
265 // Block: (256, 1, 1)
266 // source: LeetCUDA/kernels/reduce/block_all_reduce.cu
267 template <const int kNumThreads = 256>
268 __global__ void block_reduce_all(float *a, float *y, int N) {
269     int tid = threadIdx.x;
270     int idx = blockIdx.x * kNumThreads + tid;
271     constexpr int kNumWarps = (kNumThreads + kWarpSize - 1) / kWarpSize;
272     __shared__ float shared[kNumWarps];
273
274     float val = (idx < N) ? a[idx] : 0.0f;
275     int warp = tid / kWarpSize;
276     int lane = tid % kWarpSize;
277
278     val = warp_reduce_sum<kWarpSize>(val);
279     if (lane == 0)
280         shared[warp] = val;
281     __syncthreads();
282
283     val = (lane < kNumWarps) ? shared[lane] : 0.0f;
284     if (warp == 0)
285         val = warp_reduce_sum<kNumWarps>(val);
286     if (tid == 0) // tid == 0, not lane 0. 只有 block 内的一个线程负责写回全局结果, 避免重复累加
287         atomicAdd(y, val);
288 }
289
290 // ---- Dot Product: y = sum(a[i] * b[i]) ----
291 // 核心模式: elementwise 乘法 -> block reduce -> atomicAdd 全局累加
292 // Grid: ((N + 255) / 256, 1, 1)
293 // Block: (256, 1, 1)
294 // source: LeetCUDA/kernels/dot-product/dot_product.cu
295 template <const int kNumThreads = 256>
296 __global__ void dot(float *a, float *b, float *y, int N) {
297     int tid = threadIdx.x;
298     int idx = blockIdx.x * kNumThreads + tid;
299     constexpr int kNumWarps = (kNumThreads + kWarpSize - 1) / kWarpSize;
300     __shared__ float shared[kNumWarps];
301
302     float prod = (idx < N) ? a[idx] * b[idx] : 0.0f;
303     int warp = tid / kWarpSize;
304     int lane = tid % kWarpSize;
305
306     prod = warp_reduce_sum<kWarpSize>(prod);
307     if (lane == 0)
308         shared[warp] = prod;
309     __syncthreads();
310
311     prod = (lane < kNumWarps) ? shared[lane] : 0.0f;
312     if (warp == 0) // 只需要 warp 0 的线程继续 reduce 即可
313         prod = warp_reduce_sum<kNumWarps>(prod);
314     if (tid == 0) // tid == 0, not lane 0. 只有 block 内的一个线程负责写回全局结果, 避免重复累加

```

```

315     atomicAdd(y, prod);
316 }
317
318 // Dot Product + float4
319 // Grid: ((N + 255) / 256, 1, 1), 每个block处理256元素, float4向量化后每个线程处理4元素
320 // Block: (64, 1, 1), 256/4=64
321 // 注意: 该版本默认输入地址满足 float4 对齐; 最适合 N 按 4 对齐的场景
322 // source: LeetCUDA/kernels/dot-product/dot_product.cu
323 template <const int kNumThreads = 256 / 4>
324 __global__ void dot_vec4(float *a, float *b, float *y, int N) {
325     int tid = threadIdx.x;
326     int idx = (blockIdx.x * kNumThreads + tid) * 4;
327     constexpr int kNumWarps = (kNumThreads + kWarpSize - 1) / kWarpSize;
328     __shared__ float shared[kNumWarps];
329
330     float4 reg_a = FLOAT4(a[idx]);
331     float4 reg_b = FLOAT4(b[idx]);
332     float prod = (idx < N) ? (reg_a.x * reg_b.x + reg_a.y * reg_b.y +
333                             reg_a.z * reg_b.z + reg_a.w * reg_b.w)
334                       : 0.0f;
335     int warp = tid / kWarpSize;
336     int lane = tid % kWarpSize;
337
338     prod = warp_reduce_sum<kWarpSize>(prod);
339     if (lane == 0)
340         shared[warp] = prod;
341     __syncthreads();
342
343     prod = (lane < kNumWarps) ? shared[lane] : 0.0f;
344     if (warp == 0)
345         prod = warp_reduce_sum<kNumWarps>(prod);
346     if (tid == 0)
347         atomicAdd(y, prod);
348 }
349
350
351 // =====
352 // Phase 2: Elementwise Ops (逐元素操作, 演示 coalesced access + vectorize)
353 // =====
354 // 面试要点:
355 //   - 逐元素操作是最简单的 kernel, 核心考点是 memory coalescing
356 //   - float4 向量化可将内存事务数减为 1/4, 大幅提升 bandwidth utilization
357 //   - grid/block 维度设计: grid(N/threads), block(threads), 一维即可
358
359 // ---- ReLU: y = max(0, x) ----
360 // Grid: ((N + 255) / 256, 1, 1)
361 // Block: (256, 1, 1)
362 // source: LeetCUDA/kernels/relu/relu.cu
363 __global__ void relu(float *x, float *y, int N) {
364     int idx = blockIdx.x * blockDim.x + threadIdx.x;
365     if (idx < N)
366         y[idx] = fmaxf(0.0f, x[idx]);
367 }
368
369 // ReLU + float4 向量化: 每个线程处理 4 个元素, 减少 75% 的 load/store 指令
370 // block(64)x4(float4)=256 元素/block, 与基础版吞吐相同
371 // Grid: ((N + 255) / 256, 1, 1)
372 // Block: (64, 1, 1)
373 // 注意: 该版本默认地址满足 float4 对齐; 最适合 N 按 4 对齐的场景
374 // source: LeetCUDA/kernels/relu/relu.cu
375 __global__ void relu_vec4(float *x, float *y, int N) {
376     int idx = (blockIdx.x * blockDim.x + threadIdx.x) * 4;
377     if (idx < N) {

```

```

378     float4 reg_x = FLOAT4(x[idx]); // 单条 128-bit load
379     float4 reg_y;
380     reg_y.x = fmaxf(0.0f, reg_x.x);
381     reg_y.y = fmaxf(0.0f, reg_x.y);
382     reg_y.z = fmaxf(0.0f, reg_x.z);
383     reg_y.w = fmaxf(0.0f, reg_x.w);
384     FLOAT4(y[idx]) = reg_y; // 单条 128-bit store
385 }
386 }
387
388 // ---- Elementwise Add: c = a + b ----
389 // Grid: ((N + 255) / 256, 1, 1)
390 // Block: (256, 1, 1)
391 // source: LeetCUDA/kernels/elementwise/elementwise.cu
392 __global__ void elementwise_add(float *a, float *b, float *c, int N) {
393     int idx = blockIdx.x * blockDim.x + threadIdx.x;
394     if (idx < N)
395         c[idx] = a[idx] + b[idx];
396 }
397
398 // Elementwise Add + float4 向量化
399 // Grid: ((N + 255) / 256, 1, 1)
400 // Block: (64, 1, 1), block(64)*4(float4)=256 元素/block
401 // 注意: 主路径要求 4 元素对齐; 尾部不足 4 个元素时回退到标量处理
402 // source: LeetCUDA/kernels/elementwise/elementwise.cu
403 __global__ void elementwise_add_vec4(float *a, float *b, float *c, int N) {
404     int idx = 4 * (blockIdx.x * blockDim.x + threadIdx.x);
405     if ((idx + 3) < N) {
406         float4 reg_a = FLOAT4(a[idx]);
407         float4 reg_b = FLOAT4(b[idx]);
408         float4 reg_c;
409         reg_c.x = reg_a.x + reg_b.x;
410         reg_c.y = reg_a.y + reg_b.y;
411         reg_c.z = reg_a.z + reg_b.z;
412         reg_c.w = reg_a.w + reg_b.w;
413         FLOAT4(c[idx]) = reg_c;
414     } else if (idx < N) {
415         for (int i = 0; (idx + i) < N; i++) {
416             c[idx + i] = a[idx + i] + b[idx + i];
417         }
418     }
419 }
420
421 // ---- Histogram: y[a[i]]++ ----
422 // 演示 atomicAdd 的用法: 多个线程可能同时更新同一个 bin
423 // Grid: ((N + 255) / 256, 1, 1)
424 // Block: (256, 1, 1)
425 // source: LeetCUDA/kernels/histogram/histogram.cu
426 __global__ void histogram(int *a, int *y, int N) {
427     int idx = blockIdx.x * blockDim.x + threadIdx.x;
428     if (idx < N)
429         atomicAdd(&(y[a[idx]]), 1);
430 }
431
432 // ---- Merge Attn States: 合并 Split-KV Attention 的部分结果 ----
433 // 实现 FlashInfer 论文 (arXiv 2501.01005) Section 2.2 的 attention 合并逻辑。
434 //
435 // 面试要点:
436 //   - 应用场景: Split-KV attention 将序列沿 K 维分段计算 attention,
437 //     每段产出部分结果 (O_i, LSE_i), 本 kernel 将两段按指数权重加权合并。
438 //   - 数学公式 (5 步推导):
439 //     Step 1 — max 归一化 (数值稳定, 与 safe softmax 同理):
440 //         L_max = max(LSE_1, LSE_2)

```

```

441 // Step 2 — 还原指数权重 (un-normalized) :
442 //      w_1 = exp(LSE_1 - L_max), w_2 = exp(LSE_2 - L_max)
443 // Step 3 — 归一化为混合比例:
444 //      alpha = w_1 / (w_1 + w_2), beta = w_2 / (w_1 + w_2)
445 //      满足 alpha + beta = 1
446 // Step 4 — 逐元素加权合并:
447 //      O = alpha * O_1 + beta * O_2
448 // - LSE 布局 [num_heads, num_tokens] (与 FlashAttention-2惯例一致,
449 //   lse[head_idx][token_idx])
450 // - Output 布局 [num_tokens, num_heads, head_size] (token 维在最外,
451 //   展平为 [T0H0, T0H1, ..., T1H0, ...])
452 // -  $-\infty$  LSE  $\rightarrow$   $-\infty$ : 空 attention 段 (causal mask 等导致全部 score
453 //   为  $-\infty$ ) 的 LSE 可能为  $+\infty$ , 替换后  $\exp(-\infty - L_{\max}) = 0$ ,
454 //   该段权重退化为 0
455 // - 128-bit 向量化: uint4 = 4  $\times$  float, 每线程处理 4 个输出元素,
456 //   pack load  $\rightarrow$  FMA  $\rightarrow$  pack store 一条流水线
457 // - AI  $\approx$  3 FLOP / 20 bytes  $\approx$  0.15 FLOPS/Byte  $\rightarrow$  severely memory-bound
458 //
459 // 线程到数据的 3D 映射 (以 num_tokens=2, num_heads=2, head_size=8 为例):
460 // kPackSize = 16 / sizeof(float) = 4 (每线程 4 个 float = 128-bit)
461 // threads_per_head = head_size / 4 = 2
462 // total_threads = num_tokens * num_heads * threads_per_head = 8
463 //
464 // global_idx:      0      1      2      3      4      5      6      7
465 // token_head_idx:  0      0      1      1      2      2      3      3
466 //   (第几个 (token,head) 对, 行优先)
467 // pack_idx:        0      1      0      1      0      1      0      1
468 //   (该 (token,head) 对内第几个 pack)
469 // token_idx:        0      0      0      0      1      1      1      1
470 //   (token_head_idx / num_heads, token 变化慢)
471 // head_idx:         0      0      1      1      0      0      1      1
472 //   (token_head_idx % num_heads, head 变化快)
473 // pack_offset:      0      4      0      4      0      4      0      4
474 //   (pack_idx * 4, 该 pack 在 head_size 中的起始元素偏移)
475 // head_offset:      0      0      8      8      16     16     24     24
476 //   (token_idx * num_heads * head_size + head_idx * head_size,
477 //   该 (token,head) 对在 output 展平数组中的起始偏移)
478 //
479 // Grid: ((total_threads + 127) / 128, 1, 1)
480 // Block: (128, 1, 1)
481 // source: LeetCUDA/kernels/openai-triton/merge-attn-states/cuda_merge_attn_states.cu
482 __global__ void merge_attn_states(
483     float *output,           // [num_tokens, num_heads, head_size]
484     const float *prefix_output, // [num_tokens, num_heads, head_size]
485     const float *prefix_lse,   // [num_heads, num_tokens]
486     const float *suffix_output, // [num_tokens, num_heads, head_size]
487     const float *suffix_lse,   // [num_heads, num_tokens]
488     int num_tokens, int num_heads, int head_size) {
489
490     constexpr int kNumThreads = 128;
491     constexpr int kPackSize = 16 / sizeof(float); // 4 floats = 128-bit
492     using pack_t = uint4;
493
494     // 每个 (token, head) 对需要 threads_per_head 个线程覆盖 head_size 个元素
495     const int threads_per_head = head_size / kPackSize;
496     // 实际有效总线程数, 超过这个数的线程会被忽略, block是按照kNumThreads=128
497     // 来分配的, 那么最后一个block的线程数可能会超过实际需要的线程数
498     const int total_threads = num_tokens * num_heads * threads_per_head;
499
500     const int global_idx = blockIdx.x * kNumThreads + threadIdx.x;
501     if (global_idx >= total_threads)
502         return;
503

```

```

504 // global_idx → (token_head_idx, pack_idx) → (token_idx, head_idx, pack_offset)
505 // token_head_idx: 第几个 (token, head) 对, 行优先展平
506 const int token_head_idx = global_idx / threads_per_head;
507 // pack_idx: 该 (token, head) 对内第几个 pack, 0 ~ threads_per_head-1
508 const int pack_idx = global_idx % threads_per_head;
509
510 // token_head_idx 分解为 (token_idx, head_idx)
511 const int token_idx = token_head_idx / num_heads; // token 变化慢 (外维)
512 const int head_idx = token_head_idx % num_heads; // head 变化快 (内维)
513
514 // pack_offset: 该 pack 覆盖的元素在 head_size 中的起始偏移 (0, 4, 8, ...)
515 const int pack_offset = pack_idx * kPackSize;
516 // head_offset: 该 (token, head) 对在 output 展平一维数组中的起始偏移
517 const int head_offset = token_idx * num_heads * head_size + head_idx * head_size;
518
519 // 定位到当前 (token, head) 的输出段起始
520 const float *prefix_head = prefix_output + head_offset;
521 const float *suffix_head = suffix_output + head_offset;
522 float *output_head = output + head_offset;
523
524 // LSE 布局 [num_heads, num_tokens]: lse[head_idx][token_idx]
525 float p_lse = prefix_lse[head_idx * num_tokens + token_idx];
526 float s_lse = suffix_lse[head_idx * num_tokens + token_idx];
527
528 // inf → -inf: 空 attention 段 LSE 可能为 +inf, 替换后权重退化为 0
529 p_lse = isinf(p_lse) ? -INFINITY : p_lse;
530 s_lse = isinf(s_lse) ? -INFINITY : s_lse;
531
532 // Step 1: max 归一化 (与 safe softmax 同理, 防 exp 溢出)
533 const float max_lse = fmaxf(p_lse, s_lse);
534 p_lse -= max_lse;
535 s_lse -= max_lse;
536
537 // Step 2-3: 指数还原 → 混合比例 alpha, beta
538 const float p_se = expf(p_lse);
539 const float s_se = expf(s_lse);
540 const float out_se = p_se + s_se;
541 const float p_scale = p_se / out_se; // alpha = w_1 / (w_1 + w_2)
542 const float s_scale = s_se / out_se; // beta = w_2 / (w_1 + w_2)
543
544 // Step 4: 逐元素加权合并 0 = alpha * 0_1 + beta * 0_2
545 // 128-bit 向量化: uint4 load → per-element FMA → uint4 store
546 if (pack_offset < head_size) {
547     pack_t p_pack =
548         reinterpret_cast<const pack_t *>(prefix_head)[pack_idx];
549     pack_t s_pack =
550         reinterpret_cast<const pack_t *>(suffix_head)[pack_idx];
551     pack_t o_pack;
552
553     #pragma unroll
554     for (int i = 0; i < kPackSize; ++i) {
555         const float p_v = reinterpret_cast<const float *>(&p_pack)[i];
556         const float s_v = reinterpret_cast<const float *>(&s_pack)[i];
557         const float o_v = p_v * p_scale + (s_v * s_scale); // FMA
558         reinterpret_cast<float *>(&o_pack)[i] = o_v;
559     }
560
561     reinterpret_cast<pack_t *>(output_head)[pack_idx] = o_pack;
562 }
563 }
564
565 // =====
566 // Phase 3: Reduce 类 Ops — Softmax / RMS Norm / Layer Norm

```

```

567 // =====
568 // 面试要点:
569 //   - Softmax 三种实现递进: naive (溢出) → safe (2-pass, max 减法) →
570 //   online (1-pass, 增量更新)
571 //   - Online Softmax 是 FlashAttention-2的数学基础
572 //   - RMS Norm vs Layer Norm: RMS 只需 1 次 reduce, Layer Norm 需要 2 次 (mean
573 //   + variance)
574 //   - Per-token 设计: 一个 block 处理一个 token, 无需跨 block 同步
575
576 // =====
577 // Phase 3a: Softmax — 三级递进 (面试核心考点)
578 // =====
579 // 面试常问: 「Softmax 有哪些实现方式? 各有什么优缺点? 」
580 // 回答线索: naive(溢出) → safe → online
581
582 // ---- Level 1: 基础 Softmax (per-token, 无 max 减法, 数值不稳定) ----
583 // grid(S*h/h, h), block(h), 一个 block 处理一个 token
584 // 问题: x 值很大时 exp(x) 溢出为 inf
585 template <const int kNumThreads = 256>
586 // Grid: (S, 1, 1), S=batch*seq_len, DISPATCH_SOFTMAX_F32_PER_TOKEN_KERNEL
587 // Block: (H, 1, 1), 由外层 dispatch 选择 H=32/64/128/256/512/1024, 一个 block 处理一个 token
588 // source: LeetCUDA/kernels/softmax/softmax.cu
589 __global__ void softmax_per_token(float *x, float *y, int N) {
590     const int tid = threadIdx.x;
591     const int idx = blockIdx.x * blockDim.x + tid;
592
593     float exp_val = (idx < N) ? expf(x[idx]) : 0.0f;
594     float exp_sum = block_reduce_sum<kNumThreads>(exp_val);
595     if (idx < N)
596         y[idx] = exp_val / exp_sum;
597 }
598
599 // ---- Level 2: Safe Softmax (2-pass: 先 max 再 exp, 数值稳定) ----
600 // 面试重点: 为什么 Softmax 需要 Safe?
601 //   - expf 溢出阈值约 88.7: exp(88)≈1.65e38 (接近 float32 上限 3.4e38), exp(89)≈4.5e38 已溢出为 inf
602 //   - 减去 max 后: exp(x - max) ≤ exp(0) = 1.0, 永不超过 1
603 //   - 数学等价性: softmax(x) = softmax(x - c) 对任意常数 c 成立
604 //   - 代价: 2 次 block reduce (先 max, 再 sum), 但仍 O(N/B) 高效
605 // Grid: (S, 1, 1)
606 // Block: (H, 1, 1), 由外层 dispatch 选择 H=32/64/128/256/512/1024
607 // source: LeetCUDA/kernels/softmax/softmax.cu
608 template <const int kNumThreads = 256>
609 __global__ void safe_softmax_per_token(float *x, float *y, int N) {
610     const int tid = threadIdx.x;
611     const int idx = blockIdx.x * blockDim.x + tid;
612
613     // Pass 1: block reduce max — 找最大值
614     float val = (idx < N) ? x[idx] : -FLT_MAX;
615     float max_val = block_reduce_max<kNumThreads>(val);
616
617     // Pass 2: exp(x - max) → block reduce sum
618     float exp_val = (idx < N) ? expf(x[idx] - max_val) : 0.0f;
619     float exp_sum = block_reduce_sum<kNumThreads>(exp_val);
620
621     if (idx < N)
622         y[idx] = exp_val / exp_sum;
623 }
624
625 // ---- Level 3: Online Safe Softmax (FlashAttention-2的数学基础) ----
626 // 面试重点:
627 // 两种使用场景 (公式不同, 但结果等价) :
628 //   1) 单元素增量更新 (online update, 处理新元素 x_i):
629 //       m_new = max(m_old, x_i)

```

```

630 //      d_new = d_old * exp(m_old - m_new) + exp(x_i - m_new)
631 //  2) 二元合并 (binary merge, 合并两个部分累加器, warp/block reduce 使用):
632 //      m = max(m1, m2) safe softmax用的max值
633 //      d = d1*exp(m1-m) + d2*exp(m2-m) softmax用的分母值
634 //      当 m1≥m2 时退化为 d1 + d2*exp(m2-m1) (d_bigger + d_smaller*exp(m_smaller - m_bigger))
635 //  算法来源: "Online normalizer calculation for softmax" (arXiv:1805.02867)
636 //  Warp Reduce for Online Softmax — binary merge of two partial (m,d) accumulators.
637 //
638 //  不同于单元元素增量更新公式 (m_new = max(m_old, x_i); d_new = d_old*exp(m_old-m_new) + exp(x_i-m_new)) ,
639 //  warp reduce 中每次合并的是两个**已各自归一化到各自 max 的部分累加器**:
640 //  (m1,d1): max 和  $\sum \exp(x_j - m1)$ , 覆盖集合 S1
641 //  (m2,d2): max 和  $\sum \exp(x_k - m2)$ , 覆盖集合 S2
642 //
643 //  合并公式 (对称, 不依赖"新/旧"概念) :
644 //  m = max(m1, m2)
645 //  d = d1*exp(m1-m) + d2*exp(m2-m) ← m1≥m2 时退化为 d1 + d2*exp(m2-m1)
646 //
647 //  该操作满足结合律 → XOR butterfly 任意归约顺序均保证全局正确结果。
648 struct __align__(8) MD {
649     float m; // running max
650     float d; // running denominator (sum of exp(x - max))
651 };
652
653 template <const int kWarpWidth = kWarpSize>
654 __device__ __forceinline__ MD warp_reduce_md(MD md1) {
655     #pragma unroll
656     for (int mask = kWarpWidth >> 1; mask >= 1; mask >>= 1) {
657         MD md2;
658         md2.m = __shfl_xor_sync(0xffffffff, md1.m, mask, kWarpWidth);
659         md2.d = __shfl_xor_sync(0xffffffff, md1.d, mask, kWarpWidth);
660
661         MD b_m = (md1.m > md2.m) ? md1 : md2; // max
662         MD s_m = (md1.m > md2.m) ? md2 : md1;
663
664         // b_m.d 无需 rescale (其基准 max 已是全局 max) ,
665         // s_m.d 需 rescale: exp(s_m.m - b_m.m)
666         md1.d = b_m.d + s_m.d * __expf(s_m.m - b_m.m);
667         md1.m = b_m.m;
668     }
669     return md1;
670 }
671
672 // 注意: 这里默认一个 block 处理一个 token; 边界线程的 d=0 只参与归约, 不会写回 y
673 // Grid: (S, 1, 1)
674 // Block: (H, 1, 1), 由外层 dispatch 选择 H=32/64/128/256/512/1024
675 // source: LeetCUDA/kernels/softmax/softmax.cu
676 template <const int kNumThreads = 256>
677 __global__ void online_safe_softmax_per_token(const float *x, float *y, int N) {
678     int tid = threadIdx.x;
679     int idx = blockIdx.x * kNumThreads + threadIdx.x;
680     const int kNumWarps = (kNumThreads + kWarpSize - 1) / kWarpSize;
681     __shared__ MD shared[kNumWarps];
682
683     float val = (idx < N) ? x[idx] : -FLT_MAX;
684     int warp = tid / kWarpSize;
685     int lane = tid % kWarpSize;
686
687     // 初始化: 每个线程持有一个 (max, denom) 对
688     MD md;
689     md.m = idx < N ? val : -FLT_MAX;
690     md.d = idx < N ? 1.0f : 0.0f;
691
692     // 第一级规约: warp_reduce_md 在归约中自动更新 m 和 d

```

```

693 md = warp_reduce_md<kWarpSize>(md);
694
695 if (lane == 0)
696     shared[warp] = md;
697 __syncthreads();
698
699 // 第二级归约: 每个 warp 结果再做一次 warp_reduce_md (复用 block_reduce 模式)
700 md = lane < kNumWarps ? shared[lane] : MD{-FLT_MAX, 0.0f};
701 md = warp_reduce_md<kWarpSize>(md); // 用kWarpSize确保每个lane都能拿到最终结果
702
703 // 用全局 max 和 denom 做最终 softmax
704 float d_inv = __fdividef(1.0f, md.d);
705 // 边界线程即使看到 d=0 的填充值, 也不会走到写回路径
706 if (idx < N) {
707     y[idx] = __expf(val - md.m) * d_inv;
708 }
709 }
710
711 // =====
712 // Phase 3b: RMS Normalization (1-pass reduce)
713 // =====
714 // 面试要点:
715 //   - RMS Norm:  $y = (x / \text{rms}(x)) * g$ ,  $1/\text{rms}(x) = \text{rsqrt}(\text{mean}(x^2))$ 
716 //   - 只需 1 次 block reduce (sum of squares), 比 Layer Norm 少 1 次同步
717 //   - Llama 系列使用 RMS Norm
718 //   - grid(N, K/K), block(K): 一行一个 block
719 // Grid: (N, 1, 1), N=batch*seq_len, 每行一个 block
720 // Block: (128, 1, 1), kNumThreads=K=128 (K>128 时调整模板参数)
721 // source: LeetCUDA/kernels/rms-norm/rms_norm.cu
722 template <const int kNumThreads = 128>
723 __global__ void rms_norm(float *x, float *y, float g, int N, int K) {
724     int tid = threadIdx.x;
725     int idx = blockIdx.x * blockDim.x + threadIdx.x;
726     const float epsilon = 1e-5f;
727
728     __shared__ float s_variance;
729     float value = (idx < N * K) ? x[idx] : 0.0f;
730     float variance = value * value;
731     variance = block_reduce_sum<kNumThreads>(variance);
732     if (tid == 0)
733         s_variance = rsqrtf(variance / (float)K + epsilon); // 1/rms(x)
734     __syncthreads();
735     if (idx < N * K)
736         y[idx] = (value * s_variance) * g;
737 }
738
739 // RMS Norm + float4
740 // Grid: (N, 1, 1)
741 // Block: (32, 1, 1), 128/4=32; 对应一行 K 元素按 4 个一组交给 32 个线程处理
742 // 注意: 该版本默认 K 按 4 对齐, 且输入/输出地址满足 float4 对齐
743 // source: LeetCUDA/kernels/rms-norm/rms_norm.cu
744 template <const int kNumThreads = 128 / 4>
745 __global__ void rms_norm_vec4(float *x, float *y, float g, int N, int K) {
746     int tid = threadIdx.x;
747     int idx = (blockIdx.x * blockDim.x + threadIdx.x) * 4;
748     const float epsilon = 1e-5f;
749
750     __shared__ float s_variance;
751     float4 reg_x = FLOAT4(x[idx]);
752     float variance = (idx < N * K) ? (reg_x.x * reg_x.x + reg_x.y * reg_x.y +
753                                     reg_x.z * reg_x.z + reg_x.w * reg_x.w)
754                                     : 0.0f;
755     variance = block_reduce_sum<kNumThreads>(variance);

```

```

756     if (tid == 0)
757         s_variance = rsqrtf(variance / (float)K + epsilon);
758     __syncthreads();
759     float4 reg_y;
760     reg_y.x = reg_x.x * s_variance * g;
761     reg_y.y = reg_x.y * s_variance * g;
762     reg_y.z = reg_x.z * s_variance * g;
763     reg_y.w = reg_x.w * s_variance * g;
764     if (idx < N * K)
765         FLOAT4(y[idx]) = reg_y;
766 }
767
768 // =====
769 // Phase 3c: Layer Normalization (2-pass reduce)
770 // =====
771 // 面试要点:
772 //   - Layer Norm:  $y = ((x - \text{mean}) / \text{std}) * g + b$ ,  $\text{std} = \sqrt{\text{variance}}$ ,  $\text{variance} = \text{mean}((x - \text{mean})^2)$ 
773 //   - 需要 2 次 block reduce: 先 mean (sum/K), 再 variance (sum((x-mean)2)/K)
774 //   - 两次 __syncthreads 必须到位, 否则 s_mean 未对所有线程可见就计算 variance
775 // Grid: (N, 1, 1), 一行一个 block
776 // Block: (128, 1, 1), kNumThreads=K=128
777 // source: LeetCUDA/kernels/layer-norm/layer_norm.cu
778 template <const int kNumThreads = 128>
779 __global__ void layer_norm(float *x, float *y, float g, float b, int N, int K) {
780     int tid = threadIdx.x;
781     int idx = blockIdx.x * blockDim.x + threadIdx.x;
782     const float epsilon = 1e-5f;
783
784     __shared__ float s_mean;
785     __shared__ float s_variance;
786     float value = (idx < N * K) ? x[idx] : 0.0f;
787
788     // Pass 1: compute mean
789     float sum = block_reduce_sum<kNumThreads>(value);
790     if (tid == 0)
791         s_mean = sum / (float)K;
792     __syncthreads(); // 必须等待 s_mean 对所有线程可见
793
794     // Pass 2: compute variance = (x - mean)2
795     float variance = (value - s_mean) * (value - s_mean);
796     variance = block_reduce_sum<kNumThreads>(variance);
797     if (tid == 0)
798         s_variance = rsqrtf(variance / (float)K + epsilon); // 1/std
799     __syncthreads(); // 必须等待 s_variance 对所有线程可见
800
801     if (idx < N * K)
802         y[idx] = ((value - s_mean) * s_variance) * g + b;
803 }
804
805 // Layer Norm + float4
806 // Grid: (N, 1, 1)
807 // Block: (32, 1, 1), 128/4=32; 对应一行 K 元素按 4 个一组交给 32 个线程处理
808 // 注意: 该版本默认 K 按 4 对齐, 且输入/输出地址满足 float4 对齐
809 // source: LeetCUDA/kernels/layer-norm/layer_norm.cu
810 template <const int kNumThreads = 128 / 4>
811 __global__ void layer_norm_vec4(float *x, float *y, float g, float b, int N, int K) {
812     int tid = threadIdx.x;
813     int idx = (blockIdx.x * blockDim.x + threadIdx.x) * 4;
814     const float epsilon = 1e-5f;
815
816     __shared__ float s_mean;
817     __shared__ float s_variance;
818     float4 reg_x = FLOAT4(x[idx]);

```

```

819 float value = (idx < N * K) ? (reg_x.x + reg_x.y + reg_x.z + reg_x.w) : 0.0f;
820
821 float sum = block_reduce_sum<kNumThreads>(value);
822 if (tid == 0)
823     s_mean = sum / (float)K;
824 __syncthreads();
825
826 float4 reg_x_hat;
827 reg_x_hat.x = reg_x.x - s_mean;
828 reg_x_hat.y = reg_x.y - s_mean;
829 reg_x_hat.z = reg_x.z - s_mean;
830 reg_x_hat.w = reg_x.w - s_mean;
831 float variance = reg_x_hat.x * reg_x_hat.x + reg_x_hat.y * reg_x_hat.y +
832                 reg_x_hat.z * reg_x_hat.z + reg_x_hat.w * reg_x_hat.w;
833 variance = block_reduce_sum<kNumThreads>(variance);
834 if (tid == 0)
835     s_variance = rsqrtf(variance / (float)K + epsilon);
836 __syncthreads();
837
838 float4 reg_y;
839 reg_y.x = reg_x_hat.x * s_variance * g + b;
840 reg_y.y = reg_x_hat.y * s_variance * g + b;
841 reg_y.z = reg_x_hat.z * s_variance * g + b;
842 reg_y.w = reg_x_hat.w * s_variance * g + b;
843 if (idx < N * K)
844     FLOAT4(y[idx]) = reg_y;
845 }
846
847 // =====
848 // Phase 4: RoPE — 旋转位置编码 (Rotary Position Embedding)
849 // =====
850 // 面试要点:
851 //   - RoPE 数学公式: 对每对相邻维度做 2D 旋转
852 //     [x1']   [cos(θ)  -sin(θ)] [x1]
853 //     [x2'] = [sin(θ)   cos(θ)] [x2]
854 //   -  $\theta_i = 1 / (\text{theta}^{(2i/d)})$ , theta=10000.0f (Llama 风格)
855 //   - token_pos = idx / N: token 在序列中的位置
856 //   - token_idx = idx % N: token 内的维度对索引
857 //   - 输入 [seq_len, hidden_size], 输出同形状
858
859 // Grid: ((seq_len * N + 255) / 256, 1, 1)
860 // Block: (256, 1, 1)
861 // source: LeetCUDA/kernels/rope/rope.cu
862 __global__ void rope(float *x, float *out, int seq_len, int N) {
863     int idx = blockIdx.x * blockDim.x + threadIdx.x;
864     float x1 = x[idx * 2];
865     float x2 = x[idx * 2 + 1];
866     int token_pos = idx / N; // 序列位置
867     int token_idx = idx % N; // 维度对索引
868
869     // 频率计算:  $\theta_i = 1 / (10000^{(2i/d)})$ 
870     float exp_v = 1.0f / powf(10000.0f, 2 * token_idx / (N * 2.0f));
871     float sin_v = sinf(token_pos * exp_v);
872     float cos_v = cosf(token_pos * exp_v);
873
874     // 2D 旋转
875     float out1 = x1 * cos_v - x2 * sin_v;
876     float out2 = x1 * sin_v + x2 * cos_v;
877     out[idx * 2] = out1;
878     out[idx * 2 + 1] = out2;
879 }
880
881 // =====

```

```

882 // Phase 5: Mat Transpose — 矩阵转置 (Bank Conflict 专题)
883 // =====
884 // 面试要点 (Bank Conflict 专题) :
885 //   - Shared memory 有 32 个 bank, 每个 bank 4 bytes (32-bit)
886 //   - 同一 warp 的多个线程访问同一 bank 的不同地址 → bank conflict
887 //   - n-way bank conflict: n 个线程冲突 → 访问串行化为 n 次
888 //   - 解决方案: PAD (在每行末尾加 1 个元素, 打破地址对齐)
889 //
890 // 转置的四步演进:
891 //   naive: 非合并写入 (列优先写) → 每个 warp 产生 32 次内存事务
892 //   shared: 写入 smem (行优先) → 从 smem 读取 (列优先) → 合并写入 gmem
893 //   BCF: smem 布局 [kWarpSize_S*4][kWarpSize_S+PAD] = [64][17], PAD=1 加在第二维消除 bank conflict
894 //   merge_write: 进一步将 4 次 separate store 合并为 1 次 float4 store
895 // 注: 本文件仅实现 Level 1(naive) 与 Level 4(BCF+merge_write), Level 2/3 省略
896
897 // 每个线程处理 1 个元素, block(16,16)
898 // 读: x 按行优先访问, warp 的 32 线程访问 32 个连续地址 → 合并读取 ✓
899 // 写: y 按列优先写入, 32 线程跨 row 行分散 → 非合并写入 x (32 次内存事务)
900 // Grid: ((col + 15) / 16, (row + 15) / 16, 1), 每线程 1 元素
901 // Block: (16, 16, 1)
902 // source: LeetCUDA/kernels/mat-transpose/mat_transpose.cu
903 __global__ void mat_transpose(float *x, float *y, const int row, const int col) {
904     const int c = blockIdx.x * blockDim.x + threadIdx.x; // col in x
905     const int r = blockIdx.y * blockDim.y + threadIdx.y; // row in x
906     if (r < row && c < col) {
907         // x[r][c] → y[c][r]; x stride=col, y (transposed) stride=row
908         y[c * row + r] = x[r * col + c];
909     }
910 }
911
912 // Grid: ((col + 15) / 16, (row + 63) / 64, 1), 每线程 4 元素(float4)
913 // Block: (16, 16, 1)
914 // 注意: 该版本默认按 float4 打包写回; 最适合 row 能按 4 对齐的场景
915 // source: LeetCUDA/kernels/mat-transpose/mat_transpose.cu
916 __global__ void mat_transpose_padded(
917     float *x, float *y, const int row, const int col) {
918     const int tx = threadIdx.x;
919     const int ty = threadIdx.y;
920
921     constexpr int TILE = 16;
922     constexpr int PAD = 1;
923     // Bank conflict fix: 每行多 1 个元素, 打破 32-bank 对齐
924     __shared__ float tile[TILE * 4][TILE + PAD]; // 64x16
925
926     // x 空间坐标; 每线程覆盖 4 行 (actual row = x_r * 4)
927     const int x_c = blockIdx.x * TILE + tx;
928     const int x_r = blockIdx.y * TILE + ty;
929
930     if (x_r * 4 < row && x_c < col) {
931         // Step 1: 4 rows × 1 col → smem (row-major);
932 #pragma unroll
933         for (int i = 0; i < 4; ++i) {
934             tile[ty * 4 + i][tx] = x[(x_r * 4 + i) * col + x_c];
935         }
936         __syncthreads();
937
938         // Step 2: Transposed read from smem
939         float4 reg_trans;
940         reg_trans.x = tile[tx * 4 + 0][ty];
941         reg_trans.y = tile[tx * 4 + 1][ty];
942         reg_trans.z = tile[tx * 4 + 2][ty];
943         reg_trans.w = tile[tx * 4 + 3][ty];
944

```

```

945 // y 空间坐标: y_r = x 的列, y_c = x 的行
946 const int y_r = blockIdx.x * TILE + ty; // = x col
947 const int y_c = (blockIdx.y * TILE + tx) * 4; // = x row
948
949 // Coalesced write: y[y_r][y_c .. y_c+3]
950 FLOAT4(y[y_r * row + y_c]) = reg_trans;
951 }
952 }
953
954 // =====
955 // Phase 6: GEMV — 矩阵向量乘 (M or N = 1, 纯 memory-bound 算子, warp-per-row 策略)
956 // =====
957 // 面试要点:
958 // - GEMV 是典型的 memory-bound 算子:  $AI \approx O(1)$ , 瓶颈在内存带宽
959 // - 核心策略: warp-per-row (每个 warp 负责矩阵的一行)
960 // - 不同 K 值对应不同分块策略:
961 //   K=32 倍数: 一个 warp 的 32 线程恰好覆盖 K 维
962 //   K=128 倍数: 每个线程用 float4 处理 4 元素, warp 覆盖 128
963 //   K=16 < 32: 一个 warp 用不满, 用 kRowPerWarp=2 让每个 warp 处理 2 行
964 //
965 // a: MxK, x: Kx1, y: Mx1, 计算:  $y = a * x$ ; N = 1
966
967 // ---- SGEMV K32: 基础 warp-per-row ----
968 // 设计: block(32, 4), blockDim.x=kWarpSize=32 (K 需为 32 倍数时一轮覆盖, 否则内层循环 kNumWarps 次)
969 // grid(M/4), 每个 warp 负责一行
970 // K 为 32 的倍数时, warp 的 32 个线程恰好覆盖 K 维
971 // Grid: ((M + 3) / 4, 1, 1), 每 block 处理 4 行
972 // Block: (32, 4, 1), 每行 1 warp
973 // 注意: 该版本最适合 K 按 32 对齐; 当 K 更小时通常切到 K16 这类专用分支
974 // source: LeetCode/kernels/sgemv/sgemv.cu
975 __global__ void sgemv_k32(float *a, float *x, float *y, int M, int K) {
976     int tx = threadIdx.x; // 0~31
977     int ty = threadIdx.y; // 0~3
978     int lane = tx % kWarpSize; // 0~31
979     int m = blockIdx.x * blockDim.y + ty; // 全局行号
980     if (m < M) {
981         float sum = 0.0f;
982         // 沿 K 维的迭代数 = ceil(K/32), 每个 warp 要累加完整的K, 那么
983         // 每个thread就要负责累加NUM_ITERS个元素, NUM_ITERS = ceil(K/32)
984         const int NUM_ITERS = (K + kWarpSize - 1) / kWarpSize;
985 #pragma unroll
986         for (int w = 0; w < NUM_ITERS; ++w) {
987             // 假设K是32的整倍数, m * K 本行的起始地址, x: Kx1
988             int k = w * kWarpSize + lane;
989             sum += a[m * K + k] * x[k];
990         }
991         sum = warp_reduce_sum<kWarpSize>(sum);
992         // 每个 warp 处理一行, lane 0 写回结果
993         if (lane == 0)
994             y[m] = sum;
995     }
996 }
997
998 // ---- SGEMV K128: float4 向量化 ----
999 // 每个线程处理 4 个元素(float4), 一个 warp 覆盖 128 个元素
1000 // Grid: ((M + 3) / 4, 1, 1)
1001 // Block: (32, 4, 1)
1002 // 注意: 该版本最适合 K 按 128 对齐, 且 x/a 的地址满足 float4 对齐
1003 // source: LeetCode/kernels/sgemv/sgemv.cu
1004 __global__ void sgemv_k128(float *a, float *x, float *y, int M, int K) {
1005     int tx = threadIdx.x; // 0~31
1006     int ty = threadIdx.y; // 0~3
1007     int lane = tx % kWarpSize; // 0~31

```

```

1008 int m = blockDim.y * blockIdx.x + ty;
1009
1010 if (m < M) {
1011     float sum = 0.0f;
1012     // 沿 K 维的迭代数 = ceil(K/128), 每个 warp 每轮用 float4 覆盖 128 个 K 元素
1013     const int NUM_ITERS = ((K + kWarpSize - 1) / kWarpSize) + 4 - 1) / 4;
1014 #pragma unroll
1015     for (int w = 0; w < NUM_ITERS; ++w) {
1016         int k = (w * kWarpSize + lane) * 4;
1017         float4 reg_x = FLOAT4(x[k]);
1018         float4 reg_a = FLOAT4(a[m * K + k]);
1019         sum += (reg_a.x * reg_x.x + reg_a.y * reg_x.y + reg_a.z * reg_x.z +
1020                reg_a.w * reg_x.w);
1021     }
1022     sum = warp_reduce_sum<kWarpSize>(sum);
1023     if (lane == 0)
1024         y[m] = sum;
1025 }
1026 }
1027
1028 // ---- SGEMV K16: K < WarpSize, kRowPerWarp=2 ----
1029 // 面试亮点: K=16 < 32, 一个 warp 可以处理多行
1030 // kRowPerWarp=2, kNumLanePerRow=16, 前 16 个 lane 处理 row0, 后 16 个 lane 处理 row1
1031 // Grid: ((M + 7) / 8, 1, 1), NUM_ROWS=8
1032 // Block: (32, 4, 1)
1033 // 注意: 这一版是面向 K=16 的专用写法; kRowPerWarp=2 时一个 warp 同时处理 2 行
1034 // source: LeetCUDA/kernels/sgemv/sgemv.cu
1035 template <const int kRowPerWarp = 2>
1036 __global__ void sgemv_k16(float *A, float *x, float *y, int M, int K) {
1037     constexpr int kNumLanePerRow = (kWarpSize + kRowPerWarp - 1) / kRowPerWarp; // 16
1038     int tx = threadIdx.x;
1039     int ty = threadIdx.y;
1040     int lane = tx % kWarpSize;
1041     int k = lane % kNumLanePerRow; // row0: 0~15, t0~t15; row1: 0~15, t16~t31
1042     int m = (blockDim.y * blockIdx.x + ty) * kRowPerWarp + lane / kNumLanePerRow;
1043     if (m < M) {
1044         float sum = A[m * K + k] * x[k];
1045         // 按照kNumLanePerRow=16, 分2组各自做 warp reduce sum, k==0的lane写回结果
1046         sum = warp_reduce_sum<kNumLanePerRow>(sum);
1047         // 注意: 判断条件是 k == 0, 不是 lane == 0!
1048         if (k == 0)
1049             y[m] = sum;
1050     }
1051 }
1052
1053 // =====
1054 // Phase 7: GEMM — 矩阵矩阵乘 (GPU 最重要的算子, 面试核心考点)
1055 // =====
1056 // 面试要点 (GEMM 优化五层金字塔):
1057 //   Level 1 — Tiling (分块 + shared memory): 将数据从 HBM 搬到 SMEM 复用
1058 //   Level 2 — Thread Tile (寄存器分块): 每个线程计算 TM×TN
1059 //   个元素, 提高计算密度 Level 3 — Vectorize (向量化访存): float4/half2, 减少
1060 //   load/store 指令数 Level 4 — Tensor Core (MMA
1061 //   m16n8k16): 硬件矩阵乘单元, warp 级指令 Level 5 — Warp Specialization +
1062 //   TMA (WGMMMA m64n128k16): Hopper 异步执行
1063 //
1064 // 计算密度递进:
1065 //   Level 1: AI ≈ B_K / (2×sizeof) ≈ 32/8 = 4 → 仍是 memory-bound
1066 //   Level 2: AI ≈ TM×TN×B_K / (2×sizeof) ≈ 8×8×8/8 = 64 → compute-bound
1067 //   Level 4: Tensor Core 提供硬件加速的 256 FMA/cycle/warp → 大幅提升吞吐
1068
1069 // =====
1070 // Phase 7a: SGEMM (非 Tensor Core 路径)

```

```

1071 // =====
1072
1073 // ---- Level 1: SGEMM — Block Tile 32x32 + K Tile 32 ----
1074 // 最基础的 tiling 实现, 演示 shared memory 的核心用法
1075 //  $C = A \times B$ ,  $C[M, N] = A[M, K] \times B[K, N]$ 
1076 // BM=BN=32, BK=32, block(32, 32), 一个线程计算 c 的一个元素
1077 // Grid:  $((N + 31) / 32, (M + 31) / 32, 1)$ 
1078 // Block: (32, 32, 1), 1024 线程
1079 // source: LeetCUDA/kernels/sgemm/sgemm.cu
1080 __global__ void sgemm(float *a, float *b, float *c, int M, int N, int K) {
1081     constexpr int BM = 32; // vec 版: 32x4 = 128
1082     constexpr int BN = 32; // vec 版: 32x4 = 128
1083     constexpr int BK = 32;
1084     __shared__ float s_a[BM][BK], s_b[BK][BN]; // 32x32x4=4KB smem, float = 4 bytes
1085
1086     int bx = blockIdx.x;
1087     int by = blockIdx.y;
1088     int tx = threadIdx.x;
1089     int tid = threadIdx.y * blockDim.x + tx;
1090
1091     // 线程到 smem 的映射: 32x32 线程, 每个线程加载 a 和 b 各 1 个元素
1092     // 技巧: 一般来说 “/” 表示线程不是连续排布的, “%” 表示线程是连续排布的
1093     // 因此, 在需要考虑连续访问的维度使用 “%”, 比如, 连续的线程访问列方向连续的元素
1094     //  $A[M, K]$ , M的stride=K, K的stride=1 → 线程连续访问 K 维度 → 用 %,
1095     // 线程不连续访问 M 维度 → 用 /;
1096     int load_smem_a_m = tid / 32; // row 0~31 由 32 线程加载;
1097     int load_smem_a_k = tid % 32; // col 0~31 由 32 线程加载;
1098     int load_smem_b_k = tid / 32; // row 0~31 由 32 线程加载;
1099     int load_smem_b_n = tid % 32; // col 0~31 由 32 线程加载;
1100     int load_gmem_a_m = by * BM + load_smem_a_m; // gmem row;
1101     int load_gmem_b_n = bx * BN + load_smem_b_n; // gmem col;
1102
1103     float sum = 0.f; // 遍历完整的K, slice K;
1104     // 这里不用pragma unroll, 因为K不是编译器常量, 编译器无法展开循环
1105     for (int bk = 0; bk < (K + BK - 1) / BK; ++bk) {
1106         int load_gmem_a_k = bk * BK + load_smem_a_k; //  $A[M, K]$ 
1107         int load_gmem_a_addr = load_gmem_a_m * K + load_gmem_a_k;
1108         s_a[load_smem_a_m][load_smem_a_k] = a[load_gmem_a_addr];
1109         int load_gmem_b_k = bk * BK + load_smem_b_k; //  $B[K, N]$ 
1110         int load_gmem_b_addr = load_gmem_b_k * N + load_gmem_b_n;
1111         s_b[load_smem_b_k][load_smem_b_n] = b[load_gmem_b_addr];
1112         __syncthreads(); // 确保整个 smem tile 加载完毕
1113
1114 #pragma unroll
1115         for (int k = 0; k < BK; ++k) {
1116             int comp_smem_a_m = load_smem_a_m; // vec 版: 0~127, 0, 1, 2, ... (连续)
1117             int comp_smem_b_n = load_smem_b_n; // vec 版: 0~127, 0, 4, 8, ... (间隔)
1118             sum += s_a[comp_smem_a_m][k] * s_b[k][comp_smem_b_n];
1119         }
1120         __syncthreads(); // 确保 smem 不会在下一轮加载时被覆盖
1121     }
1122     int store_gmem_c_m = load_gmem_a_m; // vec 版: 0~127, 0, 1, 2, ... (连续) [128x128]
1123     int store_gmem_c_n = load_gmem_b_n; // vec 版: 0~127, 0, 4, 8, ... (间隔)
1124     int store_gmem_c_addr = store_gmem_c_m * N + store_gmem_c_n;
1125     c[store_gmem_c_addr] = sum; //  $C[M, N] = A[M, K] \times B[K, N]$ 
1126 }
1127
1128 // ---- Level 1+: SGEMM Vec4 — Block Tile 128x128 + K Tile 32 + Thread Tile 4x4 ----
1129 // 在 Level 1 基础上引入两层优化:
1130 // 1) float4 向量化加载: A/B 各用 1 条 128-bit load 取代 4 条 32-bit load
1131 // 2) Thread Tile 4x4: 每线程计算 16 个 C 元素, 提升计算/访存比 (AI 从
1132 // BK/2≈16 提升到 TM*TN*BK/2≈256), 减少线程总数带来的同步开销
1133 //  $C = A \times B$ ,  $C[M, N] = A[M, K] \times B[K, N]$ , A/B 均 row-major

```

```

1134 // BM=BN=128, BK=32, block(32, 32)=1024 线程, 每线程负责 4×4=16 个 C 元素
1135 // 1024 × 16 = 16384 = 128 × 128 ✓
1136 //
1137 // 线程到 4×4 tile 的映射 (与加载映射解耦, 独立计算更清晰):
1138 // m_tile = tid / 32 (0~31), 每 tile 4 行 → 行 [m_tile*4, m_tile*4+3], 覆盖 0~127
1139 // n_tile = tid % 32 (0~31), 每 tile 4 列 → 列 [n_tile*4, n_tile*4+3], 覆盖 0~127
1140 //
1141 // 加载映射 (每线程 4 个元素, float4):
1142 // A[128][32]: a_m = tid/8 (8 线程/行), a_k = (tid%8)*4 (4 列/线程) → 8×4=32 列 ✓
1143 // B[32][128]: b_k = tid/32 (32 线程/行), b_n = (tid%32)*4 (4 列/线程) → 32×4=128 列 ✓
1144 // row-major 下 A[m][k..k+3] 与 B[k][n..n+3] 均连续 → float4 load 合法
1145 //
1146 // △ Bank Conflict 提示 (面试加分点):
1147 // s_b[32][128] 上 warp 内 32 线程按 stride=4 访问 (tid%32 决定列 0,4,8,...,124)
1148 // → 每 4 个线程落同一 bank 不同地址 → 4-way bank conflict。生产代码可用
1149 // s_b[BK][BN+1] PAD 打散, 这里保持最简布局便于讲解。
1150 //
1151 // Grid: ((N + 127) / 128, (M + 127) / 128, 1)
1152 // Block: (32, 32, 1), 1024 线程
1153 // 假设: M/N 为 128 的倍数, K 为 32 的倍数 (与 Level 1 naive 版一致的边界约定)
1154 // source: LeetCUDA/kernels/sgemm/sgemm.cu (vec4 variant)
1155 __global__ void sgemm_vec4(float *a, float *b, float *c, int M, int N, int K) {
1156     constexpr int BM = 128;
1157     constexpr int BN = 128;
1158     constexpr int BK = 32;
1159     __shared__ float s_a[BM][BK]; // 128*32*4 = 16KB, float = 4 bytes
1160     __shared__ float s_b[BK][BN]; // 32*128*4 = 16KB
1161
1162     int bx = blockIdx.x;
1163     int by = blockIdx.y;
1164     int tx = threadIdx.x;
1165     int tid = threadIdx.y * blockDim.x + tx; // 0~1023
1166
1167     // 加载 A: 每线程加载 s_a[a_m][a_k..a_k+3] 共 4 个元素
1168     int load_smem_a_m = tid / 8; // 0~127, 8 线程/行
1169     int load_smem_a_k = (tid % 8) * 4; // 0,4,...,28
1170     // 加载 B: 每线程加载 s_b[b_k][b_n..b_n+3] 共 4 个元素
1171     int load_smem_b_k = tid / 32; // 0~31, 32 线程/行
1172     int load_smem_b_n = (tid % 32) * 4; // 0,4,...,124
1173
1174     int load_gmem_a_m = by * BM + load_smem_a_m;
1175     int load_gmem_b_n = bx * BN + load_smem_b_n;
1176
1177     // 4×4 Thread Tile 基址 (独立于加载映射), 这里compute索引的计算逻辑要和
1178     // load索引的计算逻辑分开, load/compute是可以独立索引的, 理解这点很重要。
1179     // 目标C Tile为[BM,BN]=[128x128], 有32x32线程, 则每个线程处理4x4 tile
1180     // 那么, 就可以不重不漏地覆盖[32x4,32x4]=[128x128]的大小
1181     int comp_smem_a_m_base = (tid / 32) * 4; // 0,4,8,...,124
1182     int comp_smem_b_n_base = (tid % 32) * 4; // 0,4,8,...,124
1183
1184     float sum[4][4] = {0.f};
1185     for (int bk = 0; bk < (K + BK - 1) / BK; ++bk) {
1186         int load_gmem_a_k = bk * BK + load_smem_a_k; // A [M, K]
1187         int load_gmem_a_addr = load_gmem_a_m * K + load_gmem_a_k;
1188         FLOAT4(s_a[load_smem_a_m][load_smem_a_k]) = FLOAT4(a[load_gmem_a_addr]); // s_a [BM,BK]
1189         int load_gmem_b_k = bk * BK + load_smem_b_k; // B [K, N]
1190         int load_gmem_b_addr = load_gmem_b_k * N + load_gmem_b_n;
1191         FLOAT4(s_b[load_smem_b_k][load_smem_b_n]) = FLOAT4(b[load_gmem_b_addr]); // s_b [BK,BN]
1192         __syncthreads();
1193     }
1194     #pragma unroll
1195     for (int k = 0; k < BK; ++k) {
1196         // 每次迭代加载 4 个 A 元素 + 4 个 B 元素, 再做 4×4=16 次 FMA

```

```

1197     float a_vals[4] = {s_a[comp_smem_a_m_base + 0][k],
1198                       s_a[comp_smem_a_m_base + 1][k],
1199                       s_a[comp_smem_a_m_base + 2][k],
1200                       s_a[comp_smem_a_m_base + 3][k]};
1201     float b_vals[4] = {s_b[k][comp_smem_b_n_base + 0],
1202                       s_b[k][comp_smem_b_n_base + 1],
1203                       s_b[k][comp_smem_b_n_base + 2],
1204                       s_b[k][comp_smem_b_n_base + 3]};
1205 #pragma unroll
1206     for (int i = 0; i < 4; ++i) {
1207 #pragma unroll
1208         for (int j = 0; j < 4; ++j) {
1209             sum[i][j] += a_vals[i] * b_vals[j];
1210         }
1211     }
1212 }
1213 __syncthreads();
1214 }
1215
1216 // 存储 4x4: 每行 4 个元素连续 → 可用 float4 store (要求 N 为 4 的倍数以保证对齐)
1217 int store_gmem_c_m = by * BM + comp_smem_a_m_base;
1218 int store_gmem_c_n = bx * BN + comp_smem_b_n_base;
1219 #pragma unroll
1220 for (int i = 0; i < 4; ++i) {
1221     int store_gmem_c_addr = (store_gmem_c_m + i) * N + store_gmem_c_n;
1222     float4 reg_c;
1223     reg_c.x = sum[i][0];
1224     reg_c.y = sum[i][1];
1225     reg_c.z = sum[i][2];
1226     reg_c.w = sum[i][3];
1227     FLOAT4(c[store_gmem_c_addr]) = reg_c;
1228 }
1229 }
1230
1231 // =====
1232 // Phase 7b: HGEMM — Tensor Core 路径 (MMA m16n8k16 + WGMMA m64n128k16)
1233 // =====
1234 // 面试要点:
1235 //   - MMA (Matrix Multiply-Accumulate): Ampere+ 的 Tensor Core 指令
1236 //   - m16n8k16 含义: M=16, N=8, K=16 的矩阵乘, 结果 [16x8] = [16x16]x[16x8]
1237 //   - ldmatrix: 从 shared memory 加载 16x16 的矩阵片段到寄存器 (4 条 32-bit
1238 //     寄存器)
1239 //   - Multistage Pipeline: s2/s3/s4 个 stage, 用 cp.async 异步加载下一批数据
1240 //   - Block Swizzle: 在 grid 维度做 swizzle, 改善 L2 cache locality
1241 //
1242 // ★ TN 布局详解 (面试高频考点):
1243 //   TN 命名约定: 来自 BLAS 的 op(A) × op(B) 语义
1244 //   BLAS 源自 Fortran, 默认列优先 (column-major) 存储
1245 //   N = Normal (列优先, BLAS 原生格式)
1246 //   T = Transposed (行优先, 相对 BLAS 来说是"转置过的")
1247 //   第一个字母 → A 的 op, 第二个字母 → B 的 op
1248 //   所以 TN 表示: A 是行优先 (相对 BLAS=Transposed), B 是列优先 (相对
1249 //   BLAS=Normal) 即: C = op(A) × op(B) = A^T × B? 不对! 在 row-major
1250 //   视角下: TN = A row-major [MxK], B^T row-major [NxK] (等价于 B col-major [KxN]) 在 cuBLAS
1251 //   调用中: cublasGemmEx(..., CUBLAS_OP_T, CUBLAS_OP_N, ...)
1252 //   T on A: BLAS 把 row-major 的 A 视为 A^T, 传 T 表示"转置回去"
1253 //   N on B: B 已经是 BLAS 原生的 col-major, 无需转置
1254 //
1255 // 记忆口诀: TN = A行(T) B列(N), 第一字母 A 第二字母 B
1256 //   T = row-major (行优先, 对 BLAS 来说是 transposed)
1257 //   N = col-major (列优先, BLAS native = normal)
1258 //
1259 // LeetCUDA _nn 布局对比 (A/B 均 row-major 自然存储): C[MxN] = A[MxK] × B[KxN]

```

```

1260 // - A: row-major [M, K], B: row-major [K, N]
1261 // - 按 N=col-major/T=row-major 约定, 二者 BLAS 视角均为 T → cuBLAS 等效 (T,T)
1262 // - 问题: ldmatrix 默认加载 col-major, B 是 row-major 需要 .trans
1263 //
1264 // TN 布局: C[M×N] = A[M×K] × B[K×N], B 以 B^T=[N×K] row-major 存储 (A 行优先, B 列优先)
1265 // - A [M×K]: row-major → 全局索引 A[m*K + k], smem s_a[BM][BK]
1266 // - B^T [N×K]: row-major → 全局索引 B[n*K+k] 即访问原 B 元素 (k,n) (△ 内维连续的是 K)
1267 // - 优势: B^T 已是 row-major, ldmatrix 无需 .trans, 天然匹配 MMA row.col
1268 // MMA 指令: mma.sync.aligned.m16n8k16.row.col
1269 // - row.col: A 输入 row-major, B 输入 col-major → 与 TN 布局天然匹配
1270 //
1271 // WGMMA (Warp Group MMA, Hopper+):
1272 // - warpgroup 级指令 (128 threads = 4 warps), 异步执行
1273 // - m64n128k16: 一次处理 64×128×16 的矩阵乘 (总 tile 量 131072, 是 MMA m16n8k16 的 64 倍)
1274 // - Warp Specialization: Producer(128 threads) 做 TMA 搬运, Consumer(128
1275 // threads) 做计算
1276 // - TMA: 硬件 DMA, ~零寄存器开销, 支持 1D~5D 寻址
1277 //
1278 // =====
1279 // Phase 7b-1: MMA PTX 宏定义
1280 // =====
1281 //
1282 // ---- gmem → smem: cp.async ----
1283 // cp.async.commit_group / wait_group / wait_all 语义 (PTX ISA §9.7.9.25.3) :
1284 // - commit_group: 将此前所有未提交的 cp.async 归入一个新的 async-group (per-thread) 。
1285 // - wait_group N: 阻塞直到最多 N 个 async-group 尚未完成 (即 pending ≤ N) 。
1286 // N=0 → 等所有 group 完成。与 wgmma.wait_group 语义一致。
1287 // - wait_all: 等价于 commit_group + wait_group 0。
1288 #define CP_ASYNC_COMMIT_GROUP() asm volatile("cp.async.commit_group;\n" ::)
1289 #define CP_ASYNC_WAIT_ALL() asm volatile("cp.async.wait_all;\n" ::)
1290 #define CP_ASYNC_WAIT_GROUP(n) \
1291     asm volatile("cp.async.wait_group %0;\n" :: "n"(n))
1292 //
1293 // 注意: cg 只支持 16 bytes, ca 支持 4/8/16 bytes
1294 #define CP_ASYNC_CG(dst, src, bytes) \
1295     asm volatile( \
1296         "cp.async.cg.shared.global.L2::128B [%0], [%1], %2;\n" :: "r"(dst), \
1297         "l"(src), "n"(bytes))
1298 //
1299 // ---- ldmatrix: smem → register (Tensor Core 专用) ----
1300 // ldmatrix.sync.aligned.xN.m8n8.shared.b16
1301 // 每次加载 8×8 的 half 矩阵片段到 1/2/4 条 32-bit 寄存器
1302 // aligned: 要求 128-bit 对齐, trans: 转置加载
1303 #define LDMATRIX_X4(R0, R1, R2, R3, addr) \
1304     asm volatile( \
1305         "ldmatrix.sync.aligned.x4.m8n8.shared.b16 {%0, %1, %2, %3}, [%4];\n" \
1306         : "=r"(R0), "=r"(R1), "=r"(R2), "=r"(R3) \
1307         : "r"(addr))
1308 //
1309 #define LDMATRIX_X2(R0, R1, addr) \
1310     asm volatile("ldmatrix.sync.aligned.x2.m8n8.shared.b16 {%0, %1}, [%2];\n" \
1311         : "=r"(R0), "=r"(R1) \
1312         : "r"(addr))
1313 //
1314 // ldmatrix.x2.trans: 转置加载 (用于 NN 布局中需要 col-major B 矩阵的场景)
1315 // FA 中 V[Bc,d] 为 row-major, 但 P@V 的 MMA 需要 col-major 的 B → 使用 trans
1316 #define LDMATRIX_X2_T(R0, R1, addr) \
1317     asm volatile( \
1318         "ldmatrix.sync.aligned.x2.trans.m8n8.shared.b16 {%0, %1}, [%2];\n" \
1319         : "=r"(R0), "=r"(R1) \
1320         : "r"(addr))
1321 //
1322 // ---- mma.sync.aligned.m16n8k16.row.col.f16.f16.f16.f16 ----

```

```

1323 // m16n8k16: M=16, N=8, K=16 (Ampere Tensor Core 的基本 tile)
1324 // row.col: A 是 row-major, B 是 col-major
1325 // f16.f16.f16.f16: A/B 是 f16, C/D 是 f16 (f32 累加版本用 f32.f16.f16.f32)
1326 // 2 个输出寄存器 (RD0, RD1), 4 个 A 寄存器 + 2 个 B 寄存器
1327 // C 矩阵大小 16x8=128 元素 = 128 个 half; 32 线程分担, 每线程 4 half = 2 个 uint32
1328 #define MMA16816(RD0, RD1, RA0, RA1, RA2, RA3, RB0, RB1, RC0, RC1) \
1329     asm volatile( \
1330         "mma.sync.aligned.m16n8k16.row.col.f16.f16.f16.f16 {%0, %1}, {%2, %3, " \
1331         "%4, %5}, {%6, %7}, {%8, %9};\n" \
1332         : "=r"(RD0), "=r"(RD1) \
1333         : "r"(RA0), "r"(RA1), "r"(RA2), "r"(RA3), "r"(RB0), "r"(RB1), "r"(RC0), \
1334         "r"(RC1)) \
1335
1336 // mma.sync.aligned.m16n8k16.row.col.f32.f16.f16.f32
1337 // f32 accumulator variant: 4 output registers (RD0-RD3), A/B still f16.
1338 // C matrix 16x8=128 elements, 32 threads x 4 f32 = 4 uint32 per thread.
1339 #define MMA16816F32(RD0, RD1, RD2, RD3, RA0, RA1, RA2, RA3, RB0, RB1, RC0, \
1340                     RC1, RC2, RC3) \
1341     asm volatile( \
1342         "mma.sync.aligned.m16n8k16.row.col.f32.f16.f16.f32 {%0, %1, %2, " \
1343         "%3}, {%4, %5, %6, %7}, {%8, %9}, {%10, %11, %12, %13};\n" \
1344         : "=r"(RD0), "=r"(RD1), "=r"(RD2), "=r"(RD3) \
1345         : "r"(RA0), "r"(RA1), "r"(RA2), "r"(RA3), "r"(RB0), "r"(RB1), "r"(RC0), \
1346         "r"(RC1), "r"(RC2), "r"(RC3)) \
1347
1348 // ---- MMA 辅助函数 ----
1349 // div_cceil: 整数除法向上取整
1350 #define HOST_DEVICE_INLINE __device__ __host__ inline
1351 HOST_DEVICE_INLINE int div_cceil(int a, int b) {
1352     return (a % b != 0) ? (a / b + 1) : (a / b);
1353 }
1354
1355 // =====
1356 // Phase 7b-2: HGEMM MMA — m16n8k16 + multistage pipeline + TN 布局 (统一循环版)
1357 // =====
1358 // 面试重点 — Tile Hierarchy:
1359 //   MMA Atom:          m16n8k16 (1 条 MMA 指令处理的最小 tile)
1360 //   MMA Tile (more warps): 2x4=8 个 MMA atom → [2x16, 8x4]=[32,32]
1361 //   VAL Tile (more values): 4x4=16 expand → [32x4, 32x4]=[128,128]
1362 //   实际: kMmaTileM=2, kMmaTileN=4, kValTileM=4, kValTileN=4
1363 //           → BM=16x2x4=128, BN=8x4x4=128, Warps=2x4=8, Threads=8x32=256
1364 //
1365 // ★ TN 布局 (T=A 行优先, N=B 列优先):
1366 //   - A[M][K]: row-major → 全局索引 A[m*K + k], smem s_a[BM][BK]
1367 //   - B[K][N]: col-major (等价于 B^T[N][K] row-major) → 全局索引 B[n*K+k]
1368 //   - ldmatrix A: 用 x4 (非转置), A row-major 原生匹配
1369 //   - ldmatrix B: 用 x2 (非转置), B^T row-major 逐行加载 = B 的列, 天然匹配 MMA row.col
1370 //   - MMA 指令: mma.sync.aligned.m16n8k16.row.col → 天然匹配 TN 布局
1371 //
1372 // ★ 统一循环设计 (k 从 0 开始, 消除尾端重复代码):
1373 //   - k 从 0 开始: 每次迭代 k 直接对应 tile k, sel = k % kStages (零偏移)
1374 //   - 加载语义为"预取未来": 迭代 k 加载 tile (k+kStages-1) 供后续使用
1375 //   - cp.async 条件化: 仅当 k+kStages-1 < NUM_K_TILES 时加载
1376 //   - WAIT_GROUP 自适应: 满载期用 kStages-2, 尾部排空用 0
1377 //
1378 // ★ Block Swizzle: 把 C 的逻辑 tile 布局改写为紧凑的二维发射窗口, 增加 A/B tile 的 L2 reuse 机会。
1379 //   C tile 坐标为 (by, bx), by 沿 M 方向, bx 沿 N 方向; 一个 bx 读取同一块 B^T[bx*BN:(bx+1)*BN, :]
1380 //   (TN 布局中 B^T[N][K] 为 row-major), 一个 by 读取同一块 A[by*BM:(by+1)*BM, :]。
1381 //   下图只画 4 个逻辑上相邻的 CTA; SM0~SM3 仅表示可能的发射序列, 不是硬件 SM 绑定或调度保证。
1382 //
1383 //   No Swizzle: grid = (tiles_n, tiles_m, 1), blockIdx.x = bx。
1384 //   C-Matrix (同一 by 行; CTA 先在很宽的 N 方向范围内展开):
1385 //

```

```

1386 //      N / bx → 0      1      2      3      ...
1387 //      M ↓
1388 //      by = 0      

|        |        |        |        |
|--------|--------|--------|--------|
| SM0    | SM1    | SM2    | SM3    |
| A 0,B0 | A 0,B1 | A 0,B2 | A 0,B3 |

      ...
1389 //
1390 //
1391 //
1392 // 同一 by 的 CTA 复用 A[by], 但分别读取 B^T 0、B^T 1 等不同 B tile。只有 scheduler
1393 // 推进到下一条 by 行、再次遇到相同 bx 时, 才有机会复用 B; 当 tiles_n 很大时, 这段时间内
1394 // L2 更容易被其他 B tile 挤占。
1395 //
1396 // Thread Block Swizzle: 先把 N 切成 S 个连续窗口; 一个 z slice 只含 grid_x 个 bx。
1397 // 下图用 grid_x=2 展示一个窄窗口, scheduler 更早进入下一条 by 行:
1398 //
1399 //      N / local x → 0      1
1400 //      M ↓
1401 //      by = 0      

|     |     |
|-----|-----|
| SM0 | SM1 |
| SM2 | SM3 |

      → A 0; B^T 0, B^T 1
1402 //
1403 //      by = 1      

|     |     |
|-----|-----|
| SM2 | SM3 |
|-----|-----|

      → A 1; B^T 0, B^T 1
1404 //
1405 //
1406 // 对于这个 2x2 窗口: 同一行横向复用 A (SM0/SM1、SM2/SM3), 同一列纵向复用 B
1407 // (SM0/SM2、SM1/SM3)。真实 grid_x 不必等于 2, 但缩小它会缩短再次访问相同 bx 的距离。
1408 // 此优化只提高 scheduler 在相近时间执行这些 CTA、命中 L2 的机会, 不保证 CTA 的 z 顺序、
1409 // 缓存驻留或 L2 hit。
1410 //
1411 // Launch (kBlockSwizzle=false, 当前默认路径):
1412 //   grid = (ceil(N / BN), ceil(M / BM), 1), blockIdx.x 直接就是 N-tile 编号 bx。
1413 // Launch (kBlockSwizzle=true, 需由 launch 端显式构造 3D grid):
1414 //   tiles_n = ceil(N / BN), S = ceil(N / 2048)
1415 //   grid = (ceil(tiles_n / S), ceil(M / BM), S)
1416 //   bx = blockIdx.z * grid.x + blockIdx.x, col_start = bx * BN。
1417 // 例如 N=8192、BN=128: tiles_n=64, S=4, grid.x=16; z=0/1/2/3 分别覆盖
1418 // bx=0..15/16..31/32..47/48..63, 即 columns [0,2048)/[2048,4096)/[4096,6144)/[6144,8192)。
1419 // grid.x*grid.z 可能产生 bx>=tiles_n 的冗余 CTA。当前 kernel 仍要求 M/N/K 分别按 BM/BN/BK
1420 // 对齐; ceil 只保证逻辑 tile 编号不遗漏, 并不使部分尾 tile 变为安全。
1421 // Block: (256, 1, 1), 8 warps
1422 // source: LeetCUDA/kernels/hgemm/mma/basic/hgemm_mma_stage_tn.cu
1423 // =====
1424 template <const int kMmaM = 16,                // MMA atom M dim (m16n8k16)
1425           const int kMmaN = 8,                // MMA atom N dim
1426           const int kMmaK = 16,               // MMA atom K dim, also BK tile = kMmaK
1427           const int kMmaTileM = 2,           // warps along M, 2 → warp tile M = 32
1428           const int kMmaTileN = 4,           // warps along N, 4 → warp tile N = 32
1429           const int kValTileM = 4,           // value-repeat along M, BM = 16*2*4 = 128
1430           const int kValTileN = 4,           // value-repeat along N, BN = 8*4*4 = 128
1431           const int kStages = 3,              // cp.async pipeline depth
1432           const int kBlockSwizzle = 0>       // 1 enables 3D grid swizzle for L2 locality
1433 __global__ void __launch_bounds__(256)
1434   hgemm_mma_stages_tn(half *A, half *B, half *C, int M, int N, int K) {
1435   static_assert(kBlockSwizzle == 0 || kBlockSwizzle == 1, "kBlockSwizzle must be 0 or 1");
1436   static_assert(kStages >= 2, "kStages must be >= 2 for cp.async pipeline");
1437   // Block Swizzle: 0 时 bx=blockIdx.x; 1 时将 (blockIdx.z, blockIdx.x)
1438   // 线性化为原始 N-tile 编号 bx=z*gridDim.x+x。by 始终是 M-tile 编号。
1439   const int bx = ((int)kBlockSwizzle) * blockIdx.z * gridDim.x + blockIdx.x;
1440   const int by = blockIdx.y;
1441   constexpr int BM = kMmaM * kMmaTileM * kValTileM; // 16*2*4=128
1442   constexpr int BN = kMmaN * kMmaTileN * kValTileN; // 8*4*4=128
1443   constexpr int BK = kMmaK;                          // 16
1444
1445   // Dynamic shared memory: kStages 个 stage 的 A 和 B
1446   // TN 布局: s_a[BM][BK]=[128][16](A row-major), s_b[BN][BK]=[128][16](B^T
1447   // row-major, 即 B col-major [K×N] 在 smem 中按 B^T[N×K] 存储)
1448   // 原始实现会按配置决定是否给 A/B 的 K 维加 PAD; 尤其 B 在 TN 布局下常见会额外

```

```

1449 // 加 B_PAD 来打散 bank 映射, 避免按列访问时出现明显 bank conflict。这里先保留最简 PAD=0 版本。
1450 extern __shared__ half smem[];
1451 half *s_a = smem;
1452 half *s_b = smem + kStages * BM * BK; // A 和 B 连续存放
1453 constexpr int s_a_stage_offset = BM * BK; // 128*16
1454 constexpr int s_b_stage_offset = BN * BK; // 128*16  $\Delta$  BN(128)×BK(16) = BT row-major
1455 const int tid = threadIdx.y * blockDim.x + threadIdx.x;
1456 const int warp_id = tid / kWarpSize; // 0~7
1457 const int lane_id = tid % kWarpSize; // 0~31
1458 // warp_m变化快(0->0,1->1), warp_n变化慢([0,1]->0,[2,3]->1,...), 因此,
1459 // 这种2x4的MMA(Warp) layout是按照col major的顺序来排列MMA0~MMA7的
1460 //
1461 //           N direction (warp_n)
1462 //           0      1      2      3
1463 // M direction (warp_m) 0  MMA0    MMA2    MMA4    MMA6
1464 //                       1  MMA1    MMA3    MMA5    MMA7
1465 // MMA的排布方式不是唯一的, 现在这样排是因为结合MMA/VAL Tile之后, 刚好能覆盖整个
1466 // C Tile[128,128]。只要调整MMA/VAL Tile, 这里的排布方式也可以跟着调整。要注意的
1467 // 是, MMA0-7逻辑上是可以认为是并行执行的, 各自的计算结果累计加到对应的C Tile位置上。
1468 const int warp_m = warp_id % kMmaTileM; // 0,1 (M 方向 2 个 warp) kMmaTileM = 2
1469 const int warp_n = warp_id / kMmaTileM; // 0,1,2,3 (N 方向 4 个 warp)
1470
1471 // 线程到 global memory 的映射 (用于加载 A 和 B) 共 256 个线程
1472 // TN 布局关键: A[m*K+k] 是 row-major, BT[n*K+k] 是 row-major (内维连续的是 K)
1473 int load_smem_a_m = tid / 2; // 0~127
1474 int load_smem_a_k = (tid % 2 == 0) ? 0 : 8; // 0, 8
1475 int load_smem_b_n = tid / 2; // 0~127  $\rightarrow$  BT 的 N 方向 (row-major 的行)
1476 int load_smem_b_k = (tid % 2 == 0) ? 0 : 8; // 0, 8  $\rightarrow$  BT 的 K 方向 (row-major 的列)
1477 int load_gmem_a_m = by * BM + load_smem_a_m; // C/A 全局行号 = M 方向的 tile 起始 + 线程偏移
1478 int load_gmem_b_n = bx * BN + load_smem_b_n; // C/B 全局列号 = N 方向的 tile 起始 + 线程偏移
1479 if (load_gmem_a_m >= M || load_gmem_b_n >= N)
1480     return;
1481
1482 // 8个Warps(MMAs)的排布是M方向2个, N方向4个, 则MMA Atom一次性能处理的tile大小是[2*16,4*8]=[32,32]。
1483 // CUDA中每个block能放的线程数是有上限的 (warp数量有限), 一般为4或者8个warps。为了能处理更大的C tile,
1484 // 需要把MMA Atom的tile再M方向和N方向各自重复4次, 得到[128,128]的C block tile, 这就是Value Tile的作用。
1485 // MMA Tile对应到cutlass cute中的TiledMMA的概念, Value Tile对应到cutlass cute中的PermuteMNK的概念。
1486 uint32_t RC[kValTileM][kValTileN][2] = {0}; // 初始化为 0
1487
1488 // CVTA: 一次转换 smem 基地址, 避免每次 cp.async 都做转换
1489 uint32_t smem_a_base_ptr = __cvta_generic_to_shared(s_a);
1490 uint32_t smem_b_base_ptr = __cvta_generic_to_shared(s_b);
1491
1492 // 预加载前 (kStages-1) 个 stage
1493 // TN 布局: A 的 gmem 索引用 m*K+k(row-major), BT 用 n*K+k(row-major, 即 B col-major [K×N])
1494 #pragma unroll
1495 for (int k = 0; k < (kStages - 1); ++k) {
1496     int load_gmem_a_k = k * BK + load_smem_a_k;
1497     int load_gmem_a_addr = load_gmem_a_m * K + load_gmem_a_k; // A: [m][k]
1498     uint32_t load_smem_a_ptr = (smem_a_base_ptr +
1499         (k * s_a_stage_offset + load_smem_a_m * BK + load_smem_a_k) * sizeof(half)
1500     );
1501     CP_ASYNC_CG(load_smem_a_ptr, &A[load_gmem_a_addr], 16);
1502
1503     int load_gmem_b_k = k * BK + load_smem_b_k;
1504     // BT: [n][k] row-major (即 B[k][n] col-major)  $\Delta$ 
1505     int load_gmem_b_addr = load_gmem_b_n * K + load_gmem_b_k;
1506     uint32_t load_smem_b_ptr = (smem_b_base_ptr +
1507         (k * s_b_stage_offset + load_smem_b_n * BK + load_smem_b_k) * sizeof(half)
1508     );
1509     CP_ASYNC_CG(load_smem_b_ptr, &B[load_gmem_b_addr], 16);
1510
1511     CP_ASYNC_COMMIT_GROUP();
1512 }

```

```

1512 CP_ASYNC_WAIT_GROUP(kStages - 2); // 允许有 kStages-2 个group未完成
1513 __syncthreads();
1514
1515 // 统一循环: k 从 0 开始, 每次迭代负责 tile k (加载 + 计算合并为单循环)
1516 const int NUM_K_TILES = div_ceil(K, BK);
1517 // 此处不用pragma unroll, 因为 K 不是编译期常量, 因此NUM_K_TILES 也不是编译期常量
1518 for (int k = 0; k < NUM_K_TILES; ++k) {
1519     int smem_sel = k % kStages; // 计算 tile k 的 stage
1520     int smem_sel_next = (k + kStages - 1) % kStages; // 预加载目标 stage
1521
1522     // 条件加载: 预加载 tile (k+kStages-1) 供将来使用。因为在prefetch loop中已经
1523     // 加载了(kStages-1) 个 stage, 因此这里是从 tile (k+kStages-1) 开始预加载
1524     // TN 布局: A 的 gmem 地址用 m*K+k (row-major), B^T 的 gmem 地址用 n*K+k
1525     // (row-major, 内维连续的是 K)
1526     if (k + kStages - 1 < NUM_K_TILES) {
1527         int load_gmem_a_k = (k + kStages - 1) * BK + load_smem_a_k;
1528         int load_gmem_a_addr = load_gmem_a_m * K + load_gmem_a_k; // A: row-major [m][k]
1529         int load_gmem_b_k = (k + kStages - 1) * BK + load_smem_b_k;
1530         int load_gmem_b_addr = load_gmem_b_n * K + load_gmem_b_k; // B^T: row-major [n][k]
1531
1532         uint32_t load_smem_a_ptr =
1533             (smem_a_base_ptr + (smem_sel_next * s_a_stage_offset +
1534                 load_smem_a_m * BK + load_smem_a_k) *
1535                 sizeof(half));
1536         CP_ASYNC.CG(load_smem_a_ptr, &A[load_gmem_a_addr], 16);
1537
1538         uint32_t load_smem_b_ptr =
1539             (smem_b_base_ptr + (smem_sel_next * s_b_stage_offset +
1540                 load_smem_b_n * BK + load_smem_b_k) *
1541                 sizeof(half));
1542         CP_ASYNC.CG(load_smem_b_ptr, &B[load_gmem_b_addr], 16);
1543         CP_ASYNC.COMMIT_GROUP();
1544     }
1545
1546     // ldmatrix: 从 smem_sel 加载 A 和 B 到寄存器
1547     // TN 布局关键: A 用 x4 (非转置), 因为 A 是 row-major; B 用 x2 (非转置), smem 中
1548     // B^T 为 row-major, 逐行加载即得 B 的列, 天然匹配 col-major B
1549     uint32_t RA[kValTileM][4]; // [4][4], M方向4次repeat, A regs = 4 uint32_t per MMA
1550     uint32_t RB[kValTileN][2]; // [4][2], N方向4次repeat, B regs = 2 uint32_t per MMA
1551
1552     // ldmatrix.x4: 加载 A 的 m16k16 片段 (row-major A, 非转置)
1553 #pragma unroll
1554     for (int i = 0; i < kValTileM; ++i) {
1555         // {0,1} * (16 * 4) + i * 16 = {0,64} + {0,16,32,48} = {0,16,32,48,64,80,96,112}
1556         // 其中 warp_m {0,1}; warp_m_0 offsets {0,16,32,48}; warp_m_1 offsets {64,80,96,112}
1557         int warp_smem_a_m = warp_m * (kMmaM * kValTileM) + i * kMmaM;
1558         // {0,16,32,...,112} + {0~15} = {0~127}, 按照col-major的顺序访问A的4个8x8 matrix (16x16)
1559         // ldmatrix.{...}.x4.{...} 需要warp内32个线程都参与, 每个线程提供一个有效的, 不重叠的addr
1560         int lane_smem_a_m = warp_smem_a_m + lane_id % 16; // t{0...15}=0~15, t{16...31}=0~15
1561         int lane_smem_a_k = (lane_id / 16) * 8; // 0, 8, t{0...15}=0, t{16...31}=8
1562         uint32_t lane_smem_a_ptr =
1563             (smem_a_base_ptr +
1564                 (smem_sel * s_a_stage_offset + lane_smem_a_m * BK + lane_smem_a_k) *
1565                 sizeof(half));
1566         LDMATRIX_X4(RA[i][0], RA[i][1], RA[i][2], RA[i][3], lane_smem_a_ptr);
1567     }
1568
1569     // ldmatrix.x2: 加载 B 的 k16n8 片段 (非转置)
1570     // 为什么不用 .trans? 因为 smem 中存的是 B^T row-major [N][K],
1571     // ldmatrix 逐行加载 B^T 的行 = B 的列, 天然给出 col-major B fragment → 直接匹配 MMA row.col
1572 #pragma unroll
1573     for (int j = 0; j < kValTileN; ++j) {

```

```

1575 // {0,...,3} * (8 * 4) + j * 8 = {0,32,64,96} + {0,8,16,24} = {0,8,...,120}
1576 // warp_n_0 offsets {0,8,16,24}; warp_n_1 offsets {32,40,48,56}
1577 // warp_n_2 offsets {64,72,80,88}; warp_n_3 offsets {96,104,112,120}
1578 int warp_smem_b_n = warp_n * (kMmaN * kValTileN) + j * kMmaN;
1579 // {0,8,...,120} + {0~7} = {0~127}, 按照row-major的顺序访问B^T的2个8x8 matrix (8x16)
1580 // ldmatrix {...}.x2 {...} 需要warp内前16个线程参与, 后16个线程传的addr会被忽略
1581 int lane_smem_b_n = warp_smem_b_n + lane_id % 8; // t{0...7}=0~7, t{8...15}=0~7
1582 int lane_smem_b_k = ((lane_id / 8) % 2) * 8; // t{0...7}=0, t{8...15}=8
1583 uint32_t lane_smem_b_ptr =
1584     (smem_b_base_ptr +
1585      (smem_sel * s_b_stage_offset + lane_smem_b_n * BK + lane_smem_b_k) *
1586       sizeof(half));
1587 LDMATRIX_X2(RB[j][0], RB[j][1], lane_smem_b_ptr);
1588 }
1589
1590 // MMA compute: 每个Warp(MMA) 在M方向重复kValTileM(4)次, 在N方向重复kValTileN(4)次
1591 #pragma unroll
1592 for (int i = 0; i < kValTileM; ++i) {
1593 #pragma unroll
1594     for (int j = 0; j < kValTileN; ++j) {
1595         MMA16816(RC[i][j][0], RC[i][j][1], // C fragment
1596                 RA[i][0], RA[i][1], RA[i][2], RA[i][3], // A fragment
1597                 RB[j][0], RB[j][1], // B fragment
1598                 RC[i][j][0], RC[i][j][1]);
1599     }
1600 }
1601
1602 // 自适应等待: 流水线满载期用 kStages-2, 尾部排空用 0
1603 if (k + kStages - 1 < NUM_K_TILES) {
1604     CP_ASYNC_WAIT_GROUP(kStages - 2);
1605 } else { // 对于尾部的 k, 等待所有剩余的 cp.async 完成
1606     CP_ASYNC_WAIT_GROUP(0);
1607 }
1608 __syncthreads();
1609 }
1610
1611 // Epilogue: 寄存器 → global memory (通过 warp shuffle + 128-bit store)
1612 {
1613     for (int i = 0; i < kValTileM; ++i) {
1614         // RC[kValTileM][kValTileN][2] 中 RC[...] [...] [2] 中保存了2个 uint32
1615         // 寄存器, 每个uint32 寄存器表示两个临近的fp16值: {c0,c1}, 然后RC[...] [...] [0]
1616         // 和RC[...] [...] [1]代表的是按照col-major排布的2个8x8子矩阵上 (不同物理行, 跨8x8)
1617         // 同一个位置上的元素, 也就是, 实际代表了2个不同的行的元素, 因此要分开RC0, RC1;
1618         // RC0表示第一行, 8个half, 可以用4个uint32寄存器来装; 同理, RC1表示第二行。
1619         uint32_t RC0[kValTileN][4]; // 32 bits x 4 = 128 bits = 8 half
1620         uint32_t RC1[kValTileN][4]; // 32 bits x 4 = 128 bits = 8 half
1621         // =====
1622         // MMA m16n8k16 C fragment — a single 16x8 matrix (registers per warp):
1623         // Thread t holds 4 half values → RC[0] for rows 0-7, RC[1] for rows 8-15.
1624         // Mapping: row = t/4, col-pair = t%4 (each uint32 = 2 half values).
1625         //
1626         //      cols: c0-1   c2-3   c4-5   c6-7
1627         //  r0:  T0{c0,c1}  T1{c2,c3}  T2{c4,c5}  T3{c6,c7}  ↵
1628         //  r1:  T4{c0,c1}  T5{c2,c3}  T6{c4,c5}  T7{c6,c7}  |
1629         //  r2:  T8{c0,c1}  T9{c2,c3}  T10{c4,c5} T11{c6,c7} | RC[0]
1630         //  ...
1631         //  r7:  T28{c0,c1} T29{c2,c3} T30{c4,c5} T31{c6,c7} ↵
1632         //  r8:  T0{c0,c1}  T1{c2,c3}  T2{c4,c5}  T3{c6,c7}  ↵
1633         //  r9:  T4{c0,c1}  T5{c2,c3}  T6{c4,c5}  T7{c6,c7}  |
1634         //  r10: T8{c0,c1}  T9{c2,c3}  T10{c4,c5} T11{c6,c7} | RC[1]
1635         //  ...
1636         //  r15: T28{c0,c1} T29{c2,c3} T30{c4,c5} T31{c6,c7} ↵
1637         //

```

```

1638 // Within a 4-lane group (e.g. T0-T3 for row 0):
1639 //   lane+0 holds {c0,c1}, lane+1 holds {c2,c3},
1640 //   lane+2 holds {c4,c5}, lane+3 holds {c6,c7}.
1641 // shfl 从 lane+1/+2/+3 各收 2 个 half 到 lane+0, 4 次 shuffle 凑齐
1642 // 一行 8 个 half = {c0,c1,c2,c3,c4,c5,c6,c7}, 再由 lane%4==0 128-bit store.
1643 // =====
1644 #pragma unroll
1645 for (int j = 0; j < kValTileN; ++j) {
1646     RC0[j][0] = RC[i][j][0];
1647     RC1[j][0] = RC[i][j][1];
1648     RC0[j][1] = __shfl_sync(0xffffffff, RC[i][j][0], lane_id + 1);
1649     RC0[j][2] = __shfl_sync(0xffffffff, RC[i][j][0], lane_id + 2);
1650     RC0[j][3] = __shfl_sync(0xffffffff, RC[i][j][0], lane_id + 3);
1651     RC1[j][1] = __shfl_sync(0xffffffff, RC[i][j][1], lane_id + 1);
1652     RC1[j][2] = __shfl_sync(0xffffffff, RC[i][j][1], lane_id + 2);
1653     RC1[j][3] = __shfl_sync(0xffffffff, RC[i][j][1], lane_id + 3);
1654 }
1655 // 每 4 个 lane 中只有 lane 0 做 128-bit store
1656 // lane_id / 4 → 行号映射 (每 4 个连续 lane 负责上8x8矩阵和下8x8矩阵各一行) :
1657 //
1658 // | lane_id | lane_id/4 | RC[0] 的行 | RC[1] 的行 |
1659 // |-----|-----|-----|-----|
1660 // | 0~3    | 0        | row 0      | row 8      |
1661 // | 4~7    | 1        | row 1      | row 9      |
1662 // | 8~11   | 2        | row 2      | row 10     |
1663 // | 12~15  | 3        | row 3      | row 11     |
1664 // | 16~19  | 4        | row 4      | row 12     |
1665 // | 20~23  | 5        | row 5      | row 13     |
1666 // | 24~27  | 6        | row 6      | row 14     |
1667 // | 28~31  | 7        | row 7      | row 15     |
1668 //
1669 if (lane_id % 4 == 0) {
1670     // {0,1} * (16 * 4) + i * 16 = {0,64} + {0,16,32,48} = {0,16,32,48,64,80,96,112}
1671     int store_warp_smem_c_m = warp_m * (kMmaM * kValTileM) + i * kMmaM; // smem row
1672     // 这里用 lane_id / 4 → {0,1,2,3,4,5,6,7}, 对应 RC0/RC1 (+8) 的行号, 表示2个8x8矩阵的行号
1673     int store_lane_gmem_c_m = by * BM + store_warp_smem_c_m + lane_id / 4; // gmem row
1674 #pragma unroll
1675 for (int j = 0; j < kValTileN; ++j) {
1676     // {0,...,3} * (8 * 4) + j * 8 = {0,32,64,96} + {0,8,16,24} = {0,8,...,120}
1677     int store_warp_smem_c_n = warp_n * (kMmaN * kValTileN) + j * kMmaN; // smem col
1678     int store_lane_gmem_c_n = bx * BN + store_warp_smem_c_n; // gmem col
1679     int store_gmem_c_addr_0 = store_lane_gmem_c_m * N + store_lane_gmem_c_n; // 1-th 8x8 matrix
1680     int store_gmem_c_addr_1 = (store_lane_gmem_c_m + 8) * N + store_lane_gmem_c_n; // 2-th 8x8 matrix
1681     // 128-bit store: 一次写入 8 个 half
1682     *reinterpret_cast<float4*>(&C[store_gmem_c_addr_0]) =
1683         *reinterpret_cast<float4*>(&RC0[j][0]);
1684     *reinterpret_cast<float4*>(&C[store_gmem_c_addr_1]) =
1685         *reinterpret_cast<float4*>(&RC1[j][0]);
1686 }
1687 }
1688 }
1689 }
1690 }
1691
1692 // =====
1693 // Phase 7b-3: HGEMM MMA Swizzle — m16n8k16 + multistage pipeline + TN 布局
1694 // + XOR swizzle 消除 smem bank conflict (统一循环版)
1695 // =====
1696 // 面试要点 (Swizzle 原理 — smem Bank Conflict 消除) :
1697 // - 问题: ldmatrix 以 8x8 片段 (m8n8) 从 smem 加载 16x16 的矩阵。同一 warp 中
1698 //   不同 lane 访问的地址在 bank 上产生冲突 → 串行化 (most common: 2/4-way) 。
1699 // - XOR swizzle 方案: 对列地址做 `((j>>3) ^ (i>>2)) % 2 << 3` 的 XOR 置换,
1700 //   将连续 4 行的 bank 映射打散。每 4 行为一组的 pattern 交替:

```

```

1701 //      rows 0~3: 列 0→地址偏移 0, 列 8→地址偏移 8
1702 //      rows 4~7: 列 0→地址偏移 8, 列 8→地址偏移 0 (XOR 翻转)
1703 //      rows 8~11: repeat rows 0~3 pattern
1704 //      rows 12~15: repeat rows 4~7 pattern
1705 // - 效果: 原来 n-way 的 bank conflict 降低到 1-way (fully conflict-free)。
1706 // - 优势: 无需 smem PAD (不浪费空间), 原理上完全消除特定 pattern 的 bank conflict。
1707 // - 局限性: 要求 kColStride ≤ 16 (即 BK ≤ 16), kStep ∈ {4,8}。
1708 //      对 MMA m16n8k16 的 BK=16 而言刚好满足。
1709 //
1710 // 参考: LeetCUDA/kernels/hgemm/mma/swizzle/hgemm_mma_stage_tn_swizzle.cu
1711 // 参考: https://zhuanlan.zhihu.com/p/4746910252
1712
1713 // ---- Swizzle 辅助函数 ----
1714
1715 // i: row index; j: col index.
1716 // Returns the chunk-level (8 fp16 = 16 B per chunk) column swizzle, i.e.
1717 // 0 / 8 / 16 / ... that the caller adds to the chunk-aligned base column.
1718 // The within-chunk offset (`j & 7`) is preserved by the caller.
1719 //
1720 // kColStride matches the equivalent CuTe ``Swizzle<B, M, S>`` swizzle
1721 // hardware mode used by TMA bulk-tensor descriptors:
1722 // * kColStride = 16 fp16 = 32 B/row -> Swizzle<1, 4, 3> = SWIZZLE_32B
1723 // * kColStride = 32 fp16 = 64 B/row -> Swizzle<2, 4, 3> = SWIZZLE_64B
1724 // * kColStride = 64 fp16 = 128 B/row -> Swizzle<3, 4, 3> = SWIZZLE_128B
1725 // The XOR mask is derived from the byte-address bit positions used by the
1726 // underlying CuTe swizzle pattern: bits {4..(4+B-1)} XOR bits
1727 // {7..(7+B-1)} of the absolute byte address. Translating to (i, j) with
1728 // row stride ``kColStride * 2`` bytes:
1729 // * 32B (B=1) : (j>>3) ^ (i>>2) [chunk in {0, 1}]
1730 // * 64B (B=2) : chunk(2b) ^ {(i>>1)&1, (i>>2)&1} [{0..3}]
1731 // * 128B (B=3) : chunk(3b) ^ {i&1, (i>>1)&1, (i>>2)&1} [{0..7}]
1732 // source: LeetCUDA/ffpa-attn/cuffpa/swizzle.cuh
1733 //
1734 // v1/v2 实现 (由编译宏 NOTES_V2_ENABLE_SWIZZLE_V2 门控):
1735 // swizzle_v1_impl : 原手写 XOR 分支 (permuted<kColStride,8>), 按列宽展开。
1736 // swizzle_v2_impl : 用本地 SwizzleBMS<B,M,S> 镜像 cute::Swizzle<B,M,S>::apply
1737 // 位级公式统一表达 (16/32/64); kColStride=8 回退 v1.
1738 // swizzle : 公开派发器, 宏门控, 接口与原 swizzle<kColStride> 一致。
1739 // v2 不是新算法, 而是把 v1 的手写展开重新表达为 cute 的统一位置换公式,
1740 // 便于脱离 cute 框架理解 swizzle 本质. v1/v2 bit-exact 等价。
1741 template <const int kColStride = 16, const int kStep = 8>
1742 static __host__ __device__ __forceinline__ int permuted(int i, int j) {
1743     static_assert(kColStride == 16 || kColStride == 32 || kColStride == 64 ||
1744         kColStride == 8,
1745         "kColStride must be one of {8, 16, 32, 64} (matches "
1746         "SWIZZLE_32B/64B/128B + the "
1747         "kStep=4 narrow legacy case).");
1748     static_assert(kStep == 4 || kStep == 8, "kStep must be 8 or 4.");
1749     static_assert(kColStride % kStep == 0,
1750         "kColStride must be multiple of kStep.");
1751     if constexpr (kStep == 4) {
1752         static_assert(kColStride <= 16, "kStep=4 only supports (kColStride <= 16).");
1753         return (((j >> 2) ^ (i >> 2)) % (kColStride >> 2)) << 2;
1754     } else if constexpr (kColStride == 16) {
1755         return (((j >> 3) ^ (i >> 2)) & 1) << 3; // SWIZZLE_32B
1756     } else if constexpr (kColStride == 32) {
1757         const int chunk = (j >> 3) & 3;
1758         const int xor_mask = ((i >> 1) & 1) | (((i >> 2) & 1) << 1);
1759         return (chunk ^ xor_mask) << 3; // SWIZZLE_64B
1760     } else { // kColStride == 64
1761         const int chunk = (j >> 3) & 7;
1762         const int xor_mask =
1763             (i & 1) | (((i >> 1) & 1) << 1) | (((i >> 2) & 1) << 2);

```

```

1764     return (chunk ^ xor_mask) << 3; // SWIZZLE_128B
1765 }
1766 }
1767
1768 // Manually SMEM swizzling for bank conflict free.
1769 // -----
1770 // [INFO] Assert smem store layout col_stride <= 16, prefer 16. |
1771 // [INFO] For logical_col_stride > 16, we have to permute the |
1772 // [INFO] smem store layout using col major ZigZag method: |
1773 // [INFO] e.g, --> Q smem logical layout [Br][64]. |
1774 // [INFO] --> col major ZigZag permuted --> |
1775 // [INFO] --> Q smem store layout [4][Br][16]. |
1776 // -----
1777 // -----
1778 // -----swizzle layout-----
1779 // -----logical col 0~64, step 8-----
1780 // -----smem col 0~16, step 8-----
1781 // -----
1782 // |bank |b 0~3 |b 4~7 |b 0~3 |b 4~7 |b 0~3 |b 4~7 |b 0~3 |b 4~7 |
1783 // |row 0 | 0 | 8 | 0 | 8 | 0 | 8 | 0 | 8 |
1784 // |bank |b 8~11|b12~15|b 8~11|b12~15|b 8~11|b12~15|b 8~11|b12~15|
1785 // |row 1 | 0 | 8 | 0 | 8 | 0 | 8 | 0 | 8 |
1786 // |bank |b16~19|b20~23|b16~19|b20~23|b16~19|b20~23|b16~19|b20~23|
1787 // |row 2 | 0 | 8 | 0 | 8 | 0 | 8 | 0 | 8 |
1788 // |bank |b24~27|b28~31|b24~27|b28~31|b24~27|b28~31|b24~27|b28~31|
1789 // |row 3 | 0 | 8 | 0 | 8 | 0 | 8 | 0 | 8 |
1790 // -----
1791 // |bank |b 0~3 |b 4~7 |b 0~3 |b 4~7 |b 0~3 |b 4~7 |b 0~3 |b 4~7 |
1792 // |row 4 | 8 | 0 | 8 | 0 | 8 | 0 | 8 | 0 |
1793 // |bank |b 8~11|b12~15|b 8~11|b12~15|b 8~11|b12~15|b 8~11|b12~15|
1794 // |row 5 | 8 | 0 | 8 | 0 | 8 | 0 | 8 | 0 |
1795 // |bank |b16~19|b20~23|b16~19|b20~23|b16~19|b20~23|b16~19|b20~23|
1796 // |row 6 | 8 | 0 | 8 | 0 | 8 | 0 | 8 | 0 |
1797 // |bank |b24~27|b28~31|b24~27|b28~31|b24~27|b28~31|b24~27|b28~31|
1798 // |row 7 | 8 | 0 | 8 | 0 | 8 | 0 | 8 | 0 |
1799 // -----
1800 // |bank |b 0~3 |b 4~7 |b 0~3 |b 4~7 |b 0~3 |b 4~7 |b 0~3 |b 4~7 |
1801 // |row 8 | 0 | 8 | 0 | 8 | 0 | 8 | 0 | 8 |
1802 // |bank |b 8~11|b12~15|b 8~11|b12~15|b 8~11|b12~15|b 8~11|b12~15|
1803 // |row 9 | 0 | 8 | 0 | 8 | 0 | 8 | 0 | 8 |
1804 // |bank |b16~19|b20~23|b16~19|b20~23|b16~19|b20~23|b16~19|b20~23|
1805 // |row 10| 0 | 8 | 0 | 8 | 0 | 8 | 0 | 8 |
1806 // |bank |b24~27|b28~31|b24~27|b28~31|b24~27|b28~31|b24~27|b28~31|
1807 // |row 11| 0 | 8 | 0 | 8 | 0 | 8 | 0 | 8 |
1808 // -----
1809 // |bank |b 0~3 |b 4~7 |b 0~3 |b 4~7 |b 0~3 |b 4~7 |b 0~3 |b 4~7 |
1810 // |row 12| 8 | 0 | 8 | 0 | 8 | 0 | 8 | 0 |
1811 // |bank |b 8~11|b12~15|b 8~11|b12~15|b 8~11|b12~15|b 8~11|b12~15|
1812 // |row 13| 8 | 0 | 8 | 0 | 8 | 0 | 8 | 0 |
1813 // |bank |b16~19|b20~23|b16~19|b20~23|b16~19|b20~23|b16~19|b20~23|
1814 // |row 14| 8 | 0 | 8 | 0 | 8 | 0 | 8 | 0 |
1815 // |bank |b24~27|b28~31|b24~27|b28~31|b24~27|b28~31|b24~27|b28~31|
1816 // |row 15| 8 | 0 | 8 | 0 | 8 | 0 | 8 | 0 |
1817 // -----
1818 template <const int kColStride = 16>
1819 static __host__ __device__ __forceinline__ int swizzle_v1_impl(int i, int j) {
1820     return permuted<kColStride, 8>(i, j);
1821 }
1822
1823 // Swizzle v2: cute::Swizzle<B,M,S> 位级镜像（脱离 cute 框架独立理解）
1824 //
1825 // v2 双重目的：
1826 // (1) 探索优化 v1 的可能 -- v1 按 kColStride 手写 XOR 分支展开，已被 nvcc

```

```

1827 //      strength-reduce 到极简位运算; v2 用 cute 的统一位置换公式重新表达,
1828 //      预期中性, 但统一表达式可能更利于合并到地址算术 (bonus).
1829 //      (2) 把 cute Swizzle 的核心位逻辑从 cute 框架提取出来独立理解 -- 让 notes-v2
1830 //      不依赖 cute 头即可展示 Swizzle<B,M,S> 的位级语义, 便于读者脱离 cute
1831 //      模板体系直接掌握 swizzle 原理.
1832 //
1833 // 注释叙述参考: 上方 hgemm_mma_stages_tn_cute 处的 Swizzle<B,M,S> 原理解释
1834 // (二维逻辑空间 + 列置换), 此处适配 v2 的 (B,M,S)=(1/2/3, 4, 3) 语境.
1835 //
1836 // Swizzle<B,M,S> 的核心不是改变逻辑 Tensor 的 shape, 而是把一维 byte offset
1837 // 重新解释为一个二维逻辑空间, 再把二维坐标映射回 bank-conflict-free 的物理
1838 // offset:
1839 // 1. 连续的 2^M 个元素组成二维空间中的一个基本元素 (元素宽度);
1840 // 2. 连续的 2^S 个基本元素组成一行 (列数);
1841 // 3. 二维空间包含 2^B 行 (行数, 这些行的 chunk 互相置换);
1842 // 4. 对二维坐标做列置换: icol' = irow ^ icol;
1843 // 5. 保留基本元素内部的低 M 位, 再将置换后的二维坐标编码回 offset.
1844 //
1845 // 位级实现 (与 cutlass/include/cute/swizzle.hpp 一致):
1846 // bit 布局: 0bxxxxxxxxxxxxxxxxYYxxxxxxxxZZxxxx
1847 //           ^--^ MBase: 保留不参与置换的低位数
1848 //           ^-^   ^-^ BBits: XOR 掩码位数
1849 //           ^-----^ SShift: YYY 相对 ZZZ 的位移
1850 // apply(off) = off ^ shiftr(off & yyy_msk, S)
1851 // yyy_msk    = ((1<<B)-1) << (M + max(0,S)) // YYY 掩码位置
1852 // zzz_msk    = ((1<<B)-1) << (M - min(0,S)) // ZZZ 掩码位置 (文档用)
1853 //
1854 // 模板参数物理/语义含义:
1855 // B (BBits) : 二维逻辑空间的行数位 = 2^B 行, 即 swizzle 周期覆盖 2^B 个 chunk
1856 //             (这些 chunk 互相置换); 同时是 TMA 硬件 swizzle 模式
1857 //             SWIZZLE_{32,64,128}B 的位数 (B=1/2/3).
1858 // M (MBase) : 基本元素宽度位 = 2^M 个连续元素组成一个基本元素. 对 fp16 +
1859 //             8-element chunk, 基本元素 = 16B = 2^4, 故 M=4; 地址 bit[0..M-1]
1860 //             在 apply 中保持不变 (保留).
1861 // S (SShift) : 二维空间每行列数位 = 2^S 个基本元素组成一行; 同时是 YYY 掩码
1862 //             相对 ZZZ 掩码的位移. S>0 表示 YYY 在更高位, 向右移 S 位到 ZZZ
1863 //             位置后 XOR. 本 notes-v2 三模式 S 恒为 3 (YYY 在 bit[7..7+B-1],
1864 //             即行索引 i 的低 B 位; ZZZ 在 bit[4..4+B-1], 即列 chunk 索引).
1865 //
1866 // 完整 swizzle 周期覆盖 2^(M+S+B) 个元素 (fp16 下 2^(M+S+B+1) 字节):
1867 // (B,M,S)=(1,4,3): 256 elem / 512B (SWIZZLE_32B)
1868 // (B,M,S)=(2,4,3): 512 elem / 1024B (SWIZZLE_64B)
1869 // (B,M,S)=(3,4,3): 1024 elem / 2048B (SWIZZLE_128B)
1870 template <int B, int M, int S>
1871 struct SwizzleBMS {
1872     static_assert(M >= 0, "MBase must be non-negative.");
1873     static_assert(B > 0, "BBits must be positive.");
1874     static_assert((S > 0 ? S : -S) >= B,
1875         "abs(SShift) must be >= BBits (cute::Swizzle constraint).");
1876
1877     static constexpr int bit_msk = (1 << B) - 1;
1878     static constexpr int yyy_msk = bit_msk << (M + (S > 0 ? S : 0));
1879     static constexpr int zzz_msk = bit_msk << (M - (S < 0 ? S : 0));
1880
1881     // apply: 对 byte offset 做 XOR 位置换, ZZZ ^= (YYY >> S).
1882     static __host__ __device__ __forceinline__ int apply(int offset) {
1883         if constexpr (S >= 0) {
1884             return offset ^ ((offset & yyy_msk) >> S);
1885         } else {
1886             return offset ^ ((offset & yyy_msk) << (-S));
1887         }
1888     }
1889 };

```

```

1890
1891 // swizzle_v2_impl: 用 SwizzleBMS<B,M,S>::apply 计算 chunk 级列 swizzle.
1892 //
1893 // 流程: (i,j) -> byte offset off=(i*kColStride+j)*sizeof(half) -> SwizzleBMS
1894 //      位级置换 -> 取 [M, M+B) 位作为物理 chunk 索引 -> 乘 kStep 还原为列偏移.
1895 //
1896 // kColStride -> (B,M,S) 映射表 (对应 TMA 硬件 swizzle 模式):
1897 //   kColStride=16 fp16=32B/row -> (B,M,S)=(1,4,3) = SWIZZLE_32B
1898 //   kColStride=32 fp16=64B/row -> (B,M,S)=(2,4,3) = SWIZZLE_64B
1899 //   kColStride=64 fp16=128B/row -> (B,M,S)=(3,4,3) = SWIZZLE_128B
1900 //
1901 // kColStride=8 (kStep=4 legacy): 无对应 cute 标准 TMA swizzle 模式, 回退到
1902 //   swizzle_v1_impl. 该路径当前无任何 kernel 使用, 回退保证接口契约零变更.
1903 //
1904 // 等价性: 对 (16,32,64), v2 输出与 v1 bit-exact 相同 (见 test_swizzle_equiv).
1905 //   证明: off 的 bit[4..4+B-1] 恰是 (j>>3) 低 B 位 (因 i*kColStride 在这些位
1906 //   为 0); bit[7..7+B-1] 恰是 i 低 B 位 (因 j<kColStride 不贡献高位). apply
1907 //   把 YYY(bit[7..]) XOR 到 ZZZ(bit[4..]), 取 [M,M+B)=bit[4..4+B-1] 即得
1908 //   (j>>3)^i 的低 B 位 = v1 的 chunk^xor_mask.
1909 template <const int kColStride = 16>
1910 static __host__ __device__ __forceinline__ int swizzle_v2_impl(int i, int j) {
1911     if constexpr (kColStride == 8) {
1912         return swizzle_v1_impl<kColStride>(i, j);
1913     } else {
1914         static_assert(kColStride == 16 || kColStride == 32 || kColStride == 64,
1915             "v2 supports kColStride in {16, 32, 64} (cute TMA swizzle); "
1916             "8 falls back to v1.");
1917         constexpr int B = (kColStride == 16) ? 1 : (kColStride == 32) ? 2 : 3;
1918         constexpr int M = 4;
1919         constexpr int S = 3;
1920         constexpr int kStep = 8;
1921         const int off = (i * kColStride + j) * (int)sizeof(half);
1922         const int sw = SwizzleBMS<B, M, S>::apply(off);
1923         return ((sw >> M) & ((1 << B) - 1)) * kStep;
1924     }
1925 }
1926
1927 // swizzle: 公开派发器, 由编译宏 NOTES_V2_ENABLE_SWIZZLE_V2 门控 v1/v2.
1928 //   未定义宏 (默认) -> swizzle_v1_impl (原手写 XOR 分支, 历史默认路径).
1929 //   定义宏 -> swizzle_v2_impl (cute Swizzle<B,M,S> 位级镜像).
1930 // 接口与原 swizzle<kColStride>(i,j) 完全一致, 所有 kernel 调用点无需改动.
1931 // v1/v2 等价性由 host 端 test_swizzle_equiv (--swizzle-eq-check) 验证.
1932 template <const int kColStride = 16>
1933 static __device__ __forceinline__ int swizzle(int i, int j) {
1934     #if defined(NOTES_V2_ENABLE_SWIZZLE_V2)
1935         return swizzle_v2_impl<kColStride>(i, j);
1936     #else
1937         return swizzle_v1_impl<kColStride>(i, j);
1938     #endif
1939 }
1940
1941 // =====
1942 // Phase 7b-3: HGEMM MMA — m16n8k16 + multistage pipeline + TN 布局 + XOR swizzle
1943 //      + Register Double Buffering (generic kValTileK, default BK=64)
1944 // =====
1945 // 在 Phase 7b-2 的 smem XOR swizzle 基础上增加寄存器双缓冲:
1946 //   - kValTileK: 每个 BK tile 包含 kValTileK 个 kMmaK slice (BK = kMmaK * kValTileK)
1947 //   - 统一 BK tile: 所有 k_step slice 连续存放在同一个 BK 宽的 smem tile 中
1948 //   - RA[2][kValTileM][4] / RB[2][kValTileN][2]: 双份寄存器乒乓切换 (与 kValTileK 无关)
1949 //   - ldmatrix 与 MMA 计算重叠: 加载 k_step+1 的同时用另一组寄存器做 k_step 计算
1950 //   - 支持任意 kValTileK >= 2; 默认 kValTileK=4 (BK=64), kStages=2 推荐 64KB smem
1951 //
1952 // ★ smem 布局:

```

```

1953 // s_a: [stage0][stage1]...[stage(kStages-1)], 每个 stage = BM × BK = 128 × BK
1954 // s_b: 紧接 s_a 之后
1955 // 每个 stage 内 k_step slice 列偏移 = k_step * kMmaK
1956 //
1957 // ★ 寄存器双缓冲时间线 (每 k 迭代内) :
1958 // Step 1: G→S cp.async 预取 stage(k+kStages-1) 全部 k_step (条件)
1959 // Step 2: MMA k_step=0 (用 reg[load], 已在上轮 post-wait 中 ldmatrix)
1960 // Step 3: for k_step=1..kValTileK-1: ldmatrix(列偏移 k_step*kMmaK)→reg[store], flip, MMA→reg[load]
1961 // Step 4: adaptive wait + __syncthreads
1962 // Step 5: S→R ldmatrix 预加载 stage(k+1) k_step=0 → reg[store] (条件)
1963 //
1964 // 参考: LeetCUDA/kernels/hgemm/mma/swizzle/hgemm_mma_stage_tn_swizzle_x2.cu
1965 // Grid: ((N+127)/128/S, (M+127)/128, S), S=(N+2047)/2048, 3D block swizzle
1966 // Block: (256, 1, 1), 8 warps
1967 template <const int kMmaM = 16,           // MMA atom M dim (m16n8k16)
1968           const int kMmaN = 8,           // MMA atom N dim
1969           const int kMmaK = 16,          // MMA atom K dim
1970           const int kMmaTileM = 2,        // warps along M, 2 → warp tile M = 32
1971           const int kMmaTileN = 4,        // warps along N, 4 → warp tile N = 32
1972           const int kValTileM = 4,        // value-repeat along M, BM = 16*2*4 = 128
1973           const int kValTileN = 4,        // value-repeat along N, BN = 8*4*4 = 128
1974           const int kValTileK = 4,        // MMA_K slices per BK tile, BK = kMmaK * kValTileK
1975           const int kStages = 2,          // cp.async pipeline depth (2 for kValTileK>=4 to fit smem)
1976           const int kBlockSwizzle = 0>    // 1 enables 3D grid swizzle for L2 locality
1977 __global__ void __launch_bounds__(256)
1978   hgemm_mma_stages_tn_swizzle(half *A, half *B, half *C, int M, int N, int K) {
1979   static_assert(kValTileK >= 2, "Register double buffering requires kValTileK >= 2");
1980   static_assert(kBlockSwizzle == 0 || kBlockSwizzle == 1, "kBlockSwizzle must be 0 or 1");
1981   static_assert(kStages >= 2, "kStages must be >= 2 for cp.async pipeline");
1982   // Block Swizzle: 在 grid x 维度做 swizzle, 改善 L2 cache 局部性
1983   const int bx = ((int)kBlockSwizzle) * blockDim.z * gridDim.x + blockIdx.x;
1984   const int by = blockIdx.y;
1985   constexpr int BM = kMmaM * kMmaTileM * kValTileM; // 16*2*4=128
1986   constexpr int BN = kMmaN * kMmaTileN * kValTileN; // 8*4*4=128
1987   constexpr int BK = kMmaK * kValTileK;             // 16 * kValTileK
1988
1989   // kStages stages, each with BM×BK for A, BN×BK for B
1990   // smem: kValTileK 个 kMmaK slice 连续存放在同一个 BK 宽 tile 中
1991   extern __shared__ half smem[];
1992   half *s_a = smem;
1993   half *s_b = smem + kStages * BM * BK;
1994   constexpr int s_a_stage_offset = BM * BK;
1995   constexpr int s_b_stage_offset = BN * BK;
1996
1997   const int tid = threadIdx.y * blockDim.x + threadIdx.x;
1998   const int warp_id = tid / kWarpSize;
1999   const int lane_id = tid % kWarpSize;
2000   const int warp_m = warp_id % kMmaTileM; // 0,1 (M 方向 2 个 warp) kMmaTileM = 2
2001   const int warp_n = warp_id / kMmaTileM; // 0,1,2,3 (N 方向 4 个 warp)
2002
2003   // 线程到 global memory 的映射 (用于加载 A 和 B) 共 256 个线程
2004   // TN 布局关键: A[m*K+k] 是 row-major, B^T[n*K+k] 是 row-major (内维连续的是 K)
2005   // 注意: smem_a_k 和 smem_b_k 依然使用 0/8, 在后续的 kValTileK 循环中
2006   // 会加上 k_step*kMmaK, 最终覆盖全部 BK 列
2007   int load_smem_a_m = tid / 2;           // 0~127
2008   int load_smem_a_k = (tid % 2 == 0) ? 0 : 8; // 0, 8
2009   int load_smem_b_n = tid / 2;           // 0~127 → B^T 的 N 方向 (row-major 的行)
2010   int load_smem_b_k = (tid % 2 == 0) ? 0 : 8; // 0, 8 → B^T 的 K 方向 (row-major 的列)
2011   int load_gmem_a_m = by * BM + load_smem_a_m; // C/A 全局行号 = M 方向的 tile 起始 + 线程偏移
2012   int load_gmem_b_n = bx * BN + load_smem_b_n; // C/B 全局列号 = N 方向的 tile 起始 + 线程偏移
2013   if (load_gmem_a_m >= M || load_gmem_b_n >= N)
2014     return;
2015

```

```

2016 // 8个Warps(MMAs)的排布是M方向2个, N方向4个, 则MMA Atom一次性能处理的tile大小是[2*16,4*8]=[32,32].
2017 // CUDA中每个block能放的线程数是有上限的 (warp数量有限), 一般为4或者8个warps。为了能处理更大的C tile,
2018 // 需要把MMA Atom的tile再M方向和N方向各自重复4次, 得到[128,128]的C block tile, 这就是Value Tile的作用。
2019 // MMA Tile对应到cutlass cute中的TiledMMA的概念, Value Tile对应到cutlass cute中的PermuteMNK的概念。
2020 uint32_t RC[kValTileM][kValTileN][2] = {0}; // 初始化为 0
2021
2022 uint32_t smem_a_base_ptr = __cvta_generic_to_shared(s_a);
2023 uint32_t smem_b_base_ptr = __cvta_generic_to_shared(s_b);
2024
2025 // Prefetch (kStages-1) stages: load all k_step slices for each early stage
2026 #pragma unroll
2027 for (int k = 0; k < (kStages - 1); ++k) {
2028 #pragma unroll
2029 for (int k_step = 0; k_step < kValTileK; ++k_step) {
2030 int load_gmem_a_k = k * BK + (k_step * kMmaK) + load_smem_a_k;
2031 int load_gmem_a_addr = load_gmem_a_m * K + load_gmem_a_k;
2032 int load_gmem_b_k = k * BK + (k_step * kMmaK) + load_smem_b_k;
2033 int load_gmem_b_addr = load_gmem_b_n * K + load_gmem_b_k;
2034
2035 uint32_t load_smem_a_ptr =
2036 (smem_a_base_ptr +
2037 (k * s_a_stage_offset + load_smem_a_m * BK +
2038 k_step * kMmaK +
2039 swizzle<kMmaK>(load_smem_a_m, load_smem_a_k)) *
2040 sizeof(half));
2041 CP_ASYNC.CG(load_smem_a_ptr, &A[load_gmem_a_addr], 16);
2042
2043 uint32_t load_smem_b_ptr =
2044 (smem_b_base_ptr +
2045 (k * s_b_stage_offset + load_smem_b_n * BK +
2046 k_step * kMmaK +
2047 swizzle<kMmaK>(load_smem_b_n, load_smem_b_k)) *
2048 sizeof(half));
2049 CP_ASYNC.CG(load_smem_b_ptr, &B[load_gmem_b_addr], 16);
2050 }
2051 CP_ASYNC.COMMIT_GROUP();
2052 }
2053
2054 CP_ASYNC.WAIT_GROUP(kStages - 2);
2055 __syncthreads();
2056
2057 // Register double buffers: RA[0/1] for k_step=0/1 A data, RB[0/1] for B data
2058 uint32_t RA[2][kValTileM][4];
2059 uint32_t RB[2][kValTileN][2];
2060 int reg_st_idx = 0; // write target for ldmatrix
2061 int reg_ld_idx = 1; // read source for MMA
2062
2063 // Initial ldmatrix: load stage 0, S -> R, k_step=0 (0~15列) -> reg[0]
2064 // 此时, reg_st_idx = 0, 对0位置的寄存器buffer做初始化。
2065 #pragma unroll
2066 for (int i = 0; i < kValTileM; ++i) {
2067 int warp_smem_a_m = warp_m * (kMmaM * kValTileM) + i * kMmaM;
2068 int lane_smem_a_m = warp_smem_a_m + lane_id % 16;
2069 int lane_smem_a_k = (lane_id / 16) * 8;
2070 uint32_t lane_smem_a_ptr =
2071 (smem_a_base_ptr +
2072 (0 * s_a_stage_offset + lane_smem_a_m * BK +
2073 swizzle<kMmaK>(lane_smem_a_m, lane_smem_a_k)) *
2074 sizeof(half));
2075 LDMATRIX_X4(RA[reg_st_idx][i][0], RA[reg_st_idx][i][1],
2076 RA[reg_st_idx][i][2], RA[reg_st_idx][i][3],
2077 lane_smem_a_ptr);
2078 }

```

```

2079 #pragma unroll
2080 for (int j = 0; j < kValTileN; ++j) {
2081     int warp_smem_b_n = warp_n * (kMmaN * kValTileN) + j * kMmaN;
2082     int lane_smem_b_n = warp_smem_b_n + lane_id % 8;
2083     int lane_smem_b_k = ((lane_id / 8) % 2) * 8;
2084     uint32_t lane_smem_b_ptr =
2085         (smem_b_base_ptr +
2086          (0 * s_b_stage_offset + lane_smem_b_n * BK +
2087           swizzle<kMmaK>(lane_smem_b_n, lane_smem_b_k)) *
2088           sizeof(half));
2089     LDMATRIX_X2(RB[reg_st_idx][j][0], RB[reg_st_idx][j][1],
2090                lane_smem_b_ptr);
2091 }
2092
2093 // 统一循环: k 从 0 开始, 条件 G→S / S→R / wait 处理所有边界情况
2094 // BK = kMmaK * kValTileK, 每个 k 迭代覆盖 BK 个 K 元素
2095 const int NUM_K_TILES = div_ceil(K, BK);
2096 for (int k = 0; k < NUM_K_TILES; ++k) {
2097     int smem_sel = k % kStages;
2098     int smem_sel_next = (k + kStages - 1) % kStages;
2099
2100     // G→S: 条件预取 stage(k+kStages-1) 全部 k_step slice
2101     if (k + kStages - 1 < NUM_K_TILES) {
2102 #pragma unroll
2103         for (int k_step = 0; k_step < kValTileK; ++k_step) {
2104             int load_gmem_a_k =
2105                 (k + kStages - 1) * BK + (k_step * kMmaK) + load_smem_a_k;
2106             int load_gmem_a_addr = load_gmem_a_m * K + load_gmem_a_k;
2107             int load_gmem_b_k =
2108                 (k + kStages - 1) * BK + (k_step * kMmaK) + load_smem_b_k;
2109             int load_gmem_b_addr = load_gmem_b_n * K + load_gmem_b_k;
2110
2111             uint32_t load_smem_a_ptr =
2112                 (smem_a_base_ptr +
2113                  (smem_sel_next * s_a_stage_offset + load_smem_a_m * BK +
2114                   k_step * kMmaK +
2115                  swizzle<kMmaK>(load_smem_a_m, load_smem_a_k)) *
2116                  sizeof(half));
2117             CP_ASYNC.CG(load_smem_a_ptr, &A[load_gmem_a_addr], 16);
2118
2119             uint32_t load_smem_b_ptr =
2120                 (smem_b_base_ptr +
2121                  (smem_sel_next * s_b_stage_offset + load_smem_b_n * BK +
2122                   k_step * kMmaK +
2123                  swizzle<kMmaK>(load_smem_b_n, load_smem_b_k)) *
2124                  sizeof(half));
2125             CP_ASYNC.CG(load_smem_b_ptr, &B[load_gmem_b_addr], 16);
2126         }
2127         CP_ASYNC.COMMIT_GROUP();
2128     }
2129
2130     // 内层 k_step 循环: flip → ldmatrix(k_step+1) → MMA, 乒乓交替
2131     // 初始: st=0, ld=1, reg[0]=preload k_step=0; reg[1]=未初始化
2132     // 每轮: flip→ldmatrix next k_step→reg[st], MMA reg[ld] (k_step+1 的 ldmatrix 条件化)
2133
2134 #pragma unroll
2135     for (int k_step = 0; k_step < kValTileK; ++k_step) {
2136         reg_st_idx ^= 1; // 0 -> 1; 1 -> 0
2137         reg_ld_idx ^= 1; // 1 -> 0; 0 -> 1
2138
2139         if (k_step + 1 < kValTileK) {
2140             int smem_k_offset = (k_step + 1) * kMmaK;
2141 #pragma unroll

```

```

2142     for (int i = 0; i < kValTileM; ++i) {
2143         int warp_smem_a_m = warp_m * (kMmaM * kValTileM) + i * kMmaM;
2144         int lane_smem_a_m = warp_smem_a_m + lane_id % 16;
2145         int lane_smem_a_k = (lane_id / 16) * 8;
2146         uint32_t lane_smem_a_ptr =
2147             (smem_a_base_ptr +
2148              (smem_sel * s_a_stage_offset + lane_smem_a_m * BK +
2149               smem_k_offset +
2150               swizzle<kMmaK>(lane_smem_a_m, lane_smem_a_k)) *
2151              sizeof(half));
2152         LDMATRIX_X4(RA[reg_st_idx][i][0], RA[reg_st_idx][i][1],
2153                     RA[reg_st_idx][i][2], RA[reg_st_idx][i][3],
2154                     lane_smem_a_ptr);
2155     }
2156     #pragma unroll
2157     for (int j = 0; j < kValTileN; ++j) {
2158         int warp_smem_b_n = warp_n * (kMmaN * kValTileN) + j * kMmaN;
2159         int lane_smem_b_n = warp_smem_b_n + lane_id % 8;
2160         int lane_smem_b_k = ((lane_id / 8) % 2) * 8;
2161         uint32_t lane_smem_b_ptr =
2162             (smem_b_base_ptr +
2163              (smem_sel * s_b_stage_offset + lane_smem_b_n * BK +
2164               smem_k_offset +
2165               swizzle<kMmaK>(lane_smem_b_n, lane_smem_b_k)) *
2166              sizeof(half));
2167         LDMATRIX_X2(RB[reg_st_idx][j][0], RB[reg_st_idx][j][1],
2168                     lane_smem_b_ptr);
2169     }
2170 }
2171
2172 #pragma unroll
2173 for (int i = 0; i < kValTileM; ++i) {
2174     #pragma unroll
2175     for (int j = 0; j < kValTileN; ++j) {
2176         HMMMA16816(RC[i][j][0], RC[i][j][1],
2177                    RA[reg_ld_idx][i][0], RA[reg_ld_idx][i][1],
2178                    RA[reg_ld_idx][i][2], RA[reg_ld_idx][i][3],
2179                    RB[reg_ld_idx][j][0], RB[reg_ld_idx][j][1],
2180                    RC[i][j][0], RC[i][j][1]);
2181     }
2182 }
2183 }
2184
2185 // 自适应等待: 满载期用 kStages-2, 尾部排空用 0
2186 if (k + kStages - 1 < NUM_K_TILES) {
2187     CP_ASYNC_WAIT_GROUP(kStages - 2);
2188 } else {
2189     CP_ASYNC_WAIT_GROUP(0);
2190 }
2191 __syncthreads();
2192
2193 // 从下一阶段加载 k_step=0 数据, 供下一轮 k 迭代使用, 此时 reg_st_idx=0
2194 if (k + 1 < NUM_K_TILES) {
2195     smem_sel = (k + 1) % kStages; // old smem_sel + 1
2196     #pragma unroll
2197     for (int i = 0; i < kValTileM; ++i) {
2198         int warp_smem_a_m = warp_m * (kMmaM * kValTileM) + i * kMmaM;
2199         int lane_smem_a_m = warp_smem_a_m + lane_id % 16;
2200         int lane_smem_a_k = (lane_id / 16) * 8;
2201         uint32_t lane_smem_a_ptr =
2202             (smem_a_base_ptr +
2203              (smem_sel * s_a_stage_offset + lane_smem_a_m * BK +
2204               swizzle<kMmaK>(lane_smem_a_m, lane_smem_a_k)) *

```

```

2205         sizeof(half));
2206     LDMATRIX_X4(RA[reg_st_idx][i][0], RA[reg_st_idx][i][1],
2207               RA[reg_st_idx][i][2], RA[reg_st_idx][i][3],
2208               lane_smem_a_ptr);
2209 }
2210 #pragma unroll
2211 for (int j = 0; j < kValTileN; ++j) {
2212     int warp_smem_b_n = warp_n * (kMmaN * kValTileN) + j * kMmaN;
2213     int lane_smem_b_n = warp_smem_b_n + lane_id % 8;
2214     int lane_smem_b_k = ((lane_id / 8) % 2) * 8;
2215     uint32_t lane_smem_b_ptr =
2216         (smem_b_base_ptr +
2217          (smem_sel * s_b_stage_offset + lane_smem_b_n * BK +
2218           swizzle<kMmaK>(lane_smem_b_n, lane_smem_b_k)) *
2219          sizeof(half));
2220     LDMATRIX_X2(RB[reg_st_idx][j][0], RB[reg_st_idx][j][1],
2221               lane_smem_b_ptr);
2222 }
2223 }
2224 }
2225
2226 // Epilogue: 复用 RA[2][4][4] 寄存器做 warp shuffle → 128-bit collective store
2227 // 做RC的warp shuffle正好需要[2][4][4]的寄存器空间, RA[2][4][4]正好可以复用
2228 for (int i = 0; i < kValTileM; ++i) {
2229 #pragma unroll
2230     for (int j = 0; j < kValTileN; ++j) {
2231         RA[0][j][0] = RC[i][j][0];
2232         RA[1][j][0] = RC[i][j][1];
2233         RA[0][j][1] = __shfl_sync(0xffffffff, RC[i][j][0], lane_id + 1);
2234         RA[0][j][2] = __shfl_sync(0xffffffff, RC[i][j][0], lane_id + 2);
2235         RA[0][j][3] = __shfl_sync(0xffffffff, RC[i][j][0], lane_id + 3);
2236         RA[1][j][1] = __shfl_sync(0xffffffff, RC[i][j][1], lane_id + 1);
2237         RA[1][j][2] = __shfl_sync(0xffffffff, RC[i][j][1], lane_id + 2);
2238         RA[1][j][3] = __shfl_sync(0xffffffff, RC[i][j][1], lane_id + 3);
2239     }
2240     if (lane_id % 4 == 0) {
2241         int store_warp_smem_c_m = warp_m * (kMmaM * kValTileM) + i * kMmaM;
2242         int store_lane_gmem_c_m = by * BM + store_warp_smem_c_m + lane_id / 4;
2243 #pragma unroll
2244         for (int j = 0; j < kValTileN; ++j) {
2245             int store_warp_smem_c_n = warp_n * (kMmaN * kValTileN) + j * kMmaN;
2246             int store_lane_gmem_c_n = bx * BN + store_warp_smem_c_n;
2247             int store_gmem_c_addr_0 =
2248                 store_lane_gmem_c_m * N + store_lane_gmem_c_n;
2249             int store_gmem_c_addr_1 =
2250                 (store_lane_gmem_c_m + 8) * N + store_lane_gmem_c_n;
2251             *reinterpret_cast<float4*>(&C[store_gmem_c_addr_0]) =
2252                 *reinterpret_cast<float4*>(&RA[0][j][0]);
2253             *reinterpret_cast<float4*>(&C[store_gmem_c_addr_1]) =
2254                 *reinterpret_cast<float4*>(&RA[1][j][0]);
2255         }
2256     }
2257 }
2258 }
2259
2260 // =====
2261 // Phase 7c: HGEMM CuTe — CUTLASS CuTe DSL 实现 (SM80+, Tensor Core 全自动调度)
2262 // =====
2263 // 面试要点 (CuTe vs 手写 MMA PTX) :
2264 //   - CuTe (CUTLASS Templates) 是 NVIDIA CUTLASS 3.x 的核心 DSL, 提供编译期
2265 //     Tensor 抽象, 自动推导 MMA、Copy、Swizzle 的线程-数据映射
2266 //   - 与手写 MMA PTX (Phase 7b) 的对比:
2267 //     - 手写 MMA: 手动计算 smem 地址、手动 ldmatrix/mma PTX、手动 swizzle、手动 epilogue shuffle

```

```

2268 // - CuTe: 声明 TiledMMA / TiledCopy / SmemLayout, 编译器自动推导线程-数据映射,
2269 //      cute::copy / cute::gemm 自动展开为高效指令序列
2270 // - 核心抽象 ("CuTe 五要素") :
2271 //      1. Tensor: 全局/共享/寄存器数据的逻辑视图 = (ptr, shape, stride)
2272 //      2. TiledMMA: 描述 MMA 指令 + warp/warpgroup 排布 + value tile 的 compile-time type
2273 //      3. TiledCopy: 描述数据搬运 (G→S, S→R, R→S, S→G) 的线程-元素映射
2274 //      4. Layout + Swizzle: 描述 smem 的数据排布和 bank conflict 消除
2275 //      5. Pipeline: cp.async + multistage, 由 cute::copy 自动管理
2276 //
2277 // CuTe kernel 的参数推导链 (launch wrapper 负责实例化, kernel 只接收实例化后的类型) :
2278 //      MMA Atom (SM80_16x8x16_F16F16F16F16_TN)
2279 //      → TiledMMA (EURepeat=MxNxK, ValTile=MxNxK)
2280 //      → TiledCopy (G2S/S2R/R2S/S2G, 每种有 ThrLayout + ValLayout)
2281 //      → SmemLayout (Atom + Swizzle + tile_to_shape + Stage)
2282 //
2283 // 调参口诀 (面试速记) :
2284 //      BM/BN/BK 越大 → smem 越大 → occupancy 越低, 但 K 循环更少 → 指令开销更低
2285 //      Swizzle<B,M,S> 的核心不是改变逻辑 Tensor 的 shape, 而是把一维 offset
2286 //      重新解释为一个二维逻辑空间, 再把二维坐标映射回 bank-conflict-free 的物理 offset:
2287 //      1. 连续的 2^M 个元素组成二维空间中的一个基本元素; (元素宽度)
2288 //      2. 连续的 2^S 个基本元素组成一行; (列数)
2289 //      3. 二维空间包含 2^B 行; (行数)
2290 //      4. 对二维坐标做列置换: icol' = irow ^ icol;
2291 //      5. 保留基本元素内部的低 M 位, 再将置换后的二维坐标编码回 offset。
2292 //      因此 M 描述基本元素宽度, S 描述二维空间的列数, B 描述二维空间的行数。
2293 //      CuTe 的位级实现等价于:
2294 //      offset' = offset ^ ((offset & YYY) >> S),
2295 //      YYY = ((1 << B) - 1) << (M + S)。
2296 //      对 Swizzle<3,3,3>: 每个基本元素有 2^3=8 个值, 每行有 2^3=8 个基本元素,
2297 //      二维空间有 2^3=8 行; 元素地址位 [6:8] XOR 到 [3:5], 低 3 位保持不变。
2298 //      一个完整 swizzle 周期覆盖 2^(M+S+B)=2^9=512 个元素 (FP16 下为 1024B)。
2299 //      kStage: 2=最低延迟, 3/4=更好隐藏延迟 → 权衡 smem 占用
2300 //      EURepeat: 在 M/N/K 方向重复 MMA atom 的执行单元布局, 决定 TiledMMA 的线程排布
2301 //      与逻辑 tile 组织; MMA atom越多, 需要的线程数就越多
2302 //
2303 // ★ 与手写 MMA Swizzle (Phase 7b-3) 的本质区别:
2304 //      - 手写 swizzle: 在 smem 地址计算时手动 XOR 列索引
2305 //      - CuTe Swizzle: SmemLayout 声明式指定地址位的 XOR pattern, cute::copy 自动应用
2306 //      - CuTe 优势: 类型安全, 编译器可在编译期推导映射; 布局变更通常只需修改类型定义
2307 //
2308 // ★ 本实现的 Tile 配置:
2309 //      - gmem/smem tile: BM=128, BN=256, BK=32, kStage=2; 这是 A/B tile 的目标 shape。
2310 //      - MMA atom: SM80_16x8x16_F16F16F16F16_TN; EURepeat=2x2x1, MMA_P_T=32x32x16。
2311 //      - TiledMMA 的逻辑 MMA tile 是 32x32x16, G2S/S2R copy 再把它映射到更大的 BMxBN tile;
2312 //      BN=256 不是由 "8x2x16" 直接推导出的 MMA tile, 而是本 wrapper 选择的 B tile 尺寸。
2313 //      - Smem Swizzle<3,3,3>: 512-element address period 的 XOR 重排, 针对 ldmatrix
2314 //      的访问模式降低 bank conflict; 512 是 swizzle 周期, 不等于 A atom 的逻辑元素数。
2315 //      - Threads: size(MMA{}) = 128 (4 warps × 2x2 EU repeat)。
2316 //      - Dynamic smem: Stage=2 时 A[128x32] + B[256x32] 共 49,152 bytes (约 48 KiB)。
2317 //
2318 // 参考: LeetCUDA/kernels/hgemm/cutlass/hgemm_mma_stage_tn_cute.cu
2319 // 参考: https://zhuanlan.zhihu.com/p/671419093 (CuTe Swizzle 详解)
2320 // =====
2321
2322 #if defined(NOTES_V2_ENABLE_CUTE)
2323
2324 #include <cute/tensor.hpp>
2325
2326 // =====
2327 // Phase 7c-1: CuTe HGEMM Kernel — TN 布局 + Smem Swizzle + Multistage Pipeline
2328 // =====
2329 // TN 布局: C[MxN] = A[MxK] × B^T[NxK]
2330 //      - A[M][K] row-major, B^T[N][K] row-major (等价 B col-major [KxN])

```

```

2331 // - CuTe 中通过 make_tensor(make_gmem_ptr(ptr), shape, stride) 描述
2332 //
2333 // Pipeline 流程 (stage 是 K tile 的循环缓冲槽, 不是一条完整计算链):
2334 // 1. PREFETCH: 预加载 kStage-1 个 K tile (G→S via cp.async), 填满 stage。
2335 // 2. 主循环 over K tiles:
2336 // a. 从当前 stage 做 S→R: cute::copy(...) → ldmatrix;
2337 // b. 对当前 K slice 做 MMA: cute::gemm(...) → mma.sync;
2338 // c. 在处理当前 tile 的首个 MMA slice 时, 异步提交下一个 tile 的 G→S;
2339 // d. 当前 tile 的最后一个 K slice 完成后等待并切换 smem stage。
2340 // 3. Epilogue: R→S (UniversalCopy 写入 C scratchpad) → S→G (128-bit store)。
2341 //
2342 // 面试对比 — 手写 MMA vs CuTe 代码量:
2343 // 手写: ldmatrix PTX 地址计算 + swizzle XOR + mma PTX + epilogue shuffle (~200行)
2344 // CuTe: partition + copy + gemm (~80行), 其余由 launch wrapper 的类型推导完成
2345 template <typename T,                                // element type (half)
2346         int BM, int BN, int BK,                      // block tile: BM=128, BN=256, BK=32
2347         int kStage,                                  // cp.async pipeline depth (2)
2348         typename TiledMMA,                          // MMA: SM80_16x8x16_F16F16F16F16_TN, EURepeat=2x2x1
2349         typename G2SCopyA,                          // G→S A: SM80_CP_ASYNC_CACHEGLOBAL<uint128_t>
2350         typename G2SCopyB,                          // G→S B: same as G2SCopyA
2351         typename SmemLayoutA,                      // smem A: Swizzle<3,3,3> + 8×BK atom → (BM,BK,kStage)
2352         typename SmemLayoutB,                      // smem B: same atom → (BN,BK,kStage)
2353         typename SmemLayoutC,                      // C scratchpad: Swizzle<3,3,3> + 32×32 atom, pipe=4
2354         typename S2RCopyAtomA,                    // S→R A: SM75_U32x4_LDSM_N (ldmatrix.x4)
2355         typename S2RCopyAtomB,                    // S→R B: same as S2RCopyAtomA
2356         typename R2SCopyAtomC,                    // R→S C: UniversalCopy<int> (32-bit store)
2357         typename S2GCopyAtomC,                    // S→G C: UniversalCopy<uint128_t> (128-bit wide store)
2358         typename S2GCopyC,                        // S→G TiledCopy: ThrLayout{32,4} × ValLayout{1,8}
2359         const int BlockSwizzle = 0>                // 1 enables 3D grid swizzle for L2 locality
2360 __global__ void hgemm_mma_stages_tn_cute(T *Aptr, T *Bptr, T *Dptr, int m,
2361         int n, int k) {
2362     using namespace cute;
2363     static_assert(BlockSwizzle == 0 || BlockSwizzle == 1, "BlockSwizzle must be 0 or 1");
2364     static_assert(kStage >= 2, "kStage must be >= 2 for cp.async pipeline");
2365
2366     // 动态 shared memory: KStage 个 A/B tile; C epilogue 复用当前 A stage 的空间。
2367     extern __shared__ T shm_data[];
2368     T *Ashm = shm_data;
2369     T *Bshm = shm_data + cute::cosize(SmemLayoutA{});
2370
2371     int idx = threadIdx.x;
2372     // BlockSwizzle: 0/1 控制是否启用 thread block swizzle, 改善 L2 cache 局部性
2373     int ix = ((int)BlockSwizzle) * blockIdx.z * gridDim.x + blockIdx.x;
2374     int iy = blockIdx.y; // M 方向 block 索引
2375     // 当前 kernel 只有 CTA 级边界判断, 没有 tile 内的逐元素 predication; 调用方需保证
2376     // M % BM == 0、N % BN == 0、K % BK == 0, 才能避免边界 tile 的越界访问或未写回。
2377     if (iy * BM >= m || ix * BN >= n) return;
2378
2379     // CuTe Tensor 抽象: (ptr, shape, stride) 三元组描述数据布局
2380     // make_stride(leading_dim, Int<1>{}) → row-major: 同行相邻元素步长=1, 跨行步长=leading_dim
2381     Tensor A = make_tensor(make_gmem_ptr(Aptr), make_shape(m, k),
2382         make_stride(k, Int<1>{}));
2383     Tensor B = make_tensor(make_gmem_ptr(Bptr), make_shape(n, k),
2384         make_stride(k, Int<1>{}));
2385     Tensor D = make_tensor(make_gmem_ptr(Dptr), make_shape(m, n),
2386         make_stride(n, Int<1>{}));
2387
2388     // local_tile: 按 tiler 切出当前 CTA 的逻辑 tile; 返回 Tensor 仍保留未切出的 remainder mode。
2389     // make_coord(iy, _): 固定 M/N 方向的 CTA 坐标, _ 保留 K 方向的 tile remainder:
2390     // gA: (BM, BK, num_tile_k), gB: (BN, BK, num_tile_k), gD: (BM, BN)。
2391     Tensor gA = local_tile(A, make_tile(Int<BM>{}, Int<BK>{}), make_coord(iy, _));
2392     Tensor gB = local_tile(B, make_tile(Int<BN>{}, Int<BK>{}), make_coord(ix, _));
2393     Tensor gD = local_tile(D, make_tile(Int<BM>{}, Int<BN>{}), make_coord(iy, ix));

```

```

2394
2395 // Shared memory Tensor: 按 SmemLayout 解读 smem 区域
2396 auto sA = make_tensor(make_smem_ptr(Ashm), SmemLayoutA{}); // (BM, BK, kStage)
2397 auto sB = make_tensor(make_smem_ptr(Bshm), SmemLayoutB{}); // (BN, BK, kStage)
2398
2399 // TiledMMA partition: 将 MMA 的 A/B/C tile 映射到各线程的寄存器 fragment
2400 TiledMMA tiled_mma;
2401 auto thr_mma = tiled_mma.get_slice(idx);
2402 auto tCrA = thr_mma.partition_fragment_A(gA(_, _, 0)); // (MMA, MMA_M, MMA_K)
2403 auto tCrB = thr_mma.partition_fragment_B(gB(_, _, 0)); // (MMA, MMA_N, MMA_K)
2404 auto tCrD = thr_mma.partition_fragment_C(gD); // (MMA, MMA_M, MMA_N)
2405 clear(tCrD); // 累加器清零
2406
2407 // G2S TiledCopy: 描述 global → shared memory 的数据搬运 (128-bit cp.async)。
2408 G2SCopyA g2s_tiled_copy_a;
2409 auto g2s_thr_copy_a = g2s_tiled_copy_a.get_slice(idx);
2410 auto tAgA_copy = g2s_thr_copy_a.partition_S(gA); // (CPY, CPY_M, CPY_K, num_tile_k)
2411 auto tAsA_copy = g2s_thr_copy_a.partition_D(sA); // (CPY, CPY_M, CPY_K, kStage)
2412
2413 G2SCopyB g2s_tiled_copy_b;
2414 auto g2s_thr_copy_b = g2s_tiled_copy_b.get_slice(idx);
2415 auto tBgB_copy = g2s_thr_copy_b.partition_S(gB); // (CPY, CPY_N, CPY_K, num_tile_k)
2416 auto tBsB_copy = g2s_thr_copy_b.partition_D(sB);
2417
2418 // S2R TiledCopy: 描述 shared → register 的数据搬运 (使用 ldmatrix)
2419 // make_tiled_copy_A/B: 根据 TiledMMA 自动推导 S2R copy 的线程-数据映射
2420 auto s2r_tiled_copy_a = make_tiled_copy_A(S2RCopyAtomA{}, tiled_mma);
2421 auto s2r_thr_copy_a = s2r_tiled_copy_a.get_slice(idx);
2422 auto tAsA = s2r_thr_copy_a.partition_S(sA); // (CPY, CPY_M, CPY_K, kStage)
2423 auto tCrA_view = s2r_thr_copy_a.retile_D(tCrA); // (CPY, CPY_M, CPY_K) — 与 rA 寄存器布局对齐
2424
2425 auto s2r_tiled_copy_b = make_tiled_copy_B(S2RCopyAtomB{}, tiled_mma);
2426 auto s2r_thr_copy_b = s2r_tiled_copy_b.get_slice(idx);
2427 auto tBsB = s2r_thr_copy_b.partition_S(sB);
2428 auto tCrB_view = s2r_thr_copy_b.retile_D(tCrB);
2429
2430 // PREFETCH: 预加载前 (kStage - 1) 个 K tile, 填满循环缓冲; 每次 copy 提交一个异步 G→S 操作。
2431 int itile_to_read = 0, ismem_read = 0, ismem_write = 0;
2432 #pragma unroll
2433 for (int istage = 0; istage < kStage - 1; ++istage) {
2434     cute::copy(g2s_tiled_copy_a, tAgA_copy(_, _, _, istage),
2435               tAsA_copy(_, _, _, istage));
2436     cute::copy(g2s_tiled_copy_b, tBgB_copy(_, _, _, istage),
2437               tBsB_copy(_, _, _, istage));
2438     cp_async_fence();
2439     ++itile_to_read;
2440     ++ismem_write;
2441 }
2442 cp_async_wait<kStage - 2>();
2443 __syncthreads();
2444
2445 // K 维第一层循环 — 按 BK 大小将 K 切分为 num_k_tiles 个 tile。
2446 // 外层 k_tile 遍历这些 BK-tile, 内层 k_step 遍历 tile 内的 MMA_K slice。
2447 // 当前实现使用整除, 要求 K % BK == 0; 没有为尾部 K tile 做 predication/padding。
2448 int num_k_tiles = k / BK;
2449 // 循环前预加载首轮数据: 将第一个 K tile 的第 0 个 K slice 从 smem 加载到寄存器。
2450 cute::copy(s2r_tiled_copy_a, tAsA(_, _, 0, ismem_read), tCrA_view(_, _, 0));
2451 cute::copy(s2r_tiled_copy_b, tBsB(_, _, 0, ismem_read), tCrB_view(_, _, 0));
2452
2453 #pragma unroll 1
2454 for (int k_tile = 0; k_tile < num_k_tiles; ++k_tile) {
2455     // num_k_steps: BK tile 内 K 方向的 MMA 迭代次数, 取自 fragment 的第三 mode (K-mode)。
2456     //

```

```

2457 // 为什么只显式展开 K? MMA_M 和 MMA_N 去哪了?
2458 //   tCrA=(MMA, MMA_M, MMA_K), tCrB=(MMA, MMA_N, MMA_K), tCrD=(MMA, MMA_M, MMA_N)
2459 //   - K 方向: 每个 k_step 需要从 smem 加载**不同的** A/B 数据 (ldmatrix),
2460 //     所以 K 循环必须显式写, 每次迭代做 S→R copy + cute::gemm。
2461 //   - M/N 方向: 同一个 k_step 内, 所有 M、N 位置的 MMA atom 共享
2462 //     同一批寄存器数据。cute::gemm 根据 tiled_mma 的类型信息 (EURepeat=2×2)
2463 //     在编译期自动展开为覆盖全部 (MMA_M, MMA_N) 对的 mma.sync 指令序列,
2464 //     无需手动写 M/N 循环。——这等价于手写 MMA 中的:
2465 //       for (i=0; i<kValTileM; ++i)
2466 //         for (j=0; j<kValTileN; ++j)
2467 //           HMMA16816(RC[i][j][0], RC[i][j][1], ...);
2468 //
2469 // CUTLASS 推导链 (mma_atom.hpp) :
2470 //   make_tiled_mma 将 MMA_P_T::<2> (即 kMmaPK) 存入 PermutationMnK
2471 //   → TiledMMA::permutation_mnk<2>() 返回 kMmaPK
2472 //   → thrfrg_A 对 A(M,K) 按 (permutation_mnk<0>(), permutation_mnk<2>())
2473 //     做 logical_divide, 即按 (kMmaPM, kMmaPK) tile 化 K 维:
2474 //     K 被分为 BK/kMmaPK 个 perm-K slice
2475 //     随后按 AtomShape_MnK<2> (MMA_K_atom) 做 zipped_divide
2476 //     每个 perm-K 含 kMmaEURepeatK 个 atom-K slice
2477 //   → partition_A 选当前线程后, K-mode size = BK / kMmaPK = 32 / 16 = 2
2478 // 本配置: kMmaPK=16 (MMA_K_atom=16 × kMmaEURepeatK=1), BK=32 → num_k_steps=2
2479 int num_k_steps = size<2>(tCrA);
2480
2481 #pragma unroll
2482 for (int k_step = 0; k_step < num_k_steps; ++k_step) {
2483     int k_step_next = (k_step + 1) % num_k_steps;
2484
2485     if (k_step == num_k_steps - 1) {
2486         // 等待下一个 K tile 的 smem 数据就绪, 然后切换到下一个 stage。
2487         cp_async_wait<kStage - 2>();
2488         __syncthreads();
2489         ismem_read = (ismem_read + 1) % kStage;
2490     }
2491
2492     // S→R: 加载下一组 A/B fragment 到寄存器。
2493     //
2494     // cute::copy 原生支持多 mode——理论上可以一次加载全部 K slice:
2495     //   cute::copy(s2r_tiled_copy_a, tAsA(_, _, _, ismem_read), tCrA_view);
2496     // 但这里刻意逐 k_step_next 加载, 目的是做**寄存器双缓冲**:
2497     //   当前 k_step 做 MMA 计算时, ldmatrix 预加载 k_step_next 的数据,
2498     //   让 S→R 延迟与 MMA 计算重叠。
2499     // 如果一次性加载全部 K slice, S→R 和 MMA 就会彻底串行。
2500     cute::copy(s2r_tiled_copy_a, tAsA(_, _, k_step_next, ismem_read),
2501               tCrA_view(_, _, k_step_next));
2502     cute::copy(s2r_tiled_copy_b, tBsB(_, _, k_step_next, ismem_read),
2503               tCrB_view(_, _, k_step_next));
2504
2505     if (k_step == 0) {
2506         // G→S: 提交下一个 K tile (整个 BK) 的异步拷贝。
2507         if (itile_to_read < num_k_tiles) {
2508             cute::copy(g2s_tiled_copy_a, tAgA_copy(_, _, _, itile_to_read),
2509                       tAsA_copy(_, _, _, ismem_write));
2509             cute::copy(g2s_tiled_copy_b, tBgB_copy(_, _, _, itile_to_read),
2510                       tBsB_copy(_, _, _, ismem_write));
2511             ++itile_to_read;
2512             ismem_write = (ismem_write + 1) % kStage;
2513         }
2514         cp_async_fence();
2515     }
2516 }
2517
2518 // MMA: cute::gemm 内部根据 tiled_mma 的类型信息, 自动展开为覆盖
2519 // 全部 (MMA_M, MMA_N) 对的 mma.sync 指令序列, 累加到 tCrD。

```

```

2520     cute::gemm(tiled_mma, tCrD, tCrA(_, _, k_step), tCrB(_, _, k_step), tCrD);
2521 }
2522 }
2523
2524 // Epilogue: D 寄存器 → shared memory → global memory。
2525 // 使用当前 A stage 作为 C scratchpad, 避免为 epilogue 单独分配完整的 C shared memory。
2526 // 这里是“复用 storage”, 不是把 A Tensor 转换成 C Tensor:
2527 //   sA 的逻辑 shape 是 (BM, BK, kStage)=(128,32,2), 而 sA(_,_,ismem_read)
2528 //   只选出其中一个 A-stage 的 storage 起点; sC 随后以完全独立的 SmemLayoutC
2529 //   解释这同一段地址。当前 wrapper 的固定配置下:
2530 //   一个 A stage: 128 x 32 = 4096 个 half;
2531 //   C scratchpad: 32 x 32 x 4 = 4096 个 half。
2532 //   两者容量刚好相等, 但 logical shape 和 layout 不是同一个: A 的 layout atom
2533 //   是 8x32, C 的 layout atom 是 32x32, 二者都含 Swizzle<3,3,3>, 却不能把
2534 //   C 数据再按 SmemLayoutA 读取。launch wrapper 的 static_assert 用当前配置
2535 //   检查 C scratchpad 能放进一个 A pipe; 此处只别名该单个 stage, 不是整块双缓冲 sA。
2536 //
2537 // mainloop 已结束, 不会再通过 sA 的 A/B load view 消费这个 pipe 的旧 A 数据。
2538 // 从这一行开始, 对这块地址的所有访问都通过 SmemLayoutC 派生的 sC 完成: R2S
2539 // 用它写 C, S2G 用它读 C。因此 C 的 swizzle/address mapping 在写和读两侧一致。
2540 auto sC = make_tensor(sA(_, _, ismem_read).data(), SmemLayoutC{});
2541
2542 // R2S TiledCopy: 寄存器 → shared memory (使用 UniversalCopy 做类型转换)
2543 // R2SCopyAtomC = CopyAtom<UniversalCopy<int>, T>; 以 32-bit 粒度搬运 T 数据
2544 auto r2s_tiled_copy_c = make_tiled_copy_C(R2SCopyAtomC{}, tiled_mma);
2545 auto r2s_thr_copy_c = r2s_tiled_copy_c.get_slice(idx);
2546 // tCrD 是 TiledMMA 产生的 accumulator fragment, 逻辑 shape 为
2547 // (MMA, MMA_M, MMA_N)。它的寄存器元素已经就位, 但该 layout 是为 MMA
2548 // accumulator 服务的, 并不直接按 R2S copy atom 的 source-value 顺序表达。
2549 //
2550 // retile_S 中的 S 指“这个 TiledCopy 的 source side”, 不是 shared memory。
2551 // 它基于 r2s_tiled_copy_c 的 reference layout 创建一个共享 tCrD.data() 的
2552 // zero-copy layout view, 将逻辑坐标表示为 (CPY, CPY_M, CPY_N)。这一步不读取
2553 // 或写入寄存器/共享内存, 不进行 dtype conversion, 也不交换 lane 间数据; 真正的
2554 // R2S 数据搬运发生在下面的 cute::copy(r2s_tiled_copy_c, ..., tCsC_r2s(...))。
2555 // 可将它与主循环的 s2r_thr_copy_a.retile_D(tCrA) 对照: 二者都是为了让已有
2556 // fragment 的 per-thread value ordering 匹配某个 TiledCopy 的 source/destination
2557 // 角色, 而不是另一次 memory transfer。
2558 // tCrD: (MMA, MMA_M, MMA_N) -> retile -> tCrC_r2s: (CPY, CPY_M, CPY_N)
2559 auto tCrC_r2s = r2s_thr_copy_c.retile_S(tCrD); // (CPY, CPY_M, CPY_N)
2560 // get_slice(idx) 已固定 CTA 内当前 logical copy thread; partition_D 再按
2561 // R2S TiledCopy 的 destination mapping 切分 sC。推导链:
2562 //   MMA_P_T = Tile<kMmaPM, kMmaPN, kMmaPK> = Tile<32, 32, 16>
2563 //   → TiledMMA::tile_size<0>(mma) = 32, tile_size<1>(mma) = 32
2564 //   → make_tiled_copy_C 的 tiler = (32, 32)
2565 //   → SmemLayoutC 的逻辑 MxN = (kMmaPM, kMmaPN) = (32, 32) 恰好等于该 tiler
2566 //   → partition_D: zipped_divide(sC, tiler=(32,32)) 后 M/N 维无 remainder
2567 //   → 结果 shape 的 M/N remainder = _1, _1
2568 // 最后一维 pipe=4 来自 SmemLayoutC 的第三维 kSmemLayoutCBatch, 而不是
2569 // “thread-level tensor 自动只剩 CPY”这一条规则。_1 表示该逻辑 mode 的 extent
2570 // 是 1, 并非该 mode 被删除或 C tile 没有 M/N 覆盖。
2571 auto tCsC_r2s = r2s_thr_copy_c.partition_D(sC); // (CPY, _1, _1, pipe)
2572
2573 // S2G TiledCopy: shared memory → global memory (128-bit store)
2574 // S2GCopyAtomC(CopyAtom<UniversalCopy<cute::uint128_t>, T>) -> S2GCopyC
2575 S2GCopyC s2g_tiled_copy_c;
2576 auto s2g_thr_copy_c = s2g_tiled_copy_c.get_slice(idx);
2577 // tCsC_s2g 与 tCsC_r2s 都从同一个 sC 产生, 因而保留相同的 pipe=4; 前者是
2578 // S2G source mapping, 后者是 R2S destination mapping。tCgC_s2g 则面向完整
2579 // gD=(BM,BN)=(128,256), 相对于 S2G copy tiler 仍有非平凡的 M/N repetition,
2580 // 所以它保留 CPY_M、CPY_N。这里的 CPY/CPY_M/CPY_N 都是编译期 copy-partition
2581 // 的逻辑 mode, 不应直接理解为固定的 lane 数、warp 数或物理连续维度。
2582 auto tCsC_s2g = s2g_thr_copy_c.partition_S(sC); // (CPY, _1, _1, pipe)

```

```

2583 auto tCgC_s2g = s2g_thr_copy_c.partition_D(gD); // (CPY, CPY_M, CPY_N)
2584
2585 // group_modes<B,E> 将 layout 的半开区间 [B,E) 组合成一个嵌套 mode:
2586 //   group_modes<1,3>(Tensor(CPY, CPY_M, CPY_N))
2587 //   -> Tensor(CPY, (CPY_M, CPY_N))
2588 // 它只重组 Tensor 的 layout, 不分配内存、不移动 data(), 也不改变元素访问的物理地址。
2589 // 第二个 mode 仍然保存原来 CPY_M/CPY_N 的层次关系, 只是现在可以用一个坐标
2590 // i+j 访问这个组合 mode; 因此这里的 group_modes 更准确地说是“合并 mode”,
2591 // 而不是把底层数据拷贝或无条件 flatten 成普通一维数组。
2592 // 从 type/shape 结构看, 结果确实是 Tensor(CPY, (CPY_M, CPY_N)); 但 grouped
2593 // mode 的 cardinality 是两个子 mode 的乘积, 因此 size<1>(…) 可作为一个标量
2594 // 范围遍历其全部 logical coordinate。于是 (_, i+j) 中的 i+j 是该 nested mode
2595 // 的线性 logical coordinate, 不是在 C++ 中对二维 tuple 做加法, 更不是 raw
2596 // pointer offset; 最终物理地址仍由各自 Tensor 的 grouped layout 计算。
2597 //
2598 // 这里为什么要对 tCrC_r2s 和 tCgC_s2g 同时 group?
2599 //   - tCrC_r2s: 每个线程持有的寄存器 C fragment, 原始形状为 (CPY, CPY_M, CPY_N);
2600 //   - tCgC_s2g: 该线程对应的 global-memory C 输出位置, 形状与源 Tensor 对齐;
2601 //   - 两者使用相同的 [1,3) 合并后, 源/目的 Tensor 仍保持相同的坐标空间,
2602 //     cute::copy 可以按相同的 (CPY, i+j) 坐标完成寄存器到 global 的对应搬运。
2603 //
2604 // tCsC_r2s/tCsC_s2g 没有 group 第 3 个 pipe mode, 因为它们还需要显式保留
2605 // shared-memory C scratchpad 的 pipeline 维度; step=size<3>(tCsC_r2s) 表示
2606 // 每一轮可以复用的 C shared-memory pipeline 深度。外层 i 在合并后的 C 元素空间中
2607 // 按 step 前进, 内层 j 选择当前 pipeline slot: 寄存器先写入 sC(_,0,0,j),
2608 // 再从同一个 slot 写回 global memory。
2609 // 当前 kSmemLayoutCBatch=4, 因此 step=4。代码未对 i+j 做尾部 predication,
2610 // 这个循环依赖当前静态 layout 中 grouped extent 能被 pipe depth 整除; 不应将
2611 // “每轮正好处理 4 个 fragment” 误认为任意 tile/copy 配置都自动成立的通用规则。
2612 auto tCgC_s2gx = group_modes<1, 3>(tCgC_s2g);
2613 auto tCrC_r2sx = group_modes<1, 3>(tCrC_r2s);
2614
2615 int step = size<3>(tCsC_r2s); // C scratchpad 的 pipe 数 (由 kSmemLayoutCBatch=4 指定)
2616 // 双层循环的语义 (注意 step 与 CPY_N 无关):
2617 //   group_modes<1,3> 之后, tCrC_r2sx 的 shape 为 (CPY, (CPY_M, CPY_N))。
2618 //   size<1>(tCrC_r2sx) = CPY_M * CPY_N, 即该线程需处理的 C fragment 总数。
2619 //   外循环以 step=4 为步长, 内循环 j=0..3 每次处理 1 个 fragment, 将其写入
2620 //   第 j 号 C scratchpad pipe slot, 再读出写回 global。
2621 //
2622 //   这里 step=4 是 scratchpad pipeline 深度, 不是 CPY_N。CPY_N 的值取决于
2623 //   TiledMMA 的 C-layout 如何将 (128,256) 的 gD tile 分配到 128 个线程上,
2624 //   通常 ≠ 4。循环之所以成立, 是因为 grouped mode 用线性坐标 i+j 遍历整个
2625 //   CPY_M * CPY_N 的扁平空间——它不再区分 M 和 N 方向, 也不要求 CPY_N == step。
2626 //   唯一前提是 CPY_M * CPY_N 能被 step 整除 (当前静态 layout 编译期保证)。
2627 //
2628 //   第 j 号 pipe slot 在 sC 上的物理地址由 R2S partition_D / S2G partition_S
2629 //   各自推导; 因为二者都从同一个 sC (SmemLayoutC) 出发, pipe mode 保持一致,
2630 //   写入和读取命中同一段 shared memory。
2631 #pragma unroll
2632 for (int i = 0; i < size<1>(tCrC_r2sx); i += step) {
2633     // 每轮处理 step 个 fragment, 内层 j 选 pipe slot。
2634 #pragma unroll
2635     for (int j = 0; j < step; ++j) {
2636         // 这两行是 R2R staging: make_tensor_like<T> 分配当前线程私有、拥有 storage
2637         // 的 register Tensor; 普通 cute::copy 将 accumulator fragment materialize
2638         // 成输出元素类型 T 的临时 payload。
2639         //
2640         // cute::copy (无 copy-atom 的普通版) 内部按元素做
2641         //   dst(i) = static_cast<T>(static_cast<SrcType>(src(i)))
2642         // 因此当 accumulator 与 T 类型不同时, 它自动完成 dtype 转换; 当二者相同
2643         // (如本 kernel 的 f16 accumulator + T=half), cast 退化为 no-op。
2644         //
2645         // 即使类型一致, t 还有一个不可省略的作用: layout 归一化。

```

```

2646 // tCrC_r2sx(_, i+j) 的 layout 是 retile_S → group_modes → slice 层层
2647 // 叠加的 composed layout, 而 make_tensor_like<T> + cute::copy 把它
2648 // 物化到一个拥有简单 strides 的 owning register Tensor 中。后续
2649 // r2s_tiled_copy_c 的 copy_unpack 需要按 copy-atom 的 val-layout
2650 // 拆分 source 元素, 简单 owning layout 比复杂的 composed layout 更容易
2651 // 被编译器推导内联。上游同源实现将其概括为"cope with accumulator and
2652 // output data type difference", 实际承担了类型适配和 layout 归一化两个职责。
2653 //
2654 // 这不是 retile_S 的一部分, 也不是 warp shuffle: 源/目的都在当前线程的
2655 // register 中, 代码没有显式 __shfl_sync。不能仅根据这段 C++ 断言最终 SASS
2656 // 绝不会含 shuffle; 但语义上的跨线程 regrouping 不由 R2R copy 承担, 而是
2657 // 由随后的 R2S → shared memory → S2G 路径完成。若特定 accumulator/output
2658 // type 与 R2S copy-atom contract 已直接兼容, 可以设计直接 R2S 的变体; 本例
2659 // 保持同源实现的通用 staging 写法。
2660 auto t = make_tensor_like<T>(tCrC_r2sx(_, i + j));
2661 cute::copy(tCrC_r2sx(_, i + j), t);
2662 // R2S 的 copy atom 才在此处将 thread-local payload 写入 j 号 C scratchpad
2663 // slot; SmemLayoutC 定义写入后的跨线程地址重组。
2664 //
2665 // cute::copy(r2s_tiled_copy_c, src, dst) 的调用链路:
2666 //   r2s_tiled_copy_c 是 TiledCopy, 但继承自 Copy_Atom<UniversalCopy<int>,T>
2667 //   → 匹配 copy(Copy_Atom<...> const&, src, dst) 重载 [copy.hpp:~L190]
2668 //   → src(t) 与 dst(tCsC_r2s(_,0,0,j)) 都是 rank-1 → 直接 copy_atom.call
2669 //   → copy_atom.call 内部: 若 size(src)==NumValSrc (atom 编译期 val-count),
2670 //   走 copy_unpack; 否则递归剥 mode, 最终都落到 UniversalCopy<int>::copy
2671 //   → 硬件层面就是逐 uint32_t 的 register-to-smem store [arch/copy.hpp:~L46]
2672 //
2673 // 这里没有任何运行时"查找表": t 的第 i 个元素之所以对应 dst 的第 i 个
2674 // shared memory offset, 是因为 pre-partition 阶段已经把映射烧进 layout 了:
2675 //   - t 的 layout 来自 retile_S → group_modes → slice → make_tensor_like
2676 //   → 元素顺序已按 R2S atom 的 source-side value ordering 排好
2677 //   - tCsC_r2s 的 layout 来自 r2s_thr_copy_c.partition_D(sC)
2678 //   → TiledCopy::tidfrg_D 用 LayoutCopy_TV + Tiler_MN + ValLayoutDst
2679 //   计算了当前线程每个 val 在 smem 中的物理 offset
2680 // 两者共享同一个 copy atom 的 ValLayoutSrc/ValLayoutDst 定义,
2681 // 因此逐元素 copy 天然把正确的数据写到了正确的地址。
2682 cute::copy(r2s_tiled_copy_c, t, tCsC_r2s(_, 0, 0, j));
2683 }
2684 // 这是 CTA 范围的 rendezvous: 所有线程完成当前批 R2S 写入后, S2G 才能从
2685 // sC 的重组布局读取完整数据。它承担了手写版中显式跨 lane 拼接所需的协作边界。
2686 __syncthreads();
2687
2688 // S→G: 从同一个 pipe slot 读回 global memory, 保持源/目的坐标一一对应。
2689 #pragma unroll
2690 for (int j = 0; j < step; ++j) {
2691 // S2GCopyC 的 atom 为 UniversalCopy<uint128_t>: 与 R2S 的 UniversalCopy<int>
2692 // (32-bit) 不同, 它一次搬运 128-bit (8 个 half), 映射为一条 wide store
2693 // (如 st.global.v4.f32 或 st.global.b128)。
2694 //
2695 // cute::copy(s2g_tiled_copy_c, src, dst) 的调用链路:
2696 //   s2g_tiled_copy_c 是 TiledCopy, 继承自 Copy_Atom<UniversalCopy<uint128_t>,T>
2697 //   → 匹配 copy(Copy_Atom<...> const&, src, dst) 重载 [copy.hpp:~L190]
2698 //   → src=tCsC_s2g(_,0,0,j) 与 dst=tCgC_s2gx(_,i+j) 都是 rank-1
2699 //   → 直接 copy_atom.call(src, dst)
2700 //   → copy_atom.call 内部: size(src)==NumValSrc → copy_unpack
2701 //   → 逐 128-bit chunk 调用 UniversalCopy<uint128_t>::copy
2702 //   → 硬件层面就是 shared-memory-to-global wide store [arch/copy.hpp:~L46]
2703 //
2704 // S2G 的 pre-partition 与 R2S 对称, 但 source/destination 角色互换:
2705 //   - tCsC_s2g 来自 s2g_thr_copy_c.partition_S(sC): TiledCopy::tidfrg_S
2706 //   用 LayoutCopy_TV + Tiler_MN + ValLayoutSrc 计算了当前线程每个 val
2707 //   在 sC (shared memory) 中的物理 offset。
2708 //   S2GCopyC 的 tiler = product(ThrLayout{32,4}, ValLayout{1,8})

```

```

2709 //          = (32x1, 4x8) = (32, 32) — 恰好与 R2S tiler 相同,
2710 // 因此 tCsC_s2g 也是 (CPY, _1, _1, pipe), _1,_1 来自 sC=(32,32,4) 无
2711 // M/N remainder。
2712 // - tCgC_s2gx(_, i+j) 来自 s2g_thr_copy_c.partition_D(gD) → group_modes
2713 // → slice。gD=(128,256) 相对于 tiler (32,32) 有 4x8 个 tile repetition,
2714 // 因此 tCgC_s2g 为 (CPY, CPY_M, CPY_N), group_modes<1,3> 合并 M/N
2715 // repetition 后得到 (CPY, (CPY_M,CPY_N)), 再用线性坐标 i+j 切片。
2716 // 两者共享同一个 copy atom 的 ValLayoutSrc/ValLayoutDst 定义,
2717 // 因此逐 chunk copy 天然从正确的 smem 地址读到正确的 gmem 地址。
2718 cute::copy(s2g_tiled_copy_c, tCsC_s2g(_, 0, 0, j), tCgC_s2gx(_, i + j));
2719 }
2720 // 所有线程都完成当前 pipe slots 的 S2G 读取后, 下一轮 R2S 才能覆盖这 4 个
2721 // scratchpad slots, 避免一部分线程仍在读取旧数据时被其他线程提前重写。
2722 __syncthreads();
2723 }
2724
2725 // 与 hgemm_mma_stages_tn 的手写 epilogue 对照: 手写版必须显式处理 MMA fragment
2726 // 的物理分布, 使用 RC0/RC1 暂存, 再用 __shfl_sync 从同一 warp 的相邻 lane 收齐
2727 // 8 个 half, 并由 lane_id % 4 == 0 的线程做 float4 (128-bit) global store。
2728 // CuTe 版没有把这些 lane 映射写在 kernel 源码中: retile_S 表达 accumulator
2729 // fragment 与 R2S copy 的对应, R2S/sC/S2G 通过 shared-memory scratchpad 完成
2730 // CTA 范围的数据重组, S2GCopyC 以 uint128_t 表达 wide store。两者的共同目标都是
2731 // 将分散在计算线程寄存器中的 C fragment 变为适合连续 global-memory 写回的布局;
2732 // 区别是手写版直接编排 shuffle/store, CuTe 版将映射交给 TiledMMA/TiledCopy 类型推导。
2733 }
2734
2735 // =====
2736 // Phase 7c-2: CuTe HGEMM Launch Wrapper — 类型实例化 + Grid/Block 配置
2737 // =====
2738 // CuTe 的核心设计理念: "类型即配置" (Type-level configuration)
2739 // 所有的 MMA atom、TiledCopy、SmemLayout、Swizzle 都是 **编译期类型**,
2740 // kernel 接收这些类型的实例 (通常为空 struct), 在编译期完成全部映射推导。
2741 //
2742 // 以下每个类型定义后面都标注了它在 kernel body 中的对应变量/用法, 形成 1:1 对照。
2743 //
2744 // 面试常问: 「CuTe 为什么比手写 PTX 简洁?」
2745 // 答: 手写需要:
2746 // 1) 手动计算每线程的 smem 地址偏移
2747 // 2) 手动计算 ldmatrix 的 lane 映射
2748 // 3) 手动插入 swizzle XOR 到每个地址计算点
2749 // 4) 手动做 epilogue 的 warp shuffle 编排
2750 // CuTe 的 partition + copy + gemm 通过 TiledCopy/TiledMMA 的类型信息
2751 // 在编译期自动完成上述全部推导, 生成与手写等价的 PTX 指令序列。
2752 // =====
2753 template <typename T, const int Stages = 2, const int BlockSwizzle = 0>
2754 void launch_hgemm_mma_stages_tn_cute(T *a, T *b, T *c, int M, int N, int K) {
2755     using namespace cute;
2756
2757     // — Tile 尺寸 (kernel 模板参数 BM/BN/BK/kStage 的值) —
2758     //
2759     // kernel 用法对照:
2760     // BM=128: local_tile(A, make_tile(Int<BM>{}, Int<BK>{}), ...) → gA=(BM,BK,num_k_tiles)
2761     //          local_tile(D, make_tile(Int<BM>{}, Int<BN>{}), ...) → gD=(BM,BN)
2762     // BN=256: local_tile(B, make_tile(Int<BN>{}, Int<BK>{}), ...) → gB=(BN,BK,num_k_tiles)
2763     //          local_tile(D, ...) → gD 的列方向
2764     // BK=32: num_k_tiles = k / BK; 每个 BK tile 内 num_k_steps = BK/kMmaPK = 2 个 MMA_K slice
2765     // kStage=2: sA/sB 的第三维 = (BM,BK,kStage); cp_async_wait<kStage-2>() 流水线同步
2766     // kSmemLayoutCBatch=4: step = size<3>(tCsC_r2s) → C scratchpad pipe 深度 = 4
2767     auto BM = Int<128>{};
2768     auto BN = Int<256>{};
2769     auto BK = Int<32>{};
2770     auto KStage = Int<Stages>{};
2771     auto kSmemLayoutCBatch = Int<4>{};

```

```

2772 // — SmemLayoutA / SmemLayoutB —
2773 // kernel 用法:
2774 // auto sA = make_tensor(make_smem_ptr(Ashm), SmemLayoutA{}); // (BM, BK, kStage)
2775 // auto sB = make_tensor(make_smem_ptr(Bshm), SmemLayoutB{}); // (BN, BK, kStage)
2776 // SmemLayout 由三部分组成: Swizzle<3,3,3> + atom layout(8×BK) + tile_to_shape 扩展到完整 tile+stage。
2777 //
2778 // 对当前 base layout shape=(8,32), stride=(32,1), T=half 的例子:
2779 // - M=3: 连续 2^3=8 个 half 组成一个基本元素, 即 16 bytes;
2780 // - S=3: 连续 2^3=8 个基本元素组成一行, 即 128 bytes;
2781 // - B=3: 共有 2^3=8 行。
2782 // 于是一个 8×32 的逻辑 tile 正好对应 8 行 × 128B, 覆盖全部 32 个 shared-memory bank。
2783 // Swizzle 对每个基本元素的列坐标执行 icol' = irow ^ icol,
2784 // 再将 (irow, icol') 和基本元素内部的 3 个低位编码回物理 offset。
2785 // 其位级形式为 offset' = offset ^ ((offset & (0b111 << 6)) >> 3):
2786 // 地址位 [6:8] XOR 到 [3:5], 地址位 [0:2] 不参与置换。
2787 // composition(Swizzle, base_layout) 的语义是 R(c) = Swizzle(base_layout(c)):
2788 // 逻辑坐标仍由 base_layout 给出, 只有最终的 shared-memory offset 被重排。
2789 // tile_to_shape: 通过 blocked product 重复这个带 swizzle 的 block layout,
2790 // 使结果 shape 匹配 BM×BK×kStage; 默认按目标 shape 的 mode order 重复各维。
2791 using SmemLayoutAtom = decltype(composition(
2792     Swizzle<3, 3, 3>{},
2793     make_layout(make_shape(Int<8>{}, Int<BK>{}),
2794         make_stride(Int<BK>{}, Int<1>{}))));
2795 using SmemLayoutA = decltype(tile_to_shape(
2796     SmemLayoutAtom{}, make_shape(Int<BM>{}, Int<BK>{}, Int<kStage>{})));
2797 using SmemLayoutB = decltype(tile_to_shape(
2798     SmemLayoutAtom{}, make_shape(Int<BN>{}, Int<BK>{}, Int<kStage>{})));
2799
2800 // — MMA (TiledMMA) —
2801 // kernel 用法:
2802 // TiledMMA tiled_mma;
2803 // auto thr_mma = tiled_mma.get_slice(threadIdx.x); // 每线程的 MMA slice
2804 // auto tCrA = thr_mma.partition_fragment_A(gA(_,_,0)); // (MMA, MMA_M, MMA_K)
2805 // auto tCrB = thr_mma.partition_fragment_B(gB(_,_,0)); // (MMA, MMA_N, MMA_K)
2806 // auto tCrD = thr_mma.partition_fragment_C(gD); // (MMA, MMA_M, MMA_N)
2807 // cute::gemm(tiled_mma, tCrD, tCrA(_,_,k_step), tCrB(_,_,k_step), tCrD);
2808 // MMA 还决定 S2R/R2S copy 的 tiler: make_tiled_copy_A/B/C 都依赖 tiled_mma 的
2809 // tile_size<0/1>(mma) = (32,32) 来推导线程→数据映射。
2810 //
2811 // 推导链: SM80_16x8x16_F16F16F16F16_TN → MMA_Atom
2812 // → make_tiled_mma(atom, EURepeat{2,2,1}, ValTile{32,32,16})
2813 // → TiledMMA: 128 threads = 4 warps × (2×2 EU slices), 逻辑 MMA tile = 32×32×16
2814 // TN = A row-major, B col-major; 因此传给本 kernel 的 B 指针实际指向
2815 // B^T[N,K] 的 row-major 存储 (等价于 GEMM 语义中的 B[K,N] col-major)。
2816 using mma_op = SM80_16x8x16_F16F16F16F16_TN;
2817 using mma_traits = MMA_Traits<mma_op>;
2818 using mma_atom = MMA_Atom<mma_traits>;
2819 using mma_atom_shape = mma_traits::Shape_MNK; // (Int<16>, Int<8>, Int<16>)
2820
2821 static constexpr int kMmaEURepeatM = 2;
2822 static constexpr int kMmaEURepeatN = 2;
2823 static constexpr int kMmaEURepeatK = 1;
2824 static constexpr int kMmaPM = 1 * kMmaEURepeatM * get<0>(mma_atom_shape{}); // 32
2825 static constexpr int kMmaPN = 2 * kMmaEURepeatN * get<1>(mma_atom_shape{}); // 32
2826 static constexpr int kMmaPK = 1 * kMmaEURepeatK * get<2>(mma_atom_shape{}); // 16
2827 // kMmaPK=16 → kernel 中 num_k_steps = size<2>(tCrA) = BK/kMmaPK = 32/16 = 2
2828
2829 using MMA_EU_RepeatT = decltype(make_layout(make_shape(
2830     Int<kMmaEURepeatM>{}, Int<kMmaEURepeatN>{}, Int<kMmaEURepeatK>{})));
2831 using MMA_P_T = Tile<Int<kMmaPM>, Int<kMmaPN>, Int<kMmaPK>>;
2832 using MMA = decltype(make_tiled_mma(mma_atom{}, MMA_EU_RepeatT{}, MMA_P_T{}));
2833

```

```

2835 // — G2SCopyA / G2SCopyB —
2836 // kernel 用法:
2837 // G2SCopyA g2s_tiled_copy_a; G2SCopyB g2s_tiled_copy_b;
2838 // cute::copy(g2s_tiled_copy_a, tAgA_copy(_,_,_,istage), tAsA_copy(_,_,_,istage));
2839 // cute::copy(g2s_tiled_copy_b, tBgB_copy(_,_,_,istage), tBsB_copy(_,_,_,istage));
2840 // G→S: 128-bit cp.async。ThrLayout{32,4} × ValLayout{1,8}:
2841 // 128 个线程, 每线程搬运 1×8=8 个 half = 128 bits。
2842 using g2s_copy_op = SM80_CP_ASYNC_CACHEGLOBAL<cute::uint128_t>;
2843 using g2s_copy_traits = Copy_Traits<g2s_copy_op>;
2844 using g2s_copy_atom = Copy_Atom<g2s_copy_traits, T>;
2845 using G2SCopyA = decltype(make_tiled_copy(
2846     g2s_copy_atom{},
2847     make_layout(make_shape(Int<32>{}, Int<4>{}),
2848         make_stride(Int<4>{}, Int<1>{})),
2849     make_layout(make_shape(Int<1>{}, Int<8>{}))));
2850 using G2SCopyB = G2SCopyA;
2851
2852 // — S2RCopyAtomA / S2RCopyAtomB —
2853 // kernel 用法:
2854 // auto s2r_tiled_copy_a = make_tiled_copy_A(S2RCopyAtomA{}, tiled_mma);
2855 // auto s2r_tiled_copy_b = make_tiled_copy_B(S2RCopyAtomB{}, tiled_mma);
2856 // cute::copy(s2r_tiled_copy_a, tAsA(_,_,k_step_next,ismem_read), tCrA_view(_,_,k_step_next));
2857 // cute::copy(s2r_tiled_copy_b, tBsB(_,_,k_step_next,ismem_read), tCrB_view(_,_,k_step_next));
2858 // S→R: ldmatrix (SM75_U32x4_LDSM_N)。make_tiled_copy_A/B 根据 TiledMMA 自动推导
2859 // 与 MMA fragment 对齐的 shared→register 映射, 无需手动指定 ThrLayout/ValLayout。
2860 using s2r_copy_op = SM75_U32x4_LDSM_N;
2861 using s2r_copy_traits = Copy_Traits<s2r_copy_op>;
2862 using s2r_copy_atom = Copy_Atom<s2r_copy_traits, T>;
2863 using S2RCopyAtomA = s2r_copy_atom;
2864 using S2RCopyAtomB = s2r_copy_atom;
2865
2866 // — SmemLayoutC —
2867 // kernel 用法:
2868 // auto sC = make_tensor(sA(_,_,ismem_read).data(), SmemLayoutC{});
2869 // 复用当前 A stage 的空间作为 C scratchpad
2870 // sC shape = (kMmaPM, kMmaPN, kSmemLayoutCBatch) = (32, 32, 4)
2871 // pipe 数 4 由 kSmemLayoutCBatch 指定 → step = size<3>(tCsC_r2s) = 4
2872 // 与 A/B 相同的 Swizzle<3,3,3> 规则, 但 base layout 是 32×32 (MMA tile 大小),
2873 // 再扩展为 4 个 epilogue scratchpad pipe slots。这 4 个 slots 不等于主循环
2874 // 的 kStage K-tile pipeline。
2875 using SmemLayoutAtomC = decltype(composition(
2876     Swizzle<3, 3, 3>{},
2877     make_layout(make_shape(Int<kMmaPM>{}, Int<kMmaPN>{}),
2878         make_stride(Int<kMmaPN>{}, Int<1>{}))));
2879 using SmemLayoutC = decltype(tile_to_shape(
2880     SmemLayoutAtomC{},
2881     make_shape(Int<kMmaPM>{}, Int<kMmaPN>{}, Int<kSmemLayoutCBatch>{})));
2882
2883 // — R2SCopyAtomC —
2884 // kernel 用法:
2885 // auto r2s_tiled_copy_c = make_tiled_copy_C(R2SCopyAtomC{}, tiled_mma);
2886 // cute::copy(r2s_tiled_copy_c, t, tCsC_r2s(_, 0, 0, j));
2887 // R→S: UniversalCopy<int> 以 32-bit 粒度将寄存器 payload 写入 C scratchpad。
2888 // make_tiled_copy_C 从 TiledMMA 的 tile_size<0/1>(mma)=(32,32) 推导 tiler,
2889 // 因此 R2S copy 的 tiler = (32,32), 与 SmemLayoutC 的 (32,32) 恰好匹配。
2890 using R2SCopyAtomC = Copy_Atom<UniversalCopy<int>, T>;
2891
2892 // — S2GCopyC —
2893 // kernel 用法:
2894 // S2GCopyC s2g_tiled_copy_c;
2895 // cute::copy(s2g_tiled_copy_c, tCsC_s2g(_, 0, 0, j), tCgC_s2gx(_, i+j));
2896 // S→G: UniversalCopy<uint128_t> 以 128-bit 粒度做 shared→global wide store。
2897 // ThrLayout{32,4} × ValLayout{1,8} → tiler = product((32,4),(1,8)) = (32,32),

```

```

2898 // 与 R2S tiler 相同, 因此 partition_S(sC) 得到的 pipe 维度一致。
2899 using S2GCopyAtomC = CopyAtom<UniversalCopy<cute::uint128_t>, T>;
2900 using S2GCopyC = decltype(make_tiled_copy(
2901     S2GCopyAtomC{}),
2902     make_layout(make_shape(Int<32>{}, Int<4>{}),
2903         make_stride(Int<4>{}, Int<1>{})),
2904     make_layout(make_shape(Int<1>{}, Int<8>{}))));
2905
2906 // — Grid/Block 配置 —
2907 // kernel 用法:
2908 // int idx = threadIdx.x;          // 0..127
2909 // int ix = ((int)BlockSwizzle) * blockIdx.z * gridDim.x + blockIdx.x;
2910 // int iy = blockIdx.y;           // M 方向 CTA 坐标
2911 // if (iy * BM >= m || ix * BN >= n) return; // CTA 级边界保护
2912 // 由于 kernel 没有 tile 内的逐元素 predication, 调用方需保证
2913 // M % BM == 0, N % BN == 0, K % BK == 0。
2914 // 当不启用 BlockSwizzle 时, BZ=1 退化为纯 2D grid, 行为不变。
2915 constexpr int kSwizzleStride = 2048;
2916 int BX = (N + BN - 1) / BN;
2917 int BY = (M + BM - 1) / BM;
2918 int BZ = BlockSwizzle ? (N + kSwizzleStride - 1) / kSwizzleStride : 1;
2919 BX = BlockSwizzle ? (BX + BZ - 1) / BZ : BX;
2920 dim3 block(size(MMA{})); // 128 threads (= size(TiledMMA))
2921 dim3 grid(BX, BY, BZ);
2922
2923 // — Dynamic Shared Memory 大小 —
2924 // kernel 用法:
2925 // extern __shared__ T shm_data[];
2926 // T *Ashm = shm_data;
2927 // T *Bshm = shm_data + cute::cosize(SmemLayoutA{});
2928 // cosize(SmemLayoutA) = BM*BK*kStage = 128*32*2 = 8192 个 T
2929 // cosize(SmemLayoutB) = BN*BK*kStage = 256*32*2 = 16384 个 T
2930 // 合计 A+B = 24576 个 T; C epilogue 复用一个 A stage (128*32 = 4096 个 T)。
2931 // static_assert 保证 C scratchpad (kMmaPM*kMmaPN*kSmemLayoutCBatch = 32*32*4 = 4096)
2932 // 能放进一个 A stage (128*32 = 4096)。
2933 static constexpr int shm_size_AB =
2934     cute::cosize(SmemLayoutA{}) + cute::cosize(SmemLayoutB{});
2935 static constexpr int shm_size_C = cute::cosize(SmemLayoutC{});
2936 static_assert(size<0>(SmemLayoutA{}) * size<1>(SmemLayoutA{}) >= size(SmemLayoutC{}),
2937     "C shared memory must fit within one A pipe");
2938 static constexpr int kShmSize =
2939     cute::max(shm_size_AB, shm_size_C) * sizeof(T);
2940
2941 cudaFuncSetAttribute(
2942     hgemm_mma_stages_tn_cute<T, BM, BN, BK, KStage, MMA, G2SCopyA, G2SCopyB,
2943         SmemLayoutA, SmemLayoutB, SmemLayoutC,
2944         S2RCopyAtomA, S2RCopyAtomB, R2SCopyAtomC,
2945         S2GCopyAtomC, S2GCopyC, BlockSwizzle>,
2946     cudaFuncAttributeMaxDynamicSharedMemorySize, kShmSize);
2947
2948 hgemm_mma_stages_tn_cute<T, BM, BN, BK, KStage, MMA, G2SCopyA, G2SCopyB,
2949     SmemLayoutA, SmemLayoutB, SmemLayoutC, S2RCopyAtomA,
2950     S2RCopyAtomB, R2SCopyAtomC, S2GCopyAtomC, S2GCopyC,
2951     BlockSwizzle>
2952     <<<grid, block, kShmSize>>>(a, b, c, M, N, K);
2953 }
2954
2955 #endif /* NOTES_V2_ENABLE_CUTE */
2956
2957 #if defined(NOTES_V2_ENABLE_WGMMMA) || defined(NOTES_V2_ENABLE_TMA_MMA_WS)
2958 // TMA/mbarrier helpers.
2959 //
2960 // Two implementations gated by NOTES_V2_FORCE_INLINE_ASYNC_PROXY:

```

```

2961 // - Defined (default): raw `asm volatile` PTX. ptxas keeps setmaxnreg
2962 // because raw asm carries no source annotation that ptxas would treat as
2963 // an "extern call" boundary (warning C7506 otherwise drops the hint).
2964 // - Undef'd: cuda::ptx:: / cuda::device:: C++ wrappers. Cleaner API but
2965 // ptxas drops setmaxnreg with C7506 even though the wrappers inline to
2966 // identical PTX; useful for comparing behavior / debugging.
2967 //
2968 // cuda::barrier (init/arrive/wait) is shared by both paths: it inlines to raw
2969 // mbarrier PTX and does not trigger C7506, so it needs no gating.
2970 //
2971 // Mirrors the raw-PTX style of tmp/LeetGPU/CUDA/22_GEMM/sm90_wgmma_tma_ws_pingpong.cu.
2972 #if defined(NOTES_V2_FORCE_INLINE_ASYNC_PROXY)
2973
2974 static __device__ __forceinline__ uint32_t
2975 cast_smem_ptr_to_uint(void const *ptr) {
2976     return static_cast<uint32_t>(__cvta_generic_to_shared(ptr));
2977 }
2978
2979 static __device__ __forceinline__ void tma_fence_proxy_async_shared_cta() {
2980     asm volatile("fence.proxy.async.shared::cta;\n" ::: "memory");
2981 }
2982
2983 // cp.async.bulk.tensor.2d: 2D TMA copy from global to shared::cluster.
2984 // [dst] = shared memory destination (smem addr, 32-bit)
2985 // [desc, {c0, c1}] = CUTensorMap + (minor_coord, major_coord) coords
2986 // [mbar] = mbarrier in shared memory (smem addr, 32-bit)
2987 // mbarrier::complete_tx::bytes: barrier flips phase when TMA bytes land.
2988 // L2::cache_hint omitted (EVICT_NORMAL); matches the cuda::ptx default.
2989 static __device__ __forceinline__ void tma_load_2d(
2990     void *dst, const CUTensorMap *tensor_map, int minor_coord, int major_coord,
2991     cuda::barrier<cuda::thread_scope_block> &barrier) {
2992     uint64_t gmem_int_desc = reinterpret_cast<uint64_t>(tensor_map);
2993     uint32_t smem_int_ptr = cast_smem_ptr_to_uint(dst);
2994     uint32_t smem_int_mbar =
2995         cast_smem_ptr_to_uint(reinterpret_cast<uint64_t *>(&barrier));
2996     asm volatile(
2997         "cp.async.bulk.tensor.2d.shared::cluster.global.mbarrier::complete_tx::bytes"
2998         " [%0], [%1, {%3, %4}], [%2];"
2999         :
3000         : "r"(smem_int_ptr), "l"(gmem_int_desc), "r"(smem_int_mbar),
3001           "r"(minor_coord), "r"(major_coord)
3002         : "memory");
3003 }
3004
3005 // mbarrier.arrive.expect_tx: register one arrival on the mbarrier and declare
3006 // that `bytes` of async copy traffic is expected before the phase flips.
3007 // Equivalent to cuda::ptx::mbarrier_arrive_expect_tx(release, cta, shared).
3008 static __device__ __forceinline__ void tma_arrive_expect_tx(
3009     cuda::barrier<cuda::thread_scope_block> &barrier, uint32_t bytes) {
3010     uint32_t smem_int_mbar =
3011         cast_smem_ptr_to_uint(reinterpret_cast<uint64_t *>(&barrier));
3012     asm volatile(
3013         "mbarrier.arrive.expect_tx.shared::cta.b64 __, [%0], %1;\n"
3014         :
3015         : "r"(smem_int_mbar), "r"(bytes)
3016         : "memory");
3017 }
3018
3019 #else // !NOTES_V2_FORCE_INLINE_ASYNC_PROXY: use cuda::ptx / cuda::device wrappers
3020
3021 static __device__ __forceinline__ void tma_fence_proxy_async_shared_cta() {
3022     #if CUDART_VERSION >= 13020
3023         cuda::ptx::fence_proxy_async(cuda::ptx::space_shared);

```

```

3024 #else
3025     cuda::device::experimental::fence_proxy_async_shared_cta();
3026 #endif
3027 }
3028
3029 static __device__ __forceinline__ void tma_load_2d(
3030     void *dst, const CUtensorMap *tensor_map, int minor_coord, int major_coord,
3031     cuda::barrier<cuda::thread_scope_block> &barrier) {
3032     #if CUDART_VERSION >= 13020
3033         const int32_t coords[] {minor_coord, major_coord};
3034         auto *barrier_handle = cuda::device::barrier_native_handle(barrier);
3035         cuda::ptx::cp_async_bulk_tensor(cuda::ptx::space_cluster,
3036                                         cuda::ptx::space_global, dst, tensor_map,
3037                                         coords, barrier_handle);
3038     #else
3039         cuda::device::experimental::cp_async_bulk_tensor_2d_global_to_shared(
3040             dst, tensor_map, minor_coord, major_coord, barrier);
3041     #endif
3042 }
3043
3044 static __device__ __forceinline__ void tma_arrive_expect_tx(
3045     cuda::barrier<cuda::thread_scope_block> &barrier, uint32_t bytes) {
3046     #if CUDART_VERSION >= 13020
3047         auto *barrier_handle = cuda::device::barrier_native_handle(barrier);
3048         [[maybe_unused]] auto token = cuda::ptx::mbarrier_arrive_expect_tx(
3049             cuda::ptx::sem_release, cuda::ptx::scope_cta, cuda::ptx::space_shared,
3050             barrier_handle, bytes);
3051     #else
3052         [[maybe_unused]] auto token =
3053             cuda::device::barrier_arrive_tx(barrier, 1, bytes);
3054     #endif
3055 }
3056
3057 #endif // NOTES_V2_FORCE_INLINE_ASYNC_PROXY
3058
3059 // Warpgroup-level register rebalancing for warp specialization.
3060 // setmaxnreg.dec releases registers (producer TMA path needs few),
3061 // setmaxnreg.inc requests registers (consumer MMA path needs many).
3062 // Both require all warps in a warpgroup to execute the same instruction,
3063 // and require __launch_bounds__(N, 1) so the compiler permits up to 256
3064 // regs/warp (otherwise the hint may have no effect). Supported on sm_90a/sm_120a.
3065 // __forceinline__ is mandatory: without it ptxas drops setmaxnreg with
3066 // warning C7506 "ignored to maintain compatibility into 'extern' call",
3067 // because a non-inlined call boundary forces a fixed register convention.
3068 //
3069 // On sm_120a (Blackwell, CUDA 13.2) ptxas drops setmaxnreg with C7506 even
3070 // when the PTX is fully inlined (no call.uni), because ptxas treats
3071 // cp.async.bulk.tensor (TMA) usage as an implicit extern-call boundary.
3072 // sm_90a (Hopper) is unaffected. Gate the *call sites* with
3073 // NOTES_V2_ENABLE_SETMAXNREGS so sm_120a builds stay warning-free and avoid
3074 // the register-allocation side effects of __launch_bounds__(N,1) until ptxas
3075 // is fixed. The function templates themselves are always defined.
3076 #if defined(NOTES_V2_ENABLE_SETMAXNREGS)
3077     #define NOTES_V2_REG_DEALLOC(N) warpgroup_reg_dealloc<N>()
3078     #define NOTES_V2_REG_ALLOC(N) warpgroup_reg_alloc<N>()
3079 #else
3080     #define NOTES_V2_REG_DEALLOC(N) ((void)0)
3081     #define NOTES_V2_REG_ALLOC(N) ((void)0)
3082 #endif
3083
3084 template <uint32_t kNumRegs>
3085 __device__ __forceinline__ void warpgroup_reg_dealloc()
3086 {

```

```

3087     asm volatile("setmaxnreg.dec.sync.aligned.u32 %0;\n" : : "n"(kNumRegs));
3088 }
3089
3090 template <uint32_t kNumRegs>
3091 __device__ __forceinline__ void warpgroup_reg_alloc()
3092 {
3093     asm volatile("setmaxnreg.inc.sync.aligned.u32 %0;\n" : : "n"(kNumRegs));
3094 }
3095 #endif
3096
3097 #if defined(NOTES_V2_ENABLE_WGMMMA)
3098 // =====
3099 // Phase 7d: HGEMM WGMMMA — m64n128k16 + TMA + Warp Specialization (Hopper)
3100 // =====
3101 // 面试要点 (WGMMMA vs MMA 对比) :
3102 //   - MMA: warp 级 (32 threads) , 同步执行
3103 //   - WGMMMA: warpgroup 级 (128 threads = 4 warps) , 异步执行 (fire-and-forget)
3104 //   - m64n128k16: M=64, N=128, K=16 → 一次处理 64×128×16=131K 个乘加 (MMA m16n8k16的 4×16=64倍)
3105 //   - TMA (Tensor Memory Accelerator): 硬件 DMA, 2D 寻址, 零寄存器开销
3106 //   - Warp Specialization: Producer 做 TMA, Consumer 做 WGMMMA, 通过 barrier 同步
3107 //   - 128B swizzle: shared memory 的 128B swizzle 模式, 避免 bank conflict
3108
3109 // ---- WGMMMA 辅助函数 ----
3110 // wgmma.commit_group / wait_group 语义 (PTX ISA §9.7.15.7) :
3111 //   - commit_group: 将此前所有未提交的 wgmma.mma_async 归入一个新的 wgmma-group
3112 //     (per-warpgroup, 故需 .sync.aligned 确保所有线程同步执行) 。
3113 //   - wait_group N: 阻塞直到最多 N 个 wgmma-group 尚未完成 (pending ≤ N) 。
3114 //     N=0 → 等所有 group 完成。★ 与 cp.async.wait_group 语义一致 (PTX ISA §9.7.9.25.3.3) ,
3115 //     都是 "wait until only N or fewer groups are pending"。
3116 #define WGMMMA_FENCE() asm volatile("wgmma.fence.sync.aligned;\n" ::: "memory")
3117 #define WGMMMA_COMMIT_GROUP() \
3118     asm volatile("wgmma.commit_group.sync.aligned;\n" ::: "memory") \
3119 #define WGMMMA_WAIT_GROUP(n) \
3120     asm volatile("wgmma.wait_group.sync.aligned %0;\n" :: "n"(n) : "memory")
3121
3122 // SMEM_DESC_ENCODE: 将 byte 偏移编码为 WGMMMA descriptor 位域中的 14-bit 值。
3123 // 编码规则 (PTX ISA §9.7.15.5.1.2.2) :
3124 //   encoded = (x & 0x3FFFF) >> 4
3125 // 其中 0x3FFFF = 2^18 - 1 = 262143, 只保留低 18 bit。
3126 // >>4 是因为 shared memory 地址/偏移必须 16B 对齐 (2^4) , 低 4 bit 恒为 0,
3127 // 可省略以扩大可表示范围。
3128 // 解码: decoded = encoded << 4 (回到原始 byte 值) 。
3129 #define SMEM_DESC_ENCODE(x) (((uint64_t)(x)) & 0x3FFFF) >> 0x4
3130
3131 // make_smem_desc: 构造 WGMMMA (warpgroup MMA) 的 64-bit shared memory 矩阵描述符。
3132 // 硬件 WGMMMA (如 wgmma.mma_async.m64n128k16) 需要 descA/descB 两个 64-bit 描述符,
3133 // 来定位 shared memory 中的 A/B tile 并告知 swizzle 模式。
3134 //
3135 // 64-bit descriptor 位域布局 (参见 PTX ISA §9.7.15.5.1.2.2 & CUTLASS GmmaDescriptor) :
3136 // -----
3137 // | 位域          | 范围      | 大小 | 含义
3138 // |-----|-----|-----|-----|
3139 // | start_address  | [0, 14)   | 14   | smem 基址编码
3140 // | (unused)       | [14, 16)  | 2    | -
3141 // | leading_byte_offset | [16, 30) | 14   | 主要维度 (K) 的字节步长; swizzle 下硬件未使用 (assumed=1)
3142 // | (unused)       | [30, 32)  | 2    | -
3143 // | stride_byte_offset | [32, 46) | 14   | 跨越维度 (M/N) 的字节步长: K-Major 下为 8-row stripe 间距
3144 // | (unused)       | [46, 49)  | 3    | -
3145 // | base_offset    | [49, 52)  | 3    | swizzle pattern 偏移
3146 // | (unused)       | [52, 62)  | 10   | -
3147 // | layout_type    | [62, 64)  | 2    | swizzle 模式
3148 // -----
3149 // layout_type: 0=None, 1=128B swizzle, 2=64B swizzle, 3=32B swizzle

```

```

3150 //
3151 // K-Major + 128B swizzle + half(16-bit) 的几何推导 (PTX ISA §9.7.15.5.1.2) :
3152 // - 128B swizzle atom 在 128-bit 单位下为 8×8
3153 // - 归一化到 half(2B) 后: atom 覆盖 8×(128/16)=64 个 K 方向元素 × 8 个 M 方向元素
3154 // - 即每个 atom = 64(K) × 8(M) 个 half = 64×8×2 = 1024 bytes
3155 // - 这是 stride_byte_offset=1024 的来源
3156 //
3157 // - leading_byte_offset 在 K-Major swizzle 下 unused (hardware assumes 1) ,
3158 // 此处填 16 仅为占位, 编码后为 1。
3159 //
3160 // - base_offset=0 (bits [49,52] 未显式置位) , 表示 smem ptr 已 1024B 对齐。
3161 // 若未对齐需计算 (pattern_start_addr >> 7) & 0x7。
3162 //
3163 // 参考: CUTLASS GmmaDescriptor (cute/arch/mma_sm90_desc.hpp)
3164 // 注: descA/descB 共用此函数; 对 B 而言物理存储为 [BN, BK], 详见 WgmmaSMem 注释。
3165 __device__ inline uint64_t make_smem_desc(half *ptr) {
3166 // __cvta_generic_to_shared: 将通用地址空间中的指针转换为 shared memory 地址
3167 // (一个 32-bit 的 smem byte 偏移量, 相对于当前 CTA 的 shared memory 基址)。
3168 uint32_t addr = static_cast<uint32_t>(__cvta_generic_to_shared(ptr));
3169
3170 // 从全零开始: 所有位域初始化为 0。
3171 uint64_t desc = 0x0000000000000000;
3172
3173 // — bits [0, 14): start_address —
3174 // SMEM_DESC_ENCODE(addr) 将 addr 右移 4 位 (丢弃低 4 个 0 bit) ,
3175 // 只保留 14 位。硬件解码时左移 4 位恢复完整的 16B-aligned 地址。
3176 // 可寻址范围: 2^(14+4) = 2^18 = 256K 个 half = 512 KB 共享内存。
3177 desc |= SMEM_DESC_ENCODE(addr);
3178
3179 // — bits [16, 30): leading_byte_offset —
3180 // 原始值 16 (字节), 编码后为 (16 & 0x3FFFF) >> 4 = 1。
3181 // << 16 将编码值放到 bits [16, 30)。
3182 // K-Major + 128B swizzle 下该字段硬件未使用 (assumed to be 1, 参见 PTX ISA §9.7.15.5.1.2.1.1) ,
3183 // 此处填 16 仅为占位, 使编码值为 1 满足硬件预期。
3184 // 若为 MN-Major 或 INTERLEAVE 布局, LBO 有实际含义:
3185 // 对 INTERLEAVE: 同一 8×2 brick 内第一列到第二列的字节偏移。
3186 // 对 MN-Major swizzle: 偏移从首 (swizzle-byte-size/16) 行到下一组行。
3187 desc |= SMEM_DESC_ENCODE((uint64_t)16) << 16;
3188
3189 // — bits [32, 46): stride_byte_offset —
3190 // 原始值 1024 (字节), 编码后为 (1024 & 0x3FFFF) >> 4 = 64。
3191 // << 32 将编码值放到 bits [32, 46)。
3192 // K-Major 下定义为"从首 8 行到次 8 行的字节偏移" (PTX ISA §9.7.15.5.1.2.1.2) 。
3193 // 1024 的来源 (K-Major + 128B swizzle + half) :
3194 // 一个 128B swizzle atom 覆盖 64(K) × 8(M) 个 half。
3195 // 从 1 个 8-row stripe 到下一个 stripe 需要跨越 8 × 64 × 2 = 1024 bytes。
3196 // 若为 MN-Major: SB0 是从首列到下一列的偏移 (8 列一组) 。
3197 desc |= SMEM_DESC_ENCODE((uint64_t)1024) << 32;
3198
3199 // — bits [62, 64): layout_type (swizzle mode) —
3200 // 1llu << 62 = 二进制 01 在 bits [62, 64) = layout_type = 1。
3201 // 1 = 128B swizzle mode。
3202 // 其他取值: 0=None, 2=64B swizzle, 3=32B swizzle。
3203 // 128B swizzle 将 smem 中连续的行按 128B 粒度进行 XOR 重映射,
3204 // 确保同一 warp 内各线程访问不同 bank 的同一偏移时不会产生 bank conflict。
3205 desc |= 1llu << 62;
3206
3207 return desc;
3208 }
3209
3210 // ---- WGMMMA PTX 指令宏 ----
3211 // wgmma.mma_async.sync.aligned.m64n128k16.f16.f16.f16
3212 // m64n128k16: M=64, N=128, K=16。一次 WGMMMA 计算 D[64×128] += A[64×16] × B[16×128]。

```

```

3213 // f16.f16.f16: A=f16, B=f16, D=f16 (累加器也是 f16 精度)。
3214 // 每线程 32 个 uint32 输出: D=64×128=8192 half, warpgroup 128 线程分担,
3215 // 每线程 64 half = 32 uint32。
3216 // descA/descB: shared memory 描述符 (由 make_smem_desc 生成, 64-bit 编码)。
3217 // Scaled: 0=清零累加器再写入, 1=与累加器中的旧值累加 (K 维迭代必须用 1)。
3218 // ScaleA/ScaleB: 1=正常符号, -1=翻转符号 (此处始终用 1)。
3219 // TransA/TransB: 0=K-major (K 维连续), 1=MN-major (M/N 维连续)。本实现始终用 0。
3220 // 参考: PTX ISA §9.7.15.4 (wgmma.mma_async)
3221 #define WGMMA_M64N128K16_F16F16F16(d, sA, sB, Scaled, ScaleA, ScaleB, TransA, \
3222                                     TransB) \
3223 { \
3224     uint64_t desc_a = make_smem_desc(&(sA)[0]); \
3225     uint64_t desc_b = make_smem_desc(&(sB)[0]); \
3226     asm volatile( \
3227         "{\n" \
3228         "wgmma.mma_async.sync.aligned.m64n128k16.f16.f16.f16 " \
3229         "{%0,  %1,  %2,  %3,  %4,  %5,  %6,  %7,  " \
3230         " %8,  %9,  %10, %11, %12, %13, %14, %15, " \
3231         " %16, %17, %18, %19, %20, %21, %22, %23, " \
3232         " %24, %25, %26, %27, %28, %29, %30, %31}, " \
3233         " %32, " \
3234         " %33, " \
3235         " %34, %35, %36, %37, %38;\n" \
3236         "}\n" \
3237         : "+r"((d)[0][0]), "+r"((d)[0][1]), "+r"((d)[0][2]), "+r"((d)[0][3]), \
3238         "+r"((d)[1][0]), "+r"((d)[1][1]), "+r"((d)[1][2]), "+r"((d)[1][3]), \
3239         "+r"((d)[2][0]), "+r"((d)[2][1]), "+r"((d)[2][2]), "+r"((d)[2][3]), \
3240         "+r"((d)[3][0]), "+r"((d)[3][1]), "+r"((d)[3][2]), "+r"((d)[3][3]), \
3241         "+r"((d)[4][0]), "+r"((d)[4][1]), "+r"((d)[4][2]), "+r"((d)[4][3]), \
3242         "+r"((d)[5][0]), "+r"((d)[5][1]), "+r"((d)[5][2]), "+r"((d)[5][3]), \
3243         "+r"((d)[6][0]), "+r"((d)[6][1]), "+r"((d)[6][2]), "+r"((d)[6][3]), \
3244         "+r"((d)[7][0]), "+r"((d)[7][1]), "+r"((d)[7][2]), "+r"((d)[7][3]) \
3245         : "l"(desc_a), "l"(desc_b), "n"(int32_t(Scaled)), \
3246         "n"(int32_t(ScaleA)), "n"(int32_t(ScaleB)), "n"(int32_t(TransA)), \
3247         "n"(int32_t(TransB))); \
3248 } \
3249
3250 // ---- TMA Shared Memory Layout ----
3251 // Multi-stage pipeline: kStages 个 stage, 每个 stage 存储 A[BM×BK] + B[BK×BN]
3252 //
3253 // 每个 stage 包含两块 smem:
3254 //   A tile: [BM, BK] row-major → BK×BM 个 half, 地址连续
3255 //   B tile: TMA 按 [BN, BK] 物理写入 (详见下方 ★), BN×BK 个 half
3256 //
3257 // ★ 关键区别 — WGMMA 的 B 物理存储 vs 逻辑视图:
3258 //
3259 //   上层数据: 源矩阵 BT [N, K] row-major (TN 布局, B 的列以行优先形式存储)。
3260 //   物理存储: TMA 将 BT 的 tile 写入 smem, 物理布局为 [BN, BK] row-major
3261 //               (BN=128 行, BK=64 列, 每行 64 个连续的 half)。
3262 //   TMA 的 smem_box_shape = (BK=64, BN=128) 定义了每个 tile 的 shape,
3263 //   TMA 按行写入, 行内 BK 个 half 连续 → 物理上就是 [BN][BK]。
3264 //   逻辑视图: WGMMA 指令 (wgmma.mma_async.m64n128k16, PTX ISA §9.7.15.4)
3265 //               通过 imm-trans-b=0 指定 B 为 K-major, 即逻辑上按 [BK, BN] 寻址。
3266 //               实际 [BN, BK] → 逻辑 [BK, BN] 的"转置"是通过 descB 中的 128B
3267 //               swizzle (layout_type=1) 在硬件地址重映射层完成的, 无需软件转置。
3268 //
3269 //   stride_byte_offset=1024 的来源 (K-major + 128B swizzle + f16, PTX ISA §9.7.15.5.1.2.1.2) :
3270 //       128B swizzle atom = 8(N) × 8(K) × 128-bit = 8 rows × 64 个 half。
3271 //       stride_byte_offset = 从当前 8-row stripe 到下一个 8-row stripe 的字节偏移
3272 //           = 8 rows × (BK=64) halves × 2 bytes = 1024。
3273 //       这个值恰好等于物理 [BN, BK] 布局中连续 8 行的跨度 ← 进一步证实物理存储是 [BN, BK]。
3274 //
3275 //   与 MMA(TN) 的对比:

```

```

3276 // MMA (TN): smem 存 B^T [N, K] row-major, ldmatrix 按列解出 col-major B fragment。
3277 // WGMMMA: smem 物理存 [BN, BK] (TMA 直写), WGMMMA 通过 swizzle 硬件以 K-major [BK, BN] 逻辑视图读取。
3278 // 简言之: swizzle 桥接了 "物理 row-major [BN, BK]" 和 "逻辑 K-major [BK, BN]" 的差异。
3279 template <int BM, int BN, int BK, int QSIZE> struct WgmmaSMem {
3280     alignas(128) half A[BM * BK * QSIZE]; // A tile: row-major [BM, BK]
3281     // B tile 标记:
3282     // - 物理存储: TMA 从 B^T[N,K] row-major 搬入, 物理上是 [BN, BK] row-major (BN 行, BK 列)
3283     // - 逻辑视图: WGMMMA 通过 descB 的 128B swizzle (PTX ISA §9.7.15.5.1.2) 将地址重映射,
3284     //   以 K-major [BK, BN] 逻辑视图供 wgmma.mma_async (imm-trans-b=0, PTX ISA §9.7.15.4) 读取
3285     // - B[BN * BK * QSIZE] 与 B[BK * BN * QSIZE] 数值等价 (BN×BK = BK×BN),
3286     //   但写 BN×BK 更能体现物理存储形态
3287     alignas(128) half B[BN * BK * QSIZE];
3288 };
3289
3290 // ---- WGMMMA Kernel: Warp Specialization + TMA ----
3291 // 面试重点 — Warp Specialization (Hopper 最核心的编程模型变化):
3292 //
3293 // 背景: 传统 GEMM kernel 中, 所有线程同步地"加载→计算→存回",
3294 // 数据搬运和计算无法重叠。Hopper 的 TMA + WGMMMA 支持将工作拆分为两个
3295 // warpgroup, 让数据搬运和矩阵乘完全异步执行:
3296 //
3297 // WG0 (128 threads, Producer): 仅 thread 0 提交 TMA 2D 拷贝指令,
3298 //   将 A/B tile 从 HBM 搬到 shared memory。
3299 // WG1 (128 threads, Consumer): 所有 128 个线程参与 WGMMMA 矩阵乘。
3300 //
3301 // Producer 和 Consumer 通过 cuda::barrier (CTA 级别) 同步:
3302 // - full[stage]: Producer 发信号表示 stage stage 的数据已就绪, 可被使用
3303 // - empty[stage]: Consumer 发信号表示 stage stage 的使用完毕, 可以被覆盖
3304 //
3305 // 本节重点理解:
3306 // 1) 为什么 Producer 只需要 thread 0? TMA 是硬件 DMA 指令, 一次提交
3307 //   即可搬运整个 2D tile, 无需所有线程参与。
3308 // 2) barrier 的 arrive count = 129: 128 个 Consumer 线程 + 1 个 Producer 提交线程。
3309 // 3) 多 stage pipeline 使 Consumer 计算 stage stage 的同时,
3310 //   Producer 可以搬运 stage (stage+1), 隐藏 HBM→SMEM 延迟。
3311 //
3312 // Tile Hierarchy (与 MMA m16n8k16 kernel 对比):
3313 // WGMMMA Atom: m64n128k16 (一次处理 64×128×16, 是 MMA 的 64 倍)
3314 // K Tile: BK=64, 每个 K tile 包含 BK/kWgmmaK=4 个 WGMMMA atom
3315 // M Tile: BM=128, 每个 M tile 包含 BM/kWgmmaM=2 个 WGMMMA atom
3316 // N Tile: BN=128, 单个 kWgmmaN=128 即可覆盖, 无需在 N 方向分块
3317 // Block Tile: C[128,128] = A[128,64] × B[64,128]
3318 // Threads: 256 = 2 warpgroups × 128 threads/warpgroup
3319 // Producer(WG0): 128 threads (仅 thread 0 做 TMA 提交)
3320 // Consumer(WG1): 128 threads = 4 warps (全部参与 WGMMMA 和写回)
3321 //
3322 // kStages=3: 3 级流水线。Consumer 滞后 Producer 最多 2 步, 确保 HBM→SMEM
3323 // 的延迟被计算完全掩盖。
3324 //
3325 // Grid: ((N+127)/128/S, (M+127)/128, S), S=(N+2047)/2048, 3D block swizzle
3326 // - grid.z = S 个 swizzle 分区, 将连续 block 打散到不同 N 区域改善 L2 命中
3327 // Block: (256, 1, 1), 2 warpgroups
3328 // source: LeetCUDA/kernels/hgemm/wgmma/hgemm_wgmma_fp16acc_stages_tn.cu
3329 template <const int kWgmmaM = 64, // WGMMMA atom M dim (m64n128k16)
3330           const int kWgmmaN = 128, // WGMMMA atom N dim
3331           const int kWgmmaK = 16, // WGMMMA atom K dim
3332           const int BM = 128, // block tile M, 2 × kWgmmaM
3333           const int BN = 128, // block tile N, 1 × kWgmmaN
3334           const int BK = 64, // block tile K, 4 × kWgmmaK
3335           const int kNumThreads = 256, // 2 warpgroups × 128 threads
3336           const int kStages = 3, // TMA pipeline depth (full/empty barriers)
3337           const int kBlockSwizzle = 0> // 1 enables 3D grid swizzle for L2 locality
3338 __global__ void __launch_bounds__(kNumThreads, 1)

```

```

3339     hgemm_wgmma_stages_tn(
3340         int M, int N, int K, half *C,
3341         const CUtensorMap *__restrict__ tensorMapA,
3342         const CUtensorMap *__restrict__ tensorMapB) {
3343     static_assert(kBlockSwizzle == 0 || kBlockSwizzle == 1, "kBlockSwizzle must be 0 or 1");
3344     // 注意: tensorMapA/tensorMapB 需要由 host 侧按当前 tile 布局预先创建;
3345     // 对 row-major [M, K] 矩阵, TMA shape 参数写的是 (K, M) 而不是 (M, K),
3346     // 也就是TMA descriptor中把连续的维度写在最内层, 非连续的维度写在最外层。
3347     // 这是 TMA descriptor 最容易背错的地方之一。notes 这里只保留 kernel 主体,
3348     // 不展开宿主侧 create_tensor_map 细节。
3349
3350     // Block Swizzle: 在 grid x 维度做 swizzle, 改善 L2 cache 局部性
3351     // bx = blockIdx.z * gridDim.x + blockIdx.x, 将相邻 block 打散到不同 N 区域
3352     const int bx = ((int)kBlockSwizzle) * blockIdx.z * gridDim.x + blockIdx.x;
3353     const int by = blockIdx.y;
3354     constexpr int kConsumerThreads = kNumThreads / 2; // 128 threads = 1 warpgroup
3355     // kNumConsumers = (kNumThreads / 128) - 1 = 2 - 1 = 1 (1 个 consumer WG)
3356     // 这里的 -1 是因为 2 个 warpgroup 中 1 个是 producer, 剩余都是 consumer
3357     constexpr int kNumConsumers = (kNumThreads / kConsumerThreads) - 1; // 1 consumer WG
3358     // kWarpgroupM = BM / kNumConsumers: 每个 consumer warpgroup 负责的 M 行数
3359     // 当只有一个 consumer 时, 它负责全部 BM=128 行
3360     constexpr int kWarpgroupM = BM / kNumConsumers; // 128
3361
3362     // 边界检查: 确保当前 block 不超出 M/N 范围
3363     if (bx >= div_ceil(N, BN) || by >= div_ceil(M, BM))
3364         return;
3365
3366     // ---- Shared Memory 分配 ----
3367     // 动态 shared memory (由 host 侧通过 kernel launch 的 smem 参数指定大小)
3368     // TMA + WGMMMA 仅需 __align__(128), 而非 __align__(1024)。原因:
3369     // make_smem_desc() 构造 WGMMMA descriptor 时, start_address 字段
3370     // 完整编码了 __cvta_generic_to_shared 的绝对 32-bit 偏移量,
3371     // 硬件根据该绝对地址自动推导并补偿 swizzle phase 偏移 (等价于
3372     // descriptor 内置的 base_offset 机制)。因此即使 smem 基址未落到
3373     // 1024B 边界, WGMMMA 硬件也能正确读取 TMA 写入的数据。
3374     // 对比 hgemm_tma_mma_ws_tn: 消费者使用 ldmatrix + 手写 swizzle<64>()
3375     // , 该函数无法感知 smem 绝对地址, 硬编码了 phase=0 的假设, 因此必须
3376     // __align__(1024) 来保证 phase 确实为 0。从健壮性角度看本 kernel 也
3377     // 应该用 __align__(1024); 这里使用 128 也可以正常运行。
3378     extern __shared__ __align__(1024) uint8_t smem_tma_wgmma_ws[];
3379     WgmmaSMem<BM, BN, BK, kStages> &s =
3380         *reinterpret_cast<WgmmaSMem<BM, BN, BK, kStages>*>(smem_tma_wgmma_ws);
3381     half *s_a = s.A;
3382     half *s_b = s.B;
3383
3384     // ---- cuda::barrier 初始化 ----
3385     //
3386     // ★ Barrier 机制速查 (arrive / wait 核心语义):
3387     //
3388     // cuda::barrier 是一个 CTA 级同步原语, 基于 **phase (奇偶交替)** 工作:
3389     //
3390     // init(bar, N): 设置 arrive_count = N。
3391     // 含义: 每 phase 需要 N 个线程各调用一次 arrive(), barrier 才翻转 phase。
3392     //
3393     // arrive(): 线程声明"我已到达此 barrier 点"。
3394     // - 非阻塞, 立即返回一个 token (包含当前 phase 值)。
3395     // - 不关心"谁"到达, 只关心"到达次数"是否达到 arrive_count。
3396     //
3397     // wait(token): 线程阻塞, 直到 barrier 的当前 phase ≠ token 中的 phase。
3398     // - 即: 等待 phase 翻转。
3399     // - Phase 翻转条件: (1) arrive 次数达到 arrive_count
3400     // (2) 若使用了 barrier_arrive_tx, 还需 TMA 字节全部写完
3401     //

```

```

3402 // 典型用法: b.wait(b.arrive())
3403 // - arrive() 注册自己到达, 拿到 token (当前 phase = P)
3404 // - wait(token) 阻塞直到 phase 翻转 (P → P+1)
3405 // - 如果自己是第 N 个到达者 → phase 立即翻转 → wait 立即返回 (不阻塞)
3406 // - 如果自己不是最后一个 → wait 阻塞, 直到第 N 个到达者触发翻转
3407 //
3408 // ★ 本 kernel 的 Pipeline 同步协议 (kStages=3, arrive_count=129):
3409 //
3410 // Producer (1 thread)          Consumer (128 threads)
3411 // -----
3412 //                                C0: arrive(empty[*]) ×128 [init: 标记所有 stage 为空]
3413 // ┌ for each k_tile: ─┐          ┌ for each k_tile: ─┐
3414 // | P1: arrive+wait(empty[q]) |   | C1: arrive+wait(full[q]) |
3415 // |   ↓ 等 stage q 被消费完   |   |   ↓ 等 TMA 数据就绪   |
3416 // | P2: TMA(A[q]) + TMA(B[q]) |   | C2: WGMMMA 计算       |
3417 // |   ↓ 异步拷贝             |   | C3: wait WGMMMA 完成    |
3418 // | P3: arrive_tx(full[q])    |   | C4: arrive(empty[q])    |
3419 // |   ↑ 通知: stage q 数据就绪 |   |   ↑ 通知: stage q 已消费完 |
3420 // └──────────────────┘          └──────────────────┘
3421 //
3422 // 关键不变式 (invariant):
3423 // - Producer 不会覆盖 Consumer 正在读的 stage
3424 // - Consumer 不会读 Producer 还没写完的 stage
3425 // - 通过 full/empty 两个 barrier 的 phase 交替来保证
3426 //
3427 // 每个 stage 有两个 barrier:
3428 //   full[stage]: TMA 数据就绪信号。Producer 发 (arrive_tx), Consumer 等 (wait)。
3429 //   empty[stage]: Stage 空闲信号。Consumer 发 (arrive), Producer 等 (wait)。
3430 //
3431 // arrive_count = 128 (consumer) + 1 (producer) = 129:
3432 // 每 phase, 128 个 consumer 线程 + 1 个 producer 线程都要 arrive 一次。
3433 #pragma nv_diag_suppress static_var_with_dynamic_init
3434 __shared__ cuda::barrier<cuda::thread_scope_block> full[kStages];
3435 __shared__ cuda::barrier<cuda::thread_scope_block> empty[kStages];
3436 #pragma nv_diag_default static_var_with_dynamic_init
3437
3438 // K 方向总 tile 数。要求 K 能被 BK 整除, 否则尾 tile 被丢弃。
3439 const int NUM_K_TILES = div_ceil(K, BK);
3440 const int wg_idx = threadIdx.x / kConsumerThreads; // 0=Producer, 1=Consumer
3441 const int wg_tid = threadIdx.x % kConsumerThreads; // 0~127 within warpgroup
3442
3443 // 初始化 barriers (仅 thread 0 执行)
3444 if (threadIdx.x == 0) {
3445     for (int i = 0; i < kStages; ++i) {
3446         // arrive_count = 129: 128 consumer + 1 producer
3447         // init 是 CUDA barrier 的 hidden friend 函数 (定义在 cuda::barrier 类体内),
3448         // 只能通过 ADL (Argument-Dependent Lookup) 调用。因为第一个参数 &full[i] 的类型是
3449         // cuda::barrier<cuda::thread_scope_block>*, 编译器通过 ADL 在 cuda 命名空间中自动找到它。
3450         init(&full[i], kConsumerThreads + 1); // 128 consumer + 1 producer
3451         init(&empty[i], kConsumerThreads + 1); // same
3452     }
3453     // fence_proxy_async_shared_cta: 确保 barrier 在 smem 中的初始化
3454     // 对 async proxy (TMA/WGMMMA) 可见。
3455     tma_fence_proxy_async_shared_cta();
3456 }
3457 __syncthreads();
3458
3459 // =====
3460 // ★ Warp Specialization 并发模型 — 为什么只有 if/else 没有 while?
3461 // =====
3462 //
3463 // 256 个线程在同一个 SM 上**并发执行**。if/else 只是分工, 不是先后顺序:
3464 // - WG0 (threadIdx.x 0~127): 全部走 if 分支, 做 Producer。

```

```

3465 // - WG1 (threadIdx.x 128~255): 全部走 else 分支, 做 Consumer。
3466 //
3467 // SM 的 warp scheduler 会交替调度来自 WG0 和 WG1 的 warp, 两者**同时**推进:
3468 // Producer: for (k_iter = 0 .. NUM_K_TILES) { P1→P2→P3 } ← 自己的循环
3469 // Consumer: for (k_iter = 0 .. NUM_K_TILES) { C1→C2→C3→C4 } ← 自己的循环
3470 //
3471 // 两个 warpgroups 各自独立迭代 K-tiles, 通过 full[] / empty[] barrier 同步:
3472 // - Producer 领先 Consumer 太多 → wait(empty[q]) 阻塞, 等 Consumer 消费完
3473 // - Consumer 领先 Producer 太多 → wait(full[q]) 阻塞, 等 TMA 搬运完
3474 //
3475 // 这是生产者-消费者模型的 GPU 实现: 不需要共享循环变量, barrier 的 phase
3476 // 交替天然保证了流水线串行化 (at most kStages-1 steps ahead)。
3477 // =====
3478 // Producer Warpgroup (WG0, threadIdx.x 0~127)
3479 // 职责: 提交 TMA 2D 拷贝, 将 A/B tile 从 HBM 异步搬运到 SMEM。
3480 // 只有 wg_tid==0 执行实际拷贝提交, 其余 127 个线程空闲。
3481 // =====
3482 if (wg_idx == 0) {
3483     NOTES_V2_REG_DEALLOC(40);
3484     if (wg_tid == 0) {
3485         // stage: 当前操作的 stage 索引 (round-robin 0 -> 1 -> 2 -> 0 -> ...)
3486         int stage = 0;
3487         for (int k = 0; k < NUM_K_TILES; ++k, ++stage) {
3488             if (stage == kStages)
3489                 stage = 0;
3490
3491             // Step P1: 等待 stage stage 变为"空" (可被覆盖写入)
3492             //
3493             // 模式 empty[stage].wait(empty[stage].arrive()):
3494             // - arrive(): Producer (1 个线程) 在 empty barrier 上注册到达,
3495             //   获得当前 phase 的 token。如果 Consumer 已经在 C4 步骤积累了
3496             //   128 次 arrive (来自上一轮迭代或 C0 初始化), 则加上这次共 129 次
3497             //   → phase 立即翻转。
3498             // - wait(token): 阻塞直到 barrier phase ≠ token 中的 phase。
3499             //   由于 arrive() 是第 129 次到达, phase 在 arrive 时已翻转,
3500             //   wait 看到 phase 已变 → 立即返回 (首轮不阻塞)。
3501             //
3502             // 语义: Producer 说"我准备好写 stage stage 了, Consumer 用完没有? "
3503             //   如果 Consumer 还没用完 → phase 未翻转 → wait 阻塞等待。
3504             //   如果 Consumer 已用完 → phase 已翻转 → wait 立即返回。
3505             empty[stage].wait(empty[stage].arrive());
3506
3507             // Step P2: 提交 TMA 2D 拷贝指令
3508             // cp_async_bulk_tensor_2d_global_to_shared:
3509             //   参数1(dst): smem 目标地址 (当前 stage 的 A/B tile 起始位置)
3510             //   参数2(tensorMap): Host 预创建的 TMA descriptor (描述源矩阵的 shape/stride/dtype)
3511             //   参数3/4(coords): (k_offset, m_offset) 或 (k_offset, n_offset) 的全局坐标
3512             //   参数5(barrier): 拷贝完成后自动 arrive 到此 barrier
3513             //
3514             // TMA 是硬件 DMA 引擎: 一次指令提交即可搬运整个 2D tile,
3515             // 无需线程逐元素搬运, 零寄存器开销。
3516             // TMA 2D 加载 A tile: coords = (k_offset, m_offset)
3517             tma_load_2d(&s_a[stage * BK * BM], tensorMapA, k * BK, by * BM,
3518                 full[stage]);
3519
3520             // TMA 2D 加载 B tile: coords = (k_offset, n_offset)
3521             tma_load_2d(&s_b[stage * BK * BN], tensorMapB, k * BK, bx * BN,
3522                 full[stage]);
3523
3524             // Step P3: 通知 Consumer: stage stage 的 TMA 数据已就绪
3525             //
3526             // barrier_arrive_tx(bar, arrive_count_update, byte_count):
3527             // - 在 bar 上注册 1 次到达 (arrive_count_update=1)

```

```

3528 // - 同时声明预期有 byte_count 字节将通过 async copy (TMA) 写入 smem
3529 // - Phase 翻转条件:
3530 //     (a) 总 arrive 次数达到 129 (128 consumer + 1 producer)
3531 //     (b) 所有声明的 async 字节已写入 smem
3532 // 两个条件都满足后 phase 才翻转, Consumer 的 wait() 才返回。
3533 // - 即: Consumer 不会在 TMA 数据完整到达前就开始读 smem。
3534 tma_arrive_expect_tx(full[stage], (BK * BN + BK * BM) * sizeof(half));
3535 }
3536 }
3537 }
3538 // =====
3539 // Consumer Warpgroup (WG1, threadIdx.x 128~255)
3540 // 职责: 等待 TMA 数据就绪 -> 发射 WGMMMA 做矩阵乘 -> 积累结果 -> 写回 C
3541 // 所有 128 个线程 (4 warps) 全部参与。
3542 // =====
3543 else {
3544     // Step C0: Consumer 初始化 — 标记所有 stage 为"空" (可被 Producer 写入)
3545     //
3546     // 所有 128 个 Consumer 线程对每个 stage 的 empty barrier 调用 arrive()。
3547     // 这是 Pipeline 的"预热"步骤——没有它, Producer 的 empty[stage].wait()
3548     // 在第一轮会永远阻塞 (因为 Producer 的 1 次 arrive 不足以凑够 129)。
3549     //
3550     // 注意: 此时每个 empty[i] 只有 128 次 arrive, 未达 129, phase 不翻转。
3551     // Producer 后续的 empty[stage].arrive() 作为第 129 次, 触发 phase 翻转。
3552     NOTES_V2_REG_ALLOC(232);
3553
3554     for (int i = 0; i < kStages; ++i) {
3555         [[maybe_unused]] auto token = empty[i].arrive();
3556     }
3557
3558     // 累加器寄存器声明
3559     // d[kWarpgroupM / kWgmmaM][kWgmmaN / 16][4]:
3560     // - d[0][*][*]: M 方向第 1 个 WGMMMA atom (rows 0~63)
3561     // - d[1][*][*]: M 方向第 2 个 WGMMMA atom (rows 64~127)
3562     // - d[*][g][*]: N 方向第 g 组 16 列
3563     // - d[*][*][0..3]: 4 条 uint32 寄存器, 共 8 个 half (覆盖 16x16 子块)
3564     // 每个线程总共 2 * 8 * 4 = 2 * 32 = 64 uint32 = 128 half (每次WGMMMA Atom 32 uint32 输出)
3565     // 128 线程 * 128 half = 16384 half = 128 * 128 = BM*BN (刚好覆盖整个 C tile)
3566     uint32_t d[kWarpgroupM / kWgmmaM][kWgmmaN / 16][4] = {};
3567
3568     int stage = 0;
3569     // K 维外循环: 沿 K tile 迭代 (BK=64, 每个 K tile 做 4 次 WGMMMA 累加)
3570     for (int k = 0; k < NUM_K_TILES; ++k, ++stage) {
3571         if (stage == kStages)
3572             stage = 0;
3573
3574         // Step C1: 等待 TMA 数据就绪 (full 信号)
3575         //
3576         // 模式 full[stage].wait(full[stage].arrive()):
3577         // - 128 个 Consumer 线程各调用 arrive(), 共 128 次到达。
3578         // - 当前 phase 累积: 128/129, phase 尚未翻转。
3579         // - wait(token) 阻塞, 直到 Producer 的 barrier_arrive_tx (Step P3)
3580         //   贡献第 129 次到达 + TMA 字节全部写完 -> phase 翻转 -> wait 返回。
3581         //
3582         // 语义: Consumer 说"我准备好读 stage stage 了, 数据到了没有?"
3583         full[stage].wait(full[stage].arrive());
3584
3585         // Step C2: 发射 WGMMMA 指令序列
3586         //
3587         // WGMMMA 是异步指令 (fire-and-forget), 发射后立即返回, 不等待计算完成。
3588         // 标准流程: FENCE -> 发射 WGMMMA -> COMMIT -> WAIT。
3589         //
3590         // WGMMMA_FENCE (wgmma.fence.sync.aligned) :

```

```

3591 // - 确保 TMA 写入 smem 的数据对 async proxy (WGMMMA) 可见
3592 // - 确保累加器寄存器 d[] 已准备好接收 WGMMMA 输出
3593 // - 本质是 proxy 间的内存序 (memory ordering), 不是传统 __syncthreads
3594 //
3595 // 每个 WGMMMA_M64N128K16_F16F16F16 指令:
3596 // D[64×128] += A[64×16] × B[16×128]
3597 // A/B 均为 K-major (imm-trans=0), 由 descA/descB 描述 smem 中的布局。
3598 // ScaleD=1: 累加。K 维有 BK/kWgmmaK=4 次迭代, 每次都用 ScaleD=1。
3599 WGMMMA_FENCE();
3600
3601 // M 维迭代: BM/kWgmmaM = 128/64 = 2 个 WGMMMA atom
3602 // 每个 atom 处理 64 行 M 方向数据。
3603 #pragma unroll
3604 for (int m = 0; m < kWarpgroupM / kWgmmaM; ++m) {
3605     // wgmma_sA 指向当前 stage 中 A tile 的第 m 个 64*BK 子块
3606     // s_a 布局: [kStages][BM][BK] -> stage * BK*BM + BK * (m*64)
3607     half *wgmma_sA = s_a + stage * BK * BM + BK * m * kWgmmaM;
3608
3609     // K 维迭代: BK/kWgmmaK = 64/16 = 4 次 WGMMMA (累加)
3610     // 每次处理 K=16 维的矩阵乘, 4 次累加后覆盖完整的 BK=64。
3611     #pragma unroll
3612     for (int k_step = 0; k_step < BK / kWgmmaK; ++k_step) {
3613         // 第 k_step 次 WGMMMA:
3614         // A: wgmma_sA + k_step * kWgmmaK (A 的 K 维起始位置)
3615         // B: s_b + stage * BK * BN + k_step * kWgmmaK (B 的 K 维起始位置)
3616         // 注意 B 的 smem 布局是 [BK, BN] row-major, K-major 读取时从
3617         // 第 k_it*16 行开始取 16 行, 每行 BN=128 列。
3618         // ScaleD=1: 累加 (K 维迭代需要累积结果)
3619         WGMMMA_M64N128K16_F16F16F16(d[m], wgmma_sA + k_step * kWgmmaK,
3620                                     s_b + stage * BK * BN + k_step * kWgmmaK,
3621                                     1, // ScaleD=1: accumulate (不清零)
3622                                     1, 1, 0, 0);
3623     }
3624 }
3625
3626 // Step C3: 提交并等待 WGMMMA 完成
3627 //
3628 // WGMMMA_COMMIT_GROUP (wgmma.commit_group.sync.aligned):
3629 // 将自上一次 COMMIT 以来发射的所有 WGMMMA 归为一组, 提交到 async proxy 执行。
3630 // 类比 cp.async.commit_group, 但 scope 是 warpgroup 而非 per-thread。
3631 //
3632 // WGMMMA_WAIT_GROUP(0) (wgmma.wait_group.sync.aligned 0):
3633 // 阻塞直到最多 N 个 group 尚未完成 (pending ≤ N)。N=0 即等待所有 group。
3634 // ★ 语义与 cp.async.wait_group 完全一致 (PTX ISA §9.7.15.7.3 vs §9.7.9.25.3.3) :
3635 // 两者都是 "wait until only N or fewer of the most recent groups are pending"。
3636 // 此处用 0 是因为 Consumer 在本次迭代中只 commit 了 1 个 group,
3637 // 必须等它全部完成才能进入下一步 (读下一 stage 的数据)。
3638 WGMMMA_COMMIT_GROUP();
3639 WGMMMA_WAIT_GROUP(0);
3640
3641 // Step C4: 释放 stage stage — 通知 Producer 可以覆盖写入
3642 //
3643 // 128 个 Consumer 线程在 empty[stage] 上 arrive(), 为 **下一 phase** 积累到达次数。
3644 // 这 128 次 arrive 不会立即翻转 phase (还需 Producer 在下一轮的 1 次 arrive)。
3645 //
3646 // 语义: Consumer 说"stage stage 我已经读完了, 你可以放心覆盖了。"
3647 [[maybe_unused]] auto token = empty[stage].arrive();
3648 }
3649
3650 // =====
3651 // Epilogue: 将寄存器中的累加结果写回 global memory C
3652 // =====
3653 //

```

```

3654 // WGMMMA m64n128k16.f16.f16.f16 的输出寄存器映射:
3655 //
3656 // 对 warpgroup 内每个线程, 输出 64 个 half = 32 个 uint32。
3657 // 这些 half 分布在 d[2][8][4] 数组中:
3658 // d[m_it][g][0]: (row, col) 和 (row, col+2) 位置的 2 个 half
3659 // d[m_it][g][1]: (row+8, col) 和 (row+8, col+2) 位置的 2 个 half
3660 // d[m_it][g][2]: (row, col+8) 和 (row, col+10) 位置的 2 个 half
3661 // d[m_it][g][3]: (row+8, col+8) 和 (row+8, col+10) 位置的 2 个 half
3662 //
3663 // 其中:
3664 // row = warp * 16 + lane / 4 (0~63, 4 warps * 16 rows)
3665 // col = g * 16 + 2 * (lane % 4) (0~126, step 2, 8 g's * 16 cols)
3666 //
3667 // 每个线程覆盖一个 16x16 子块中的 16 个 half:
3668 // +-----+-----+
3669 // | reg[0] | reg[2] | <- rows [row, row+8)
3670 // |(col,+2)|(col+8)|
3671 // +-----+-----+
3672 // | reg[1] | reg[3] | <- rows [row+8, row+16)
3673 // |(col,+2)|(col+8)|
3674 // +-----+-----+
3675 // 每个 reg 包含 2 个连续 half (col 和 col+2), 用 uint32 存储。
3676 //
3677 // 三维遍历:
3678 // m_it=0: rows 0~63, m_it=1: rows 64~127
3679 // g=0..7: 每个覆盖 16 列, 8*16=128 列 ok
3680 // 每个线程写 4 uint32 * 2 half/uint32 * 8 g * 2 m_it = 128 half。
3681 // 128 线程 * 128 half = 16384 half = 128 * 128 ok
3682
3683 const int lane = wg_tid % 32;
3684 const int warp = wg_tid / 32;
3685 // row: 当前线程在当前 WGMMMA atom 中负责的起始行。
3686 // warp 0~3 各负责 16 行: rows [warp*16, warp*16+15]。
3687 // lane/4: 每 4 个连续 lane 负责同一行 (因为 WGMMMA fragment 中每行有 4 个 uint32,
3688 // 分别覆盖列对 {c0,c1}、{c2,c3}、{c8,c9}、{c10,c11}, 由 lane%4 区分)。
3689 // 因此 lane/4 给出该行在 16-row block 内的行号 (0~7 对应 rows 0~7,
3690 // 同一 warp 中 lane/4==0 的有 lane 0,1,2,3, 都负责 row 0)。
3691 const int row = warp * 16 + lane / 4;
3692 // block_C: 当前 C tile 在 global memory 中的起始地址
3693 // by*BM 是 M 方向偏移, bx*BN 是 N 方向偏移
3694 half *block_C = C + by * BM * N + bx * BN;
3695
3696 // 2 * (8 * 4) = 2 * 32 = 64 uint32 = 128 half
3697 // NOTE: 这里可以考虑通过 R->S, S->G 的方式做一次 128 bits写回。
3698 #pragma unroll
3699 for (int m = 0; m < kWarpgroupM / kWgmmaM; ++m) {
3700     int yo = m * kWgmmaM; // M 方向行偏移 (0 或 64)
3701     #pragma unroll
3702     for (int g = 0; g < kWgmmaN / 16; ++g) {
3703         int col = g * 16 + 2 * (lane % 4); // N 方向列偏移 (0,2,4,...,14 then 16,18,...)
3704         // 一次 uint32 store 写入 2 个 half (连续列 col 和 col+2)
3705         // 左上象限: (row, col)
3706         *reinterpret_cast<uint32_t*>(&block_C[(row + yo) * N + col]) = d[m][g][0];
3707         // 左下象限: (row+8, col)
3708         *reinterpret_cast<uint32_t*>(&block_C[(row + yo + 8) * N + col]) = d[m][g][1];
3709         // 右上象限: (row, col+8)
3710         *reinterpret_cast<uint32_t*>(&block_C[(row + yo) * N + col + 8]) = d[m][g][2];
3711         // 右下象限: (row+8, col+8)
3712         *reinterpret_cast<uint32_t*>(&block_C[(row + yo + 8) * N + col + 8]) = d[m][g][3];
3713     }
3714 }
3715 }
3716 }

```

```

3717 #endif /* NOTES_V2_ENABLE_WGMMA */
3718
3719
3720 #if defined(NOTES_V2_ENABLE_TMA_MMA_WS)
3721 // SM120 不支持 WGMMA, 但支持相同的 TMA 生产者协议和 warp 级 mma.sync。
3722 // 保持 128B TMA swizzle 并显式声明物理布局供 ldmatrix 消费者使用。
3723 template <int BM, int BN, int BK, int QSIZE> struct TmaMmaWSSMem {
3724     static_assert(BK == 64, "The 128B swizzle helper below is specialized for BK=64");
3725     half A[BM * BK * QSIZE];
3726     half B[BN * BK * QSIZE];
3727 };
3728
3729 // 消费者 MMA 层级映射参考 hgemmmma_stages_tn_swizzle。与那个
3730 // 八 warp kernel 不同, 本 warp-specialized 消费者仅拥有四个 warp。
3731 template <const int kMmaM = 16,           // MMA atom M dim (m16n8k16)
3732           const int kMmaN = 8,           // MMA atom N dim
3733           const int kMmaK = 16,          // MMA atom K dim
3734           const int kMmaTileM = 2,       // consumer warps along M, warp tile M = 32
3735           const int kMmaTileN = 2,       // consumer warps along N, warp tile N = 16
3736           const int kValTileM = 4,       // value-repeat along M, BM = 16*2*4 = 128
3737           const int kValTileN = 8,       // value-repeat along N, BN = 8*2*8 = 128
3738           const int kValTileK = 4,       // MMA_K slices per BK tile, BK = 16*4 = 64
3739           const int kStages = 2,         // TMA full/empty pipeline depth
3740           const int kNumThreads = 256,   // 128 producer + 128 consumer threads
3741           const int kBlockSwizzle = 0>   // 1 enables 3D grid swizzle for L2 locality
3742 __global__ void __launch_bounds__(kNumThreads)
3743     hgemmmma_ws_tn(
3744         int M, int N, int K, half *C,
3745         const CUtensorMap * __restrict__ tensorMapA,
3746         const CUtensorMap * __restrict__ tensorMapB) {
3747
3748     constexpr int BM = kMmaM * kMmaTileM * kValTileM;
3749     constexpr int BN = kMmaN * kMmaTileN * kValTileN;
3750     constexpr int BK = kMmaK * kValTileK;
3751     static_assert(kMmaM == 16 && kMmaN == 8 && kMmaK == 16, "This kernel uses mma.sync.m16n8k16");
3752     static_assert(kMmaTileM * kMmaTileN == 4, "The consumer warpgroup has exactly four warps");
3753     static_assert(BM == 128 && BN == 128 && BK == 64, "TMA desc and 128B swizzle require 128x128x64");
3754     static_assert(kNumThreads == 256, "Use one producer and one consumer warpgroup");
3755     static_assert(kBlockSwizzle == 0 || kBlockSwizzle == 1, "kBlockSwizzle must be 0 or 1");
3756
3757     const int bx = kBlockSwizzle * blockIdx.z * gridDim.x + blockIdx.x;
3758     const int by = blockIdx.y;
3759     constexpr int kConsumerThreads = kNumThreads / 2;
3760     static_assert(kConsumerThreads == kMmaTileM * kMmaTileN * kWarpSize,
3761         "Consumer threads must cover the MMA warp grid");
3762
3763     if (bx >= div_ceil(N, BN) || by >= div_ceil(M, BM))
3764         return;
3765
3766     // TMA CU_TENSOR_MAP_SWIZZLE_128B 要求 smem 基地址 1024 字节对齐,
3767     // 以确保硬件 swizzle phase 从零开始。消费者 swizzle<64>()
3768     // 假设零 phase; 其他对齐 (如 128) 会导致 phase 偏移, 产生不匹配的
3769     // 物理地址, 破坏整个输出。
3770     //
3771     // 为什么 hgemmmma_stages_tn 只需要 __align__(128), 而本 kernel
3772     // 必须 __align__(1024)? 核心区别在于消费者如何感知 swizzle phase:
3773     //
3774     // WGMMA 消费者: 通过 make_smem_desc() 构造 64-bit WGMMA descriptor,
3775     // 其中 bits [49,52) 的 base_offset 字段编码了 smem 基址相对于
3776     // 1024B 边界的偏移量 ((addr >> 7) & 0x7)。硬件在读取 smem 时会
3777     // 用这个 base_offset 自动补偿 swizzle phase, 因此 WGMMA 路径不
3778     // 需要 1024 字节对齐也能得到正确数据。
3779     //

```

```

3780 // 本 kernel 消费者：使用 ldmatrix + 手写 swizzle<64>() 来计算
3781 // smem 物理地址。这个函数是纯软件公式，没有任何 base_offset
3782 // 补偿机制。它假设 smem 基址恰好落在 1024B 边界上 (phase=0)。
3783 // 如果基址只满足 128B 对齐，TMA 硬件会以非零 phase 写入数据，
3784 // 而 ldmatrix 以零 phase 读取 → 物理地址彻底错位 → 全输出错误。
3785 //
3786 // 简言之：WGMMMA 用硬件 descriptor base_offset 自动解决 phase 偏移；
3787 // ldmatrix 路径没有等价机制，必须靠 1024B 对齐来保证 phase=0。
3788 extern __shared__ __align__(1024) uint8_t smem_tma_mma_ws[];
3789 TmaMmaWSSMem<BM, BN, BK, kStages> &s =
3790     *reinterpret_cast<TmaMmaWSSMem<BM, BN, BK, kStages> *>(smem_tma_mma_ws);
3791 half *s_a = s.A;
3792 half *s_b = s.B;
3793
3794 #pragma nv_diag_suppress static_var_with_dynamic_init
3795 __shared__ cuda::barrier<cuda::thread_scope_block> full[kStages];
3796 __shared__ cuda::barrier<cuda::thread_scope_block> empty[kStages];
3797 #pragma nv_diag_default static_var_with_dynamic_init
3798
3799 const int NUM_K_TILES = div_ceil(K, BK);
3800 const int wg_idx = threadIdx.x / kConsumerThreads;
3801 const int wg_tid = threadIdx.x % kConsumerThreads;
3802
3803 if (threadIdx.x == 0) {
3804     for (int stage = 0; stage < kStages; ++stage) {
3805         init(&full[stage], kConsumerThreads + 1);
3806         init(&empty[stage], kConsumerThreads + 1);
3807     }
3808     tma_fence_proxy_async_shared_cta();
3809 }
3810 __syncthreads();
3811
3812 if (wg_idx == 0) {
3813     // 生产者 Warpgroup (wg_idx==0)
3814     NOTES_V2_REG_DEALLOC(40);
3815     if (wg_tid == 0) {
3816         int stage = 0;
3817         for (int k = 0; k < NUM_K_TILES; ++k, ++stage) {
3818             if (stage == kStages)
3819                 stage = 0;
3820
3821             empty[stage].wait(empty[stage].arrive());
3822
3823             tma_load_2d(&s_a[stage * BM * BK], tensorMapA, k * BK, by * BM,
3824                 full[stage]);
3825             tma_load_2d(&s_b[stage * BN * BK], tensorMapB, k * BK, bx * BN,
3826                 full[stage]);
3827             tma_arrive_expect_tx(full[stage], (BM * BK + BN * BK) * sizeof(half));
3828         }
3829     }
3830 } else {
3831     // 消费者 Warpgroup (wg_idx==1)
3832     // 初始化：标记所有 stage 为"空" (可被 Producer 写入)
3833     NOTES_V2_REG_ALLOC(232);
3834
3835     for (int stage = 0; stage < kStages; ++stage) {
3836         [[maybe_unused]] auto token = empty[stage].arrive();
3837     }
3838
3839     const int warp_id = wg_tid / kWarpSize;
3840     const int lane_id = wg_tid % kWarpSize;
3841     // WG1 内形成 2x2 warp grid, warp_m / warp_n 分别表示 warp 在 M/N 方向的索引。
3842     const int warp_m = warp_id % kMmaTileM; // kMmaTileM = 2, {0,1}

```

```

3843 const int warp_n = warp_id / kMmaTileM; // kMmaTileN = 2, {0,1}
3844 uint32_t RA[kValTileM][4];
3845 uint32_t RB[kValTileN][2];
3846 uint32_t RC[kValTileM][kValTileN][2] = {0};
3847
3848 const uint32_t smem_a_base_ptr = __cvta_generic_to_shared(s_a);
3849 const uint32_t smem_b_base_ptr = __cvta_generic_to_shared(s_b);
3850 int stage = 0;
3851 for (int k = 0; k < NUM_K_TILES; ++k, ++stage) {
3852     if (stage == kStages)
3853         stage = 0;
3854
3855     full[stage].wait(full[stage].arrive());
3856     tma_fence_proxy_async_shared_cta();
3857
3858     // 与 cp.async pipeline (hgemm_mma_stages_tn 等) 不同, 这里 ldmatrix +
3859     // mma.sync 之前/后 不需要 __syncthreads, 原因在于 TMA + async proxy 的
3860     // 同步机制与 warp specialization 的线程分工:
3861     //
3862     // 1. 生产者-消费者解耦: TMA 拷贝由 producer (wg_idx==0 的单个线程)
3863     //    通过 tma_load_2d + tma_arrive_expect_tx 提交, 而非所有 256 个线程
3864     //    各自做 cp.async。cp.async 需要 __syncthreads 确保所有线程的异步
3865     //    拷贝都完成后再进入计算; 而 TMA 路径用 cuda::barrier 的 full/empty
3866     //    协议完成 CTA 级同步——full[stage].wait() 返回时, producer 已通过
3867     //    expect_tx 声明了完整的传输字节数, 且硬件保证这些字节已全部写入 smem。
3868     //
3869     // 2. async proxy fence 替代 __syncthreads: fence_proxy_async(space_shared)
3870     //    在 async proxy (TMA 写入通道) 和后续 smem load (ldmatrix 读取通道)
3871     //    之间建立内存序。它确保 fence 之前的所有 async proxy 写操作 (TMA)
3872     //    对 fence 之后的所有 smem 读操作 (ldmatrix) 可见。这是一个轻量级
3873     //    proxy 间 ordering, 不需要阻塞线程, 也不会让 warp 等待其他 warp。
3874     //
3875     // 3. 消费者内部 warp 间无依赖: WG1 内的 4 个 warp (2x2 grid) 各自独立
3876     //    读取 smem 中互不重叠的 A/B 区域 (warp_m 和 warp_n 分别索引不同的
3877     //    行区间)。它们之间不需要读写共享数据, 因此不需要 __syncthreads 来
3878     //    对齐 warp 间的执行进度。mma.sync 本身是 warp 级同步指令, 保证同一
3879     //    warp 内 32 个线程的 ldmatrix 和 mma 操作正确协作。
3880
3881     // 注意: 这里已经没有NUM_K_TILES循环了, 因为K loop load已经被WG0生产者处理了
3882 #pragma unroll
3883     for (int k_step = 0; k_step < kValTileK; ++k_step) {
3884
3885 #pragma unroll
3886         for (int i = 0; i < kValTileM; ++i) { // kValTileM = 4
3887             // kMmaM * kValTileM = 16*4 = 64, 每个消费者 warp 在 M 方向覆盖 64 行。
3888             const int warp_smem_a_m = warp_m * (kMmaM * kValTileM) + i * kMmaM;
3889             const int lane_smem_a_m = warp_smem_a_m + lane_id % 16;
3890             // 与 hgemm_mma_stages_tn_swizzle 不同: k_step * kMmaK 必须放在
3891             // lane_smem_a_k 中传给 swizzle<64>(), 而非像 swizzle kernel
3892             // 那样作为 smem_k_offset 加在 swizzle 外部。原因:
3893             //   swizzle<64> 作用于完整 BK=64, swizzle 周期覆盖全部 64 列,
3894             //   k_step=0 和 k_step=1 的 chunk 会被 XOR 交叉混合 → k_step
3895             //   偏移必须在 swizzle 内部参与 chunk 计算。
3896             //   swizzle<kMmaK> 作用于 kMmaK=16, 每个 kMmaK slice 独立
3897             //   swizzle, slice 之间互不跨越 → k_step * kMmaK 可以加在外部。
3898             const int lane_smem_a_k = (k_step * kMmaK) + (lane_id / 16) * 8;
3899             const uint32_t lane_smem_a_ptr = smem_a_base_ptr +
3900                 (stage * BM * BK + lane_smem_a_m * BK +
3901                  swizzle<64>(lane_smem_a_m, lane_smem_a_k)) *
3902                 sizeof(half);
3903             LDMATRIX_X4(RA[i][0], RA[i][1], RA[i][2], RA[i][3], lane_smem_a_ptr);
3904         }
3905     }

```

```

3906 #pragma unroll
3907     for (int j = 0; j < kValTileN; ++j) { // kValTileN = 8
3908         // kMmaN * kValTileN = 8*8 = 64, 每个消费者 warp 在 B^T 的 N 方向覆盖 64 行。
3909         const int warp_smem_b_n = warp_n * (kMmaN * kValTileN) + j * kMmaN;
3910         const int lane_smem_b_n = warp_smem_b_n + lane_id % 8;
3911         const int lane_smem_b_k = (k_step * kMmaK) + ((lane_id / 8) % 2) * 8;
3912         const uint32_t lane_smem_b_ptr = smem_b_base_ptr +
3913             (stage * BN * BK + lane_smem_b_n * BK +
3914              swizzle<64>(lane_smem_b_n, lane_smem_b_k)) *
3915             sizeof(half);
3916         LDMATRIX_X2(RB[j][0], RB[j][1], lane_smem_b_ptr);
3917     }
3918
3919 #pragma unroll
3920     for (int i = 0; i < kValTileM; ++i) {
3921 #pragma unroll
3922         for (int j = 0; j < kValTileN; ++j) {
3923             HMMMA16816(RC[i][j][0], RC[i][j][1],
3924                        RA[i][0], RA[i][1], RA[i][2], RA[i][3],
3925                        RB[j][0], RB[j][1],
3926                        RC[i][j][0], RC[i][j][1]);
3927         }
3928     }
3929 }
3930 [[maybe_unused]] auto token = empty[stage].arrive();
3931 }
3932
3933 // Epilogue: 复用 RA[0] 和 RA[1] (已在寄存器中) 作为 shuffle 缓冲,
3934 // 与 hgemm_mma_stages_tn_swizzle 的 epilogue 模式一致。四个相邻
3935 // lane 各持有两个 uint32 fragment; shuffle 汇聚为每行一个 float4,
3936 // 由 lane 0 发出对齐的 128-bit store.
3937 #pragma unroll
3938     for (int i = 0; i < kValTileM; ++i) {
3939         const int store_warp_smem_c_m = warp_m * (kMmaM * kValTileM) + i * kMmaM;
3940         const int store_lane_gmem_c_m = by * BM + store_warp_smem_c_m + lane_id / 4;
3941 #pragma unroll
3942         for (int j = 0; j < kValTileN; ++j) {
3943             const int store_warp_smem_c_n = warp_n * (kMmaN * kValTileN) + j * kMmaN;
3944             RA[0][0] = RC[i][j][0];
3945             RA[1][0] = RC[i][j][1];
3946             RA[0][1] = __shfl_sync(0xffffffff, RC[i][j][0], lane_id + 1);
3947             RA[0][2] = __shfl_sync(0xffffffff, RC[i][j][0], lane_id + 2);
3948             RA[0][3] = __shfl_sync(0xffffffff, RC[i][j][0], lane_id + 3);
3949             RA[1][1] = __shfl_sync(0xffffffff, RC[i][j][1], lane_id + 1);
3950             RA[1][2] = __shfl_sync(0xffffffff, RC[i][j][1], lane_id + 2);
3951             RA[1][3] = __shfl_sync(0xffffffff, RC[i][j][1], lane_id + 3);
3952             if (lane_id % 4 == 0) {
3953                 const int store_lane_gmem_c_n = bx * BN + store_warp_smem_c_n;
3954                 const int store_gmem_c_addr_0 =
3955                     store_lane_gmem_c_m * N + store_lane_gmem_c_n;
3956                 const int store_gmem_c_addr_1 =
3957                     (store_lane_gmem_c_m + 8) * N + store_lane_gmem_c_n;
3958                 *reinterpret_cast<float4*>(&C[store_gmem_c_addr_0]) =
3959                     *reinterpret_cast<float4*>(&RA[0][0]);
3960                 *reinterpret_cast<float4*>(&C[store_gmem_c_addr_1]) =
3961                     *reinterpret_cast<float4*>(&RA[1][0]);
3962             }
3963         }
3964     }
3965 }
3966 }
3967
3968 #endif /* NOTES_V2_ENABLE_TMA_MMA_WS */

```

```

3969
3970 #if defined(NOTES_V2_ENABLE_WGMMMA) || defined(NOTES_V2_ENABLE_TMA_MMA_WS)
3971 // ---- Host-side TMA Tensor Map helpers (WGMMMA test) ----
3972 // 面试要点 (TMA descriptor 创建 — cuTensorMapEncodeTiled) :
3973 //   - TMA descriptor 描述 global memory 中矩形的 shape/stride/dtype,
3974 //     以及硬件搬运的 box (tile) 大小和 smem swizzle 模式
3975 //   - 关键易错点: TMA shape 参数写的是 (W,H) 而不是 (H,W)!
3976 //     对 row-major [M,K] 矩阵, TMA shape = (K,M), minor=K, major=M
3977 //   - cuTensorMapEncodeTiled 是 CUDA Driver API, 需 #include <cuda.h> 并链接 -lcuda
3978 //   - smem_box 维度顺序也是 (minor, major) = (BK, BM) 或 (BK, BN)
3979 //   - Swizzle 模式通常选 CU_TENSOR_MAP_SWIZZLE_128B 配合 WGMMMA 的 128B swizzle
3980 //
3981 // * gmem_prob_stride + 1 的含义:
3982 //   语法层面 — 数组名退化 + 指针算术:
3983 //     gmem_prob_stride 类型为 uint64_t[5], 在表达式中退化为 uint64_t*
3984 //     (指向首元素 gmem_prob_stride[0])。+ 1 偏移 1 个 uint64_t, 指向
3985 //     gmem_prob_stride[1], 等价于 &gmem_prob_stride[1]。
3986 //   语义层面 — 跳过隐式的最内维 stride:
3987 //     cuTensorMapEncodeTiled 的 globalStrides 参数只接收 tensorRank-1 个
3988 //     stride 值。最内维 (fastest-changing dimension) 的 stride 是隐式的,
3989 //     固定等于 sizeof(element_type), 不需要显式传入。
3990 //     对于 tensorRank=2 的 FP16 矩阵:
3991 //       dim 0 (内维, K) : stride = sizeof(half)    ← 隐式, API 不需要
3992 //       dim 1 (外维, M) : stride = sizeof(half)*K  ← 需要传给 API
3993 //     gmem_prob_stride 数组的布局是内维在前:
3994 //       [0] = sizeof(half)                        ← 内维, 不传
3995 //       [1] = sizeof(half) * BlockMinorSize * blocks_width ← 外维, 传给 API
3996 //     所以 + 1 的作用是跳过 [0], 让 API 从 [1] 开始读取, 正好得到外维 stride。
3997 //
3998 // * smem_box_stride 为什么不需要 +1?
3999 //   与 globalStrides 不同, smemBoxStrides 参数接收完整的 tensorRank 个 stride,
4000 //   最内维 stride 也必须显式传入 (不隐式)。
4001 //   smem_box_stride[5] = {1, 1, 1, 1, 1} 表示 smem tile 中所有维度元素
4002 //   都是连续存放的, 维度间没有 padding/stride。
4003 //   对于 tensorRank=2: strides[0]=1 (内维 stride), strides[1]=1 (外维 stride),
4004 //   即 smem 中第 (i,j) 元素的偏移 = i*1 + j*1 = i+j (元素连续排列)。
4005 //   所以这里直接传 smem_box_stride (即 &smem_box_stride[0]), API 会读到全部
4006 //   两个 stride 值, 不需要跳过任何一个。
4007 //
4008 // 参考: CUDA Programming Guide §TMA, PTX ISA §9.7.15.5, CUTLASS GmmaDescriptor
4009
4010 template <int BlockMajorSize, int BlockMinorSize>
4011 __host__ static inline void create_tensor_map(CUtensorMap *tma_map,
4012                                               half *gmem_ptr,
4013                                               int blocks_height,
4014                                               int blocks_width) {
4015     void *gmem_address = (void *)gmem_ptr;
4016     uint64_t gmem_prob_shape[5] = {(uint64_t)BlockMinorSize * blocks_width,
4017                                     (uint64_t)BlockMajorSize * blocks_height,
4018                                     1, 1, 1};
4019     uint64_t gmem_prob_stride[5] = {
4020         sizeof(half), sizeof(half) * BlockMinorSize * blocks_width, 0, 0, 0};
4021     uint32_t smem_box_shape[5] = {uint32_t(BlockMinorSize),
4022                                   uint32_t(BlockMajorSize), 1, 1, 1};
4023     uint32_t smem_box_stride[5] = {1, 1, 1, 1, 1};
4024     CUresult result = cuTensorMapEncodeTiled(
4025         tma_map, CU_TENSOR_MAP_DATA_TYPE_FLOAT16, 2, gmem_address,
4026         gmem_prob_shape, gmem_prob_stride + 1, smem_box_shape, smem_box_stride,
4027         CU_TENSOR_MAP_INTERLEAVE_NONE, CU_TENSOR_MAP_SWIZZLE_128B,
4028         CU_TENSOR_MAP_L2_PROMOTION_NONE, CU_TENSOR_MAP_FLOAT_OOB_FILL_NONE);
4029     if (result != CUDA_SUCCESS)
4030         printf("cuTensorMapEncodeTiled failed: %d\n", (int)result);
4031 }

```

```

4032 // BlockMajorSize = major dim (outer), BlockMinorSize = minor dim (inner, contiguous).
4033 // For row-major [Major, Minor] matrices.
4034 // Default <128, 64> matches HGEMM A/B^T and FA Q; FA K/V use <64, 64>.
4035 template <int BlockMajorSize = 128, int BlockMinorSize = 64>
4036 __host__ __static inline CUTensorMap *allocate_and_create_tensor_map(
4037     half *src, int blocks_height, int blocks_width) {
4038     CUTensorMap *tma_map_d;
4039     cudaMalloc(&tma_map_d, sizeof(CUTensorMap));
4040     CUTensorMap tma_map_host;
4041     create_tensor_map<BlockMajorSize, BlockMinorSize>(&tma_map_host, src,
4042                                                         blocks_height,
4043                                                         blocks_width);
4044     cudaMemcpy(tma_map_d, &tma_map_host, sizeof(CUTensorMap),
4045                cudaMemcpyHostToDevice);
4046     return tma_map_d;
4047 }
4048 }
4049 #endif /* NOTES_V2_ENABLE_WGMMA || NOTES_V2_ENABLE_TMA_MMA_WS */
4050
4051 // =====
4052 // Phase 8: FlashAttention-2 + MMA m16n8k16)
4053 // =====
4054 // 面试要点 (FlashAttention-2算法) :
4055 // 1. 核心问题: 标准 Attention 的  $O(N^2)$  中间矩阵 ( $S=QK^T$ ) 必须写入 HBM,
4056 //    但 HBM 带宽是瓶颈 → FlashAttention-2用 tiling + online softmax 避免写回
4057 // 2. FA 三板斧:
4058 //    a) Tiling: Q 分块 [Br,d], K/V 沿 seqLen 分块 [Bc,d]
4059 //    b) Online Softmax: 迭代更新 m(行max) 和 l(行sum), 无需全局同步
4060 //    c) Recomputation(反向): 反向传播时重新计算 S/P, 而非存储中间矩阵
4061 // 3. Split-Q 设计: 所有 warp 共享同一块 K, 各 warp 处理 Q 的不同行片段
4062 //    - warp_KV=0 (所有 warp 共享 K), warp_QP=warp_id (各 warp 不同 Q 行)
4063 //    - 优点: 减少 warp 间通信和 shuffle
4064 // 4. Online rescaling 公式 (FA 核心, arXiv:2307.08691) :
4065 //    for each K,V tile:
4066 //        S_cur = Q @ K^T // 未缩放, 存入 R_S
4067 //        m_new = max(m_old, row_max(S_cur * scale))
4068 //        P_cur = exp(S_cur * scale - m_new) // ← 写回 R_S 寄存器!
4069 //        l_new = exp(m_old - m_new) * l_old + row_sum(P_cur)
4070 //        O_new = diag(exp(m_old - m_new)) * O_old + P_cur @ V
4071 //    O_final = O_new / l_final
4072 // 5. 为什么 R_S 可以直接用作 P@V 的 A 矩阵?
4073 //    - R_S 经过 softmax 后, 存储的是  $P = \exp(S - m)$ , 数据仍然是 half 精度
4074 //    - 当前实现依赖 m16n8k16 这一路径下约定好的 fragment 布局, 使 softmax
4075 //      后的 P 可以继续留在 R_S 中供后面的 P@V 直接消费
4076 //    - 这是此实现的寄存器布局复用技巧, 不要背成“所有 MMA A/C fragment
4077 //      都天然同构”的通用结论
4078 //
4079 // 本实现参考: FlashAttention-2 (Dao et al., arXiv:2307.08691)
4080 // 从 LeetCUDA flash-attn/mma/basic/flash_attn_mma_split_q.cu 提取
4081 // Grid: ((N + 63) / 64, B * H, 1), Br=64
4082 // Block: (128, 1, 1), kNumThreads=kWarpSize*kMmaTileSeqLenQ*kMmaTileSeqLenK=128
4083 // source: LeetCUDA/kernels/flash-attn/mma/basic/flash_attn_mma_split_q.cu
4084
4085 // =====
4086 // FlashAttention-2 Split-Q Kernel (完整实现)
4087 // =====
4088 // Q,K,V,0: [batch_size, num_heads, seq_len, head_dim], [B,H,N,d]
4089 //
4090 // Tile 设计 (以 kHeadDim=64 为例) :
4091 // Br = kMmaAtomM * kMmaTileSeqLenQ * kValTileSeqLenQ = 16*{4,8}*1 = {64,128}
4092 // Bc = kMmaAtomN * kMmaTileSeqLenK * kValTileSeqLenK = 8*1*8 = 64
4093 // Warp 布局: 4 warps, warp_QP 0~3 各处理 16 行, warp_KV=0 共享 K
4094 //

```

```

4095 // 执行流程:
4096 // 1) 预加载 Q[Br,d] 到 smem (只加载一次, split-Q 的核心优势)
4097 // 2) 外循环: 沿 K seqlen 分块迭代 (Tc = seqlen/Bc)
4098 // 3) 3a: cp.async 加载当前 V + 预加载下一个 K (多 stage pipeline)
4099 // 4) 3b: Q@K^T — 沿 head dim 内循环 → ldmatrix Q/K → HMMMA16816
4100 // 5) 3c: Online Safe Softmax — warp reduce max/row → exp(S*scale - max)
4101 // → 关键: 将 P 写回 R_S 寄存器 (替换 S), R_S 现在存储 P matrix
4102 // 6) 3d: P@V — 沿 V_Bc 内循环 → ldmatrix V (transposed) → 直接用 R_S 做 A
4103 // → 当前实现依赖同一组寄存器布局约定, 避免为 P@V 额外重组 P fragment
4104 // 7) 3e: Online rescaling — 0_new = exp(m_old-m_new)*0_old + P@V
4105 // 8) 最终 rescale: 0_final = (1/l_final) * 0_final
4106 // 9) Epilogue: warp shuffle + 128-bit collective store
4107 //
4108 // ★ Pad 与手动 XOR swizzle 的 profile 结论 (SM120, B=1,H=32,N=4096,D=64) :
4109 // - XOR 能消除目标 ldmatrix lane pattern 的 bank conflict, 但收益是局部的;
4110 // XOR 本身及 `2` 的计算不是主要瓶颈, nvcc 已将该式化简为 bit operations。
4111 // - compact Q/K/V XOR 改变了 cp.async 的 shared destination pattern: LDGSTS
4112 // wavefronts 从 pad 的 30.15M 增至 68.16M (2.26x), long-scoreboard、
4113 // LG-throttle、MIO-throttle 分别约为 pad 的 3.25x、2.74x、1.60x。
4114 // - compact XOR 虽节省约 5 KiB smem, 但没有提高此 kernel 的 occupancy; 最终
4115 // 约 120.0 TFLOPS, 显著低于 Q/K/V kPad=8 的 166.6 TFLOPS。
4116 // - 所以当前 kernel 默认对 Q/K/V 都用 kPad=8。保留 swizzle 路径是为了学习和消融: 评价
4117 // shared layout 必须同时观察 ldmatrix reads 与 cp.async/LDGSTS writes, 不能只看
4118 // bank-conflict counter, 也不能只优化 XOR 地址算术。
4119 // ---- 寄存器填充辅助函数 (FlashAttention 用, 前移以供 TMA_MMA_WS kernel 使用) ----
4120 template <typename T, int M, const int N, const int K = 2>
4121 __device__ inline void fill_3D_regs(T (&R)[M][N][K], T val) {
4122 #pragma unroll
4123     for (int i = 0; i < M; ++i)
4124 #pragma unroll
4125         for (int j = 0; j < N; ++j)
4126 #pragma unroll
4127             for (int k = 0; k < K; ++k)
4128                 R[i][j][k] = val;
4129 }
4130
4131 template <typename T, int M, const int N = 2>
4132 __device__ inline void fill_2D_regs(T (&R)[M][N], T val) {
4133 #pragma unroll
4134     for (int i = 0; i < M; ++i)
4135 #pragma unroll
4136         for (int j = 0; j < N; ++j)
4137             R[i][j] = val;
4138 }
4139
4140 template <
4141     const int kHeadDim,          // head dim: 32, 64, 128
4142     const int kMmaAtomM,         // 16 (MMA instruction M dimension)
4143     const int kMmaAtomN,         // 8 (MMA instruction N dimension)
4144     const int kMmaAtomK,         // 16 (MMA instruction K dimension)
4145     const int kMmaAccF32,        // 0=f16 acc, 1=f32 acc (mma.sync.f32.f16.f16.f32)
4146     const int kMmaTileSeqLenQ,   // MMA tiles along Q's M dim, 4 → Br=16*4=64
4147     const int kMmaTileSeqLenK,   // MMA tiles along K's N dim, 1 → Bc basis=8
4148     const int kMmaTileSeqLenP,   // MMA tiles for P@V M dim, must equal kMmaTileSeqLenQ
4149     const int kMmaTileHeadDimV,  // MMA tiles for P@V N dim (head dim direction)
4150     const int kValTileSeqLenQ,   // value tiles along Q's M, 1 → Br per warp=16
4151     const int kValTileSeqLenK,   // value tiles along K's N, 8 → Bc_warp=8*8=64
4152     const int kValTileSeqLenP,   // value tiles for P@V M dim, 1
4153     const int kValTileHeadDimV,  // value tiles for P@V N dim, kHeadDim/(8*kMmaTileHeadDimV)
4154     const int kStagesK,          // pipeline stages for K: >= 1; NO stages required for Q/V
4155     const int kPadQ,             // Q row padding; 0 selects compact XOR swizzle
4156     const int kPadK,             // K row padding; 0 selects compact XOR swizzle
4157     const int kPadV>

```

```

4158 __global__ void __launch_bounds__(kWarpSize * kMmaTileSeqLenQ * kMmaTileSeqLenK)
4159 flash_attn_mma_stages_split_q(
4160     half *Q, half *K, half *V, half *O, int N, int H) {
4161     static_assert(kStagesK >= 1, "kStagesK must be >= 1");
4162     static_assert(kMmaAccF32 == 0 || kMmaAccF32 == 1, "kMmaAccF32 must be 0 or 1");
4163     static_assert(kPadQ >= 0 && kPadK >= 0 && kPadV >= 0,
4164         "Q/K/V padding must be non-negative");
4165     constexpr int kNRegs = kMmaAccF32 ? 4 : 2;
4166     constexpr bool kSwizzleQ = kPadQ == 0;
4167     constexpr bool kSwizzleK = kPadK == 0;
4168     constexpr bool kSwizzleV = kPadV == 0;
4169     constexpr int Br = kMmaAtomM * kMmaTileSeqLenQ * kValTileSeqLenQ; // 16*8*1=128
4170     constexpr int Bc = kMmaAtomN * kMmaTileSeqLenK * kValTileSeqLenK; // 8*1*8=64
4171     constexpr int kNumThreads = kWarpSize * kMmaTileSeqLenQ * kMmaTileSeqLenK; // 32*8*1=256
4172     const int Tc = (N + Bc - 1) / Bc;
4173     // 原始实现默认 seqLen 与 Bc 对齐; 最后一个不完整 tile 需要额外 pad/边界处理。
4174     // 这里保留 ceil 写法是为了说明 tile 划分方式, 不等于当前实现已经完整处理了尾 tile。
4175     const float scale = 1.0f / sqrtf((float)kHeadDim);
4176
4177     // Block indexing
4178     const int Nb_id = blockIdx.y / H; // batch id
4179     const int Nh_id = blockIdx.y % H; // head id
4180     const int Q_tile_id = blockIdx.x; // tile id along Q's M dim (Br)
4181     const int tid = threadIdx.x;
4182     const int warp_id = tid / kWarpSize;
4183     const int lane_id = tid % kWarpSize;
4184     const int warp_QP = warp_id; // Split-Q: 每个 warp 处理不同的 Q 行片段
4185     const int warp_KV = 0; // 所有 warp 共享 K (减少跨 warp 通信)
4186
4187     // Global memory base offsets for this (batch, head)
4188     // 这里默认 Q/K/V 共享同一 per-head 基址布局, 对应 self-attention 场景
4189     const int Q_gmem_offset = (Nb_id * H * N + Nh_id * N) * kHeadDim;
4190     const int K_gmem_offset = Q_gmem_offset;
4191     const int V_gmem_offset = Q_gmem_offset;
4192     const int O_gmem_offset = Q_gmem_offset;
4193
4194     // Thread-to-smem mapping for cooperative load
4195     int load_smem_Q_Br = tid / (kNumThreads / Br);
4196     int load_smem_Q_d = (tid % (kNumThreads / Br)) * (kHeadDim / (kNumThreads / Br));
4197     int load_smem_K_Bc = tid / (kNumThreads / Bc);
4198     int load_smem_K_d = (tid % (kNumThreads / Bc)) * (kHeadDim / (kNumThreads / Bc));
4199     int load_smem_V_Bc = tid / (kNumThreads / Bc);
4200     int load_smem_V_d = (tid % (kNumThreads / Bc)) * (kHeadDim / (kNumThreads / Bc));
4201
4202     int load_gmem_Q_Br = Q_tile_id * Br + load_smem_Q_Br;
4203     if (load_gmem_Q_Br >= N)
4204         return;
4205
4206     // Shared memory layout
4207     // Q/K/V independently use padded row-major when kPad* > 0 and compact XOR
4208     // swizzle when kPad* == 0. Padding and XOR are never combined per operand.
4209     // The swizzled physical layout is [col / 16][row][16]; swizzle<16>() selects
4210     // the 0/8 phase inside the final 16-half tile.
4211     extern __shared__ half smem[];
4212     constexpr int Q_tile_size = Br * (kHeadDim + kPadQ);
4213     constexpr int K_tile_size = Bc * (kHeadDim + kPadK);
4214     constexpr int kSmemStrideQ = kHeadDim + kPadQ;
4215     constexpr int kSmemStrideK = kHeadDim + kPadK;
4216     constexpr int kSmemStrideV = kHeadDim + kPadV;
4217     half *Q_tile_smem = smem;
4218     half *K_tile_smem = Q_tile_smem + Q_tile_size;
4219     half *V_tile_smem = K_tile_smem + kStagesK * K_tile_size;
4220

```

```

4221 uint32_t smem_Q_base_ptr = __cvta_generic_to_shared(Q_tile_smem);
4222 uint32_t smem_K_base_ptr = __cvta_generic_to_shared(K_tile_smem);
4223 uint32_t smem_V_base_ptr = __cvta_generic_to_shared(V_tile_smem);
4224
4225 // Online Softmax persistent state
4226 // lane_block_row_max_old[i][r]: running max for row r of warp tile i
4227 // lane_block_row_sum_old[i][r]: running denominator l for row r of warp tile i
4228 float lane_block_row_max_old[kValTileSeqLenQ][2];
4229 float lane_block_row_sum_old[kValTileSeqLenQ][2];
4230 fill_2D_regs<float, kValTileSeqLenQ, 2>(lane_block_row_max_old, -INFINITY);
4231 fill_2D_regs<float, kValTileSeqLenQ, 2>(lane_block_row_sum_old, 0.0f);
4232
4233 uint32_t R_Q[kValTileSeqLenQ][4]; // Q regs
4234 uint32_t R_K[kValTileSeqLenK][2]; // K regs
4235 uint32_t R_V[kValTileHeadDimV][2]; // V regs
4236 // R_S / R_O / R_D 都按 mma.sync.aligned.m16n8k16 的 fragment 约定存储。
4237 // f16 acc: 每 tile 2 个 uint32 (4 half); f32 acc: 每 tile 4 个 uint32 (4 f32)。
4238 // 后续 softmax、P@V、online rescale 都直接围绕这组 fragment 布局做寄存器内变换。
4239 uint32_t R_S[kValTileSeqLenQ][kValTileSeqLenK][kNRegs]; // S=Q@K^T / P=softmax(S)
4240 uint32_t R_O[kValTileSeqLenP][kValTileHeadDimV][kNRegs]; // O for current tile
4241 uint32_t R_D[kValTileSeqLenP][kValTileHeadDimV][kNRegs]; // O accumulator (final output)
4242
4243 fill_3D_regs<uint32_t, kValTileSeqLenQ, kValTileSeqLenK, kNRegs>(R_S, 0);
4244 fill_3D_regs<uint32_t, kValTileSeqLenP, kValTileHeadDimV, kNRegs>(R_D, 0);
4245
4246 // =====
4247 // Step 1: 加载 Q[Br, d] 到 shared memory (整个外循环只加载一次)
4248 // =====
4249 {
4250     int load_gmem_Q_addr =
4251         Q_gmem_offset + load_gmem_Q_Br * kHeadDim + load_smem_Q_d;
4252     if constexpr (kSwizzleQ) {
4253 #pragma unroll
4254         for (int i = 0; i < (kHeadDim / (kNumThreads / Br)); i += 8) {
4255             int col = load_smem_Q_d + i;
4256             uint32_t ptr = smem_Q_base_ptr +
4257                 ((col / kMmaAtomK) * Br * kMmaAtomK +
4258                 load_smem_Q_Br * kMmaAtomK +
4259                 swizzle<kMmaAtomK>(load_smem_Q_Br, col % kMmaAtomK)) *
4260                 sizeof(half);
4261             CP_ASYNC.CG(ptr, &Q[load_gmem_Q_addr + i], 16);
4262         }
4263     } else {
4264         uint32_t load_smem_Q_ptr =
4265             smem_Q_base_ptr +
4266             (load_smem_Q_Br * kMemStrideQ + load_smem_Q_d) * sizeof(half);
4267 #pragma unroll
4268         for (int i = 0; i < (kHeadDim / (kNumThreads / Br)); i += 8) {
4269             CP_ASYNC.CG(load_smem_Q_ptr + i * 2, &Q[load_gmem_Q_addr + i], 16);
4270         }
4271     }
4272     CP_ASYNC.COMMIT_GROUP();
4273 }
4274
4275 // =====
4276 // Step 2: 预加载前 (kStagesK-1) 个 K tile (多 stage pipeline 预热)
4277 // 注意: Q 由 blockIdx.x 固定到当前 Q tile; 而 K/V 的 seqLen 遍历始终从 tile 0 开始,
4278 // 后续在外循环里不断递增到 tile 1/2/3/.../Tc-1。
4279 // =====
4280 if constexpr (kStagesK > 1) {
4281 #pragma unroll
4282     for (int stage = 0; stage < (kStagesK - 1); ++stage) {
4283         int load_gmem_K_Bc = stage * Bc + load_smem_K_Bc;

```

```

4284     int load_gmem_K_addr =
4285         K_gmem_offset + load_gmem_K_Bc * kHeadDim + load_smem_K_d;
4286     if constexpr (kSwizzleK) {
4287 #pragma unroll
4288         for (int i = 0; i < (kHeadDim / (kNumThreads / Bc)); i += 8) {
4289             int col = load_smem_K_d + i;
4290             uint32_t ptr = smem_K_base_ptr +
4291                 (stage * K_tile_size +
4292                  (col / kMmaAtomK) * Bc * kMmaAtomK +
4293                  load_smem_K_Bc * kMmaAtomK +
4294                  swizzle<kMmaAtomK>(load_smem_K_Bc, col % kMmaAtomK)) *
4295                 sizeof(half);
4296             CP_ASYNC_CG(ptr, &K[load_gmem_K_addr + i], 16);
4297         }
4298     } else {
4299         uint32_t load_smem_K_ptr =
4300             smem_K_base_ptr +
4301             (stage * K_tile_size + load_smem_K_Bc * kSmemStrideK +
4302              load_smem_K_d) *
4303             sizeof(half);
4304 #pragma unroll
4305         for (int i = 0; i < (kHeadDim / (kNumThreads / Bc)); i += 8) {
4306             CP_ASYNC_CG(load_smem_K_ptr + i * 2, &K[load_gmem_K_addr + i], 16);
4307         }
4308     }
4309     CP_ASYNC_COMMIT_GROUP();
4310 }
4311 CP_ASYNC_WAIT_GROUP(kStagesK - 2);
4312 __syncthreads();
4313 }
4314
4315 // =====
4316 // Step 3: 外循环 — 沿 K seqlen 迭代 (Tc = ceil(seqlen/Bc))
4317 // 每次迭代处理一个 K[Bc,d] + V[Bc,d] tile
4318 // =====
4319 #pragma unroll 1
4320 for (int tile_K_seqlen = 0; tile_K_seqlen < Tc; ++tile_K_seqlen) {
4321     int smem_sel = tile_K_seqlen % kStagesK;
4322     int smem_sel_next = (tile_K_seqlen + (kStagesK - 1)) % kStagesK;
4323
4324     // ---- 3a: 异步加载 K/V tile (多 stage pipeline) ----
4325     // kStagesK>1: V 加载当前 tile, K 预取下一 tile (pipeline)
4326     // kStagesK=1: V 和 K 都加载当前 tile (smem_sel==smem_sel_next==0),
4327     // 无 pipeline 预取, 每轮 QK 前必须 wait 当前 K 就绪。
4328     // Load current V tile (no pipeline for V — one stage is enough)
4329     {
4330         int load_gmem_V_Bc = tile_K_seqlen * Bc + load_smem_V_Bc;
4331         int load_gmem_V_addr =
4332             V_gmem_offset + load_gmem_V_Bc * kHeadDim + load_smem_V_d;
4333         if constexpr (kSwizzleV) {
4334 #pragma unroll
4335             for (int i = 0; i < (kHeadDim / (kNumThreads / Bc)); i += 8) {
4336                 int col = load_smem_V_d + i;
4337                 uint32_t ptr = smem_V_base_ptr +
4338                     ((col / kMmaAtomK) * Bc * kMmaAtomK +
4339                     load_smem_V_Bc * kMmaAtomK +
4340                     swizzle<kMmaAtomK>(load_smem_V_Bc, col % kMmaAtomK)) *
4341                     sizeof(half);
4342                 CP_ASYNC_CG(ptr, &V[load_gmem_V_addr + i], 16);
4343             }
4344         } else {
4345             uint32_t load_smem_V_ptr =
4346                 smem_V_base_ptr +

```

```

4347         (load_smem_V_Bc * kSmemStrideV + load_smem_V_d) * sizeof(half);
4348 #pragma unroll
4349     for (int i = 0; i < (kHeadDim / (kNumThreads / Bc)); i += 8) {
4350         CP_ASYNC_CG(load_smem_V_ptr + i * 2, &V[load_gmem_V_addr + i], 16);
4351     }
4352 }
4353 CP_ASYNC_COMMIT_GROUP();
4354 }
4355
4356 if constexpr (kStagesK > 1) {
4357     // Prefetch next K tile (pipelined)
4358     if ((tile_K_seqlen + 1) < Tc) {
4359         int load_gmem_K_Bc = (tile_K_seqlen + 1) * Bc + load_smem_K_Bc;
4360         int load_gmem_K_addr =
4361             K_gmem_offset + load_gmem_K_Bc * kHeadDim + load_smem_K_d;
4362         if constexpr (kSwizzleK) {
4363             #pragma unroll
4364             for (int i = 0; i < (kHeadDim / (kNumThreads / Bc)); i += 8) {
4365                 int col = load_smem_K_d + i;
4366                 uint32_t ptr = smem_K_base_ptr +
4367                     (smem_sel_next * K_tile_size +
4368                     (col / kMmaAtomK) * Bc * kMmaAtomK +
4369                     load_smem_K_Bc * kMmaAtomK +
4370                     swizzle<kMmaAtomK>(load_smem_K_Bc, col % kMmaAtomK)) *
4371                     sizeof(half);
4372                 CP_ASYNC_CG(ptr, &K[load_gmem_K_addr + i], 16);
4373             }
4374         } else {
4375             uint32_t load_smem_K_ptr =
4376                 smem_K_base_ptr +
4377                 (smem_sel_next * K_tile_size +
4378                 load_smem_K_Bc * kSmemStrideK +
4379                 load_smem_K_d) *
4380                 sizeof(half);
4381             #pragma unroll
4382             for (int i = 0; i < (kHeadDim / (kNumThreads / Bc)); i += 8) {
4383                 CP_ASYNC_CG(load_smem_K_ptr + i * 2, &K[load_gmem_K_addr + i], 16);
4384             }
4385         }
4386         CP_ASYNC_COMMIT_GROUP();
4387     }
4388 } else {
4389     // kStagesK==1: 加载当前 K tile (smem_sel_next == smem_sel == 0) 。
4390     // Step 2 预加载循环 (kStagesK-1=0) 不执行, 所以每轮 3a 必须加载当前 K。
4391     {
4392         int load_gmem_K_Bc = tile_K_seqlen * Bc + load_smem_K_Bc;
4393         int load_gmem_K_addr =
4394             K_gmem_offset + load_gmem_K_Bc * kHeadDim + load_smem_K_d;
4395         if constexpr (kSwizzleK) {
4396             #pragma unroll
4397             for (int i = 0; i < (kHeadDim / (kNumThreads / Bc)); i += 8) {
4398                 int col = load_smem_K_d + i;
4399                 uint32_t ptr = smem_K_base_ptr +
4400                     (smem_sel * K_tile_size +
4401                     (col / kMmaAtomK) * Bc * kMmaAtomK +
4402                     load_smem_K_Bc * kMmaAtomK +
4403                     swizzle<kMmaAtomK>(load_smem_K_Bc, col % kMmaAtomK)) *
4404                     sizeof(half);
4405                 CP_ASYNC_CG(ptr, &K[load_gmem_K_addr + i], 16);
4406             }
4407         } else {
4408             uint32_t load_smem_K_ptr =
4409                 smem_K_base_ptr +

```

```

4410         (smem_sel * K_tile_size +
4411         load_smem_K_Bc * kSmemStrideK +
4412         load_smem_K_d) *
4413         sizeof(half);
4414 #pragma unroll
4415     for (int i = 0; i < (kHeadDim / (kNumThreads / Bc)); i += 8) {
4416         CP_ASYNC_CG(load_smem_K_ptr + i * 2, &K[load_gmem_K_addr + i], 16);
4417     }
4418 }
4419 CP_ASYNC_COMMIT_GROUP();
4420 }
4421 }
4422
4423 // 3b: Q@K^T = S[Br, Bc] — 沿 head dim (d/kMmaAtomK=16) 内循环
4424 // kStagesK==1: 3a 刚提交当前 K 的 cp.async, 必须 wait 后才能 ldmatrix.
4425 // kStagesK>1: K 已在上一轮预取并 wait (Step 2 或 3e 末尾的 wait), 无需再 wait.
4426 if constexpr (kStagesK == 1) {
4427     CP_ASYNC_WAIT_GROUP(0);
4428     __syncthreads();
4429 }
4430 fill_3D_regs<uint32_t, kValTileSeqLenQ, kValTileSeqLenK, kNRegs>(R_S, 0);
4431 #pragma unroll
4432 for (int tile_K_d = 0; tile_K_d < (kHeadDim / kMmaAtomK); ++tile_K_d) {
4433     // ldmatrix.x4: 加载 Q 的 m16k16 片段到 R_Q
4434     #pragma unroll
4435     for (int i = 0; i < kValTileSeqLenQ; ++i) {
4436         int warp_smem_Q_Br =
4437             warp_QP * (kMmaAtomM * kValTileSeqLenQ) + i * kMmaAtomM;
4438         int lane_smem_Q_Br =
4439             warp_smem_Q_Br + lane_id % 16; // ldmatrix uses 16 lanes
4440         int lane_smem_Q_d = tile_K_d * kMmaAtomK + (lane_id / 16) * 8; // 0, 8
4441         uint32_t lane_smem_Q_ptr;
4442         if constexpr (kSwizzleQ) {
4443             lane_smem_Q_ptr = smem_Q_base_ptr +
4444                 ((lane_smem_Q_d / kMmaAtomK) * Br * kMmaAtomK +
4445                 lane_smem_Q_Br * kMmaAtomK +
4446                 swizzle<kMmaAtomK>(lane_smem_Q_Br,
4447                 lane_smem_Q_d % kMmaAtomK)) *
4448                 sizeof(half);
4449         } else {
4450             lane_smem_Q_ptr = smem_Q_base_ptr +
4451                 (lane_smem_Q_Br * kSmemStrideQ + lane_smem_Q_d) * sizeof(half);
4452         }
4453         LDMATRIX_X4(R_Q[i][0], R_Q[i][1], R_Q[i][2], R_Q[i][3],
4454             lane_smem_Q_ptr);
4455     }
4456
4457     // ldmatrix.x2: 加载 K 的 k16n8 片段到 R_K
4458     // K[Bc,d] row-major = K^T[d,Bc] col-major (NT 布局的 B 矩阵)
4459     #pragma unroll
4460     for (int j = 0; j < kValTileSeqLenK; ++j) {
4461         int warp_smem_K_Bc =
4462             warp_KV * (kMmaAtomN * kValTileSeqLenK) + j * kMmaAtomN;
4463         int lane_smem_K_Bc =
4464             warp_smem_K_Bc + lane_id % 8; // ldmatrix B uses 8 lanes
4465         int lane_smem_K_d =
4466             tile_K_d * kMmaAtomK + ((lane_id / 8) % 2) * 8; // 0, 8
4467         uint32_t lane_smem_K_ptr;
4468         if constexpr (kSwizzleK) {
4469             lane_smem_K_ptr = smem_K_base_ptr +
4470                 (smem_sel * K_tile_size +
4471                 (lane_smem_K_d / kMmaAtomK) * Bc * kMmaAtomK +
4472                 lane_smem_K_Bc * kMmaAtomK +

```

```

4473         swizzle<kMmaAtomK>(lane_smem_K_Bc,
4474                             lane_smem_K_d % kMmaAtomK)) *
4475         sizeof(half);
4476     } else {
4477         lane_smem_K_ptr = smem_K_base_ptr +
4478             (smem_sel * K_tile_size + lane_smem_K_Bc * kSmemStrideK +
4479              lane_smem_K_d) *
4480             sizeof(half);
4481     }
4482     LDMATRIX_X2(R_K[j][0], R_K[j][1], lane_smem_K_ptr);
4483 }
4484
4485 // MMA: S[tile] += Q[tile] @ K^T[tile]
4486 #pragma unroll
4487 for (int i = 0; i < kValTileSeqLenQ; ++i) {
4488     #pragma unroll
4489     for (int j = 0; j < kValTileSeqLenK; ++j) {
4490         if constexpr (kMmaAccF32) {
4491             HMMA16816F32(R_S[i][j][0], R_S[i][j][1], R_S[i][j][2], R_S[i][j][3],
4492                          R_Q[i][0], R_Q[i][1], R_Q[i][2], R_Q[i][3],
4493                          R_K[j][0], R_K[j][1],
4494                          R_S[i][j][0], R_S[i][j][1], R_S[i][j][2], R_S[i][j][3]);
4495         } else {
4496             HMMA16816(R_S[i][j][0], R_S[i][j][1], R_Q[i][0], R_Q[i][1], R_Q[i][2],
4497                      R_Q[i][3], R_K[j][0], R_K[j][1], R_S[i][j][0],
4498                      R_S[i][j][1]);
4499         }
4500     }
4501 }
4502 } // end loop over d
4503 __syncthreads();
4504
4505 // =====
4506 // 3c: Online Safe Softmax — row-wise max + exp + sum, then store P back to R_S
4507 // =====
4508 // MMA C fragment layout for m16n8k16 (PTX ISA 对应的线程寄存器分布):
4509 // - R_S[i][j][0] 对应当前 16x8 tile 的 rows 0~7 里的两个 half 值 {c0,c1}
4510 // - R_S[i][j][1] 对应 rows 8~15 里的两个 half 值 {c2,c3}
4511 // - lane 0~3 持有 row 0 的片段, lane 4~7 持有 row 1, ..., lane 28~31 持有 row 7
4512 // - 对于 rows 8~15 也是同样的 lane 分组, 只是读取的是 reg[1]
4513 // 这就是为什么后面做 row max / row sum 时, warp 内真正参与同一行归约的是
4514 // {0,4,8,...,28} 这一类 4-lane 子组, 而不是整 warp 32 个线程。
4515 // Each (i, j) pair = one 16x8 MMA tile; there are kValTileSeqLenQ x
4516 // kValTileSeqLenK tiles.
4517
4518 float lane_row_max_new[kValTileSeqLenQ][2];
4519 float lane_row_sum_new[kValTileSeqLenQ][2];
4520 fill_2D_regs<float, kValTileSeqLenQ, 2>(lane_row_max_new, -INFINITY);
4521 fill_2D_regs<float, kValTileSeqLenQ, 2>(lane_row_sum_new, 0.0f);
4522
4523 // Pass 1: Thread-level reduce max across all columns (kValTileSeqLenK tiles)
4524 #pragma unroll
4525 for (int i = 0; i < kValTileSeqLenQ; ++i) {
4526     #pragma unroll
4527     for (int j = 0; j < kValTileSeqLenK; ++j) {
4528         float tmp_max_0, tmp_max_1;
4529         if constexpr (kMmaAccF32) {
4530             float *t_fptr_S = reinterpret_cast<float *>(&R_S[i][j][0]);
4531             tmp_max_0 = max(t_fptr_S[0], t_fptr_S[1]) * scale;
4532             tmp_max_1 = max(t_fptr_S[2], t_fptr_S[3]) * scale;
4533         } else {
4534             float2 t_reg_S_0 =
4535                 __half22float2(HALF2(R_S[i][j][0])); // rows 0~7: {c0, c1}

```

```

4536     float2 t_reg_S_1 =
4537         __half22float2(HALF2(R_S[i][j][1])); // rows 8~15: {c2, c3}
4538     tmp_max_0 = max(t_reg_S_0.x, t_reg_S_0.y) * scale;
4539     tmp_max_1 = max(t_reg_S_1.x, t_reg_S_1.y) * scale;
4540 }
4541 lane_row_max_new[i][0] = max(lane_row_max_new[i][0], tmp_max_0);
4542 lane_row_max_new[i][1] = max(lane_row_max_new[i][1], tmp_max_1);
4543 }
4544 // Warp-level reduce max (kWarpWidth = 4 for Q@K^T — only lanes
4545 // {0,4,8,...,28} hold valid data)
4546 lane_row_max_new[i][0] =
4547     warp_reduce_max<4, float>(lane_row_max_new[i][0]);
4548 lane_row_max_new[i][1] =
4549     warp_reduce_max<4, float>(lane_row_max_new[i][1]);
4550 }
4551
4552 // Pass 2: Compute P = exp(S*scale - m_new), store back to R_S
4553 // 面试关键点: 这里将 P 写回 R_S 寄存器!
4554 // 为什么可以? 当前实现依赖 m16n8k16 这一路径下约定好的 fragment 布局,
4555 // 使 softmax 后的 P 能继续留在 R_S 中供后面的 P@V 直接消费, 无需额外重组。
4556 #pragma unroll
4557 for (int i = 0; i < kValTileSeqLenQ; ++i) {
4558     // m_new = max(m_old, m_cur)
4559     float block_row_max_new_0 =
4560         max(lane_block_row_max_old[i][0], lane_row_max_new[i][0]);
4561     float block_row_max_new_1 =
4562         max(lane_block_row_max_old[i][1], lane_row_max_new[i][1]);
4563
4564     #pragma unroll
4565     for (int j = 0; j < kValTileSeqLenK; ++j) {
4566         if constexpr (kMmaAccF32) {
4567             float *t_fptr_S = reinterpret_cast<float *>(&R_S[i][j][0]);
4568             half *t_hptr_S = reinterpret_cast<half *>(&R_S[i][j][0]);
4569             t_fptr_S[0] = __expf(__fmaf_rn(t_fptr_S[0], scale, -block_row_max_new_0));
4570             t_fptr_S[1] = __expf(__fmaf_rn(t_fptr_S[1], scale, -block_row_max_new_0));
4571             t_fptr_S[2] = __expf(__fmaf_rn(t_fptr_S[2], scale, -block_row_max_new_1));
4572             t_fptr_S[3] = __expf(__fmaf_rn(t_fptr_S[3], scale, -block_row_max_new_1));
4573             lane_row_sum_new[i][0] += (t_fptr_S[0] + t_fptr_S[1]);
4574             lane_row_sum_new[i][1] += (t_fptr_S[2] + t_fptr_S[3]);
4575             t_hptr_S[0] = __float2half_rn(t_fptr_S[0]);
4576             t_hptr_S[1] = __float2half_rn(t_fptr_S[1]);
4577             t_hptr_S[2] = __float2half_rn(t_fptr_S[2]);
4578             t_hptr_S[3] = __float2half_rn(t_fptr_S[3]);
4579         } else {
4580             float2 t_reg_S_0 = __half22float2(HALF2(R_S[i][j][0]));
4581             float2 t_reg_S_1 = __half22float2(HALF2(R_S[i][j][1]));
4582             t_reg_S_0.x =
4583                 __expf(__fmaf_rn(t_reg_S_0.x, scale, -block_row_max_new_0));
4584             t_reg_S_0.y =
4585                 __expf(__fmaf_rn(t_reg_S_0.y, scale, -block_row_max_new_0));
4586             t_reg_S_1.x =
4587                 __expf(__fmaf_rn(t_reg_S_1.x, scale, -block_row_max_new_1));
4588             t_reg_S_1.y =
4589                 __expf(__fmaf_rn(t_reg_S_1.y, scale, -block_row_max_new_1));
4590             lane_row_sum_new[i][0] += (t_reg_S_0.x + t_reg_S_0.y);
4591             lane_row_sum_new[i][1] += (t_reg_S_1.x + t_reg_S_1.y);
4592             HALF2(R_S[i][j][0]) = __float22half2_rn(t_reg_S_0);
4593             HALF2(R_S[i][j][1]) = __float22half2_rn(t_reg_S_1);
4594         }
4595     }
4596 }
4597 // Warp-level reduce sum (kWarpWidth = 4, same as max)
4598 lane_row_sum_new[i][0] =

```

```

4599     warp_reduce_sum<4, float>(lane_row_sum_new[i][0]);
4600     lane_row_sum_new[i][1] =
4601         warp_reduce_sum<4, float>(lane_row_sum_new[i][1]);
4602 }
4603 __syncthreads();
4604
4605 // =====
4606 // 3d: P@V — P[Br,Bc] @ V[Bc,d] = O[Br,d]
4607 // =====
4608 // Wait for V to be ready before computing P@V
4609 if constexpr (kStagesK > 1) {
4610     if ((tile_K_seqlen + 1) < Tc) {
4611         CP_ASYNC_WAIT_GROUP(1);
4612     } else {
4613         CP_ASYNC_WAIT_GROUP(0);
4614     }
4615 } else {
4616     CP_ASYNC_WAIT_GROUP(0);
4617 }
4618 __syncthreads();
4619
4620 fill_3D_regs<uint32_t, kValTileSeqLenP, kValTileHeadDimV, kNRegs>(R_0, 0);
4621
4622 // tile_V_Bc: iterate over chunks of Bc/K=16 columns in P matrix
4623 // Bc=kMmaAtomK=16 → 1 iteration for kHeadDim≤64 configurations
4624 #pragma unroll
4625 for (int tile_V_Bc = 0; tile_V_Bc < (Bc / kMmaAtomK); ++tile_V_Bc) {
4626     // ldmatrix.x2.trans: load V[Bc,d] with transposition
4627     // V is row-major [Bc,d], but NN matmul needs B matrix in col-major → use
4628     // transposed ldmatrix
4629 #pragma unroll
4630 for (int j = 0; j < kValTileHeadDimV; ++j) {
4631     int warp_smem_V_d =
4632         warp_KV * (kMmaAtomN * kValTileHeadDimV) + j * kMmaAtomN;
4633     int lane_smem_V_Bc = tile_V_Bc * kMmaAtomK + lane_id % 16;
4634     int lane_smem_V_d = warp_smem_V_d;
4635     uint32_t lane_smem_V_ptr;
4636     if constexpr (kSwizzleV) {
4637         lane_smem_V_ptr = smem_V_base_ptr +
4638             ((lane_smem_V_d / kMmaAtomK) * Bc * kMmaAtomK +
4639              lane_smem_V_Bc * kMmaAtomK +
4640              swizzle<kMmaAtomK>(lane_smem_V_Bc,
4641                               lane_smem_V_d % kMmaAtomK)) *
4642             sizeof(half);
4643     } else {
4644         lane_smem_V_ptr = smem_V_base_ptr +
4645             (lane_smem_V_Bc * kSmemStrideV + lane_smem_V_d) * sizeof(half);
4646     }
4647     LDMATRIX_X2_T(R_V[j][0], R_V[j][1], lane_smem_V_ptr);
4648 }
4649
4650 // P matrix layout in R_S[i][j][2]:
4651 // MMA = m16n8k16, Br=16x4=64, Bc=8x8=64, layout: 4 warps
4652 // | 64x64 | warp_KV 0 |
4653 // | warp_QP 0 | MMA 0 ... MMA 0 (x8) |
4654 // | warp_QP 1 | MMA 1 ... MMA 1 (x8) |
4655 // | warp_QP 2 | MMA 2 ... MMA 2 (x8) |
4656 // | warp_QP 3 | MMA 3 ... MMA 3 (x8) |
4657 // tile_V_Bc selects which 16-column slice of P to use:
4658 // tile_V_Bc=0 → cols 0:16 → MMA indices 0,1 → w=0
4659 // tile_V_Bc=1 → cols 16:32 → MMA indices 2,3 → w=2
4660 // tile_V_Bc=2 → cols 32:48 → MMA indices 4,5 → w=4
4661 // tile_V_Bc=3 → cols 48:64 → MMA indices 6,7 → w=6

```

```

4662 // 对应的 MMA A fragment 布局可以这样记:
4663 // - rows 0~7: lane 0~3 -> {a0,a1} 与 {a4,a5}, lane 4~7 -> 下一行, 依次类推
4664 // - rows 8~15: lane 0~3 -> {a2,a3} 与 {a6,a7}, lane 4~7 -> 下一行, 依次类推
4665 // 当前实现正是利用这一路径下 A fragment 与前面生成的 P fragment 可以直接对接,
4666 // 才能把 R_S 中的 P 直接喂给 HMMA16816 做 P@V; 复习时不要把它背成对所有 MMA
4667 // fragment 都无条件成立的通用结论。layout转换逻辑:
4668 // C layout: 8 x (16 x 8) layout -> A layout: 4 x (16 x 16) layout
4669 int w = tile_V_Bc * 2;
4670 #pragma unroll
4671 for (int i = 0; i < kValTileSeqLenP; ++i) {
4672 #pragma unroll
4673 for (int j = 0; j < kValTileHeadDimV; ++j) {
4674 if constexpr (kMmaAccF32) {
4675     HMMA16816F32(R_0[i][j][0], R_0[i][j][1], R_0[i][j][2], R_0[i][j][3],
4676                 R_S[i][w][0], R_S[i][w][1], R_S[i][w + 1][0], R_S[i][w + 1][1],
4677                 R_V[j][0], R_V[j][1],
4678                 R_0[i][j][0], R_0[i][j][1], R_0[i][j][2], R_0[i][j][3]);
4679 } else {
4680     HMMA16816(R_0[i][j][0], R_0[i][j][1],
4681               R_S[i][w][0], R_S[i][w][1], R_S[i][w + 1][0], R_S[i][w + 1][1],
4682               R_V[j][0], R_V[j][1],
4683               R_0[i][j][0], R_0[i][j][1]);
4684 }
4685 }
4686 }
4687 } // end for tile_V_Bc
4688 __syncthreads();
4689
4690 // =====
4691 // 3e: Online rescaling — 0_new = exp(m_old - m_new) * 0_old + P@V
4692 // =====
4693 // 公式来源: FA2 paper Eq.(7-8), 使用 exp(m_old - m_new) 做 0 与 l 的 rescale
4694 #pragma unroll
4695 for (int i = 0; i < kValTileSeqLenP; ++i) {
4696     float block_row_max_new_0 = lane_row_max_new[i][0];
4697     float block_row_max_new_1 = lane_row_max_new[i][1];
4698     float block_row_sum_new_0 = lane_row_sum_new[i][0];
4699     float block_row_sum_new_1 = lane_row_sum_new[i][1];
4700
4701     float block_row_max_old_0 = lane_block_row_max_old[i][0];
4702     float block_row_max_old_1 = lane_block_row_max_old[i][1];
4703
4704     block_row_max_new_0 = max(block_row_max_old_0, block_row_max_new_0);
4705     block_row_max_new_1 = max(block_row_max_old_1, block_row_max_new_1);
4706
4707     // Handle first iteration: m_old = -inf, need to use m_new directly
4708     block_row_max_old_0 =
4709         (tile_K_seqlen > 0 ? block_row_max_old_0 : block_row_max_new_0);
4710     block_row_max_old_1 =
4711         (tile_K_seqlen > 0 ? block_row_max_old_1 : block_row_max_new_1);
4712
4713     float rescale_o_factor_0 =
4714         __expf(block_row_max_old_0 - block_row_max_new_0);
4715     float rescale_o_factor_1 =
4716         __expf(block_row_max_old_1 - block_row_max_new_1);
4717
4718     // Rescale 0_old + Add P@V in one fused step
4719 #pragma unroll
4720 for (int j = 0; j < kValTileHeadDimV; ++j) {
4721     if constexpr (kMmaAccF32) {
4722         float *t_fptr_0 = reinterpret_cast<float*>(&R_0[i][j][0]);
4723         float *t_fptr_D = reinterpret_cast<float*>(&R_D[i][j][0]);
4724         t_fptr_D[0] = __fmaf_rn(rescale_o_factor_0, t_fptr_D[0], t_fptr_0[0]);

```

```

4725     t_fptr_D[1] = __fmaf_rn(rescale_o_factor_0, t_fptr_D[1], t_fptr_0[1]);
4726     t_fptr_D[2] = __fmaf_rn(rescale_o_factor_1, t_fptr_D[2], t_fptr_0[2]);
4727     t_fptr_D[3] = __fmaf_rn(rescale_o_factor_1, t_fptr_D[3], t_fptr_0[3]);
4728 } else {
4729     float2 t_reg_0_0 = __half22float2(HALF2(R_0[i][j][0]));
4730     float2 t_reg_0_1 = __half22float2(HALF2(R_0[i][j][1]));
4731     float2 t_reg_D_0 = __half22float2(HALF2(R_D[i][j][0]));
4732     float2 t_reg_D_1 = __half22float2(HALF2(R_D[i][j][1]));
4733     t_reg_D_0.x = __fmaf_rn(rescale_o_factor_0, t_reg_D_0.x, t_reg_0_0.x);
4734     t_reg_D_0.y = __fmaf_rn(rescale_o_factor_0, t_reg_D_0.y, t_reg_0_0.y);
4735     t_reg_D_1.x = __fmaf_rn(rescale_o_factor_1, t_reg_D_1.x, t_reg_0_1.x);
4736     t_reg_D_1.y = __fmaf_rn(rescale_o_factor_1, t_reg_D_1.y, t_reg_0_1.y);
4737     HALF2(R_D[i][j][0]) = __float22half2_rn(t_reg_D_0);
4738     HALF2(R_D[i][j][1]) = __float22half2_rn(t_reg_D_1);
4739 }
4740 }
4741
4742 // Update l: l_new = exp(m_old - m_new) * l_old + row_sum(P)
4743 float block_row_sum_old_0 = lane_block_row_sum_old[i][0];
4744 float block_row_sum_old_1 = lane_block_row_sum_old[i][1];
4745 lane_block_row_sum_old[i][0] = __fmaf_rn(
4746     rescale_o_factor_0, block_row_sum_old_0, block_row_sum_new_0);
4747 lane_block_row_sum_old[i][1] = __fmaf_rn(
4748     rescale_o_factor_1, block_row_sum_old_1, block_row_sum_new_1);
4749
4750 // Update m
4751 lane_block_row_max_old[i][0] = block_row_max_new_0;
4752 lane_block_row_max_old[i][1] = block_row_max_new_1;
4753 }
4754
4755 // Wait for next K tile to be ready in smem before next iteration
4756 if constexpr (kStagesK > 1) {
4757     if ((tile_K_seqlen + 1) < Tc) {
4758         CP_ASYNC_WAIT_GROUP(0);
4759     }
4760     __syncthreads();
4761 }
4762 } // end outer loop over K seqlen
4763 __syncthreads();
4764
4765 // =====
4766 // Step 4: 最终 rescale — 0_final = (1/l_final) * 0_final
4767 // =====
4768 #pragma unroll
4769 for (int i = 0; i < kValTileSeqLenP; ++i) {
4770     float rescale_factor_0 = __frcp_rn(lane_block_row_sum_old[i][0]);
4771     float rescale_factor_1 = __frcp_rn(lane_block_row_sum_old[i][1]);
4772 #pragma unroll
4773 for (int j = 0; j < kValTileHeadDimV; ++j) {
4774     if constexpr (kMmaAccF32) {
4775         float *t_fptr_D = reinterpret_cast<float*>(&R_D[i][j][0]);
4776         half *t_hptr_D = reinterpret_cast<half*>(&R_D[i][j][0]);
4777         t_hptr_D[0] = __float2half_rn(rescale_factor_0 * t_fptr_D[0]);
4778         t_hptr_D[1] = __float2half_rn(rescale_factor_0 * t_fptr_D[1]);
4779         t_hptr_D[2] = __float2half_rn(rescale_factor_1 * t_fptr_D[2]);
4780         t_hptr_D[3] = __float2half_rn(rescale_factor_1 * t_fptr_D[3]);
4781     } else {
4782         float2 t_reg_D_0 = __half22float2(HALF2(R_D[i][j][0]));
4783         float2 t_reg_D_1 = __half22float2(HALF2(R_D[i][j][1]));
4784         t_reg_D_0.x = rescale_factor_0 * t_reg_D_0.x;
4785         t_reg_D_0.y = rescale_factor_0 * t_reg_D_0.y;
4786         t_reg_D_1.x = rescale_factor_1 * t_reg_D_1.x;
4787         t_reg_D_1.y = rescale_factor_1 * t_reg_D_1.y;

```

```

4788     HALF2(R_D[i][j][0]) = __float2half2_rn(t_reg_D_0);
4789     HALF2(R_D[i][j][1]) = __float2half2_rn(t_reg_D_1);
4790 }
4791 }
4792 }
4793
4794 // =====
4795 // Step 5: Epilogue — Collective store via warp shuffle + 128-bit store
4796 // =====
4797 // 利用 warp shuffle 将分散在各 lane 的寄存器数据收集到 lane 0~3,
4798 // 然后用 LDST128BITS (st.global.v4.f32) 一次性写入 16 bytes
4799 #pragma unroll
4800 for (int i = 0; i < kValTileSeqLenP; ++i) {
4801     #pragma unroll
4802     for (int j = 0; j < kValTileHeadDimV; ++j) {
4803         uint32_t R_Z[2][4];
4804         R_Z[0][0] = R_D[i][j][0];
4805         R_Z[1][0] = R_D[i][j][1];
4806         R_Z[0][1] = __shfl_sync(0xffffffff, R_D[i][j][0], lane_id + 1, 4);
4807         R_Z[0][2] = __shfl_sync(0xffffffff, R_D[i][j][0], lane_id + 2, 4);
4808         R_Z[0][3] = __shfl_sync(0xffffffff, R_D[i][j][0], lane_id + 3, 4);
4809         R_Z[1][1] = __shfl_sync(0xffffffff, R_D[i][j][1], lane_id + 1, 4);
4810         R_Z[1][2] = __shfl_sync(0xffffffff, R_D[i][j][1], lane_id + 2, 4);
4811         R_Z[1][3] = __shfl_sync(0xffffffff, R_D[i][j][1], lane_id + 3, 4);
4812
4813         // st.global.v4.f32: 128-bit store, 4 lanes × 32-bit
4814         if (lane_id % 4 == 0) {
4815             int store_warp_regs_0_Br =
4816                 warp_QP * (kMmaAtomM * kValTileSeqLenP) + i * kMmaAtomM;
4817             int store_lane_gmem_0_Br =
4818                 Q_tile_id * Br + store_warp_regs_0_Br + lane_id / 4;
4819             int store_warp_regs_0_d =
4820                 warp_KV * (kMmaAtomN * kValTileHeadDimV) + j * kMmaAtomN;
4821             int store_lane_gmem_0_d = store_warp_regs_0_d;
4822             int store_gmem_0_addr_0 = 0_gmem_offset +
4823                 (store_lane_gmem_0_Br + 0) * kHeadDim +
4824                 store_lane_gmem_0_d;
4825             int store_gmem_0_addr_1 = 0_gmem_offset +
4826                 (store_lane_gmem_0_Br + 8) * kHeadDim +
4827                 store_lane_gmem_0_d;
4828             // LDST128BITS = reinterpret_cast<float4*>
4829             *reinterpret_cast<float4*>(&0[store_gmem_0_addr_0]) =
4830                 *reinterpret_cast<float4*>(&R_Z[0][0]);
4831             *reinterpret_cast<float4*>(&0[store_gmem_0_addr_1]) =
4832                 *reinterpret_cast<float4*>(&R_Z[1][0]);
4833         }
4834     }
4835 }
4836 }
4837
4838 #if defined(NOTES_V2_ENABLE_TMA_MMA_WS)
4839 // =====
4840 // Phase 8b: FlashAttention-2 TMA + MMA Warp Specialization (SM120)
4841 // =====
4842 // 面试要点 (FA TMA MMA WS vs cp.async FA) :
4843 //   - Producer (WG0, 128T) 用 TMA 128B SWIZZLE 搬运 Q/K/V tile 到 smem
4844 //   - Consumer (WG1, 256T) 用 ldmatrix + mma.sync.m16n8k16 做 Q@K^T / P@V
4845 //   - 与 hgemm_tma_mma_ws_tn 的对比:
4846 //     * HGEMM 只做一次 GEMM, K 维迭代; FA 做 QK^T 和 PV 两次 GEMM, KV seqlen 迭代
4847 //     * HGEMM 的 A/B 都 staged; FA 中 Q 只 load 一次 (split-Q), K staged, V 单 buffer
4848 //
4849 // ★ split-Q 语义 (关键, 避免误解) :
4850 //   - grid = (seqlen/Br, B*H), 每个 block 处理一个 **block-level Q tile [Br, d]**

```

```

4851 //      (不是全局 Q)。对 Q 的 seqlen 做 block-level 并行。
4852 //      - 每个 block 遍历**完整的 KV seqlen** (Tc = seqlen/Bc 次外循环)。
4853 //      - 因此每个 block 的 Q tile 只需 **load 一次到 smem**，在整个 KV 遍历中复用
4854 //      → 只需 full_Q barrier (无 empty_Q)。
4855 //
4856 // ★ Pipeline 同步协议 (arrive_count = 257 = 256 consumer + 1 producer) :
4857 //      - Q: full_Q — Producer 发 arrive_tx, Consumer 在主循环前 wait 一次
4858 //      - K: full_K[kStagesK] / empty_K[kStagesK] — 多 stage pipeline
4859 //      - V: full_V / empty_V — 单 buffer 串行 (QK 前发 TMA → QK 重叠 → PV 前等 → PV 后释放)
4860 //
4861 // ★ V ldmatrix.x2.trans 正确性 (已论证无风险) :
4862 //      TMA 128B SWIZZLE 是 1-1 映射, ldmatrix 用 swizzle<64>(row,col) 计算的物理地址
4863 //      = TMA 写入的物理地址 (smem 1024B 对齐保证 phase=0)。ldmatrix.x2.trans 的转置
4864 //      语义在寄存器层面工作, 与 smem 物理布局无关 → 必然正确。
4865 //
4866 // ★ Shared Memory 占用分析 (D=64/128, kStagesK=1/2/3/4, kStagesV=1/2) :
4867 //      smem = (Q[Br,D] + K[kStagesK,Bc,D] + V[kStagesV,Bc,D]) * sizeof(half)
4868 //            = D * (Br + kStagesK*Bc + kStagesV*Bc) * 2
4869 //            = D * (128 + (kStagesK+kStagesV)*64) * 2      (Br=128, Bc=64)
4870 //
4871 //      | kStagesK | kStagesV | D=64 bytes | D=128 bytes |
4872 //      |-----|-----|-----|-----|
4873 //      | 1      | 1      | 32 KB      | 64 KB      |
4874 //      | 2      | 1      | 40 KB      | 80 KB      |
4875 //      | 2      | 2      | 48 KB      | 96 KB      |
4876 //      | 3      | 1      | 48 KB      | 96 KB      |
4877 //      | 3      | 2      | 56 KB      | 112 KB     |
4878 //      | 4      | 1      | 56 KB      | 112 KB     |
4879 //
4880 //      SM120 (RTX PRO 5000) optin smem 上限 ~100KB:
4881 //      - D=64: S=1/2/3/4 全部可行 (32-56 KB)
4882 //      - D=128: S=1/2/3 可行 (64-96 KB); S=4 (112 KB) 超出 optin 上限, 会被
4883 //        check_smem_feasible 兜底判为 SMEM SKIP。D=128 推荐 S=2 (甜点: 80 KB,
4884 //        2 blocks/SM, pipeline 深度足够隐藏 TMA 延迟)。
4885 //
4886 // v1 限制: kHeadDim=64 or 128; seqlen % Br == 0 且 seqlen % Bc == 0 (不处理尾 tile)
4887 //
4888 // =====
4889 // FA TMA WS smem offset helper (D=64/128 通用)
4890 // =====
4891 // 背景: CU_TENSOR_MAP_SWIZZLE_128B 硬件要求 box innermost dim ≤ 128B = 64 half
4892 // (见 /usr/local/cuda/include/cuda.h 的 cuTensorMapEncodeTiled 文档)。因此 D=128
4893 // 不能用单个 TMA box=(128, Br) 覆盖整行。
4894 //
4895 // 方案: D=128 时 box 固定为 (64, Br), 沿 head_dim 方向连续发 kTmaChunks 次 TMA
4896 // (minor_coord = c*64), 写入 chunk-major smem 布局 [kTmaChunks, Br, 64]:
4897 //      - chunk c 的 smem 起始偏移 = c * Br * 64
4898 //      - chunk c 内部按 [Br, 64] row-major + SWIZZLE_128B 存放 (64 half = 128B 周期)
4899 //
4900 // 本 helper 计算 (row, col) 在该 chunk-major 布局下的 swizzled smem 元素偏移:
4901 //      chunk = col / 64      ∈ [0, kTmaChunks)
4902 //      col_in = col % 64     ∈ [0, 64)
4903 //      offset = chunk * Br * 64 + row * 64 + swizzle<64>(row, col_in)
4904 //
4905 // D=64 退化: kTmaChunks=1, chunk=0, offset = row*64 + swizzle<64>(row, col)
4906 //      与原公式 (row*64 + swizzle<64>(row, col)) 完全一致 → 无回归。
4907 template <int kHeadDim, int Br>
4908 __device__ __forceinline__ int swizzle_fa(int row, int col) {
4909     static_assert(kHeadDim == 64 || kHeadDim == 128, "D=64 or 128 only");
4910     if constexpr (kHeadDim == 64) {
4911         return row * 64 + swizzle<64>(row, col);
4912     } else { // kHeadDim == 128
4913         const int chunk = col >> 6; // col / 64 ∈ {0, 1}

```

```

4914     const int col_in = col & 63;          // col % 64
4915     return chunk * (Br * 64) + row * 64 + swizzle<64>(row, col_in);
4916 }
4917 }
4918
4919 template <
4920     const int kHeadDim,          // head dim: v1 only 64
4921     const int kMmaAtomM,        // 16 (MMA instruction M dim)
4922     const int kMmaAtomN,        // 8 (MMA instruction N dim)
4923     const int kMmaAtomK,        // 16 (MMA instruction K dim)
4924     const int kMmaAccF32,       // 0=f16 acc, 1=f32 acc (mma.sync.f32.f16.f16.f32)
4925     const int kMmaTileSeqLenQ,  // MMA tiles along Q's M dim, 8 → Br=16*8=128
4926     const int kMmaTileSeqLenK,  // MMA tiles along K's N dim, 1 → Bc basis=8
4927     const int kMmaTileSeqLenP,  // MMA tiles for P@V M dim, == kMmaTileSeqLenQ
4928     const int kMmaTileHeadDimV, // MMA tiles for P@V N dim (head dim direction)
4929     const int kValTileSeqLenQ,  // value tiles along Q's M, 1 → Br per warp=16
4930     const int kValTileSeqLenK,  // value tiles along K's N, 8 → Bc_warp=8*8=64
4931     const int kValTileSeqLenP,  // value tiles for P@V M dim, 1
4932     const int kValTileHeadDimV, // value tiles for P@V N dim, kHeadDim/(8*kMmaTileHeadDimV)
4933     const int kStagesK,         // K pipeline depth (>=1)
4934     const int kStagesV,         // V pipeline depth (>=1; 1=single buffer, >=2=pipelined)
4935     const int kNumThreads>     // 384, 128 producer + 256 consumer
4936 __global__ void __launch_bounds__(kNumThreads, 1)
4937     flash_attn_tma_mma_ws_stages_split_q(
4938         half *Q, half *K, half *V, half *O, int N, int H,
4939         const UTensorMap *__restrict__ tensorMapQ,
4940         const UTensorMap *__restrict__ tensorMapK,
4941         const UTensorMap *__restrict__ tensorMapV) {
4942     static_assert(kHeadDim == 64 || kHeadDim == 128,
4943         "v1 supports kHeadDim == 64 or 128");
4944     static_assert(kNumThreads == 384, "128 producer + 256 consumer");
4945     static_assert(kMmaAccF32 == 0 || kMmaAccF32 == 1, "kMmaAccF32 must be 0 or 1");
4946     static_assert(kStagesK >= 1, "kStagesK must be >= 1");
4947     static_assert(kStagesV >= 1, "kStagesV must be >= 1");
4948     constexpr int kNRegs = kMmaAccF32 ? 4 : 2;
4949
4950     constexpr int Br = kMmaAtomM * kMmaTileSeqLenQ * kValTileSeqLenQ; // 128
4951     constexpr int Bc = kMmaAtomN * kMmaTileSeqLenK * kValTileSeqLenK; // 64
4952     // BK for QK: kHeadDim=64, split into kHeadDim/kMmaAtomK=4 MMA K slices
4953     // BK for PV: Bc=64, split into Bc/kMmaAtomK=4 MMA K slices (P rows)
4954     constexpr int kConsumerThreads = 256; // 8 warps
4955     constexpr int kProducerThreads = 128; // 4 warps, only thread 0 issues TMA
4956     constexpr int kQTileBytes = Br * kHeadDim * sizeof(half); // 16 KB
4957     constexpr int kKTileBytes = Bc * kHeadDim * sizeof(half); // 8 KB
4958     constexpr int kVTileBytes = Bc * kHeadDim * sizeof(half); // 8 KB
4959     // TMA box innermost 固定 64 half (128B), 满足 CU_TENSOR_MAP_SWIZZLE_128B 的
4960     // box innermost ≤ 128B 硬约束。D=128 时沿 head_dim 发 kTmaChunks 次 TMA,
4961     // 写入 chunk-major smem 布局 [kTmaChunks, Br, 64]。
4962     constexpr int kTmaBoxMinor = 64;
4963     constexpr int kTmaChunks = kHeadDim / kTmaBoxMinor;
4964
4965     // Block indexing — split-Q: blockIdx.x indexes Q tile along seqlen
4966     const int Nb_id = blockIdx.y / H;
4967     const int Nh_id = blockIdx.y % H;
4968     const int Q_tile_id = blockIdx.x;
4969     const int Tc = (N + Bc - 1) / Bc; // number of KV tiles along seqlen
4970     const float scale = 1.0f / sqrtf((float)kHeadDim);
4971
4972     // Per-head gmem base offset (self-attention: Q=K=V share layout)
4973     const int QKV_gmem_offset = (Nb_id * H * N + Nh_id * N) * kHeadDim;
4974     const int O_gmem_offset = QKV_gmem_offset;
4975     // TMA descriptor covers the entire [B*H*N, D] gmem as one 2D matrix; the
4976     // producer must offset major_coord by this head/batch base so each block

```

```

4977 // loads its own (batch, head) Q/K/V tile instead of always reading head 0.
4978 const int qkv_major_base = (Nb_id * H + Nh_id) * N;
4979
4980 // ---- Shared memory layout ----
4981 // TMA CU_TENSOR_MAP_SWIZZLE_128B requires 1024B-aligned smem base so the
4982 // hardware swizzle phase starts at zero. Consumer swizzle<64>() assumes
4983 // zero phase (no base_offset compensation like WGMMMA descriptor).
4984 // Q: [Br, kHeadDim] = [128, 64] = 16 KB (1024B aligned ✓)
4985 // K: [kStagesK, Bc, kHeadDim] = [kStagesK, 64, 64] (8 KB per stage)
4986 // V: [kStagesV, Bc, kHeadDim] = [kStagesV, 64, 64] (8 KB per stage)
4987 extern __shared__ __align__(1024) uint8_t smem_fa_tma_ws[];
4988 half *Q_smem = reinterpret_cast<half*>(smem_fa_tma_ws);
4989 half *K_smem = Q_smem + Br * kHeadDim;
4990 half *V_smem = K_smem + kStagesK * Bc * kHeadDim;
4991
4992 const uint32_t smem_Q_base_ptr = __cvta_generic_to_shared(Q_smem);
4993 const uint32_t smem_K_base_ptr = __cvta_generic_to_shared(K_smem);
4994 const uint32_t smem_V_base_ptr = __cvta_generic_to_shared(V_smem);
4995
4996 // ---- Barriers ----
4997 // arrive_count = kConsumerThreads + 1 = 257 (256 consumer arrives + 1 producer arrive_tx)
4998 #pragma nv_diag_suppress static_var_with_dynamic_init
4999 __shared__ cuda::barrier<cuda::thread_scope_block> full_Q;
5000 __shared__ cuda::barrier<cuda::thread_scope_block> full_K[kStagesK];
5001 __shared__ cuda::barrier<cuda::thread_scope_block> empty_K[kStagesK];
5002 __shared__ cuda::barrier<cuda::thread_scope_block> full_V[kStagesV];
5003 __shared__ cuda::barrier<cuda::thread_scope_block> empty_V[kStagesV];
5004 #pragma nv_diag_default static_var_with_dynamic_init
5005
5006 // Thread partition: first kProducerThreads (0~127) = Producer WG,
5007 // remaining kConsumerThreads (128~383) = Consumer WG.
5008 // WARN: Must use kProducerThreads (not kConsumerThreads) as the divisor, otherwise
5009 // the split is wrong: 384/256=1.5 would give Producer 256 threads and
5010 // Consumer only 128, but barriers expect 256 consumer arrives → deadlock.
5011 const int wg_idx = (threadIdx.x < kProducerThreads) ? 0 : 1;
5012 const int wg_tid = (threadIdx.x < kProducerThreads) ? threadIdx.x :
5013                 (threadIdx.x - kProducerThreads);
5014
5015 if (threadIdx.x == 0) {
5016     init(&full_Q, kConsumerThreads + 1);
5017     for (int s = 0; s < kStagesV; ++s) {
5018         init(&full_V[s], kConsumerThreads + 1);
5019         init(&empty_V[s], kConsumerThreads + 1);
5020     }
5021     for (int s = 0; s < kStagesK; ++s) {
5022         init(&full_K[s], kConsumerThreads + 1);
5023         init(&empty_K[s], kConsumerThreads + 1);
5024     }
5025     tma_fence_proxy_async_shared_cta();
5026 }
5027 __syncthreads();
5028
5029 // =====
5030 // Producer Warpgroup (WG0, threadIdx.x 0~127)
5031 // Only wg_tid == 0 issues TMA. Split-Q: Q tile loaded once, K staged, V single buffer.
5032 // D=128 时每个 tile 沿 head_dim 发 kTmaChunks 次 TMA (box innermost=64 half) ,
5033 // 共享同一 barrier: N 次 cp.async.bulk.tensor 各 reduce chunk_bytes 到 tx count,
5034 // 1 次 mbarrier_arrive_expect_tx 声明总字节 + producer thread arrive.
5035 // =====
5036 if (wg_idx == 0) {
5037     NOTES_V2_REG_DEALLOC(40);
5038     if (wg_tid == 0) {
5039         // Step P0: Load block-level Q tile [Br, d] once. D=128 时发 kTmaChunks 次。

```

```

5040 // minor_coord = c * 64, major_coord = qkv_major_base + Q_tile_id * Br
5041 // smem dst = Q_smem + c * Br * 64 (chunk-major 布局)
5042 for (int c = 0; c < kTmaChunks; ++c) {
5043     tma_load_2d(Q_smem + c * Br * kTmaBoxMinor, tensorMapQ,
5044                 c * kTmaBoxMinor, qkv_major_base + Q_tile_id * Br, full_Q);
5045 }
5046 tma_arrive_expect_tx(full_Q, kQTileBytes);
5047
5048 // Step P1: Prefetch first (kStagesK-1) K tiles.
5049 // K tile coords: minor = c*64, major = qkv_major_base + s*Bc
5050 for (int s = 0; s < kStagesK - 1; ++s) {
5051     empty_K[s].wait(empty_K[s].arrive());
5052     for (int c = 0; c < kTmaChunks; ++c) {
5053         tma_load_2d(K_smem + s * Bc * kHeadDim + c * Bc * kTmaBoxMinor,
5054                     tensorMapK, c * kTmaBoxMinor,
5055                     qkv_major_base + s * Bc, full_K[s]);
5056     }
5057     tma_arrive_expect_tx(full_K[s], kKTileBytes);
5058 }
5059
5060 // Step P1b: Prefetch first (kStagesV-1) V tiles (only if kStagesV > 1).
5061 // kStagesV=1: no prefetch; V[k] loaded in P2 overlaps only QK gemm.
5062 // kStagesV>=2: V prefetched here; in P2 V[k+kStagesV-1] overlaps
5063 // QK+softmax+PV of current iter → true V pipeline.
5064 if constexpr (kStagesV > 1) {
5065     for (int s = 0; s < kStagesV - 1; ++s) {
5066         empty_V[s].wait(empty_V[s].arrive());
5067         for (int c = 0; c < kTmaChunks; ++c) {
5068             tma_load_2d(V_smem + s * Bc * kHeadDim + c * Bc * kTmaBoxMinor,
5069                         tensorMapV, c * kTmaBoxMinor,
5070                         qkv_major_base + s * Bc, full_V[s]);
5071         }
5072         tma_arrive_expect_tx(full_V[s], kVTileBytes);
5073     }
5074 }
5075
5076 // Step P2: Main loop over KV seqlen tiles
5077 for (int k = 0; k < Tc; ++k) {
5078     [[maybe_unused]] const int stage = k % kStagesK;
5079
5080     // Issue V[k+kStagesV-1] TMA to stage_next_v (pipelined).
5081     // kStagesV=1: loads V[k] to stage 0, overlaps with QK of same iter.
5082     // kStagesV>=2: loads V[k+kStagesV-1], overlaps with QK+softmax+PV.
5083     {
5084         const int v_tile = k + kStagesV - 1;
5085         if (v_tile < Tc) {
5086             const int stage_next_v = v_tile % kStagesV;
5087             empty_V[stage_next_v].wait(empty_V[stage_next_v].arrive());
5088             for (int c = 0; c < kTmaChunks; ++c) {
5089                 tma_load_2d(V_smem + stage_next_v * Bc * kHeadDim +
5090                             c * Bc * kTmaBoxMinor,
5091                             tensorMapV, c * kTmaBoxMinor,
5092                             qkv_major_base + v_tile * Bc,
5093                             full_V[stage_next_v]);
5094             }
5095             tma_arrive_expect_tx(full_V[stage_next_v], kVTileBytes);
5096         }
5097     }
5098
5099     // Prefetch K[k+kStagesK-1] (pipelined).
5100     if (k + kStagesK - 1 < Tc) {
5101         const int stage_next = (k + kStagesK - 1) % kStagesK;
5102         empty_K[stage_next].wait(empty_K[stage_next].arrive());

```

```

5103     for (int c = 0; c < kTmaChunks; ++c) {
5104         tma_load_2d(K_smem + stage_next * Bc * kHeadDim +
5105             c * Bc * kTmaBoxMinor,
5106             tensorMapK, c * kTmaBoxMinor,
5107             qkv_major_base + (k + kStagesK - 1) * Bc,
5108             full_K[stage_next]);
5109     }
5110     tma_arrive_expect_tx(full_K[stage_next], kKTileBytes);
5111 }
5112 }
5113 }
5114 }
5115 // =====
5116 // Consumer Warpgroup (WG1, threadIdx.x 128~383, 256 threads = 8 warps)
5117 // All 256 threads participate in ldmatrix + mma.sync.
5118 // =====
5119 else {
5120     const int warp_id = wg_tid / kWarpSize; // 0~7
5121     const int lane_id = wg_tid % kWarpSize; // 0~31
5122     // Split-Q warp layout: 8 warps form 8x1 grid (warp_QP = 0~7, warp_KV = 0)
5123     // Br = 16 * 8 * 1 = 128 → 8 warps each handle 16 rows of Q
5124     const int warp_QP = warp_id;
5125     const int warp_KV = 0;
5126
5127     // Step C0: init — mark all K stages and V stages as "empty" (consumable by producer)
5128     // Consumer register budget per Triton flash_attn_v2 maxnreg strategy on
5129     // Blackwell warp_specialize: D=128 → 168, otherwise → 80.
5130     if constexpr (kHeadDim == 128) {
5131         NOTES_V2_REG_ALLOC(168);
5132     } else {
5133         NOTES_V2_REG_ALLOC(80);
5134     }
5135
5136     for (int s = 0; s < kStagesK; ++s) {
5137         [[maybe_unused]] auto token = empty_K[s].arrive();
5138     }
5139     for (int s = 0; s < kStagesV; ++s) {
5140         [[maybe_unused]] auto token = empty_V[s].arrive();
5141     }
5142
5143     // Wait for Q tile ready once (split-Q: Q smem is reused across all KV iters)
5144     full_Q.wait(full_Q.arrive());
5145
5146     // Online softmax persistent state (per-warp, per-row-pair)
5147     float lane_block_row_max_old[kValTileSeqLenQ][2];
5148     float lane_block_row_sum_old[kValTileSeqLenQ][2];
5149     fill_2D_regs<float, kValTileSeqLenQ, 2>(lane_block_row_max_old, -INFINITY);
5150     fill_2D_regs<float, kValTileSeqLenQ, 2>(lane_block_row_sum_old, 0.0f);
5151
5152     uint32_t R_Q[kValTileSeqLenQ][4];
5153     uint32_t R_K[kValTileSeqLenK][2];
5154     uint32_t R_V[kValTileHeadDimV][2];
5155     uint32_t R_S[kValTileSeqLenQ][kValTileSeqLenK][kNRegs];
5156     uint32_t R_O[kValTileSeqLenP][kValTileHeadDimV][kNRegs];
5157     uint32_t R_D[kValTileSeqLenP][kValTileHeadDimV][kNRegs];
5158
5159     fill_3D_regs<uint32_t, kValTileSeqLenP, kValTileHeadDimV, kNRegs>(R_D, 0);
5160
5161     // ---- Main loop over KV seqlen ----
5162     for (int tile_K_seqlen = 0; tile_K_seqlen < Tc; ++tile_K_seqlen) {
5163         const int stage = tile_K_seqlen % kStagesK;
5164         const int stage_v = tile_K_seqlen % kStagesV;

```

```

5166 // Wait for K[stage] ready
5167 full_K[stage].wait(full_K[stage].arrive());
5168 tma_fence_proxy_async_shared_cta();
5169
5170 // ---- 3b: Q @ K^T = S[Br, Bc] ----
5171 // Q smem reused (split-Q). K from stage `stage`.
5172 // smem addr (TMA 128B swizzle, swizzle<64>)
5173 fill_3D_regs<uint32_t, kValTileSeqLenQ, kValTileSeqLenK, kNRegs>(R_S, 0);
5174 #pragma unroll
5175 for (int tile_K_d = 0; tile_K_d < (kHeadDim / kMmaAtomK); ++tile_K_d) {
5176 // ldmatrix.x4: load Q m16k16 fragment (Q row-major, non-transpose)
5177 #pragma unroll
5178 for (int i = 0; i < kValTileSeqLenQ; ++i) {
5179     const int warp_smem_Q_Br =
5180         warp_QP * (kMmaAtomM * kValTileSeqLenQ) + i * kMmaAtomM;
5181     const int lane_smem_Q_Br = warp_smem_Q_Br + lane_id % 16;
5182     const int lane_smem_Q_d = tile_K_d * kMmaAtomK + (lane_id / 16) * 8;
5183     const uint32_t lane_smem_Q_ptr =
5184         smem_Q_base_ptr +
5185         swizzle_fa<kHeadDim, Br>(lane_smem_Q_Br, lane_smem_Q_d) *
5186             sizeof(half);
5187     LDMATRIX_X4(R_Q[i][0], R_Q[i][1], R_Q[i][2], R_Q[i][3],
5188                 lane_smem_Q_ptr);
5189 }
5190
5191 // ldmatrix.x2: load K k16n8 fragment (K row-major = K^T col-major)
5192 #pragma unroll
5193 for (int j = 0; j < kValTileSeqLenK; ++j) {
5194     const int warp_smem_K_Bc =
5195         warp_KV * (kMmaAtomN * kValTileSeqLenK) + j * kMmaAtomN;
5196     const int lane_smem_K_Bc = warp_smem_K_Bc + lane_id % 8;
5197     const int lane_smem_K_d =
5198         tile_K_d * kMmaAtomK + ((lane_id / 8) % 2) * 8;
5199     const uint32_t lane_smem_K_ptr =
5200         smem_K_base_ptr +
5201         (stage * Bc * kHeadDim +
5202          swizzle_fa<kHeadDim, Bc>(lane_smem_K_Bc, lane_smem_K_d)) *
5203             sizeof(half);
5204     LDMATRIX_X2(R_K[j][0], R_K[j][1], lane_smem_K_ptr);
5205 }
5206
5207 #pragma unroll
5208 for (int i = 0; i < kValTileSeqLenQ; ++i) {
5209 #pragma unroll
5210     for (int j = 0; j < kValTileSeqLenK; ++j) {
5211         if constexpr (kMmaAccF32) {
5212             HMMA16816F32(R_S[i][j][0], R_S[i][j][1], R_S[i][j][2], R_S[i][j][3],
5213                          R_Q[i][0], R_Q[i][1], R_Q[i][2], R_Q[i][3],
5214                          R_K[j][0], R_K[j][1],
5215                          R_S[i][j][0], R_S[i][j][1], R_S[i][j][2], R_S[i][j][3]);
5216         } else {
5217             HMMA16816(R_S[i][j][0], R_S[i][j][1], R_Q[i][0], R_Q[i][1],
5218                      R_Q[i][2], R_Q[i][3], R_K[j][0], R_K[j][1],
5219                      R_S[i][j][0], R_S[i][j][1]);
5220         }
5221     }
5222 }
5223 }
5224
5225 // Release K[stage] for producer reuse as early as possible: QK^T GEMM has
5226 // consumed all K smem data; the subsequent softmax (3c) and PV GEMM (3d)
5227 // only touch registers (R_S, R_0) and V smem, so K[stage] is free to be
5228 // overwritten by the producer's next TMA prefetch.

```

```

5229 {
5230     [[maybe_unused]] auto token = empty_K[stage].arrive();
5231 }
5232
5233 // ---- 3c: Online Safe Softmax — row max + exp + sum, P back to R_S ----
5234 // Softmax only touches R_S (registers), so V TMA latency can be overlapped
5235 // with this compute. Defer full_V.wait() until just before P@V (3d).
5236 float lane_row_max_new[kValTileSeqLenQ][2];
5237 float lane_row_sum_new[kValTileSeqLenQ][2];
5238 fill_2D_regs<float, kValTileSeqLenQ, 2>(lane_row_max_new, -INFINITY);
5239 fill_2D_regs<float, kValTileSeqLenQ, 2>(lane_row_sum_new, 0.0f);
5240
5241 // Pass 1: row-wise max across kValTileSeqLenK tiles (warp_reduce_max<4>)
5242 #pragma unroll
5243 for (int i = 0; i < kValTileSeqLenQ; ++i) {
5244     #pragma unroll
5245     for (int j = 0; j < kValTileSeqLenK; ++j) {
5246         float tmp_max_0, tmp_max_1;
5247         if constexpr (kMmaAccF32) {
5248             float *t_fptr_S = reinterpret_cast<float *>(&R_S[i][j][0]);
5249             tmp_max_0 = max(t_fptr_S[0], t_fptr_S[1]) * scale;
5250             tmp_max_1 = max(t_fptr_S[2], t_fptr_S[3]) * scale;
5251         } else {
5252             float2 t_reg_S_0 = __half22float2(HALF2(R_S[i][j][0]));
5253             float2 t_reg_S_1 = __half22float2(HALF2(R_S[i][j][1]));
5254             tmp_max_0 = max(t_reg_S_0.x, t_reg_S_0.y) * scale;
5255             tmp_max_1 = max(t_reg_S_1.x, t_reg_S_1.y) * scale;
5256         }
5257         lane_row_max_new[i][0] = max(lane_row_max_new[i][0], tmp_max_0);
5258         lane_row_max_new[i][1] = max(lane_row_max_new[i][1], tmp_max_1);
5259     }
5260     lane_row_max_new[i][0] =
5261         warp_reduce_max<4, float>(lane_row_max_new[i][0]);
5262     lane_row_max_new[i][1] =
5263         warp_reduce_max<4, float>(lane_row_max_new[i][1]);
5264 }
5265
5266 // Pass 2: P = exp(S*scale - m_new), store back to R_S, accumulate row sums
5267 #pragma unroll
5268 for (int i = 0; i < kValTileSeqLenQ; ++i) {
5269     float block_row_max_new_0 =
5270         max(lane_block_row_max_old[i][0], lane_row_max_new[i][0]);
5271     float block_row_max_new_1 =
5272         max(lane_block_row_max_old[i][1], lane_row_max_new[i][1]);
5273
5274     #pragma unroll
5275     for (int j = 0; j < kValTileSeqLenK; ++j) {
5276         if constexpr (kMmaAccF32) {
5277             float *t_fptr_S = reinterpret_cast<float *>(&R_S[i][j][0]);
5278             half *t_hptr_S = reinterpret_cast<half *>(&R_S[i][j][0]);
5279             t_fptr_S[0] = __expf(__fmaf_rn(t_fptr_S[0], scale, -block_row_max_new_0));
5280             t_fptr_S[1] = __expf(__fmaf_rn(t_fptr_S[1], scale, -block_row_max_new_0));
5281             t_fptr_S[2] = __expf(__fmaf_rn(t_fptr_S[2], scale, -block_row_max_new_1));
5282             t_fptr_S[3] = __expf(__fmaf_rn(t_fptr_S[3], scale, -block_row_max_new_1));
5283             lane_row_sum_new[i][0] += (t_fptr_S[0] + t_fptr_S[1]);
5284             lane_row_sum_new[i][1] += (t_fptr_S[2] + t_fptr_S[3]);
5285             t_hptr_S[0] = __float2half_rn(t_fptr_S[0]);
5286             t_hptr_S[1] = __float2half_rn(t_fptr_S[1]);
5287             t_hptr_S[2] = __float2half_rn(t_fptr_S[2]);
5288             t_hptr_S[3] = __float2half_rn(t_fptr_S[3]);
5289         } else {
5290             float2 t_reg_S_0 = __half22float2(HALF2(R_S[i][j][0]));
5291             float2 t_reg_S_1 = __half22float2(HALF2(R_S[i][j][1]));

```

```

5292         t_reg_S_0.x = __expf(__fmaf_rn(t_reg_S_0.x, scale, -block_row_max_new_0));
5293         t_reg_S_0.y = __expf(__fmaf_rn(t_reg_S_0.y, scale, -block_row_max_new_0));
5294         t_reg_S_1.x = __expf(__fmaf_rn(t_reg_S_1.x, scale, -block_row_max_new_1));
5295         t_reg_S_1.y = __expf(__fmaf_rn(t_reg_S_1.y, scale, -block_row_max_new_1));
5296         lane_row_sum_new[i][0] += (t_reg_S_0.x + t_reg_S_0.y);
5297         lane_row_sum_new[i][1] += (t_reg_S_1.x + t_reg_S_1.y);
5298         HALF2(R_S[i][j][0]) = __float22half2_rn(t_reg_S_0);
5299         HALF2(R_S[i][j][1]) = __float22half2_rn(t_reg_S_1);
5300     }
5301 }
5302 lane_row_sum_new[i][0] =
5303     warp_reduce_sum<4, float>(lane_row_sum_new[i][0]);
5304 lane_row_sum_new[i][1] =
5305     warp_reduce_sum<4, float>(lane_row_sum_new[i][1]);
5306 }
5307
5308 // ---- 3d: P @ V = 0[Br, d] ----
5309 // Wait for V[stage_v] ready just before P@V.
5310 // kStagesV=1: V was loaded in same iter, TMA overlaps QK+softmax.
5311 // kStagesV>=2: V was loaded in previous iter, TMA overlaps QK+softmax+PV.
5312 full_V[stage_v].wait(full_V[stage_v].arrive());
5313 tma_fence_proxy_async_shared_cta();
5314
5315 // ldmatrix.x2.trans: V row-major → col-major B fragment for NN matmul
5316 fill_3D_regs<uint32_t, kValTileSeqLenP, kValTileHeadDimV, kNRegs>(R_0, 0);
5317 #pragma unroll
5318 for (int tile_V_Bc = 0; tile_V_Bc < (Bc / kMmaAtomK); ++tile_V_Bc) {
5319 #pragma unroll
5320     for (int j = 0; j < kValTileHeadDimV; ++j) {
5321         const int warp_smem_V_d =
5322             warp_KV * (kMmaAtomN * kValTileHeadDimV) + j * kMmaAtomN;
5323         const int lane_smem_V_Bc = tile_V_Bc * kMmaAtomK + lane_id % 16;
5324         const int lane_smem_V_d = warp_smem_V_d;
5325         const uint32_t lane_smem_V_ptr =
5326             smem_V_base_ptr +
5327             (stage_v * Bc * kHeadDim +
5328              swizzle_fa<kHeadDim, Bc>(lane_smem_V_Bc, lane_smem_V_d)) *
5329             sizeof(half);
5330         LDMATRIX_X2_T(R_V[j][0], R_V[j][1], lane_smem_V_ptr);
5331     }
5332
5333     // P fragment (R_S) directly feeds A of P@V MMA (layout reuse trick)
5334     const int w = tile_V_Bc * 2;
5335 #pragma unroll
5336     for (int i = 0; i < kValTileSeqLenP; ++i) {
5337 #pragma unroll
5338         for (int j = 0; j < kValTileHeadDimV; ++j) {
5339             if constexpr (kMmaAccF32) {
5340                 HMMA16816F32(R_0[i][j][0], R_0[i][j][1], R_0[i][j][2], R_0[i][j][3],
5341                             R_S[i][w][0], R_S[i][w][1], R_S[i][w + 1][0], R_S[i][w + 1][1],
5342                             R_V[j][0], R_V[j][1],
5343                             R_0[i][j][0], R_0[i][j][1], R_0[i][j][2], R_0[i][j][3]);
5344             } else {
5345                 HMMA16816(R_0[i][j][0], R_0[i][j][1],
5346                           R_S[i][w][0], R_S[i][w][1],
5347                           R_S[i][w + 1][0], R_S[i][w + 1][1],
5348                           R_V[j][0], R_V[j][1],
5349                           R_0[i][j][0], R_0[i][j][1]);
5350             }
5351         }
5352     }
5353 }
5354

```

```

5355 // Release V[stage_v] for producer reuse: PV GEMM has consumed all
5356 // V smem data; the subsequent online rescaling (3e) only touches registers
5357 // (R_0, R_D, R_S, lane_block_*), so V[stage_v] is free to be overwritten.
5358 {
5359     [[maybe_unused]] auto token = empty_V[stage_v].arrive();
5360 }
5361
5362 // ---- 3e: Online rescaling —  $O_{\text{new}} = \exp(m_{\text{old}} - m_{\text{new}}) * O_{\text{old}} + P@V$  ----
5363 #pragma unroll
5364 for (int i = 0; i < kValTileSeqLenP; ++i) {
5365     float block_row_max_new_0 = lane_row_max_new[i][0];
5366     float block_row_max_new_1 = lane_row_max_new[i][1];
5367     float block_row_sum_new_0 = lane_row_sum_new[i][0];
5368     float block_row_sum_new_1 = lane_row_sum_new[i][1];
5369     float block_row_max_old_0 = lane_block_row_max_old[i][0];
5370     float block_row_max_old_1 = lane_block_row_max_old[i][1];
5371
5372     block_row_max_new_0 = max(block_row_max_old_0, block_row_max_new_0);
5373     block_row_max_new_1 = max(block_row_max_old_1, block_row_max_new_1);
5374
5375     // First iteration: m_old = -inf, use m_new directly
5376     block_row_max_old_0 = (tile_K_seqlen > 0 ? block_row_max_old_0
5377                               : block_row_max_new_0);
5378     block_row_max_old_1 = (tile_K_seqlen > 0 ? block_row_max_old_1
5379                               : block_row_max_new_1);
5380
5381     float rescale_o_factor_0 =
5382         __expf(block_row_max_old_0 - block_row_max_new_0);
5383     float rescale_o_factor_1 =
5384         __expf(block_row_max_old_1 - block_row_max_new_1);
5385
5386 #pragma unroll
5387 for (int j = 0; j < kValTileHeadDimV; ++j) {
5388     if constexpr (kMmaAccF32) {
5389         float *t_fptr_0 = reinterpret_cast<float*>(&R_0[i][j][0]);
5390         float *t_fptr_D = reinterpret_cast<float*>(&R_D[i][j][0]);
5391         t_fptr_D[0] = __fmaf_rn(rescale_o_factor_0, t_fptr_D[0], t_fptr_0[0]);
5392         t_fptr_D[1] = __fmaf_rn(rescale_o_factor_0, t_fptr_D[1], t_fptr_0[1]);
5393         t_fptr_D[2] = __fmaf_rn(rescale_o_factor_1, t_fptr_D[2], t_fptr_0[2]);
5394         t_fptr_D[3] = __fmaf_rn(rescale_o_factor_1, t_fptr_D[3], t_fptr_0[3]);
5395     } else {
5396         float2 t_reg_0_0 = __half22float2(HALF2(R_0[i][j][0]));
5397         float2 t_reg_0_1 = __half22float2(HALF2(R_0[i][j][1]));
5398         float2 t_reg_D_0 = __half22float2(HALF2(R_D[i][j][0]));
5399         float2 t_reg_D_1 = __half22float2(HALF2(R_D[i][j][1]));
5400         t_reg_D_0.x = __fmaf_rn(rescale_o_factor_0, t_reg_D_0.x, t_reg_0_0.x);
5401         t_reg_D_0.y = __fmaf_rn(rescale_o_factor_0, t_reg_D_0.y, t_reg_0_0.y);
5402         t_reg_D_1.x = __fmaf_rn(rescale_o_factor_1, t_reg_D_1.x, t_reg_0_1.x);
5403         t_reg_D_1.y = __fmaf_rn(rescale_o_factor_1, t_reg_D_1.y, t_reg_0_1.y);
5404         HALF2(R_D[i][j][0]) = __float22half2_rn(t_reg_D_0);
5405         HALF2(R_D[i][j][1]) = __float22half2_rn(t_reg_D_1);
5406     }
5407 }
5408
5409 float block_row_sum_old_0 = lane_block_row_sum_old[i][0];
5410 float block_row_sum_old_1 = lane_block_row_sum_old[i][1];
5411 lane_block_row_sum_old[i][0] =
5412     __fmaf_rn(rescale_o_factor_0, block_row_sum_old_0,
5413               block_row_sum_new_0);
5414 lane_block_row_sum_old[i][1] =
5415     __fmaf_rn(rescale_o_factor_1, block_row_sum_old_1,
5416               block_row_sum_new_1);
5417 lane_block_row_max_old[i][0] = block_row_max_new_0;

```

```

5418     lane_block_row_max_old[i][1] = block_row_max_new_1;
5419 }
5420 }
5421
5422 // ---- Step 4: Final rescale — 0_final = (1/l_final) * 0_final ----
5423 #pragma unroll
5424 for (int i = 0; i < kValTileSeqLenP; ++i) {
5425     float rescale_factor_0 = __frcp_rn(lane_block_row_sum_old[i][0]);
5426     float rescale_factor_1 = __frcp_rn(lane_block_row_sum_old[i][1]);
5427 #pragma unroll
5428     for (int j = 0; j < kValTileHeadDimV; ++j) {
5429         if constexpr (kMmaAccF32) {
5430             float *t_fptr_D = reinterpret_cast<float *>(&R_D[i][j][0]);
5431             half *t_hptr_D = reinterpret_cast<half *>(&R_D[i][j][0]);
5432             t_hptr_D[0] = __float2half_rn(rescale_factor_0 * t_fptr_D[0]);
5433             t_hptr_D[1] = __float2half_rn(rescale_factor_0 * t_fptr_D[1]);
5434             t_hptr_D[2] = __float2half_rn(rescale_factor_1 * t_fptr_D[2]);
5435             t_hptr_D[3] = __float2half_rn(rescale_factor_1 * t_fptr_D[3]);
5436         } else {
5437             float2 t_reg_D_0 = __half2float2(HALF2(R_D[i][j][0]));
5438             float2 t_reg_D_1 = __half2float2(HALF2(R_D[i][j][1]));
5439             t_reg_D_0.x = rescale_factor_0 * t_reg_D_0.x;
5440             t_reg_D_0.y = rescale_factor_0 * t_reg_D_0.y;
5441             t_reg_D_1.x = rescale_factor_1 * t_reg_D_1.x;
5442             t_reg_D_1.y = rescale_factor_1 * t_reg_D_1.y;
5443             HALF2(R_D[i][j][0]) = __float22half2_rn(t_reg_D_0);
5444             HALF2(R_D[i][j][1]) = __float22half2_rn(t_reg_D_1);
5445         }
5446     }
5447 }
5448
5449 // ---- Step 5: Epilogue — warp shuffle + 128-bit collective store ----
5450 #pragma unroll
5451 for (int i = 0; i < kValTileSeqLenP; ++i) {
5452 #pragma unroll
5453     for (int j = 0; j < kValTileHeadDimV; ++j) {
5454         uint32_t R_Z[2][4];
5455         R_Z[0][0] = R_D[i][j][0];
5456         R_Z[1][0] = R_D[i][j][1];
5457         R_Z[0][1] = __shfl_sync(0xffffffff, R_D[i][j][0], lane_id + 1, 4);
5458         R_Z[0][2] = __shfl_sync(0xffffffff, R_D[i][j][0], lane_id + 2, 4);
5459         R_Z[0][3] = __shfl_sync(0xffffffff, R_D[i][j][0], lane_id + 3, 4);
5460         R_Z[1][1] = __shfl_sync(0xffffffff, R_D[i][j][1], lane_id + 1, 4);
5461         R_Z[1][2] = __shfl_sync(0xffffffff, R_D[i][j][1], lane_id + 2, 4);
5462         R_Z[1][3] = __shfl_sync(0xffffffff, R_D[i][j][1], lane_id + 3, 4);
5463
5464         if (lane_id % 4 == 0) {
5465             const int store_warp_regs_0_Br =
5466                 warp_QP * (kMmaAtomM * kValTileSeqLenP) + i * kMmaAtomM;
5467             const int store_lane_gmem_0_Br =
5468                 Q_tile_id * Br + store_warp_regs_0_Br + lane_id / 4;
5469             const int store_warp_regs_0_d =
5470                 warp_KV * (kMmaAtomN * kValTileHeadDimV) + j * kMmaAtomN;
5471             const int store_lane_gmem_0_d = store_warp_regs_0_d;
5472             const int store_gmem_0_addr_0 =
5473                 0_gmem_offset + store_lane_gmem_0_Br * kHeadDim +
5474                 store_lane_gmem_0_d;
5475             const int store_gmem_0_addr_1 =
5476                 0_gmem_offset + (store_lane_gmem_0_Br + 8) * kHeadDim +
5477                 store_lane_gmem_0_d;
5478             *reinterpret_cast<float4 *>(&0[store_gmem_0_addr_0]) =
5479                 *reinterpret_cast<float4 *>(&R_Z[0][0]);
5480             *reinterpret_cast<float4 *>(&0[store_gmem_0_addr_1]) =

```

```

5481         *reinterpret_cast<float4 *>(&R_Z[1][0]);
5482     }
5483 }
5484 }
5485 }
5486 }
5487
5488 // =====
5489 // Phase 8c: FlashAttention-3-style TMA + MMA Warp Specialization (SM120)
5490 //   Dual 128-thread consumer warpgroups processing disjoint KV tiles,
5491 //   with per-WG independent K pipeline (kStagesK) and single V buffer (kStagesV=1).
5492 //
5493 // 与 flash_attn_tma_mma_ws_stages_split_q (Phase 8b) 的核心区别:
5494 //   - 8b: 1 producer WG (128T) + 1 consumer WG (256T, 8 warps), 全 KV 遍历
5495 //   - 8c: 1 producer WG (128T) + 2 consumer WGs (各128T, 4 warps), KV tile 奇偶拆分
5496 //
5497 // 线程布局 (384 threads):
5498 //   WG0 [0,127]: TMA producer (仅 thread 0 发 TMA)
5499 //   WG1 [128,255]: consumer_id=0, 处理偶数 KV tile (0,2,4,...)
5500 //   WG2 [256,383]: consumer_id=1, 处理奇数 KV tile (1,3,5,...)
5501 //
5502 // per-WG 独立 stage 语义 (关键改进):
5503 // -----
5504 // 每个 consumer WG 拥有独立的 K stage slots (kStagesK 个) 和 1 个 V buffer。
5505 // K/V smem 按 consumer_id 分 bank: K_smem[cid][stage], V_smem[cid]。
5506 // Barrier 也按 consumer_id 隔离: full_K[cid][s], empty_K[cid][s], full_V[cid], empty_V[cid]。
5507 // 两个 WG 绝不争抢同一 smem slot 或 barrier, producer 通过 cid = tile&1 路由 TMA。
5508 //
5509 // 这样 stage 语义从 "跨 WG 共享的 2 slot (每 WG 实际 depth=1)" 变为
5510 // "每 WG 独立拥有 kStagesK 个 K slot (每 WG 真正 depth=kStagesK)"。
5511 //
5512 // Pipeline 时序 (Sk=1, Sv=1):
5513 //   K[tile] TMA 被 softmax+PV+rescale 隐藏 (Sk=1 复用同 slot, 用完即释放);
5514 //   V[tile] TMA 被 rescale+QK[next]+softmax[next] 隐藏 (Sv=1 同理)。
5515 //
5516 // Pipeline 时序 (Sk=2, Sv=1, D=64 only):
5517 //   K[tile+2] 在 QK[tile] 之前就 prefetch 到另一个 slot, 被整个 tile compute 隐藏;
5518 //   V[tile] TMA 同 Sk=1, 被 QK+softmax 隐藏。
5519 //
5520 // 为什么是奇偶 tile 拆分 (而非前/后半区各自一半)?
5521 // -----
5522 // 关键约束: producer 按 global tile 顺序 0,1,2,3,... 连续 TMA。
5523 // 奇偶拆分让 cid = tile & 1 静态路由: tile 0,2,4,... -> WG1; tile 1,3,5,... -> WG2。
5524 // 每 WG 的下一个 tile (tile+2) 正好是 producer 当前/下一轮要加载的 tile,
5525 // 保证 producer/consumer 流水无停顿。
5526 //
5527 // 若改为前/后半区拆分: 过渡区两个 WG 可能同时需要同一 cid 的 slot,
5528 // 且 WG2 跳跃访问远端 tile 时数据尚未被 producer TMA 到 smem。
5529 //
5530 // Partial state merge (Split-KV / FlashDecoding-style reduction):
5531 // -----
5532 // 每个 WG 各自处理一半 KV tile 序列, 独立维护 "未归一化" 的 online-softmax
5533 // 中间状态 (m, l, Oacc), 合并后等价于全 KV 遍历。
5534 //
5535 // 稳定归并公式 (避免 exp 溢出):
5536 //   m      = max(m_0, m_1)           // 全局 row max
5537 //   alpha   = exp(m_0 - m) (<=1)      // WG1 rescale factor
5538 //   beta    = exp(m_1 - m) (<=1)      // WG2 rescale factor
5539 //   l       = alpha * l_0 + beta * l_1
5540 //   Oacc    = alpha * Oacc_0 + beta * Oacc_1
5541 //   O       = Oacc / l                // 最终归一化
5542 //
5543 // Tc==1 退化: WG2 无 tile, m_1=-inf, l_1=0, Oacc_1=0;

```

```

5544 // beta = exp(-inf) = 0, 退化为 WG1 结果。
5545 //
5546 // 限制 (首版):
5547 // - kStagesV == 1 (V 只需 1 buffer, TMA 被 QK+softmax 隐藏)
5548 // - kStagesK >= 1 (K pipeline depth; D=128 在 SM120 只支持 Sk=1)
5549 // - D=128 时 Sk=1 (Sk=2 需 112KB > 101KB optin 上限)
5550 // - Br=64, Bc=64, D=64/128, aligned seqlen (N % Br == 0 && N % Bc == 0)
5551 // - 仅 self-attention, 无 causal/varlen/GQA
5552 // =====
5553 template <
5554     const int kHeadDim,
5555     const int kMmaAtomM, const int kMmaAtomN, const int kMmaAtomK,
5556     const int kMmaAccF32,
5557     const int kMmaTileSeqLenQ, const int kMmaTileSeqLenK,
5558     const int kMmaTileSeqLenP, const int kMmaTileHeadDimV,
5559     const int kValTileSeqLenQ, const int kValTileSeqLenK,
5560     const int kValTileSeqLenP, const int kValTileHeadDimV,
5561     const int kStagesK, const int kStagesV,
5562     const int kNumThreads>
5563 __global__ void __launch_bounds__(kNumThreads, 1)
5564     flash_attn_3_tma_ws_stages_split_q(
5565         half *Q, half *K, half *V, half *O, int N, int H,
5566         const UTensorMap *__restrict__ tensorMapQ,
5567         const UTensorMap *__restrict__ tensorMapK,
5568         const UTensorMap *__restrict__ tensorMapV) {
5569     static_assert(kHeadDim == 64 || kHeadDim == 128,
5570         "v1 supports kHeadDim == 64 or 128");
5571     static_assert(kNumThreads == 384, "128 producer + 256 consumer (2 WGs)");
5572     static_assert(kMmaAccF32 == 0 || kMmaAccF32 == 1, "kMmaAccF32 must be 0 or 1");
5573     static_assert(kStagesV == 1, "per-WG V pipeline depth must be 1");
5574     static_assert(kStagesK >= 1, "kStagesK must be >= 1");
5575     constexpr int kNRegs = kMmaAccF32 ? 4 : 2;
5576
5577     constexpr int Br = kMmaAtomM * kMmaTileSeqLenQ * kValTileSeqLenQ; // 64
5578     constexpr int Bc = kMmaAtomN * kMmaTileSeqLenK * kValTileSeqLenK; // 64
5579     constexpr int kConsumerThreadsPerWG = 128;
5580     constexpr int kProducerThreads = 128;
5581     constexpr int kNumConsumerWGs = 2;
5582     constexpr int kQTileBytes = Br * kHeadDim * sizeof(half);
5583     constexpr int kKTileBytes = Bc * kHeadDim * sizeof(half);
5584     constexpr int kVTileBytes = Bc * kHeadDim * sizeof(half);
5585     constexpr int kTmaBoxMinor = 64;
5586     constexpr int kTmaChunks = kHeadDim / kTmaBoxMinor;
5587
5588     const int Nb_id = blockIdx.y / H;
5589     const int Nh_id = blockIdx.y % H;
5590     const int Q_tile_id = blockIdx.x;
5591     const int Tc = (N + Bc - 1) / Bc;
5592     const float scale = 1.0f / sqrtf((float)kHeadDim);
5593
5594     const int QKV_gmem_offset = (Nb_id * H * N + Nh_id * N) * kHeadDim;
5595     const int O_gmem_offset = QKV_gmem_offset;
5596     const int qkv_major_base = (Nb_id * H + Nh_id) * N;
5597
5598     // ---- Shared memory layout (per-WG independent K/V banks) ----
5599     // [Q_shared: Br*D]
5600     // [K[cid=0][s=0..Sk-1]: Sk*Bc*D] [K[cid=1][s=0..Sk-1]]
5601     // [V[cid=0]: Bc*D] [V[cid=1]: Bc*D]
5602     extern __shared__ __align__(1024) uint8_t smem_fa3_tma_ws[];
5603     half *Q_smem = reinterpret_cast<half *>(smem_fa3_tma_ws);
5604     half *K_smem_base = Q_smem + Br * kHeadDim;
5605     // K: [numConsumerWGs][kStagesK][Bc*kHeadDim], linearized
5606     // K_smem_base + cid * (kStagesK * Bc * kHeadDim) + stg * (Bc * kHeadDim)

```

```

5607     half *V_smem_base = K_smem_base + kNumConsumerWGs * kStagesK * Bc * kHeadDim;
5608     // V: [numConsumerWGs][Bc*kHeadDim]
5609     //   V_smem_base + cid * (Bc * kHeadDim)
5610
5611     const uint32_t smem_Q_base_ptr = __cvta_generic_to_shared(Q_smem);
5612     const uint32_t smem_K_base_ptr = __cvta_generic_to_shared(K_smem_base);
5613     const uint32_t smem_V_base_ptr = __cvta_generic_to_shared(V_smem_base);
5614
5615     #pragma nv_diag_suppress static_var_with_dynamic_init
5616     __shared__ cuda::barrier<cuda::thread_scope_block> full_Q;
5617     // per-WG K barriers: [cid][stage]
5618     __shared__ cuda::barrier<cuda::thread_scope_block> full_K[kNumConsumerWGs][kStagesK];
5619     __shared__ cuda::barrier<cuda::thread_scope_block> empty_K[kNumConsumerWGs][kStagesK];
5620     // per-WG V barriers: [cid] (kStagesV=1)
5621     __shared__ cuda::barrier<cuda::thread_scope_block> full_V[kNumConsumerWGs];
5622     __shared__ cuda::barrier<cuda::thread_scope_block> empty_V[kNumConsumerWGs];
5623     #pragma nv_diag_default static_var_with_dynamic_init
5624
5625     const bool is_producer = threadIdx.x < kProducerThreads;
5626     const int consumer_id = is_producer ? 0
5627         : (threadIdx.x - kProducerThreads) / kConsumerThreadsPerWG;
5628     const int wg_tid = is_producer ? threadIdx.x
5629         : (threadIdx.x - kProducerThreads) % kConsumerThreadsPerWG;
5630
5631     if (threadIdx.x == 0) {
5632         init(&full_Q, 256 + 1); // 2 consumer WGs * 128 + 1 producer
5633         for (int cid = 0; cid < kNumConsumerWGs; ++cid) {
5634             for (int s = 0; s < kStagesK; ++s) {
5635                 init(&full_K[cid][s], kConsumerThreadsPerWG + 1);
5636                 init(&empty_K[cid][s], kConsumerThreadsPerWG + 1);
5637             }
5638             init(&full_V[cid], kConsumerThreadsPerWG + 1);
5639             init(&empty_V[cid], kConsumerThreadsPerWG + 1);
5640         }
5641         tma_fence_proxy_async_shared_cta();
5642     }
5643     __syncthreads();
5644
5645     // Helper: K_smem ptr for (cid, stg)
5646     // K layout: K_smem_base + cid*(Sk*Bc*D) + stg*(Bc*D)
5647     // D=128 chunk-major: each tile's 2 chunks at stg*Bc*D + c*Bc*64
5648
5649     // =====
5650     // Producer Warpgroup (WG0, threadIdx.x 0~127)
5651     // Single global tile loop over all KV tiles (0..Tc-1).
5652     // cid = tile & 1 routes TMA to the owning consumer WG's smem bank.
5653     //
5654     // Per-tile load order: V first, then K (prefetch next).
5655     //   - V[tile] TMA: blocked until consumer releases V[cid] (after PV).
5656     //     But K[tile] signal already sent in warmup/prev iter, so consumer
5657     //     can do QK[tile] while producer waits for V release.
5658     //   - K[tile+2] prefetch (Sk>=2): blocked until consumer releases K[cid][stg_next]
5659     //     (after QK). QK is first compute step, always done before producer reaches here.
5660     // =====
5661     if (is_producer) {
5662         NOTES_V2_REG_DEALLOC(40);
5663         if (wg_tid == 0) {
5664             // P0: Load Q tile once
5665             for (int c = 0; c < kTmaChunks; ++c) {
5666                 tma_load_2d(Q_smem + c * Br * kTmaBoxMinor, tensorMapQ,
5667                     c * kTmaBoxMinor, qkv_major_base + Q_tile_id * Br, full_Q);
5668             }
5669             tma_arrive_expect_tx(full_Q, kQTileBytes);

```

```

5670 // P1: Warmup - prefetch first K tile for each consumer WG (Sk-1 tiles)
5671 // tile 0 -> cid=0, tile 1 -> cid=1
5672
5673 if constexpr (kStagesK > 1) {
5674     for (int cid = 0; cid < kNumConsumerWGs; ++cid) {
5675         const int tile0 = cid; // first tile for this WG
5676         if (tile0 < Tc) {
5677             const int stg = 0; // first slot
5678             empty_K[cid][stg].wait(empty_K[cid][stg].arrive());
5679             for (int c = 0; c < kTmaChunks; ++c) {
5680                 tma_load_2d(K_smem_base + cid * (kStagesK * Bc * kHeadDim) +
5681                     stg * Bc * kHeadDim + c * Bc * kTmaBoxMinor,
5682                     tensorMapK, c * kTmaBoxMinor,
5683                     qkv_major_base + tile0 * Bc, full_K[cid][stg]);
5684             }
5685             tma_arrive_expect_tx(full_K[cid][stg], kKTileBytes);
5686         }
5687     }
5688 }
5689
5690 // P2: Main loop over all global KV tiles
5691 // K 优先于 V: K 的 release 依赖 QK (tile 第一步 compute, 早完成),
5692 // V 的 release 依赖 PV (tile 最后一步 compute, 晚完成)。
5693 // 先发 K 让 consumer 尽早开始 QK, V 的 TMA 在 K 之后发出,
5694 // 被 QK+softmax 的 compute 时间隐藏。
5695 for (int tile = 0; tile < Tc; ++tile) {
5696     const int cid = tile & 1;
5697
5698     // Step 1: Load/prefetch K first (release depends on QK, which is early)
5699     if constexpr (kStagesK == 1) {
5700         // Sk=1: load K[tile] to the single slot (reuse after consumer QK release)
5701         empty_K[cid][0].wait(empty_K[cid][0].arrive());
5702         for (int c = 0; c < kTmaChunks; ++c) {
5703             tma_load_2d(K_smem_base + cid * (kStagesK * Bc * kHeadDim) +
5704                 c * Bc * kTmaBoxMinor,
5705                 tensorMapK, c * kTmaBoxMinor,
5706                 qkv_major_base + tile * Bc, full_K[cid][0]);
5707         }
5708         tma_arrive_expect_tx(full_K[cid][0], kKTileBytes);
5709     } else {
5710         // Sk>=2: prefetch K[tile+2] to next slot (true ping-pong pipeline!)
5711         const int local_tile = (tile - cid) / 2;
5712         const int stg_next = (local_tile + 1) % kStagesK;
5713         const int next_tile = tile + 2;
5714         if (next_tile < Tc) {
5715             empty_K[cid][stg_next].wait(empty_K[cid][stg_next].arrive());
5716             for (int c = 0; c < kTmaChunks; ++c) {
5717                 tma_load_2d(K_smem_base + cid * (kStagesK * Bc * kHeadDim) +
5718                     stg_next * Bc * kHeadDim + c * Bc * kTmaBoxMinor,
5719                     tensorMapK, c * kTmaBoxMinor,
5720                     qkv_major_base + next_tile * Bc,
5721                     full_K[cid][stg_next]);
5722             }
5723             tma_arrive_expect_tx(full_K[cid][stg_next], kKTileBytes);
5724         }
5725     }
5726
5727     // Step 2: Load V[tile] (release depends on PV, which is late)
5728     empty_V[cid].wait(empty_V[cid].arrive());
5729     for (int c = 0; c < kTmaChunks; ++c) {
5730         tma_load_2d(V_smem_base + cid * Bc * kHeadDim + c * Bc * kTmaBoxMinor,
5731             tensorMapV, c * kTmaBoxMinor,
5732             qkv_major_base + tile * Bc, full_V[cid]);

```

```

5733     }
5734     tma_arrive_expect_tx(full_V[cid], kVTileBytes);
5735 }
5736 }
5737 }
5738 // =====
5739 // Consumer Warpgroups (WG1 consumer_id=0, WG2 consumer_id=1)
5740 // Each WG processes every other KV tile (consumer_id, consumer_id+2, ...).
5741 // K stage index: local_iter % kStagesK (Sk=1: always 0; Sk=2: 0,1,0,1,...)
5742 // V stage index: always consumer_id (only 1 V buffer per WG)
5743 // =====
5744 else {
5745     const int warp_id = wg_tid / kWarpSize; // 0~3
5746     const int lane_id = wg_tid % kWarpSize; // 0~31
5747     const int warp_QP = warp_id;
5748     const int warp_KV = 0;
5749
5750     if constexpr (kHeadDim == 128) {
5751         NOTES_V2_REG_ALLOC(168);
5752     } else {
5753         NOTES_V2_REG_ALLOC(80);
5754     }
5755
5756     // C0: init - mark OWN K/V slots as empty (consumer arrives, producer is 129th)
5757     for (int s = 0; s < kStagesK; ++s) {
5758         [[maybe_unused]] auto tk = empty_K[consumer_id][s].arrive();
5759     }
5760     {
5761         [[maybe_unused]] auto tv = empty_V[consumer_id].arrive();
5762     }
5763
5764     // Wait for Q tile (shared by both WGs)
5765     full_Q.wait(full_Q.arrive());
5766
5767     // Partial online-softmax state (per-WG, independent)
5768     float lane_block_row_max_old[kValTileSeqLenQ][2];
5769     float lane_block_row_sum_old[kValTileSeqLenQ][2];
5770     fill_2D_regs<float, kValTileSeqLenQ, 2>(lane_block_row_max_old, -INFINITY);
5771     fill_2D_regs<float, kValTileSeqLenQ, 2>(lane_block_row_sum_old, 0.0f);
5772
5773     uint32_t R_Q[kValTileSeqLenQ][4];
5774     uint32_t R_K[kValTileSeqLenK][2];
5775     uint32_t R_V[kValTileHeadDimV][2];
5776     uint32_t R_S[kValTileSeqLenQ][kValTileSeqLenK][kNRegs];
5777     uint32_t R_O[kValTileSeqLenP][kValTileHeadDimV][kNRegs];
5778     uint32_t R_D[kValTileSeqLenP][kValTileHeadDimV][kNRegs];
5779     fill_3D_regs<uint32_t, kValTileSeqLenP, kValTileHeadDimV, kNRegs>(R_D, 0);
5780
5781     // K smem offset for this WG: base + cid*(Sk*Bc*D)
5782     // stg within: local_iter % kStagesK
5783     // V smem offset for this WG: V_smem_base + cid*(Bc*D)
5784     const uint32_t smem_K_wg_base = smem_K_base_ptr +
5785         consumer_id * (kStagesK * Bc * kHeadDim) * sizeof(half);
5786     const uint32_t smem_V_wg_base = smem_V_base_ptr +
5787         consumer_id * (Bc * kHeadDim) * sizeof(half);
5788
5789     int local_iter = 0;
5790     for (int tile = consumer_id; tile < Tc; tile += 2) {
5791         const int k_stg = local_iter % kStagesK;
5792
5793         // K smem offset for this stage: wg_base + stg*(Bc*D)
5794         const uint32_t smem_K_stage_ptr = smem_K_wg_base +
5795             k_stg * Bc * kHeadDim * sizeof(half);

```

```

5796 // Wait for K[consumer_id][k_stg] ready
5797 full_K[consumer_id][k_stg].wait(full_K[consumer_id][k_stg].arrive());
5798 tma_fence_proxy_async_shared_cta();
5800
5801 // ---- Q @ K^T = S[Br, Bc] ----
5802 // K early release: 在最后一个 tile_K_d 的 ldmatrix_K 完成后立即 release K,
5803 // 不等整个 QK MMA 循环结束。K 数据已全部读入 R_K 寄存器, smem 可被
5804 // producer 复用。这对 Sk=1 尤其重要: producer 可更早开始下一个 K TMA。
5805 fill_3D_regs<uint32_t, kValTileSeqLenQ, kValTileSeqLenK, kNRegs>(R_S, 0);
5806 #pragma unroll
5807 for (int tile_K_d = 0; tile_K_d < (kHeadDim / kMmaAtomK); ++tile_K_d) {
5808 #pragma unroll
5809 for (int i = 0; i < kValTileSeqLenQ; ++i) {
5810     const int warp_smem_Q_Br =
5811         warp_QP * (kMmaAtomM * kValTileSeqLenQ) + i * kMmaAtomM;
5812     const int lane_smem_Q_Br = warp_smem_Q_Br + lane_id % 16;
5813     const int lane_smem_Q_d = tile_K_d * kMmaAtomK + (lane_id / 16) * 8;
5814     const uint32_t lane_smem_Q_ptr =
5815         smem_Q_base_ptr +
5816         swizzle_fa<kHeadDim, Br>(lane_smem_Q_Br, lane_smem_Q_d) *
5817             sizeof(half);
5818     LDMATRIX_X4(R_Q[i][0], R_Q[i][1], R_Q[i][2], R_Q[i][3],
5819                 lane_smem_Q_ptr);
5820 }
5821
5822 #pragma unroll
5823 for (int j = 0; j < kValTileSeqLenK; ++j) {
5824     const int warp_smem_K_Bc =
5825         warp_KV * (kMmaAtomN * kValTileSeqLenK) + j * kMmaAtomN;
5826     const int lane_smem_K_Bc = warp_smem_K_Bc + lane_id % 8;
5827     const int lane_smem_K_d =
5828         tile_K_d * kMmaAtomK + ((lane_id / 8) % 2) * 8;
5829     // K smem: stage_ptr + swizzle_fa offset (no stage*Bc*D since stage_ptr already includes it)
5830     const uint32_t lane_smem_K_ptr =
5831         smem_K_stage_ptr +
5832         swizzle_fa<kHeadDim, Bc>(lane_smem_K_Bc, lane_smem_K_d) *
5833             sizeof(half);
5834     LDMATRIX_X2(R_K[j][0], R_K[j][1], lane_smem_K_ptr);
5835 }
5836
5837 // Early release K after last ldmatrix_K: K data now fully in registers
5838 if (tile_K_d == (kHeadDim / kMmaAtomK) - 1) {
5839     [[maybe_unused]] auto token = empty_K[consumer_id][k_stg].arrive();
5840 }
5841
5842 #pragma unroll
5843 for (int i = 0; i < kValTileSeqLenQ; ++i) {
5844 #pragma unroll
5845 for (int j = 0; j < kValTileSeqLenK; ++j) {
5846     if constexpr (kMmaAccF32) {
5847         HMMA16816F32(R_S[i][j][0], R_S[i][j][1], R_S[i][j][2], R_S[i][j][3],
5848                     R_Q[i][0], R_Q[i][1], R_Q[i][2], R_Q[i][3],
5849                     R_K[j][0], R_K[j][1],
5850                     R_S[i][j][0], R_S[i][j][1], R_S[i][j][2], R_S[i][j][3]);
5851     } else {
5852         HMMA16816(R_S[i][j][0], R_S[i][j][1], R_Q[i][0], R_Q[i][1],
5853                 R_Q[i][2], R_Q[i][3], R_K[j][0], R_K[j][1],
5854                 R_S[i][j][0], R_S[i][j][1]);
5855     }
5856 }
5857 }
5858 }

```

```

5859 // K already released above (after last ldmatrix_K)
5860
5861
5862 // ---- Online Safe Softmax ----
5863 // This compute overlaps with producer's V[tile] TMA (which waits for
5864 // empty_V from previous PV, or is already in-flight)
5865 float lane_row_max_new[kValTileSeqLenQ][2];
5866 float lane_row_sum_new[kValTileSeqLenQ][2];
5867 fill_2D_regs<float, kValTileSeqLenQ, 2>(lane_row_max_new, -INFINITY);
5868 fill_2D_regs<float, kValTileSeqLenQ, 2>(lane_row_sum_new, 0.0f);
5869
5870 #pragma unroll
5871 for (int i = 0; i < kValTileSeqLenQ; ++i) {
5872 #pragma unroll
5873 for (int j = 0; j < kValTileSeqLenK; ++j) {
5874     float tmp_max_0, tmp_max_1;
5875     if constexpr (kMmaAccF32) {
5876         float *t_fptr_S = reinterpret_cast<float *>(&R_S[i][j][0]);
5877         tmp_max_0 = max(t_fptr_S[0], t_fptr_S[1]) * scale;
5878         tmp_max_1 = max(t_fptr_S[2], t_fptr_S[3]) * scale;
5879     } else {
5880         float2 t_reg_S_0 = __half22float2(HALF2(R_S[i][j][0]));
5881         float2 t_reg_S_1 = __half22float2(HALF2(R_S[i][j][1]));
5882         tmp_max_0 = max(t_reg_S_0.x, t_reg_S_0.y) * scale;
5883         tmp_max_1 = max(t_reg_S_1.x, t_reg_S_1.y) * scale;
5884     }
5885     lane_row_max_new[i][0] = max(lane_row_max_new[i][0], tmp_max_0);
5886     lane_row_max_new[i][1] = max(lane_row_max_new[i][1], tmp_max_1);
5887 }
5888 lane_row_max_new[i][0] =
5889     warp_reduce_max<4, float>(lane_row_max_new[i][0]);
5890 lane_row_max_new[i][1] =
5891     warp_reduce_max<4, float>(lane_row_max_new[i][1]);
5892 }
5893
5894 #pragma unroll
5895 for (int i = 0; i < kValTileSeqLenQ; ++i) {
5896     float block_row_max_new_0 =
5897         max(lane_block_row_max_old[i][0], lane_row_max_new[i][0]);
5898     float block_row_max_new_1 =
5899         max(lane_block_row_max_old[i][1], lane_row_max_new[i][1]);
5900
5901 #pragma unroll
5902 for (int j = 0; j < kValTileSeqLenK; ++j) {
5903     if constexpr (kMmaAccF32) {
5904         float *t_fptr_S = reinterpret_cast<float *>(&R_S[i][j][0]);
5905         half *t_hptr_S = reinterpret_cast<half *>(&R_S[i][j][0]);
5906         t_fptr_S[0] = __expf(__fmaf_rn(t_fptr_S[0], scale, -block_row_max_new_0));
5907         t_fptr_S[1] = __expf(__fmaf_rn(t_fptr_S[1], scale, -block_row_max_new_0));
5908         t_fptr_S[2] = __expf(__fmaf_rn(t_fptr_S[2], scale, -block_row_max_new_1));
5909         t_fptr_S[3] = __expf(__fmaf_rn(t_fptr_S[3], scale, -block_row_max_new_1));
5910         lane_row_sum_new[i][0] += (t_fptr_S[0] + t_fptr_S[1]);
5911         lane_row_sum_new[i][1] += (t_fptr_S[2] + t_fptr_S[3]);
5912         t_hptr_S[0] = __float2half_rn(t_fptr_S[0]);
5913         t_hptr_S[1] = __float2half_rn(t_fptr_S[1]);
5914         t_hptr_S[2] = __float2half_rn(t_fptr_S[2]);
5915         t_hptr_S[3] = __float2half_rn(t_fptr_S[3]);
5916     } else {
5917         float2 t_reg_S_0 = __half22float2(HALF2(R_S[i][j][0]));
5918         float2 t_reg_S_1 = __half22float2(HALF2(R_S[i][j][1]));
5919         t_reg_S_0.x = __expf(__fmaf_rn(t_reg_S_0.x, scale, -block_row_max_new_0));
5920         t_reg_S_0.y = __expf(__fmaf_rn(t_reg_S_0.y, scale, -block_row_max_new_0));
5921         t_reg_S_1.x = __expf(__fmaf_rn(t_reg_S_1.x, scale, -block_row_max_new_1));

```

```

5922         t_reg_S_1.y = __expf(__fmaf_rn(t_reg_S_1.y, scale, -block_row_max_new_1));
5923         lane_row_sum_new[i][0] += (t_reg_S_0.x + t_reg_S_0.y);
5924         lane_row_sum_new[i][1] += (t_reg_S_1.x + t_reg_S_1.y);
5925         HALF2(R_S[i][j][0]) = __float22half2_rn(t_reg_S_0);
5926         HALF2(R_S[i][j][1]) = __float22half2_rn(t_reg_S_1);
5927     }
5928 }
5929 lane_row_sum_new[i][0] =
5930     warp_reduce_sum<4, float>(lane_row_sum_new[i][0]);
5931 lane_row_sum_new[i][1] =
5932     warp_reduce_sum<4, float>(lane_row_sum_new[i][1]);
5933 }
5934
5935 // ---- P @ V = 0[Br, d] ----
5936 // Wait for V[consumer_id] ready
5937 full_V[consumer_id].wait(full_V[consumer_id].arrive());
5938 tma_fence_proxy_async_shared_cta();
5939
5940 fill_3D_regs<uint32_t, kValTileSeqLenP, kValTileHeadDimV, kNRegs>(R_0, 0);
5941 #pragma unroll
5942 for (int tile_V_Bc = 0; tile_V_Bc < (Bc / kMmaAtomK); ++tile_V_Bc) {
5943 #pragma unroll
5944     for (int j = 0; j < kValTileHeadDimV; ++j) {
5945         const int warp_smem_V_d =
5946             warp_KV * (kMmaAtomN * kValTileHeadDimV) + j * kMmaAtomN;
5947         const int lane_smem_V_Bc = tile_V_Bc * kMmaAtomK + lane_id % 16;
5948         const int lane_smem_V_d = warp_smem_V_d;
5949         // V smem: V_wg_base + swizzle_fa offset
5950         const uint32_t lane_smem_V_ptr =
5951             smem_V_wg_base +
5952             swizzle_fa<kHeadDim, Bc>(lane_smem_V_Bc, lane_smem_V_d) *
5953                 sizeof(half);
5954         LDMATRIX_X2_T(R_V[j][0], R_V[j][1], lane_smem_V_ptr);
5955     }
5956
5957     // Early release V after last ldmatrix_V: V data now fully in registers.
5958     // This lets producer start next V TMA during the remaining PV MMA +
5959     // online rescaling, giving more overlap window than releasing after PV.
5960     if (tile_V_Bc == (Bc / kMmaAtomK) - 1) {
5961         [[maybe_unused]] auto token = empty_V[consumer_id].arrive();
5962     }
5963
5964     const int w = tile_V_Bc * 2;
5965 #pragma unroll
5966     for (int i = 0; i < kValTileSeqLenP; ++i) {
5967 #pragma unroll
5968         for (int j = 0; j < kValTileHeadDimV; ++j) {
5969             if constexpr (kMmaAccF32) {
5970                 HMMA16816F32(R_0[i][j][0], R_0[i][j][1], R_0[i][j][2], R_0[i][j][3],
5971                     R_S[i][w][0], R_S[i][w][1], R_S[i][w + 1][0], R_S[i][w + 1][1],
5972                     R_V[j][0], R_V[j][1],
5973                     R_0[i][j][0], R_0[i][j][1], R_0[i][j][2], R_0[i][j][3]);
5974             } else {
5975                 HMMA16816(R_0[i][j][0], R_0[i][j][1],
5976                     R_S[i][w][0], R_S[i][w][1],
5977                     R_S[i][w + 1][0], R_S[i][w + 1][1],
5978                     R_V[j][0], R_V[j][1],
5979                     R_0[i][j][0], R_0[i][j][1]);
5980             }
5981         }
5982     }
5983 }
5984

```

```

5985 // V already released above (after last ldmatrix_V)
5986
5987 // ---- Online rescaling -  $O_{new} = \exp(m_{old} - m_{new}) * O_{old} + P@V$  ----
5988 #pragma unroll
5989 for (int i = 0; i < kValTileSeqLenP; ++i) {
5990     float block_row_max_new_0 = lane_row_max_new[i][0];
5991     float block_row_max_new_1 = lane_row_max_new[i][1];
5992     float block_row_sum_new_0 = lane_row_sum_new[i][0];
5993     float block_row_sum_new_1 = lane_row_sum_new[i][1];
5994     float block_row_max_old_0 = lane_block_row_max_old[i][0];
5995     float block_row_max_old_1 = lane_block_row_max_old[i][1];
5996
5997     block_row_max_new_0 = max(block_row_max_old_0, block_row_max_new_0);
5998     block_row_max_new_1 = max(block_row_max_old_1, block_row_max_new_1);
5999
6000     // First iteration per WG:  $m_{old} = -inf$ , use  $m_{new}$  directly
6001     block_row_max_old_0 = (local_iter > 0 ? block_row_max_old_0
6002                                : block_row_max_new_0);
6003     block_row_max_old_1 = (local_iter > 0 ? block_row_max_old_1
6004                                : block_row_max_new_1);
6005
6006     float rescale_o_factor_0 =
6007         __expf(block_row_max_old_0 - block_row_max_new_0);
6008     float rescale_o_factor_1 =
6009         __expf(block_row_max_old_1 - block_row_max_new_1);
6010
6011 #pragma unroll
6012 for (int j = 0; j < kValTileHeadDimV; ++j) {
6013     if constexpr (kMmaAccF32) {
6014         float *t_fptr_0 = reinterpret_cast<float *>(&R_0[i][j][0]);
6015         float *t_fptr_D = reinterpret_cast<float *>(&R_D[i][j][0]);
6016         t_fptr_D[0] = __fmaf_rn(rescale_o_factor_0, t_fptr_D[0], t_fptr_0[0]);
6017         t_fptr_D[1] = __fmaf_rn(rescale_o_factor_0, t_fptr_D[1], t_fptr_0[1]);
6018         t_fptr_D[2] = __fmaf_rn(rescale_o_factor_1, t_fptr_D[2], t_fptr_0[2]);
6019         t_fptr_D[3] = __fmaf_rn(rescale_o_factor_1, t_fptr_D[3], t_fptr_0[3]);
6020     } else {
6021         float2 t_reg_0_0 = __half22float2(HALF2(R_0[i][j][0]));
6022         float2 t_reg_0_1 = __half22float2(HALF2(R_0[i][j][1]));
6023         float2 t_reg_D_0 = __half22float2(HALF2(R_D[i][j][0]));
6024         float2 t_reg_D_1 = __half22float2(HALF2(R_D[i][j][1]));
6025         t_reg_D_0.x = __fmaf_rn(rescale_o_factor_0, t_reg_D_0.x, t_reg_0_0.x);
6026         t_reg_D_0.y = __fmaf_rn(rescale_o_factor_0, t_reg_D_0.y, t_reg_0_0.y);
6027         t_reg_D_1.x = __fmaf_rn(rescale_o_factor_1, t_reg_D_1.x, t_reg_0_1.x);
6028         t_reg_D_1.y = __fmaf_rn(rescale_o_factor_1, t_reg_D_1.y, t_reg_0_1.y);
6029         HALF2(R_D[i][j][0]) = __float22half2_rn(t_reg_D_0);
6030         HALF2(R_D[i][j][1]) = __float22half2_rn(t_reg_D_1);
6031     }
6032 }
6033
6034 float block_row_sum_old_0 = lane_block_row_sum_old[i][0];
6035 float block_row_sum_old_1 = lane_block_row_sum_old[i][1];
6036 lane_block_row_sum_old[i][0] =
6037     __fmaf_rn(rescale_o_factor_0, block_row_sum_old_0,
6038               block_row_sum_new_0);
6039 lane_block_row_sum_old[i][1] =
6040     __fmaf_rn(rescale_o_factor_1, block_row_sum_old_1,
6041               block_row_sum_new_1);
6042 lane_block_row_max_old[i][0] = block_row_max_new_0;
6043 lane_block_row_max_old[i][1] = block_row_max_new_1;
6044 }
6045 ++local_iter;
6046 }
6047

```

```

6048 // =====
6049 // Final merge of two WG partial states (split-KV reduction)
6050 //
6051 // 安全 alias dynamic smem 的前提:
6052 // 1) producer 已退出 mainloop (所有 TMA 已 arrive_expect_tx);
6053 // 2) 两个 consumer 都已 release 自己的 K/V stage (QK 后 release K,
6054 //    PV 后 release V), 且 mainloop 结束后不再访问任何 K/V smem;
6055 // 3) Q_smem 在 mainloop 内只读, 此后也不再访问。
6056 // 故此时整个 dynamic smem (Q/K/V 区) 生命周期结束, 可安全重命名为
6057 // merge scratch, 不增加 launch smem_bytes。第一次 __syncthreads
6058 // 保证所有 TMA/ldmatrix 完成后才开始写 scratch。
6059 // 完整的 split-KV merge 数学推导见 kernel 头部注释。
6060 // =====
6061 __syncthreads();
6062
6063 // Merge scratch 布局 (按 wg_tid 索引, 每 WG 128 thread):
6064 // [merge_rd]: 128 * kRdPerThread 个 uint32 <- WG2 未归一化 R_D fragment
6065 // [merge_ml]: 128 * kValTileSeqLenQ 个 float4 <- WG2 的 [m0,m1,l0,l1] interleaved
6066 // WG1 用自己的 R_D/m/l 作为 (0acc_0, m_0, l_0), 读取 WG2 写出的作为
6067 // (0acc_1, m_1, l_1), 按 stable merge 公式归并后由 WG1 写最终 0。
6068 // m/l interleaved 为 float4 布局, 写入/读取各一次 128-bit store/load。
6069 constexpr int kRdPerThread = kValTileSeqLenP * kValTileHeadDimV * kNRegs;
6070 constexpr int kRdPackPerThread = kRdPerThread / 4; // float4 个数 (kNRegs 始终偶数)
6071 constexpr int kMlPackPerThread = kValTileSeqLenQ; // 每 thread 的 float4 个数
6072 uint32_t *merge_rd = reinterpret_cast<uint32_t *>(smem_fa3_tma_ws);
6073 float4 *merge_rd_pack = reinterpret_cast<float4 *>(merge_rd);
6074 float4 *merge_ml_pack = merge_rd_pack + 128 * kRdPackPerThread;
6075
6076 // WG2 写出未归一化 partial (R_D, m, l) 到 scratch, 按 wg_tid 索引。
6077 // R_D: 以 float4 (128-bit) 连续写入, kRdPackPerThread 次。
6078 // m/l: interleaved 为 float4 [m0, m1, l0, l1], 每次 128-bit store。
6079 // 按 wg_tid 索引保证 WG1 的同一 lane 读到对应的 WG2 partial:
6080 // R_D[i][j][k] fragment 对应的 Q rows 与 lane_block_row_max/sum[i][0/1]
6081 // 的两组 rows (rows 0-7 / rows 8-15) 严格对齐, 故 WG1/WG2 同一
6082 // wg_tid 的 partial 覆盖相同 Q 行, 可直接逐元素归并。
6083 // Tc==1 时 WG2 未进入循环, m=-inf/l=0/R_D=0, merge 自然退化。
6084 // 写完后第二次 __syncthreads 让 WG1 看到 WG2 的全部写入。
6085 if (consumer_id == 1) {
6086     // R_D: 128-bit stores
6087     float4 *rd_dst = merge_rd_pack + wg_tid * kRdPackPerThread;
6088 #pragma unroll
6089     for (int idx = 0; idx < kRdPackPerThread; ++idx)
6090         rd_dst[idx] = reinterpret_cast<float4 *>(&R_D[0][0][0])[idx];
6091     // m/l interleaved: [m0, m1, l0, l1] per i, 128-bit store
6092 #pragma unroll
6093     for (int i = 0; i < kValTileSeqLenQ; ++i) {
6094         float4 ml;
6095         ml.x = lane_block_row_max_old[i][0];
6096         ml.y = lane_block_row_max_old[i][1];
6097         ml.z = lane_block_row_sum_old[i][0];
6098         ml.w = lane_block_row_sum_old[i][1];
6099         merge_ml_pack[wg_tid * kMlPackPerThread + i] = ml;
6100     }
6101 }
6102 __syncthreads();
6103
6104 // WG1 merges WG2's partial into its own, normalizes, and writes 0
6105 if (consumer_id == 0) {
6106     // 把 WG2 的未归一化 R_D (0acc_1) 载入 R_0 寄存器复用为临时 buffer。
6107     // R_D: 以 float4 (128-bit) 连续读取。
6108     // 后续 merge 用 R_D[i][j] 作为 0acc_0, R_0[i][j] 作为 0acc_1, 逐元素加权后
6109     // 结果写回 R_D, 再做最终 0=0acc/l 归一化。
6110     float4 *rd_src = merge_rd_pack + wg_tid * kRdPackPerThread;

```

```

6111 #pragma unroll
6112     for (int idx = 0; idx < kRdPackPerThread; ++idx)
6113         reinterpret_cast<float4*>(&R_0[0][0][0])[idx] = rd_src[idx];
6114
6115     // 稳定归并 (实现层面, 完整数学见头部注释):
6116     //   - m0/l0 = WG1 自己的 partial (寄存器), m1/l1 = WG2 的 partial (scratch)
6117     //   - [i][0] 对应 Q rows [i*16 .. i*16+7], [i][1] 对应 rows [i*16+8 .. +15]
6118     //   - alpha=exp(m0-m)<=1, beta=exp(m1-m)<=1: 用全局 max 做 pivot 避免 exp 溢出
6119     //   - F32Acc: 直接对 float fragment 逐元素加权求和
6120     //   - F16Acc: 把 packed half2 提升到 float2 计算, 再 pack 回 half2
6121     //       (避免 half 精度下 alpha/beta 加权累加的精度损失)
6122     //   - Tc==1: m1=-inf -> beta=exp(-inf)=0, 0acc/0acc_1 项归零, 退化为 WG1 结果
6123     //   m=max(m0,m1), a=exp(m0-m), b=exp(m1-m),
6124     //   l=a*l0+b*l1, 0acc=a*0acc0+b*0acc1
6125 #pragma unroll
6126     for (int i = 0; i < kValTileSeqLenQ; ++i) {
6127         // 128-bit load: [m1_0, m1_1, l1_0, l1_1] interleaved
6128         float4 mll = merge_ml_pack[wg_tid * kMlPackPerThread + i];
6129         float m1_0 = mll.x;
6130         float m1_1 = mll.y;
6131         float l1_0 = mll.z;
6132         float l1_1 = mll.w;
6133
6134         float m0_0 = lane_block_row_max_old[i][0];
6135         float m0_1 = lane_block_row_max_old[i][1];
6136         float l0_0 = lane_block_row_sum_old[i][0];
6137         float l0_1 = lane_block_row_sum_old[i][1];
6138
6139         float m_0 = fmaxf(m0_0, m1_0);
6140         float m_1 = fmaxf(m0_1, m1_1);
6141         float alpha_0 = __expf(m0_0 - m_0);
6142         float alpha_1 = __expf(m0_1 - m_1);
6143         float beta_0 = __expf(m1_0 - m_0);
6144         float beta_1 = __expf(m1_1 - m_1);
6145
6146         lane_block_row_max_old[i][0] = m_0;
6147         lane_block_row_max_old[i][1] = m_1;
6148         lane_block_row_sum_old[i][0] = alpha_0 * l0_0 + beta_0 * l1_0;
6149         lane_block_row_sum_old[i][1] = alpha_1 * l0_1 + beta_1 * l1_1;
6150
6151 #pragma unroll
6152         for (int j = 0; j < kValTileHeadDimV; ++j) {
6153             if constexpr (kMmaAccF32) {
6154                 float *d0 = reinterpret_cast<float*>(&R_D[i][j][0]);
6155                 float *d1 = reinterpret_cast<float*>(&R_0[i][j][0]);
6156                 d0[0] = alpha_0 * d0[0] + beta_0 * d1[0];
6157                 d0[1] = alpha_0 * d0[1] + beta_0 * d1[1];
6158                 d0[2] = alpha_1 * d0[2] + beta_1 * d1[2];
6159                 d0[3] = alpha_1 * d0[3] + beta_1 * d1[3];
6160             } else {
6161                 float2 d0_0 = __half22float2(HALF2(R_D[i][j][0]));
6162                 float2 d0_1 = __half22float2(HALF2(R_D[i][j][1]));
6163                 float2 d1_0 = __half22float2(HALF2(R_0[i][j][0]));
6164                 float2 d1_1 = __half22float2(HALF2(R_0[i][j][1]));
6165                 d0_0.x = alpha_0 * d0_0.x + beta_0 * d1_0.x;
6166                 d0_0.y = alpha_0 * d0_0.y + beta_0 * d1_0.y;
6167                 d0_1.x = alpha_1 * d0_1.x + beta_1 * d1_1.x;
6168                 d0_1.y = alpha_1 * d0_1.y + beta_1 * d1_1.y;
6169                 HALF2(R_D[i][j][0]) = __float22half2_rn(d0_0);
6170                 HALF2(R_D[i][j][1]) = __float22half2_rn(d0_1);
6171             }
6172         }
6173     }

```

```

6174
6175 // ---- Final rescale: 0 = 0acc / l ----
6176 #pragma unroll
6177 for (int i = 0; i < kValTileSeqLenP; ++i) {
6178     float rescale_factor_0 = __frcp_rn(lane_block_row_sum_old[i][0]);
6179     float rescale_factor_1 = __frcp_rn(lane_block_row_sum_old[i][1]);
6180 #pragma unroll
6181     for (int j = 0; j < kValTileHeadDimV; ++j) {
6182         if constexpr (kMmaAccF32) {
6183             float *t_fptr_D = reinterpret_cast<float *>(&R_D[i][j][0]);
6184             half *t_hptr_D = reinterpret_cast<half *>(&R_D[i][j][0]);
6185             t_hptr_D[0] = __float2half_rn(rescale_factor_0 * t_fptr_D[0]);
6186             t_hptr_D[1] = __float2half_rn(rescale_factor_0 * t_fptr_D[1]);
6187             t_hptr_D[2] = __float2half_rn(rescale_factor_1 * t_fptr_D[2]);
6188             t_hptr_D[3] = __float2half_rn(rescale_factor_1 * t_fptr_D[3]);
6189         } else {
6190             float2 t_reg_D_0 = __half2float2(HALF2(R_D[i][j][0]));
6191             float2 t_reg_D_1 = __half2float2(HALF2(R_D[i][j][1]));
6192             t_reg_D_0.x = rescale_factor_0 * t_reg_D_0.x;
6193             t_reg_D_0.y = rescale_factor_0 * t_reg_D_0.y;
6194             t_reg_D_1.x = rescale_factor_1 * t_reg_D_1.x;
6195             t_reg_D_1.y = rescale_factor_1 * t_reg_D_1.y;
6196             HALF2(R_D[i][j][0]) = __float22half2_rn(t_reg_D_0);
6197             HALF2(R_D[i][j][1]) = __float22half2_rn(t_reg_D_1);
6198         }
6199     }
6200 }
6201
6202 // ---- Epilogue: warp shuffle + 128-bit collective store ----
6203 #pragma unroll
6204 for (int i = 0; i < kValTileSeqLenP; ++i) {
6205     #pragma unroll
6206     for (int j = 0; j < kValTileHeadDimV; ++j) {
6207         uint32_t R_Z[2][4];
6208         R_Z[0][0] = R_D[i][j][0];
6209         R_Z[1][0] = R_D[i][j][1];
6210         R_Z[0][1] = __shfl_sync(0xffffffff, R_D[i][j][0], lane_id + 1, 4);
6211         R_Z[0][2] = __shfl_sync(0xffffffff, R_D[i][j][0], lane_id + 2, 4);
6212         R_Z[0][3] = __shfl_sync(0xffffffff, R_D[i][j][0], lane_id + 3, 4);
6213         R_Z[1][1] = __shfl_sync(0xffffffff, R_D[i][j][1], lane_id + 1, 4);
6214         R_Z[1][2] = __shfl_sync(0xffffffff, R_D[i][j][1], lane_id + 2, 4);
6215         R_Z[1][3] = __shfl_sync(0xffffffff, R_D[i][j][1], lane_id + 3, 4);
6216
6217         if (lane_id % 4 == 0) {
6218             const int store_warp_regs_0_Br =
6219                 warp_QP * (kMmaAtomM * kValTileSeqLenP) + i * kMmaAtomM;
6220             const int store_lane_gmem_0_Br =
6221                 Q_tile_id * Br + store_warp_regs_0_Br + lane_id / 4;
6222             const int store_warp_regs_0_d =
6223                 warp_KV * (kMmaAtomN * kValTileHeadDimV) + j * kMmaAtomN;
6224             const int store_lane_gmem_0_d = store_warp_regs_0_d;
6225             const int store_gmem_0_addr_0 =
6226                 0_gmem_offset + store_lane_gmem_0_Br * kHeadDim +
6227                 store_lane_gmem_0_d;
6228             const int store_gmem_0_addr_1 =
6229                 0_gmem_offset + (store_lane_gmem_0_Br + 8) * kHeadDim +
6230                 store_lane_gmem_0_d;
6231             *reinterpret_cast<float4 *>(&0[store_gmem_0_addr_0]) =
6232                 *reinterpret_cast<float4 *>(&R_Z[0][0]);
6233             *reinterpret_cast<float4 *>(&0[store_gmem_0_addr_1]) =
6234                 *reinterpret_cast<float4 *>(&R_Z[1][0]);
6235         }
6236     }

```

```

6237     }
6238 }
6239 }
6240 }
6241 #endif // END NOTES_V2_ENABLE_TMA_MMA_WS
6242
6243 // =====
6244 // 以下是测试代码，验证 Phase 1 - Phase 8 的kernel的正确性，不评估性能。
6245 // =====
6246
6247 // Bench / test mode globals (placed early so all functions can reference them)
6248 static bool g_verbose = false;
6249 static bool g_bench_hgemm = false;
6250 static bool g_bench_hgemm_all = false;
6251 static bool g_bench_fa = false;
6252 static bool g_bench_all = false;
6253 static bool g_fa_skip_check = false;
6254 static bool g_swizzle_eq_check = false;
6255 enum class FALayout {
6256     All, Pad, SwizzleQ, SwizzleK, SwizzleV, SwizzleQK, SwizzleQV, SwizzleKV, Swizzle
6257 };
6258 static FALayout g_fa_layout = FALayout::Pad;
6259 static int g_bench_M = 4096, g_bench_N = 4096, g_bench_K = 4096;
6260 static int g_bench_B = 1, g_bench_H = 48, g_bench_Nfa = 8192, g_bench_D = 128;
6261 static int g_warmup = 2, g_repeat = 3;
6262
6263 static bool check_smem_feasible(const void *kernel_func, size_t dyn_smem_bytes) {
6264     int device = 0;
6265     int max_smem = 0;
6266     cudaFuncAttributes attrs{};
6267     cudaGetDevice(&device);
6268     cudaDeviceGetAttribute(&max_smem, cudaDevAttrMaxSharedMemoryPerBlockOptin, device);
6269     cudaFuncGetAttributes(&attrs, kernel_func);
6270     return dyn_smem_bytes + attrs.sharedSizeBytes <= (size_t)max_smem;
6271 }
6272
6273 static inline void check(cudaError_t err, const char *msg) {
6274     if (err != cudaSuccess) {
6275         fprintf(stderr, "[ERROR] %s: %s\n", msg, cudaGetErrorString(err));
6276         exit(EXIT_FAILURE);
6277     }
6278 }
6279
6280 static void test_block_reduce(int N) {
6281
6282     srand(42);
6283     float *h_a = (float *)malloc((size_t)N * sizeof(float));
6284     for (int i = 0; i < N; i++)
6285         h_a[i] = ((float)rand() / RAND_MAX) * 2.0f - 1.0f;
6286
6287     // CPU reference: sum of all elements
6288     double ref = 0.0;
6289     for (int i = 0; i < N; i++) ref += (double)h_a[i];
6290
6291     float *d_a, *d_y;
6292     check(cudaMalloc(&d_a, (size_t)N * sizeof(float)), "blockreduce alloc A");
6293     check(cudaMalloc(&d_y, sizeof(float)), "blockreduce alloc Y");
6294
6295     check(cudaMemcpy(d_a, h_a, (size_t)N * sizeof(float), cudaMemcpyHostToDevice), "blockreduce H2D A");
6296     check(cudaMemset(d_y, 0, sizeof(float)), "blockreduce zero Y");
6297
6298     dim3 block(128);
6299     dim3 grid((N + 127) / 128);

```

```

6300 block_reduce_all<128><<<grid, block>>>(d_a, d_y, N);
6301 check(cudaGetLastError(), "blockreduce launch");
6302 check(cudaDeviceSynchronize(), "blockreduce sync");
6303
6304 float result;
6305 check(cudaMemcpy(&result, d_y, sizeof(float), cudaMemcpyDeviceToHost), "blockreduce D2H");
6306
6307 float err = fabsf(result - (float)ref);
6308 printf("| %-41s | %.6e | %-4s | %-19s |\n", "BlockReduce", err,
6309         err < 1e-2f ? "PASS" : "FAIL", "None");
6310
6311 free(h_a);
6312 cudaFree(d_a); cudaFree(d_y);
6313 }
6314
6315
6316 static void test_dot(int N) {
6317
6318     srand(42);
6319     float *h_a = (float *)malloc((size_t)N * sizeof(float));
6320     float *h_b = (float *)malloc((size_t)N * sizeof(float));
6321     for (int i = 0; i < N; i++) {
6322         h_a[i] = ((float)rand() / RAND_MAX) * 2.0f - 1.0f;
6323         h_b[i] = ((float)rand() / RAND_MAX) * 2.0f - 1.0f;
6324     }
6325
6326     // CPU reference
6327     double ref = 0.0;
6328     for (int i = 0; i < N; i++) ref += (double)h_a[i] * (double)h_b[i];
6329
6330     float *d_a, *d_b, *d_y;
6331     check(cudaMalloc(&d_a, (size_t)N * sizeof(float)), "dot alloc A");
6332     check(cudaMalloc(&d_b, (size_t)N * sizeof(float)), "dot alloc B");
6333     check(cudaMalloc(&d_y, sizeof(float)), "dot alloc Y");
6334
6335     check(cudaMemcpy(d_a, h_a, (size_t)N * sizeof(float), cudaMemcpyHostToDevice), "dot H2D A");
6336     check(cudaMemcpy(d_b, h_b, (size_t)N * sizeof(float), cudaMemcpyHostToDevice), "dot H2D B");
6337     check(cudaMemset(d_y, 0, sizeof(float)), "dot zero Y");
6338
6339     dim3 block(128);
6340     dim3 grid((N + 127) / 128);
6341     dot<128><<<grid, block>>>(d_a, d_b, d_y, N);
6342     check(cudaGetLastError(), "dot launch");
6343     check(cudaDeviceSynchronize(), "dot sync");
6344
6345     float result;
6346     check(cudaMemcpy(&result, d_y, sizeof(float), cudaMemcpyDeviceToHost), "dot D2H");
6347
6348     float err = fabsf(result - (float)ref);
6349     printf("| %-41s | %.6e | %-4s | %-19s |\n", "Dot", err,
6350         err < 1e-2f ? "PASS" : "FAIL", "None");
6351
6352     // ---- Dot Vec4 ----
6353     check(cudaMemset(d_y, 0, sizeof(float)), "dot_vec4 zero Y");
6354     dim3 block_v4(32);
6355     dot_vec4<32><<<grid, block_v4>>>(d_a, d_b, d_y, N);
6356     check(cudaGetLastError(), "dot_vec4 launch");
6357     check(cudaDeviceSynchronize(), "dot_vec4 sync");
6358
6359     check(cudaMemcpy(&result, d_y, sizeof(float), cudaMemcpyDeviceToHost), "dot_vec4 D2H");
6360     float err_v4 = fabsf(result - (float)ref);
6361     printf("| %-41s | %.6e | %-4s | %-19s |\n", "Dot-Vec4", err_v4, err_v4 < 1e-2f ? "PASS" : "FAIL", "None");
6362

```

```

6363     free(h_a); free(h_b);
6364     cudaFree(d_a); cudaFree(d_b); cudaFree(d_y);
6365 }
6366
6367
6368 static void test_relu(int N) {
6369     srand(42);
6370     float *h_x = (float *)malloc((size_t)N * sizeof(float));
6371     float *h_y = (float *)malloc((size_t)N * sizeof(float));
6372     for (int i = 0; i < N; i++)
6373         h_x[i] = ((float)rand() / RAND_MAX) * 2.0f - 1.0f;
6374
6375     float *d_x, *d_y;
6376     check(cudaMalloc(&d_x, (size_t)N * sizeof(float)), "relu alloc X");
6377     check(cudaMalloc(&d_y, (size_t)N * sizeof(float)), "relu alloc Y");
6378     check(cudaMemcpy(d_x, h_x, (size_t)N * sizeof(float), cudaMemcpyHostToDevice), "relu H2D");
6379
6380     for (int i = 0; i < N; i++) h_y[i] = fmaxf(0.0f, h_x[i]);
6381     dim3 block256(256);
6382     dim3 grid256((N + 255) / 256);
6383     relu<<<grid256, block256>>>(d_x, d_y, N);
6384     check(cudaGetLastError(), "relu launch");
6385     check(cudaDeviceSynchronize(), "relu sync");
6386     check(cudaMemcpy(h_y, d_y, (size_t)N * sizeof(float), cudaMemcpyDeviceToHost), "relu D2H");
6387     float max_err = 0.0f;
6388     for (int i = 0; i < N; i++) {
6389         float expected = fmaxf(0.0f, h_x[i]);
6390         float err = fabsf(h_y[i] - expected);
6391         if (err > max_err) max_err = err;
6392     }
6393     printf("| %-41s | %.6e | %-4s | %-19s |\n", "ReLU", max_err, max_err < 1e-4f ? "PASS" : "FAIL", "None");
6394
6395     dim3 block64(64);
6396     relu_vec4<<<grid256, block64>>>(d_x, d_y, N);
6397     check(cudaGetLastError(), "relu_vec4 launch");
6398     check(cudaDeviceSynchronize(), "relu_vec4 sync");
6399     check(cudaMemcpy(h_y, d_y, (size_t)N * sizeof(float), cudaMemcpyDeviceToHost), "relu_vec4 D2H");
6400     max_err = 0.0f;
6401     for (int i = 0; i < N; i++) {
6402         float expected = fmaxf(0.0f, h_x[i]);
6403         float err = fabsf(h_y[i] - expected);
6404         if (err > max_err) max_err = err;
6405     }
6406     printf("| %-41s | %.6e | %-4s | %-19s |\n", "ReLU-Vec4", max_err, max_err < 1e-4f ? "PASS" : "FAIL", "None");
6407
6408     free(h_x); free(h_y);
6409     cudaFree(d_x); cudaFree(d_y);
6410 }
6411
6412
6413 static void test_elementwise(int N) {
6414     srand(42);
6415     float *h_a = (float *)malloc((size_t)N * sizeof(float));
6416     float *h_b = (float *)malloc((size_t)N * sizeof(float));
6417     for (int i = 0; i < N; i++) {
6418         h_a[i] = ((float)rand() / RAND_MAX) * 2.0f - 1.0f;
6419         h_b[i] = ((float)rand() / RAND_MAX) * 2.0f - 1.0f;
6420     }
6421
6422     float *d_a, *d_b, *d_c;
6423     check(cudaMalloc(&d_a, (size_t)N * sizeof(float)), "eadd alloc A");
6424     check(cudaMalloc(&d_b, (size_t)N * sizeof(float)), "eadd alloc B");

```

```

6425 check(cudaMalloc(&d_c, (size_t)N * sizeof(float)), "eadd alloc C");
6426 check(cudaMemcpy(d_a, h_a, (size_t)N * sizeof(float), cudaMemcpyHostToDevice), "eadd H2D A");
6427 check(cudaMemcpy(d_b, h_b, (size_t)N * sizeof(float), cudaMemcpyHostToDevice), "eadd H2D B");
6428
6429 dim3 block256(256);
6430 dim3 grid256((N + 255) / 256);
6431 elementwise_add<<<grid256, block256>>>(d_a, d_b, d_c, N);
6432 check(cudaGetLastError(), "eadd launch");
6433 check(cudaDeviceSynchronize(), "eadd sync");
6434 float *h_c = (float *)malloc((size_t)N * sizeof(float));
6435 check(cudaMemcpy(h_c, d_c, (size_t)N * sizeof(float), cudaMemcpyDeviceToHost), "eadd D2H");
6436 float max_err = 0.0f;
6437 for (int i = 0; i < N; i++) {
6438     float err = fabsf(h_c[i] - (h_a[i] + h_b[i]));
6439     if (err > max_err) max_err = err;
6440 }
6441 printf("| %-41s | %.6e | %-4s | %-19s |\n", "ElemwiseAdd", max_err, max_err < 1e-4f ? "PASS" : "FAIL", "
    None");
6442
6443 dim3 block64(64);
6444 check(cudaMemset(d_c, 0, (size_t)N * sizeof(float)), "eadd_vec4 zero C");
6445 elementwise_add_vec4<<<grid256, block64>>>(d_a, d_b, d_c, N);
6446 check(cudaGetLastError(), "eadd_vec4 launch");
6447 check(cudaDeviceSynchronize(), "eadd_vec4 sync");
6448 check(cudaMemcpy(h_c, d_c, (size_t)N * sizeof(float), cudaMemcpyDeviceToHost), "eadd_vec4 D2H");
6449 max_err = 0.0f;
6450 for (int i = 0; i < N; i++) {
6451     float err = fabsf(h_c[i] - (h_a[i] + h_b[i]));
6452     if (err > max_err) max_err = err;
6453 }
6454 printf("| %-41s | %.6e | %-4s | %-19s |\n", "ElemwiseAdd-Vec4", max_err, max_err < 1e-4f ? "PASS" : "FAIL"
    , "None");
6455
6456 free(h_a); free(h_b); free(h_c);
6457 cudaFree(d_a); cudaFree(d_b); cudaFree(d_c);
6458 }
6459
6460
6461 static void test_histogram(int N) {
6462     srand(42);
6463     int BINS = 16;
6464     int *h_hist = (int *)calloc(BINS, sizeof(int));
6465     int *h_hist_ref = (int *)calloc(BINS, sizeof(int));
6466     int *h_idx = (int *)malloc((size_t)N * sizeof(int));
6467     for (int i = 0; i < N; i++) h_idx[i] = rand() % BINS;
6468     for (int i = 0; i < N; i++) h_hist_ref[h_idx[i]]++;
6469
6470     int *d_idx, *d_hist;
6471     check(cudaMalloc(&d_idx, (size_t)N * sizeof(int)), "hist alloc idx");
6472     check(cudaMalloc(&d_hist, BINS * sizeof(int)), "hist alloc hist");
6473     check(cudaMemcpy(d_idx, h_idx, (size_t)N * sizeof(int), cudaMemcpyHostToDevice), "hist H2D idx");
6474     check(cudaMemset(d_hist, 0, BINS * sizeof(int)), "hist zero");
6475
6476     dim3 block256(256);
6477     dim3 grid256((N + 255) / 256);
6478     histogram<<<grid256, block256>>>(d_idx, d_hist, N);
6479     check(cudaGetLastError(), "histogram launch");
6480     check(cudaDeviceSynchronize(), "histogram sync");
6481     check(cudaMemcpy(h_hist, d_hist, BINS * sizeof(int), cudaMemcpyDeviceToHost), "hist D2H");
6482
6483     float max_err = 0.0f;
6484     for (int i = 0; i < BINS; i++) {
6485         float err = fabsf((float)(h_hist[i] - h_hist_ref[i]));

```

```

6486     if (err > max_err) max_err = err;
6487 }
6488 printf("| %-41s | %.6e | %-4s | %-19s |\n", "Histogram", max_err, max_err < 1e-4f ? "PASS" : "FAIL", "None
    ");
6489
6490 free(h_hist); free(h_hist_ref); free(h_idx);
6491 cudaFree(d_idx); cudaFree(d_hist);
6492 }
6493
6494
6495 static void test_merge_attn_states(int num_tokens, int num_heads,
6496                                   int head_size) {
6497
6498     size_t out_size = (size_t)num_tokens * num_heads * head_size * sizeof(float);
6499     size_t lse_size = (size_t)num_heads * num_tokens * sizeof(float);
6500
6501     float *h_prefix_out = (float *)malloc(out_size);
6502     float *h_suffix_out = (float *)malloc(out_size);
6503     float *h_prefix_lse = (float *)malloc(lse_size);
6504     float *h_suffix_lse = (float *)malloc(lse_size);
6505     float *h_output_ref = (float *)malloc(out_size);
6506
6507     srand(42);
6508     for (int i = 0; i < num_tokens * num_heads * head_size; i++) {
6509         h_prefix_out[i] = ((float)rand() / RAND_MAX) * 2.0f - 1.0f;
6510         h_suffix_out[i] = ((float)rand() / RAND_MAX) * 2.0f - 1.0f;
6511     }
6512     for (int i = 0; i < num_heads * num_tokens; i++) {
6513         h_prefix_lse[i] = ((float)rand() / RAND_MAX) * 20.0f - 10.0f;
6514         h_suffix_lse[i] = ((float)rand() / RAND_MAX) * 20.0f - 10.0f;
6515     }
6516
6517     // CPU reference: 与 kernel 完全相同的浮点计算
6518     for (int t = 0; t < num_tokens; t++) {
6519         for (int h = 0; h < num_heads; h++) {
6520             float p_lse = h_prefix_lse[h * num_tokens + t];
6521             float s_lse = h_suffix_lse[h * num_tokens + t];
6522             p_lse = isinf(p_lse) ? -INFINITY : p_lse;
6523             s_lse = isinf(s_lse) ? -INFINITY : s_lse;
6524
6525             float max_lse = fmaxf(p_lse, s_lse);
6526             p_lse -= max_lse;
6527             s_lse -= max_lse;
6528             float p_se = expf(p_lse);
6529             float s_se = expf(s_lse);
6530             float p_scale = p_se / (p_se + s_se);
6531             float s_scale = s_se / (p_se + s_se);
6532
6533             int head_off = t * num_heads * head_size + h * head_size;
6534             for (int d = 0; d < head_size; d++) {
6535                 h_output_ref[head_off + d] =
6536                     h_prefix_out[head_off + d] * p_scale +
6537                     h_suffix_out[head_off + d] * s_scale;
6538             }
6539         }
6540     }
6541
6542     float *d_prefix_out, *d_suffix_out, *d_prefix_lse, *d_suffix_lse, *d_output;
6543     check(cudaMalloc(&d_prefix_out, out_size), "merge_attn alloc p_out");
6544     check(cudaMalloc(&d_suffix_out, out_size), "merge_attn alloc s_out");
6545     check(cudaMalloc(&d_prefix_lse, lse_size), "merge_attn alloc p_lse");
6546     check(cudaMalloc(&d_suffix_lse, lse_size), "merge_attn alloc s_lse");
6547     check(cudaMalloc(&d_output, out_size), "merge_attn alloc output");

```

```

6548
6549 check(cudaMemcpy(d_prefix_out, h_prefix_out, out_size,
6550                 cudaMemcpyHostToDevice),
6551         "merge_attn H2D p_out");
6552 check(cudaMemcpy(d_suffix_out, h_suffix_out, out_size,
6553                 cudaMemcpyHostToDevice),
6554         "merge_attn H2D s_out");
6555 check(cudaMemcpy(d_prefix_lse, h_prefix_lse, lse_size,
6556                 cudaMemcpyHostToDevice),
6557         "merge_attn H2D p_lse");
6558 check(cudaMemcpy(d_suffix_lse, h_suffix_lse, lse_size,
6559                 cudaMemcpyHostToDevice),
6560         "merge_attn H2D s_lse");
6561
6562 int threads_per_head = head_size / 4;
6563 int total_threads = num_tokens * num_heads * threads_per_head;
6564 dim3 block(128);
6565 dim3 grid((total_threads + 127) / 128);
6566 merge_attn_states<<<grid, block>>>(
6567     d_output, d_prefix_out, d_prefix_lse, d_suffix_out, d_suffix_lse,
6568     num_tokens, num_heads, head_size);
6569 check(cudaGetLastError(), "merge_attn launch");
6570 check(cudaDeviceSynchronize(), "merge_attn sync");
6571
6572 float *h_output = (float *)malloc(out_size);
6573 check(cudaMemcpy(h_output, d_output, out_size, cudaMemcpyDeviceToHost),
6574         "merge_attn D2H");
6575
6576 float max_err = 0.0f;
6577 for (int i = 0; i < num_tokens * num_heads * head_size; i++) {
6578     float err = fabsf(h_output[i] - h_output_ref[i]);
6579     if (err > max_err) max_err = err;
6580 }
6581 printf("| %-41s | %.6e | %-4s | %-19s |\n", "MergeAttnStates", max_err,
6582        max_err < 1e-4f ? "PASS" : "FAIL", "None");
6583
6584 // 边界测试: +inf LSE → 权重退化为 0 (空 attention 段)
6585 h_prefix_lse[0] = INFINITY; // head 0, token 0 的 LSE = +inf
6586 h_suffix_lse[0] = 0.0f;
6587 check(cudaMemcpy(d_prefix_lse, h_prefix_lse, lse_size,
6588                 cudaMemcpyHostToDevice),
6589         "merge_attn H2D inf lse");
6590 merge_attn_states<<<grid, block>>>(
6591     d_output, d_prefix_out, d_prefix_lse, d_suffix_out, d_suffix_lse,
6592     num_tokens, num_heads, head_size);
6593 check(cudaGetLastError(), "merge_attn inf launch");
6594 check(cudaDeviceSynchronize(), "merge_attn inf sync");
6595 check(cudaMemcpy(h_output, d_output, out_size, cudaMemcpyDeviceToHost),
6596         "merge_attn inf D2H");
6597
6598 // token 0, head 0 的所有元素应等于 suffix_output (prefix 权重  $\alpha=0$ )
6599 float inf_err = 0.0f;
6600 for (int d = 0; d < head_size; d++) {
6601     float err = fabsf(h_output[d] - h_suffix_out[d]);
6602     if (err > inf_err) inf_err = err;
6603 }
6604 printf("| %-41s | %.6e | %-4s | %-19s |\n", "MergeAttnStates-inf", inf_err,
6605        inf_err < 1e-4f ? "PASS" : "FAIL", "None");
6606
6607 free(h_prefix_out);
6608 free(h_suffix_out);
6609 free(h_prefix_lse);
6610 free(h_suffix_lse);

```

```

6611 free(h_output_ref);
6612 free(h_output);
6613 cudaFree(d_prefix_out);
6614 cudaFree(d_suffix_out);
6615 cudaFree(d_prefix_lse);
6616 cudaFree(d_suffix_lse);
6617 cudaFree(d_output);
6618 }
6619
6620
6621 static void test_softmax(int N) {
6622     // Use N = blockDim.x so one block processes all elements as one token.
6623     constexpr int kNumThreads = 256;
6624     if (N != kNumThreads) {
6625         printf(" OnlineSafeSoftmax: N must be %d (got %d), skipping.\n", kNumThreads, N);
6626         return;
6627     }
6628
6629     srand(42);
6630     float *h_x = (float *)malloc((size_t)N * sizeof(float));
6631     float *h_y_ref = (float *)malloc((size_t)N * sizeof(float));
6632     for (int i = 0; i < N; i++)
6633         h_x[i] = ((float)rand() / RAND_MAX) * 10.0f - 5.0f; // [-5, 5]
6634
6635     // CPU reference: softmax(x_i) = exp(x_i - max) / sum(exp(x_j - max))
6636     float max_val = -FLT_MAX;
6637     for (int i = 0; i < N; i++) if (h_x[i] > max_val) max_val = h_x[i];
6638     double sum_exp = 0.0;
6639     for (int i = 0; i < N; i++) sum_exp += (double)expf(h_x[i] - max_val);
6640     for (int i = 0; i < N; i++)
6641         h_y_ref[i] = expf(h_x[i] - max_val) / (float)sum_exp;
6642
6643     float *d_x, *d_y;
6644     check(cudaMalloc(&d_x, (size_t)N * sizeof(float)), "softmax alloc X");
6645     check(cudaMalloc(&d_y, (size_t)N * sizeof(float)), "softmax alloc Y");
6646
6647     check(cudaMemcpy(d_x, h_x, (size_t)N * sizeof(float), cudaMemcpyHostToDevice), "softmax H2D X");
6648
6649     dim3 block(256);
6650     dim3 grid(1); // one token covering all N elements
6651     online_safe_softmax_per_token<<<grid, block>>>(d_x, d_y, N);
6652     check(cudaGetLastError(), "softmax launch");
6653     check(cudaDeviceSynchronize(), "softmax sync");
6654
6655     float *h_y = (float *)malloc((size_t)N * sizeof(float));
6656     check(cudaMemcpy(h_y, d_y, (size_t)N * sizeof(float), cudaMemcpyDeviceToHost), "softmax D2H");
6657
6658     float max_err = 0.0f;
6659     for (int i = 0; i < N; i++) {
6660         float err = fabsf(h_y[i] - h_y_ref[i]);
6661         if (err > max_err) max_err = err;
6662     }
6663     printf("| %-41s | %.6e | %-4s | %-19s |\n", "OnlineSafeSoftmax", max_err, max_err < 1e-4f ? "PASS" : "FAIL", "None");
6664
6665     // ---- Safe Softmax ----
6666     safe_softmax_per_token<<<grid, block>>>(d_x, d_y, N);
6667     check(cudaGetLastError(), "safe_softmax launch");
6668     check(cudaDeviceSynchronize(), "safe_softmax sync");
6669     check(cudaMemcpy(h_y, d_y, (size_t)N * sizeof(float), cudaMemcpyDeviceToHost), "safe_softmax D2H");
6670     max_err = 0.0f;
6671     for (int i = 0; i < N; i++) {
6672         float err = fabsf(h_y[i] - h_y_ref[i]);

```

```

6673     if (err > max_err) max_err = err;
6674 }
6675 printf("| %-41s | %.6e | %-4s | %-19s |\n", "SafeSoftmax", max_err, max_err < 1e-4f ? "PASS" : "FAIL", "
    None");
6676
6677 // ---- Naive Softmax ----
6678 softmax_per_token<<<grid, block>>>(d_x, d_y, N);
6679 check(cudaGetLastError(), "naive_softmax launch");
6680 check(cudaDeviceSynchronize(), "naive_softmax sync");
6681 check(cudaMemcpy(h_y, d_y, (size_t)N * sizeof(float), cudaMemcpyDeviceToHost), "naive_softmax D2H");
6682 max_err = 0.0f;
6683 for (int i = 0; i < N; i++) {
6684     float err = fabsf(h_y[i] - h_y_ref[i]);
6685     if (err > max_err) max_err = err;
6686 }
6687 printf("| %-41s | %.6e | %-4s | %-19s |\n", "NaiveSoftmax", max_err, max_err < 1e-4f ? "PASS" : "FAIL", "
    None");
6688
6689 free(h_x); free(h_y); free(h_y_ref);
6690 cudaFree(d_x); cudaFree(d_y);
6691 }
6692
6693
6694 static void test_rms_norm(int N, int K) {
6695
6696     srand(42);
6697     float *h_x = (float *)malloc((size_t)N * K * sizeof(float));
6698     float *h_y_ref = (float *)malloc((size_t)N * K * sizeof(float));
6699     for (int i = 0; i < N * K; i++)
6700         h_x[i] = ((float)rand() / RAND_MAX) * 2.0f - 1.0f;
6701     float g = 1.5f; // gain
6702
6703     // CPU reference: y = (x / rms(x)) * g
6704     float epsilon = 1e-5f;
6705     for (int n = 0; n < N; n++) {
6706         double sum_sq = 0.0;
6707         for (int k = 0; k < K; k++) sum_sq += (double)h_x[n * K + k] * (double)h_x[n * K + k];
6708         float rms = sqrtf((float)sum_sq / (float)K + epsilon);
6709         for (int k = 0; k < K; k++)
6710             h_y_ref[n * K + k] = (h_x[n * K + k] / rms) * g;
6711     }
6712
6713     float *d_x, *d_y;
6714     check(cudaMalloc(&d_x, (size_t)N * K * sizeof(float)), "rmsnorm alloc X");
6715     check(cudaMalloc(&d_y, (size_t)N * K * sizeof(float)), "rmsnorm alloc Y");
6716     check(cudaMemcpy(d_x, h_x, (size_t)N * K * sizeof(float), cudaMemcpyHostToDevice), "rmsnorm H2D X");
6717
6718     dim3 block(128);
6719     dim3 grid(N);
6720     rms_norm<<<grid, block>>>(d_x, d_y, g, N, K);
6721     check(cudaGetLastError(), "rmsnorm launch");
6722     check(cudaDeviceSynchronize(), "rmsnorm sync");
6723
6724     float *h_y = (float *)malloc((size_t)N * K * sizeof(float));
6725     check(cudaMemcpy(h_y, d_y, (size_t)N * K * sizeof(float), cudaMemcpyDeviceToHost), "rmsnorm D2H");
6726
6727     float max_err = 0.0f;
6728     for (int i = 0; i < N * K; i++) {
6729         float err = fabsf(h_y[i] - h_y_ref[i]);
6730         if (err > max_err) max_err = err;
6731     }
6732     printf("| %-41s | %.6e | %-4s | %-19s |\n", "RMSNorm", max_err, max_err < 1e-4f ? "PASS" : "FAIL", "None")
    ;

```

```

6733 // ---- RMS Norm Vec4 ----
6734 dim3 block_rv4(32);
6735 rms_norm_vec4<<<grid, block_rv4>>>(d_x, d_y, g, N, K);
6736 check(cudaGetLastError(), "rmsnorm_vec4 launch");
6737 check(cudaDeviceSynchronize(), "rmsnorm_vec4 sync");
6738 check(cudaMemcpy(h_y, d_y, (size_t)N * K * sizeof(float), cudaMemcpyDeviceToHost), "rmsnorm_vec4 D2H");
6739 max_err = 0.0f;
6740 for (int i = 0; i < N * K; i++) {
6741     float err = fabsf(h_y[i] - h_y_ref[i]);
6742     if (err > max_err) max_err = err;
6743 }
6744 printf("| %-41s | %.6e | %-4s | %-19s |\n", "RMSNorm-Vec4", max_err, max_err < 1e-4f ? "PASS" : "FAIL", "
6745     None");
6746
6747 free(h_x); free(h_y); free(h_y_ref);
6748 cudaFree(d_x); cudaFree(d_y);
6749 }
6750
6751 static void test_layer_norm(int N, int K) {
6752
6753     srand(42);
6754     float *h_x = (float *)malloc((size_t)N * K * sizeof(float));
6755     float *h_y_ref = (float *)malloc((size_t)N * K * sizeof(float));
6756     for (int i = 0; i < N * K; i++)
6757         h_x[i] = ((float)rand() / RAND_MAX) * 2.0f - 1.0f;
6758     float g = 1.5f, b = 0.3f; // gain and bias
6759
6760     // CPU reference: y = ((x - mean) / std) * g + b
6761     float epsilon = 1e-5f;
6762     for (int n = 0; n < N; n++) {
6763         double sum = 0.0;
6764         for (int k = 0; k < K; k++) sum += (double)h_x[n * K + k];
6765         float mean = (float)sum / (float)K;
6766         double sum_sq = 0.0;
6767         for (int k = 0; k < K; k++) {
6768             float diff = h_x[n * K + k] - mean;
6769             sum_sq += (double)diff * (double)diff;
6770         }
6771         float std = sqrtf((float)sum_sq / (float)K + epsilon);
6772         for (int k = 0; k < K; k++)
6773             h_y_ref[n * K + k] = ((h_x[n * K + k] - mean) / std) * g + b;
6774     }
6775
6776     float *d_x, *d_y;
6777     check(cudaMalloc(&d_x, (size_t)N * K * sizeof(float)), "layernorm alloc X");
6778     check(cudaMalloc(&d_y, (size_t)N * K * sizeof(float)), "layernorm alloc Y");
6779     check(cudaMemcpy(d_x, h_x, (size_t)N * K * sizeof(float), cudaMemcpyHostToDevice), "layernorm H2D X");
6780
6781     dim3 block(128);
6782     dim3 grid(N);
6783     layer_norm<<<grid, block>>>(d_x, d_y, g, b, N, K);
6784     check(cudaGetLastError(), "layernorm launch");
6785     check(cudaDeviceSynchronize(), "layernorm sync");
6786
6787     float *h_y = (float *)malloc((size_t)N * K * sizeof(float));
6788     check(cudaMemcpy(h_y, d_y, (size_t)N * K * sizeof(float), cudaMemcpyDeviceToHost), "layernorm D2H");
6789
6790     float max_err = 0.0f;
6791     for (int i = 0; i < N * K; i++) {
6792         float err = fabsf(h_y[i] - h_y_ref[i]);
6793         if (err > max_err) max_err = err;
6794     }

```

```

6795 }
6796 printf("| %-41s | %.6e | %-4s | %-19s |\n", "LayerNorm", max_err, max_err < 1e-4f ? "PASS" : "FAIL", "None
    ");
6797
6798 // ---- Layer Norm Vec4 ----
6799 dim3 block_lv4(32);
6800 layer_norm_vec4<<<grid, block_lv4>>>(d_x, d_y, g, b, N, K);
6801 check(cudaGetLastError(), "layernorm_vec4 launch");
6802 check(cudaDeviceSynchronize(), "layernorm_vec4 sync");
6803 check(cudaMemcpy(h_y, d_y, (size_t)N * K * sizeof(float), cudaMemcpyDeviceToHost), "layernorm_vec4 D2H");
6804 max_err = 0.0f;
6805 for (int i = 0; i < N * K; i++) {
6806     float err = fabsf(h_y[i] - h_y_ref[i]);
6807     if (err > max_err) max_err = err;
6808 }
6809 printf("| %-41s | %.6e | %-4s | %-19s |\n", "LayerNorm-Vec4", max_err, max_err < 1e-4f ? "PASS" : "FAIL",
    "None");
6810
6811 free(h_x); free(h_y); free(h_y_ref);
6812 cudaFree(d_x); cudaFree(d_y);
6813 }
6814
6815
6816 static void test_rope(int seq_len, int N) {
6817
6818     int total_pairs = seq_len * N;
6819     int total_elems = total_pairs * 2;
6820     size_t size = (size_t)total_elems * sizeof(float);
6821
6822     float *h_x = (float *)malloc(size);
6823     float *h_y_ref = (float *)malloc(size);
6824
6825     srand(42);
6826     for (int i = 0; i < total_elems; i++)
6827         h_x[i] = ((float)rand() / RAND_MAX) * 2.0f - 1.0f;
6828
6829     // CPU reference: 2D rotation for each pair
6830     for (int idx = 0; idx < total_pairs; idx++) {
6831         int token_pos = idx / N;
6832         int token_idx = idx % N;
6833         float x1 = h_x[idx * 2];
6834         float x2 = h_x[idx * 2 + 1];
6835         float theta = 1.0f / powf(10000.0f, 2.0f * token_idx / (N * 2.0f));
6836         float angle = (float)token_pos * theta;
6837         float cos_v = cosf(angle);
6838         float sin_v = sinf(angle);
6839         h_y_ref[idx * 2] = x1 * cos_v - x2 * sin_v;
6840         h_y_ref[idx * 2 + 1] = x1 * sin_v + x2 * cos_v;
6841     }
6842
6843     float *d_x, *d_y;
6844     check(cudaMalloc(&d_x, size), "rope alloc X");
6845     check(cudaMalloc(&d_y, size), "rope alloc Y");
6846     check(cudaMemcpy(d_x, h_x, size, cudaMemcpyHostToDevice), "rope H2D");
6847
6848     dim3 block(256);
6849     dim3 grid((total_pairs + 255) / 256);
6850     rope<<<grid, block>>>(d_x, d_y, seq_len, N);
6851     check(cudaGetLastError(), "rope launch");
6852     check(cudaDeviceSynchronize(), "rope sync");
6853
6854     float *h_y = (float *)malloc(size);
6855     check(cudaMemcpy(h_y, d_y, size, cudaMemcpyDeviceToHost), "rope D2H");

```

```

6856 float max_err = 0.0f;
6857 for (int i = 0; i < total_elems; i++) {
6858     float err = fabsf(h_y[i] - h_y_ref[i]);
6859     if (err > max_err) max_err = err;
6860 }
6861 printf("| %-41s | %.6e | %-4s | %-19s |\n", "RoPE", max_err, max_err < 1e-4f ? "PASS" : "FAIL", "None");
6862
6863 free(h_x);
6864 free(h_y);
6865 free(h_y_ref);
6866 cudaFree(d_x);
6867 cudaFree(d_y);
6868 }
6869
6870
6871
6872 static void test_mat_transpose(int row, int col) {
6873
6874     size_t size_in = (size_t)row * col * sizeof(float);
6875     size_t size_out = (size_t)col * row * sizeof(float);
6876
6877     float *h_x = (float *)malloc(size_in);
6878     float *h_y_ref = (float *)malloc(size_out);
6879
6880     srand(42);
6881     for (int i = 0; i < row * col; i++)
6882         h_x[i] = ((float)rand() / RAND_MAX) * 2.0f - 1.0f;
6883
6884     // CPU reference: y[j][i] = x[i][j] (row-major)
6885     for (int i = 0; i < row; i++)
6886         for (int j = 0; j < col; j++)
6887             h_y_ref[j * row + i] = h_x[i * col + j];
6888
6889     float *d_x, *d_y;
6890     check(cudaMalloc(&d_x, size_in), "mattrans alloc X");
6891     check(cudaMalloc(&d_y, size_out), "mattrans alloc Y");
6892     check(cudaMemcpy(d_x, h_x, size_in, cudaMemcpyHostToDevice), "mattrans H2D");
6893
6894     dim3 block(16, 16);
6895     dim3 grid((col + 15) / 16, (row + 15) / 16);
6896     mat_transpose<<<grid, block>>>(d_x, d_y, row, col);
6897     check(cudaGetLastError(), "mattrans launch");
6898     check(cudaDeviceSynchronize(), "mattrans sync");
6899
6900     float *h_y = (float *)malloc(size_out);
6901     check(cudaMemcpy(h_y, d_y, size_out, cudaMemcpyDeviceToHost), "mattrans D2H");
6902
6903     float max_err = 0.0f;
6904     for (int i = 0; i < col * row; i++) {
6905         float err = fabsf(h_y[i] - h_y_ref[i]);
6906         if (err > max_err) max_err = err;
6907     }
6908     printf("| %-41s | %.6e | %-4s | %-19s |\n", "MatTranspose", max_err, max_err < 1e-6f ? "PASS" : "FAIL", "None");
6909
6910     free(h_x);
6911     free(h_y);
6912     free(h_y_ref);
6913     cudaFree(d_x);
6914     cudaFree(d_y);
6915 }
6916
6917

```

```

6918 static void test_mat_transpose_padded(int row, int col) {
6919
6920     size_t size_in = (size_t)row * col * sizeof(float);
6921     size_t size_out = (size_t)col * row * sizeof(float);
6922
6923     float *h_x = (float *)malloc(size_in);
6924     float *h_y_ref = (float *)malloc(size_out);
6925
6926     srand(42);
6927     for (int i = 0; i < row * col; i++)
6928         h_x[i] = ((float)rand() / RAND_MAX) * 2.0f - 1.0f;
6929
6930     // CPU reference: y[j][i] = x[i][j] (row-major)
6931     for (int i = 0; i < row; i++)
6932         for (int j = 0; j < col; j++)
6933             h_y_ref[j * row + i] = h_x[i * col + j];
6934
6935     float *d_x, *d_y;
6936     check(cudaMalloc(&d_x, size_in), "mattrans_padded alloc X");
6937     check(cudaMalloc(&d_y, size_out), "mattrans_padded alloc Y");
6938     check(cudaMemcpy(d_x, h_x, size_in, cudaMemcpyHostToDevice), "mattrans_padded H2D");
6939
6940     dim3 block(16, 16);
6941     dim3 grid((col + 15) / 16, (row + 63) / 64);
6942     mat_transpose_padded<<<grid, block>>>(d_x, d_y, row, col);
6943     check(cudaGetLastError(), "mattrans_padded launch");
6944     check(cudaDeviceSynchronize(), "mattrans_padded sync");
6945
6946     float *h_y = (float *)malloc(size_out);
6947     check(cudaMemcpy(h_y, d_y, size_out, cudaMemcpyDeviceToHost), "mattrans_padded D2H");
6948
6949     float max_err = 0.0f;
6950     for (int i = 0; i < col * row; i++) {
6951         float err = fabsf(h_y[i] - h_y_ref[i]);
6952         if (err > max_err) max_err = err;
6953     }
6954     printf("| %-41s | %.6e | %-4s | %-19s |\n", "MatTransposePadded", max_err, max_err < 1e-6f ? "PASS" : "
        FAIL", "None");
6955
6956     free(h_x);
6957     free(h_y);
6958     free(h_y_ref);
6959     cudaFree(d_x);
6960     cudaFree(d_y);
6961 }
6962
6963
6964 static void test_sgemv(int M, int K) {
6965
6966     srand(42);
6967     float *h_a = (float *)malloc((size_t)M * K * sizeof(float));
6968     float *h_x = (float *)malloc((size_t)K * sizeof(float));
6969     float *h_y_ref = (float *)malloc((size_t)M * sizeof(float));
6970     for (int i = 0; i < M * K; i++) h_a[i] = ((float)rand() / RAND_MAX) * 2.0f - 1.0f;
6971     for (int i = 0; i < K; i++) h_x[i] = ((float)rand() / RAND_MAX) * 2.0f - 1.0f;
6972
6973     // CPU reference: y[m] = sum_k A[m,k] * x[k]
6974     for (int m = 0; m < M; m++) {
6975         double sum = 0.0;
6976         for (int k = 0; k < K; k++) sum += (double)h_a[m * K + k] * (double)h_x[k];
6977         h_y_ref[m] = (float)sum;
6978     }
6979

```

```

6980 float *d_a, *d_x, *d_y;
6981 check(cudaMalloc(&d_a, (size_t)M * K * sizeof(float)), "sgemv alloc A");
6982 check(cudaMalloc(&d_x, (size_t)K * sizeof(float)), "sgemv alloc X");
6983 check(cudaMalloc(&d_y, (size_t)M * sizeof(float)), "sgemv alloc Y");
6984
6985 check(cudaMemcpy(d_a, h_a, (size_t)M * K * sizeof(float), cudaMemcpyHostToDevice), "sgemv H2D A");
6986 check(cudaMemcpy(d_x, h_x, (size_t)K * sizeof(float), cudaMemcpyHostToDevice), "sgemv H2D X");
6987
6988 dim3 block(32, 4);
6989 dim3 grid((M + 3) / 4);
6990 sgemv_k128<<<grid, block>>>(d_a, d_x, d_y, M, K);
6991 check(cudaGetLastError(), "sgemv launch");
6992 check(cudaDeviceSynchronize(), "sgemv sync");
6993
6994 float *h_y = (float *)malloc((size_t)M * sizeof(float));
6995 check(cudaMemcpy(h_y, d_y, (size_t)M * sizeof(float), cudaMemcpyDeviceToHost), "sgemv D2H");
6996
6997 float max_err = 0.0f;
6998 for (int m = 0; m < M; m++) {
6999     float err = fabsf(h_y[m] - h_y_ref[m]);
7000     if (err > max_err) max_err = err;
7001 }
7002 printf("| %-41s | %.6e | %-4s | %-19s |\n", "SGEMV-K128", max_err, max_err < 1e-2f ? "PASS" : "FAIL", "
7003     None");
7004
7005 // ---- SGEMV K32 ----
7006 check(cudaMemset(d_y, 0, M * sizeof(float)), "sgemv_k32 zero Y");
7007 sgemv_k32<<<grid, block>>>(d_a, d_x, d_y, M, K);
7008 check(cudaGetLastError(), "sgemv_k32 launch");
7009 check(cudaDeviceSynchronize(), "sgemv_k32 sync");
7010
7011 check(cudaMemcpy(h_y, d_y, (size_t)M * sizeof(float), cudaMemcpyDeviceToHost), "sgemv_k32 D2H");
7012
7013 max_err = 0.0f;
7014 for (int m = 0; m < M; m++) {
7015     float err = fabsf(h_y[m] - h_y_ref[m]);
7016     if (err > max_err) max_err = err;
7017 }
7018 printf("| %-41s | %.6e | %-4s | %-19s |\n", "SGEMV-K32", max_err, max_err < 1e-2f ? "PASS" : "FAIL", "None
7019     ");
7020
7021 // ---- SGEMV K16 ----
7022 free(h_a); free(h_x); free(h_y); free(h_y_ref);
7023 cudaFree(d_a); cudaFree(d_x); cudaFree(d_y);
7024
7025 int K16 = 16;
7026 h_a = (float *)malloc((size_t)M * K16 * sizeof(float));
7027 h_x = (float *)malloc((size_t)K16 * sizeof(float));
7028 h_y_ref = (float *)malloc((size_t)M * sizeof(float));
7029 for (int i = 0; i < M * K16; i++) h_a[i] = ((float)rand() / RAND_MAX) * 2.0f - 1.0f;
7030 for (int i = 0; i < K16; i++) h_x[i] = ((float)rand() / RAND_MAX) * 2.0f - 1.0f;
7031
7032 for (int m = 0; m < M; m++) {
7033     double sum = 0.0;
7034     for (int k = 0; k < K16; k++) sum += (double)h_a[m * K16 + k] * (double)h_x[k];
7035     h_y_ref[m] = (float)sum;
7036 }
7037
7038 check(cudaMalloc(&d_a, (size_t)M * K16 * sizeof(float)), "sgemv_k16 alloc A");
7039 check(cudaMalloc(&d_x, (size_t)K16 * sizeof(float)), "sgemv_k16 alloc X");
7040 check(cudaMalloc(&d_y, (size_t)M * sizeof(float)), "sgemv_k16 alloc Y");
7041
7042 check(cudaMemcpy(d_a, h_a, (size_t)M * K16 * sizeof(float), cudaMemcpyHostToDevice), "sgemv_k16 H2D A");

```

```

7041 check(cudaMemcpy(d_x, h_x, (size_t)K16 * sizeof(float), cudaMemcpyHostToDevice), "sgemv_k16 H2D X");
7042
7043 dim3 grid_k16((M + 7) / 8);
7044 sgemv_k16<<<grid_k16, block>>>(d_a, d_x, d_y, M, K16);
7045 check(cudaGetLastError(), "sgemv_k16 launch");
7046 check(cudaDeviceSynchronize(), "sgemv_k16 sync");
7047
7048 h_y = (float *)malloc((size_t)M * sizeof(float));
7049 check(cudaMemcpy(h_y, d_y, (size_t)M * sizeof(float), cudaMemcpyDeviceToHost), "sgemv_k16 D2H");
7050
7051 max_err = 0.0f;
7052 for (int m = 0; m < M; m++) {
7053     float err = fabsf(h_y[m] - h_y_ref[m]);
7054     if (err > max_err) max_err = err;
7055 }
7056 printf("| %-41s | %.6e | %-4s | %-19s |\n", "SGEMV-K16", max_err, max_err < 1e-2f ? "PASS" : "FAIL", "None");
7057
7058 free(h_a); free(h_x); free(h_y); free(h_y_ref);
7059 cudaFree(d_a); cudaFree(d_x); cudaFree(d_y);
7060 }
7061
7062
7063 static void test_sgemm(int M, int N, int K) {
7064
7065     size_t size_a = (size_t)M * K * sizeof(float);
7066     size_t size_b = (size_t)K * N * sizeof(float);
7067     size_t size_c = (size_t)M * N * sizeof(float);
7068
7069     float *h_a = (float *)malloc(size_a);
7070     float *h_b = (float *)malloc(size_b);
7071     float *h_c_ref = (float *)malloc(size_c);
7072
7073     srand(42);
7074     for (int i = 0; i < M * K; i++) h_a[i] = ((float)rand() / RAND_MAX) * 2.0f - 1.0f;
7075     for (int i = 0; i < K * N; i++) h_b[i] = ((float)rand() / RAND_MAX) * 2.0f - 1.0f;
7076
7077     float *d_a, *d_b, *d_c;
7078     check(cudaMalloc(&d_a, size_a), "sgemm alloc A");
7079     check(cudaMalloc(&d_b, size_b), "sgemm alloc B");
7080     check(cudaMalloc(&d_c, size_c), "sgemm alloc C");
7081
7082     check(cudaMemcpy(d_a, h_a, size_a, cudaMemcpyHostToDevice), "sgemm H2D A");
7083     check(cudaMemcpy(d_b, h_b, size_b, cudaMemcpyHostToDevice), "sgemm H2D B");
7084
7085     // cuBLAS reference (row-major idiom: swap M/N, swap A/B)
7086     cublasHandle_t handle;
7087     cublasCreate(&handle);
7088     float alpha = 1.0f, beta = 0.0f;
7089     cublasSgemm(handle, CUBLAS_OP_N, CUBLAS_OP_N, N, M, K,
7090                 &alpha, d_b, N, d_a, K, &beta, d_c, N);
7091     check(cudaMemcpy(h_c_ref, d_c, size_c, cudaMemcpyDeviceToHost), "sgemm D2H ref");
7092
7093     // Kernel
7094     cudaEvent_t start, stop;
7095     cudaEventCreate(&start);
7096     cudaEventCreate(&stop);
7097
7098     dim3 block(32, 32);
7099     dim3 grid((N + 31) / 32, (M + 31) / 32);
7100     sgemm<<<grid, block>>>(d_a, d_b, d_c, M, N, K);
7101     check(cudaGetLastError(), "sgemm launch");
7102     check(cudaDeviceSynchronize(), "sgemm sync");

```

```

7103 float *h_c = (float *)malloc(size_c);
7104 check(cudaMemcpy(h_c, d_c, size_c, cudaMemcpyDeviceToHost), "sgemm D2H");
7105
7106 // Verify
7107 float max_err = 0.0f;
7108 for (int i = 0; i < M * N; i++) {
7109     float err = fabsf(h_c[i] - h_c_ref[i]);
7110     if (err > max_err) max_err = err;
7111 }
7112 printf("| %-41s | %.6e | %-4s | %-19s |\n", "SGEMM", max_err, max_err < 1e-2f ? "PASS" : "FAIL", "None");
7113
7114 // ---- SGEMM Vec4 (128x128 tile, 4x4 thread tile) ----
7115
7116 dim3 grid_vec4((N + 127) / 128, (M + 127) / 128);
7117 check(cudaMemset(d_c, 0, size_c), "sgemm_vec4 zero C");
7118 sgemm_vec4<<<grid_vec4, block>>>(d_a, d_b, d_c, M, N, K);
7119 check(cudaGetLastError(), "sgemm_vec4 launch");
7120 check(cudaDeviceSynchronize(), "sgemm_vec4 sync");
7121
7122 check(cudaMemcpy(h_c, d_c, size_c, cudaMemcpyDeviceToHost), "sgemm_vec4 D2H");
7123
7124 max_err = 0.0f;
7125 for (int i = 0; i < M * N; i++) {
7126     float err = fabsf(h_c[i] - h_c_ref[i]);
7127     if (err > max_err) max_err = err;
7128 }
7129 printf("| %-41s | %.6e | %-4s | %-19s |\n", "SGEMM-Vec4", max_err, max_err < 1e-2f ? "PASS" : "FAIL", "None");
7130
7131 free(h_a); free(h_b); free(h_c); free(h_c_ref);
7132 cudaFree(d_a); cudaFree(d_b); cudaFree(d_c);
7133 cublasDestroy(handle);
7134 cudaEventDestroy(start);
7135 cudaEventDestroy(stop);
7136 }
7137
7138
7139 static void test_hgemm_mma(int M, int N, int K) {
7140
7141     size_t size_a = (size_t)M * K * sizeof(half);
7142     size_t size_b = (size_t)K * N * sizeof(half);
7143     size_t size_c = (size_t)M * N * sizeof(half);
7144
7145     half *h_a = (half *)malloc(size_a);
7146     half *h_b = (half *)malloc(size_b);
7147     half *h_c_ref = (half *)malloc(size_c);
7148
7149     srand(42);
7150     for (int i = 0; i < M * K; i++) h_a[i] = __float2half(((float)rand() / RAND_MAX) * 2.0f - 1.0f);
7151     for (int i = 0; i < K * N; i++) h_b[i] = __float2half(((float)rand() / RAND_MAX) * 2.0f - 1.0f);
7152
7153     // Kernel expects B^T [NxK] row-major layout (TN layout convention).
7154     // We store h_b as B [KxN] row-major for cuBLAS, then create B^T for kernel.
7155     size_t size_b_t = (size_t)N * K * sizeof(half);
7156     half *h_b_t = (half *)malloc(size_b_t);
7157     for (int n = 0; n < N; n++)
7158         for (int k = 0; k < K; k++)
7159             h_b_t[n * K + k] = h_b[k * N + n];
7160
7161     half *d_a, *d_b, *d_b_t, *d_c;
7162     check(cudaMalloc(&d_a, size_a), "hgemm alloc A");
7163     check(cudaMalloc(&d_b, size_b), "hgemm alloc B (cuBLAS)");

```

```

7165 check(cudaMalloc(&d_b_t, size_b_t), "hgemm alloc B_t (kernel)");
7166 check(cudaMalloc(&d_c, size_c), "hgemm alloc C");
7167
7168 check(cudaMemcpy(d_a, h_a, size_a, cudaMemcpyHostToDevice), "hgemm H2D A");
7169 check(cudaMemcpy(d_b, h_b, size_b, cudaMemcpyHostToDevice), "hgemm H2D B (cuBLAS)");
7170 check(cudaMemcpy(d_b_t, h_b_t, size_b_t, cudaMemcpyHostToDevice), "hgemm H2D B_t (kernel)");
7171
7172 // cuBLAS FP16 reference (row-major idiom: swap M/N, swap A/B)
7173 // Note: use CUBLAS_COMPUTE_16F to match the kernel's f16.f16.f16.f16 accumulation.
7174 cublasHandle_t handle;
7175 cublasCreate(&handle);
7176 half alpha_h = __float2half(1.0f), beta_h = __float2half(0.0f);
7177 cublasGemmEx(handle, CUBLAS_OP_N, CUBLAS_OP_N, N, M, K,
7178             &alpha_h, d_b, CUDA_R_16F, N, d_a, CUDA_R_16F, K,
7179             &beta_h, d_c, CUDA_R_16F, N,
7180             CUBLAS_COMPUTE_16F, CUBLAS_GEMM_DEFAULT);
7181 check(cudaMemcpy(h_c_ref, d_c, size_c, cudaMemcpyDeviceToHost), "hgemm D2H ref");
7182
7183 // MMA kernel (TN layout: A row-major, B^T row-major = B col-major)
7184 cudaEvent_t start, stop;
7185 cudaEventCreate(&start);
7186 cudaEventCreate(&stop);
7187
7188 constexpr int BM = 128, BN = 128, BK = 16, kStages = 3;
7189 size_t smem_bytes = kStages * (BM * BK + BN * BK) * sizeof(half); // 24576
7190 dim3 block(256);
7191 dim3 grid((N + BN - 1) / BN, (M + BM - 1) / BM);
7192 hgemm_mma_stages_tn<<<grid, block, smem_bytes>>>(d_a, d_b_t, d_c, M, N, K);
7193 check(cudaGetLastError(), "hgemm launch");
7194 check(cudaDeviceSynchronize(), "hgemm sync");
7195
7196 half *h_c = (half *)malloc(size_c);
7197 check(cudaMemcpy(h_c, d_c, size_c, cudaMemcpyDeviceToHost), "hgemm D2H");
7198
7199 // Verify
7200 float max_err = 0.0f;
7201 for (int i = 0; i < M * N; i++) {
7202     float err = fabsf(__half2float(h_c[i]) - __half2float(h_c_ref[i]));
7203     if (err > max_err) max_err = err;
7204 }
7205 printf("| %-41s | %.6e | %-4s | %-19s |\n", "HGEMM MMA", max_err, max_err < 1.0f ? "PASS" : "FAIL", "None"
7206 );
7207
7208 free(h_a); free(h_b); free(h_b_t); free(h_c); free(h_c_ref);
7209 cudaFree(d_a); cudaFree(d_b); cudaFree(d_b_t); cudaFree(d_c);
7210 cublasDestroy(handle);
7211 cudaEventDestroy(start);
7212 cudaEventDestroy(stop);
7213 }
7214
7215 static void test_hgemm_swizzle(int M, int N, int K) {
7216     // HGEMM MMA Swizzle — m16n8k16 + multistage pipeline + TN 布局 + XOR swizzle
7217     // + Register Double Buffering (kValTileK=4, BK=64)
7218     // TN layout: C[M×N] = A[M×K] × B^T[N×K]
7219     // Kernel: hgemm_mma_stages_tn_swizzle with default template params
7220     // (kValTileK=4, kStages=2, BK=64)
7221     // smem: kStages × (BM×BK + BN×BK) halves
7222
7223     size_t size_a = (size_t)M * K * sizeof(half);
7224     size_t size_b = (size_t)K * N * sizeof(half);
7225     size_t size_c = (size_t)M * N * sizeof(half);
7226

```

```

7227 half *h_a = (half *)malloc(size_a);
7228 half *h_b = (half *)malloc(size_b);
7229 half *h_c_ref = (half *)malloc(size_c);
7230
7231 srand(42);
7232 for (int i = 0; i < M * K; i++) h_a[i] = __float2half(((float)rand() / RAND_MAX) * 2.0f - 1.0f);
7233 for (int i = 0; i < K * N; i++) h_b[i] = __float2half(((float)rand() / RAND_MAX) * 2.0f - 1.0f);
7234
7235 // Kernel expects B^T [N×K] row-major layout (TN layout convention).
7236 size_t size_b_t = (size_t)N * K * sizeof(half);
7237 half *h_b_t = (half *)malloc(size_b_t);
7238 for (int n = 0; n < N; n++)
7239     for (int k = 0; k < K; k++)
7240         h_b_t[n * K + k] = h_b[k * N + n];
7241
7242 half *d_a, *d_b, *d_b_t, *d_c;
7243 check(cudaMalloc(&d_a, size_a), "hgemm_swizzle alloc A");
7244 check(cudaMalloc(&d_b, size_b), "hgemm_swizzle alloc B (cuBLAS)");
7245 check(cudaMalloc(&d_b_t, size_b_t), "hgemm_swizzle alloc B_t (kernel)");
7246 check(cudaMalloc(&d_c, size_c), "hgemm_swizzle alloc C");
7247
7248 check(cudaMemcpy(d_a, h_a, size_a, cudaMemcpyHostToDevice), "hgemm_swizzle H2D A");
7249 check(cudaMemcpy(d_b, h_b, size_b, cudaMemcpyHostToDevice), "hgemm_swizzle H2D B (cuBLAS)");
7250 check(cudaMemcpy(d_b_t, h_b_t, size_b_t, cudaMemcpyHostToDevice), "hgemm_swizzle H2D B_t (kernel)");
7251
7252 // cuBLAS FP16 reference
7253 cublasHandle_t handle;
7254 cublasCreate(&handle);
7255 half alpha_h = __float2half(1.0f), beta_h = __float2half(0.0f);
7256 cublasGemmEx(handle, CUBLAS_OP_N, CUBLAS_OP_N, N, M, K,
7257             &alpha_h, d_b, CUDA_R_16F, N, d_a, CUDA_R_16F, K,
7258             &beta_h, d_c, CUDA_R_16F, N,
7259             CUBLAS_COMPUTE_16F, CUBLAS_GEMM_DEFAULT);
7260 check(cudaMemcpy(h_c_ref, d_c, size_c, cudaMemcpyDeviceToHost), "hgemm_swizzle D2H ref");
7261
7262 // MMA swizzle kernel (default params: kStages=2, kValTileK=4, BK=64)
7263 constexpr int BM = 128, BN = 128, BK = 64, K_STAGE_S = 2;
7264 size_t smem_bytes = K_STAGE_S * (BM * BK + BN * BK) * sizeof(half);
7265 cudaFuncSetAttribute(
7266     (const void *)hgemm_mma_stages_tn_swizzle<16, 8, 16, 2, 4, 4, 4, 4, K_STAGE_S, 0>,
7267     cudaFuncAttributeMaxDynamicSharedMemorySize, smem_bytes);
7268 dim3 block(256);
7269 dim3 grid((N + BN - 1) / BN, (M + BM - 1) / BM);
7270 hgemm_mma_stages_tn_swizzle<<<grid, block, smem_bytes>>>(d_a, d_b_t, d_c, M, N, K);
7271 check(cudaGetLastError(), "hgemm_swizzle launch");
7272 check(cudaDeviceSynchronize(), "hgemm_swizzle sync");
7273
7274 half *h_c = (half *)malloc(size_c);
7275 check(cudaMemcpy(h_c, d_c, size_c, cudaMemcpyDeviceToHost), "hgemm_swizzle D2H");
7276
7277 // Verify
7278 float max_err = 0.0f;
7279 for (int i = 0; i < M * N; i++) {
7280     float err = fabsf(__half2float(h_c[i]) - __half2float(h_c_ref[i]));
7281     if (err > max_err) max_err = err;
7282 }
7283 printf("| %-41s | %.6e | %-4s | %-19s |\n", "HGEMM Swizzle + Reg2x", max_err, max_err < 1.0f ? "PASS" : "
7284 FAIL", "None");
7285
7286 free(h_a); free(h_b); free(h_b_t); free(h_c); free(h_c_ref);
7287 cudaFree(d_a); cudaFree(d_b); cudaFree(d_b_t); cudaFree(d_c);
7288 cublasDestroy(handle);
7289 }

```

```

7289
7290
7291 #if defined(NOTES_V2_ENABLE_CUTE)
7292 static void test_hgemm_cute(int M, int N, int K) {
7293     // HGEMM CuTe — SM80_16x8x16_F16F16F16F16_TN + Swizzle<3,3,3> + kStage=2
7294     // TN layout: C[M×N] = A[M×K] × BT[N×K]
7295     // Kernel: hgemm_mma_stages_tn_cute via launch_hgemm_mma_stages_tn_cute
7296     // Tile: BM=128, BN=256, BK=32, 128 threads/block
7297
7298     size_t size_a = (size_t)M * K * sizeof(half);
7299     size_t size_b = (size_t)K * N * sizeof(half);
7300     size_t size_c = (size_t)M * N * sizeof(half);
7301
7302     half *h_a = (half *)malloc(size_a);
7303     half *h_b = (half *)malloc(size_b);
7304     half *h_c_ref = (half *)malloc(size_c);
7305
7306     srand(42);
7307     for (int i = 0; i < M * K; i++)
7308         h_a[i] = __float2half(((float)rand() / RAND_MAX) * 2.0f - 1.0f);
7309     for (int i = 0; i < K * N; i++)
7310         h_b[i] = __float2half(((float)rand() / RAND_MAX) * 2.0f - 1.0f);
7311
7312     // CuTe kernel expects BT [N×K] row-major (TN layout).
7313     size_t size_b_t = (size_t)N * K * sizeof(half);
7314     half *h_b_t = (half *)malloc(size_b_t);
7315     for (int n = 0; n < N; n++)
7316         for (int k = 0; k < K; k++)
7317             h_b_t[n * K + k] = h_b[k * N + n];
7318
7319     half *d_a, *d_b, *d_b_t, *d_c;
7320     check(cudaMalloc(&d_a, size_a), "hgemm_cute alloc A");
7321     check(cudaMalloc(&d_b, size_b), "hgemm_cute alloc B (cuBLAS)");
7322     check(cudaMalloc(&d_b_t, size_b_t), "hgemm_cute alloc B_t (kernel)");
7323     check(cudaMalloc(&d_c, size_c), "hgemm_cute alloc C");
7324
7325     check(cudaMemcpy(d_a, h_a, size_a, cudaMemcpyHostToDevice), "hgemm_cute H2D A");
7326     check(cudaMemcpy(d_b, h_b, size_b, cudaMemcpyHostToDevice), "hgemm_cute H2D B (cuBLAS)");
7327     check(cudaMemcpy(d_b_t, h_b_t, size_b_t, cudaMemcpyHostToDevice), "hgemm_cute H2D B_t (kernel)");
7328
7329     // cuBLAS FP16 reference (row-major idiom: swap M/N, swap A/B)
7330     cublasHandle_t handle;
7331     cublasCreate(&handle);
7332     half alpha_h = __float2half(1.0f), beta_h = __float2half(0.0f);
7333     cublasGemmEx(handle, CUBLAS_OP_N, CUBLAS_OP_N, N, M, K, &alpha_h, d_b,
7334                  CUDA_R_16F, N, d_a, CUDA_R_16F, K, &beta_h, d_c, CUDA_R_16F, N,
7335                  CUBLAS_COMPUTE_16F, CUBLAS_GEMM_DEFAULT);
7336     check(cudaMemcpy(h_c_ref, d_c, size_c, cudaMemcpyDeviceToHost), "hgemm_cute D2H ref");
7337
7338     // CuTe kernel (Stages=2, TN layout)
7339     launch_hgemm_mma_stages_tn_cute<half, 2, 0>(d_a, d_b_t, d_c, M, N, K);
7340     check(cudaGetLastError(), "hgemm_cute launch");
7341     check(cudaDeviceSynchronize(), "hgemm_cute sync");
7342
7343     half *h_c = (half *)malloc(size_c);
7344     check(cudaMemcpy(h_c, d_c, size_c, cudaMemcpyDeviceToHost), "hgemm_cute D2H");
7345
7346     // Verify
7347     float max_err = 0.0f;
7348     for (int i = 0; i < M * N; i++) {
7349         float err = fabsf(__half2float(h_c[i]) - __half2float(h_c_ref[i]));
7350         if (err > max_err) max_err = err;
7351     }

```

```

7352 printf("| %-41s | %.6e | %-4s | %-19s |\n", "HGEMM CuTe Swizzle + Reg2x",
7353         max_err, max_err < 1.0f ? "PASS" : "FAIL", "None");
7354
7355 free(h_a); free(h_b); free(h_b_t); free(h_c); free(h_c_ref);
7356 cudaFree(d_a); cudaFree(d_b); cudaFree(d_b_t); cudaFree(d_c);
7357 cublasDestroy(handle);
7358 }
7359 #endif /* NOTES_V2_ENABLE_CUTE */
7360
7361
7362 #if defined(NOTES_V2_ENABLE_WGMMMA)
7363 static void test_hgemm_wgmma(int M, int N, int K) {
7364     // HGEMM WGMMMA — m64n128k16 + TMA + Warp Specialization (Hopper SM90+)
7365     // TN layout: C[M×N] = A[M×K] × BT[N×K]
7366     // Kernel: hgemm_wgmma_stages_tn with default template params
7367
7368     constexpr int BM = 128, BN = 128, BK = 64, kStages = 3, kNumThreads = 256;
7369
7370     // M, K must be divisible by tile dims
7371     if (M % BM != 0 || N % BN != 0 || K % BK != 0) {
7372         printf("| %-41s | %-12s | %-4s | %-19s |\n",
7373             "HGEMM WGMMMA", "SKIP", "SKIP", "None");
7374         return;
7375     }
7376
7377     size_t size_a = (size_t)M * K * sizeof(half);
7378     size_t size_b = (size_t)K * N * sizeof(half);
7379     size_t size_c = (size_t)M * N * sizeof(half);
7380
7381     half *h_a = (half *)malloc(size_a);
7382     half *h_b = (half *)malloc(size_b);
7383     half *h_c_ref = (half *)malloc(size_c);
7384
7385     srand(42);
7386     for (int i = 0; i < M * K; i++)
7387         h_a[i] = __float2half(((float)rand() / RAND_MAX) * 2.0f - 1.0f);
7388     for (int i = 0; i < K * N; i++)
7389         h_b[i] = __float2half(((float)rand() / RAND_MAX) * 2.0f - 1.0f);
7390
7391     // BT [N×K] row-major for TN layout (same as hgemm_mma kernel)
7392     size_t size_b_t = (size_t)N * K * sizeof(half);
7393     half *h_b_t = (half *)malloc(size_b_t);
7394     for (int n = 0; n < N; n++)
7395         for (int k = 0; k < K; k++)
7396             h_b_t[n * K + k] = h_b[k * N + n];
7397
7398     half *d_a, *d_b, *d_b_t, *d_c;
7399     check(cudaMalloc(&d_a, size_a), "wgmma alloc A");
7400     check(cudaMalloc(&d_b, size_b), "wgmma alloc B (cuBLAS)");
7401     check(cudaMalloc(&d_b_t, size_b_t), "wgmma alloc B_t (kernel)");
7402     check(cudaMalloc(&d_c, size_c), "wgmma alloc C");
7403
7404     check(cudaMemcpy(d_a, h_a, size_a, cudaMemcpyHostToDevice), "wgmma H2D A");
7405     check(cudaMemcpy(d_b, h_b, size_b, cudaMemcpyHostToDevice),
7406         "wgmma H2D B (cuBLAS)");
7407     check(cudaMemcpy(d_b_t, h_b_t, size_b_t, cudaMemcpyHostToDevice),
7408         "wgmma H2D B_t (kernel)");
7409
7410     // cuBLAS FP16 reference (row-major idiom, same as hgemm_mma)
7411     cublasHandle_t handle;
7412     cublasCreate(&handle);
7413     half alpha_h = __float2half(1.0f), beta_h = __float2half(0.0f);
7414     cublasGemmEx(handle, CUBLAS_OP_N, CUBLAS_OP_N, N, M, K, &alpha_h, d_b,

```

```

7415         CUDA_R_16F, N, d_a, CUDA_R_16F, K, &beta_h, d_c, CUDA_R_16F, N,
7416         CUBLAS_COMPUTE_16F, CUBLAS_GEMM_DEFAULT);
7417     check(cudaMemcpy(h_c_ref, d_c, size_c, cudaMemcpyDeviceToHost),
7418           "wgmma D2H ref");
7419
7420     // Create TMA tensor maps for A and B^T
7421     // A[MxK] row-major: TMA box=(BK=64, BM=128), global shape=(K, M)
7422     // B^T[NxK] row-major: TMA box=(BK=64, BN=128), global shape=(K, N)
7423     CUTensorMap *tma_a =
7424         allocate_and_create_tensor_map(d_a, M / BM, K / BK);
7425     CUTensorMap *tma_b =
7426         allocate_and_create_tensor_map(d_b_t, N / BN, K / BK);
7427
7428     // Launch WGMMMA kernel
7429     // kBlockSwizzle=false → 2D grid, no swizzle
7430     size_t smem_bytes =
7431         kStages * (BM * BK + BN * BK) * sizeof(half); // 3*16384*2 = 96KB
7432     cudaFuncSetAttribute(
7433         hgemm_wgmma_stages_tn<64, 128, 16, BM, BN, BK, kNumThreads, kStages,
7434         false>,
7435         cudaFuncAttributeMaxDynamicSharedMemorySize, smem_bytes);
7436
7437     dim3 block(kNumThreads);
7438     dim3 grid((N + BN - 1) / BN, (M + BM - 1) / BM);
7439     hgemm_wgmma_stages_tn<64, 128, 16, BM, BN, BK, kNumThreads, kStages, false>
7440         <<<grid, block, smem_bytes>>>(M, N, K, d_c, tma_a, tma_b);
7441     check(cudaGetLastError(), "wgmma launch");
7442     check(cudaDeviceSynchronize(), "wgmma sync");
7443
7444     half *h_c = (half *)malloc(size_c);
7445     check(cudaMemcpy(h_c, d_c, size_c, cudaMemcpyDeviceToHost), "wgmma D2H");
7446
7447     // Verify
7448     float max_err = 0.0f;
7449     for (int i = 0; i < M * N; i++) {
7450         float err = fabsf(__half2float(h_c[i]) - __half2float(h_c_ref[i]));
7451         if (err > max_err)
7452             max_err = err;
7453     }
7454     printf("| %-41s | %.6e | %-4s | %-19s |\n", "HGEMM TMA WGMMMA WS (3-stage)", max_err,
7455           max_err < 1.0f ? "PASS" : "FAIL", "None");
7456
7457     free(h_a);
7458     free(h_b);
7459     free(h_b_t);
7460     free(h_c);
7461     free(h_c_ref);
7462     cudaFree(d_a);
7463     cudaFree(d_b);
7464     cudaFree(d_b_t);
7465     cudaFree(d_c);
7466     cudaFree(tma_a);
7467     cudaFree(tma_b);
7468     cublasDestroy(handle);
7469 }
7470 #endif /* NOTES_V2_ENABLE_WGMMMA */
7471
7472 #if defined(NOTES_V2_ENABLE_TMA_MMA_WS)
7473 template <int kStages, int kBlockSwizzle = 0>
7474 static bool launch_hgemm_tma_mma_ws(int M, int N, int K, half *d_a,
7475                                     half *d_b_t, half *d_c,
7476                                     CUTensorMap *tma_a, CUTensorMap *tma_b) {
7477     constexpr int kMmaM = 16, kMmaN = 8, kMmaK = 16;

```

```

7478 constexpr int kMmaTileM = 2, kMmaTileN = 2;
7479 constexpr int kValTileM = 4, kValTileN = 8, kValTileK = 4;
7480 constexpr int BM = kMmaM * kMmaTileM * kValTileM;
7481 constexpr int BN = kMmaN * kMmaTileN * kValTileN;
7482 constexpr int BK = kMmaK * kValTileK;
7483 constexpr int kNumThreads = 256;
7484 constexpr size_t payload_bytes =
7485     kStages * (BM * BK + BN * BK) * sizeof(half);
7486 constexpr size_t smem_bytes = payload_bytes;
7487 using Kernel = void (*)(int, int, int, half *, const CUTensorMap *,
7488     const CUTensorMap *);
7489 Kernel kernel = hgemm_tma_mma_ws_tn<
7490     kMmaM, kMmaN, kMmaK, kMmaTileM, kMmaTileN, kValTileM, kValTileN,
7491     kValTileK, kStages, kNumThreads, kBlockSwizzle>;
7492
7493 int device = 0;
7494 int max_smem = 0;
7495 cudaFuncAttributes attributes{};
7496 check(cudaGetDevice(&device), "tma_mma_ws get device");
7497 check(cudaDeviceGetAttribute(&max_smem,
7498     cudaDevAttrMaxSharedMemoryPerBlockOptin,
7499     device),
7500     "tma_mma_ws max shared memory");
7501 check(cudaFuncGetAttributes(&attributes, kernel),
7502     "tma_mma_ws function attributes");
7503 if (smem_bytes + attributes.sharedSizeBytes > size_t(max_smem)) {
7504     if (g_verbose)
7505         printf("| %-41s | %-12s | %-4s | %-19s |\n",
7506             kStages == 2 ? "HGEMM TMA MMA WS (S=2, SW=0)"
7507             : "HGEMM TMA MMA WS (S=3, SW=0)",
7508             "SMEM SKIP", "SKIP", "None");
7509     return false;
7510 }
7511
7512 check(cudaFuncSetAttribute(kernel, cudaFuncAttributeMaxDynamicSharedMemorySize,
7513     smem_bytes),
7514     "tma_mma_ws set dynamic shared memory");
7515 dim3 block(kNumThreads);
7516 constexpr int kSwizzleN = 16;
7517 const int n_tiles = N / BN;
7518 const int grid_x = kBlockSwizzle ? div_ceil(n_tiles, kSwizzleN) : n_tiles;
7519 const int grid_z = kBlockSwizzle ? kSwizzleN : 1;
7520 dim3 grid(grid_x, M / BM, grid_z);
7521 kernel<<<grid, block, smem_bytes>>>(M, N, K, d_c, tma_a, tma_b);
7522 check(cudaGetLastError(), "tma_mma_ws launch");
7523 check(cudaDeviceSynchronize(), "tma_mma_ws sync");
7524 return true;
7525 }
7526
7527 static void test_hgemm_tma_mma_ws(int M, int N, int K) {
7528     constexpr int BM = 128, BN = 128, BK = 64;
7529     if (M % BM != 0 || N % BN != 0 || K % BK != 0) {
7530         printf("| %-41s | %-12s | %-4s | %-19s |\n", "HGEMM TMA MMA WS", "SKIP",
7531             "SKIP", "None");
7532         return;
7533     }
7534
7535     const size_t size_a = size_t(M) * K * sizeof(half);
7536     const size_t size_b = size_t(K) * N * sizeof(half);
7537     const size_t size_b_t = size_t(N) * K * sizeof(half);
7538     const size_t size_c = size_t(M) * N * sizeof(half);
7539     half *h_a = static_cast<half *>(malloc(size_a));
7540     half *h_b = static_cast<half *>(malloc(size_b));

```

```

7541 half *h_b_t = static_cast<half *>(malloc(size_b_t));
7542 half *h_c = static_cast<half *>(malloc(size_c));
7543 half *h_c_ref = static_cast<half *>(malloc(size_c));
7544 srand(42);
7545 for (int i = 0; i < M * K; ++i)
7546     h_a[i] = __float2half((float(rand()) / RAND_MAX) * 2.0f - 1.0f);
7547 for (int i = 0; i < K * N; ++i)
7548     h_b[i] = __float2half((float(rand()) / RAND_MAX) * 2.0f - 1.0f);
7549 for (int n = 0; n < N; ++n)
7550     for (int k = 0; k < K; ++k)
7551         h_b_t[n * K + k] = h_b[k * N + n];
7552
7553 half *d_a, *d_b, *d_b_t, *d_c;
7554 check(cudaMalloc(&d_a, size_a), "tma_mma_ws alloc A");
7555 check(cudaMalloc(&d_b, size_b), "tma_mma_ws alloc B");
7556 check(cudaMalloc(&d_b_t, size_b_t), "tma_mma_ws alloc B_t");
7557 check(cudaMalloc(&d_c, size_c), "tma_mma_ws alloc C");
7558 check(cudaMemcpy(d_a, h_a, size_a, cudaMemcpyHostToDevice), "tma_mma_ws H2D A");
7559 check(cudaMemcpy(d_b, h_b, size_b, cudaMemcpyHostToDevice), "tma_mma_ws H2D B");
7560 check(cudaMemcpy(d_b_t, h_b_t, size_b_t, cudaMemcpyHostToDevice),
7561         "tma_mma_ws H2D B_t");
7562
7563 cublasHandle_t handle;
7564 cublasCreate(&handle);
7565 half alpha = __float2half(1.0f), beta = __float2half(0.0f);
7566 cublasGemmEx(handle, CUBLAS_OP_N, CUBLAS_OP_N, N, M, K, &alpha, d_b,
7567              CUDA_R_16F, N, d_a, CUDA_R_16F, K, &beta, d_c, CUDA_R_16F, N,
7568              CUBLAS_COMPUTE_16F, CUBLAS_GEMM_DEFAULT);
7569 check(cudaMemcpy(h_c_ref, d_c, size_c, cudaMemcpyDeviceToHost),
7570         "tma_mma_ws D2H reference");
7571
7572 CUTensorMap *tma_a = allocate_and_create_tensor_map(d_a, M / BM, K / BK);
7573 CUTensorMap *tma_b = allocate_and_create_tensor_map(d_b_t, N / BN, K / BK);
7574 for (int block_swizzle : {0, 1}) {
7575     for (int stages : {2, 3}) {
7576         const bool launched = stages == 2
7577             ? (block_swizzle
7578                 ? launch_hgemm_tma_mma_ws<2, 1>(M, N, K, d_a, d_b_t, d_c,
7579                                                  tma_a, tma_b)
7580                 : launch_hgemm_tma_mma_ws<2>(M, N, K, d_a, d_b_t, d_c,
7581                                              tma_a, tma_b))
7582             : (block_swizzle
7583                 ? launch_hgemm_tma_mma_ws<3, 1>(M, N, K, d_a, d_b_t, d_c,
7584                                                  tma_a, tma_b)
7585                 : launch_hgemm_tma_mma_ws<3>(M, N, K, d_a, d_b_t, d_c,
7586                                              tma_a, tma_b));
7587         if (!launched)
7588             continue;
7589         check(cudaMemcpy(h_c, d_c, size_c, cudaMemcpyDeviceToHost),
7590                "tma_mma_ws D2H");
7591         float max_err = 0.0f;
7592         for (int i = 0; i < M * N; ++i)
7593             max_err = fmaxf(max_err, fabsf(__half2float(h_c[i]) -
7594                                                  __half2float(h_c_ref[i])));
7595         printf("| %-41s | %.6e | %-4s | %-19s |\n",
7596                block_swizzle
7597                    ? (stages == 2 ? "HGEMM TMA MMA WS (S=2, SW=1)"
7598                                   : "HGEMM TMA MMA WS (S=3, SW=1)")
7599                    : (stages == 2 ? "HGEMM TMA MMA WS (S=2, SW=0)"
7600                                   : "HGEMM TMA MMA WS (S=3, SW=0)"),
7601                max_err, max_err < 1.0f ? "PASS" : "FAIL", "None");
7602     }
7603 }

```

```

7604     free(h_a); free(h_b); free(h_b_t); free(h_c); free(h_c_ref);
7605     cudaFree(d_a); cudaFree(d_b); cudaFree(d_b_t); cudaFree(d_c);
7606     cudaFree(tma_a); cudaFree(tma_b);
7607     cublasDestroy(handle);
7608 }
7609 #endif /* NOTES_V2_ENABLE_TMA_MMA_WS */
7610
7611
7612
7613 static void test_flash_attn(int seqlen, int head_dim) {
7614     // FlashAttention-2 with split-Q, MMA m16n8k16
7615     int B = 1, H = 8;
7616
7617     size_t sz = (size_t)B * H * seqlen * head_dim * sizeof(half);
7618
7619     srand(42);
7620     half *h_q = (half *)malloc(sz);
7621     half *h_k = (half *)malloc(sz);
7622     half *h_v = (half *)malloc(sz);
7623     for (int i = 0; i < B * H * seqlen * head_dim; i++) {
7624         h_q[i] = __float2half(((float)rand() / RAND_MAX) * 2.0f - 1.0f);
7625         h_k[i] = __float2half(((float)rand() / RAND_MAX) * 2.0f - 1.0f);
7626         h_v[i] = __float2half(((float)rand() / RAND_MAX) * 2.0f - 1.0f);
7627     }
7628
7629     // CPU reference in FP32: 0 = softmax(Q @ K^T / sqrt(d)) @ V
7630     float *ref_q = (float *)malloc(sz * 4 / sizeof(half)); // 4x for float
7631     float *ref_k = (float *)malloc(sz * 4 / sizeof(half));
7632     float *ref_v = (float *)malloc(sz * 4 / sizeof(half));
7633     float *ref_o = (float *)malloc(sz * 4 / sizeof(half));
7634     int count = B * H * seqlen * head_dim;
7635     for (int i = 0; i < count; i++) {
7636         ref_q[i] = __half2float(h_q[i]);
7637         ref_k[i] = __half2float(h_k[i]);
7638         ref_v[i] = __half2float(h_v[i]);
7639     }
7640
7641     float scale = 1.0f / sqrtf((float)head_dim);
7642     for (int bi = 0; bi < B * H; bi++) {
7643         for (int qi = 0; qi < seqlen; qi++) {
7644             // S[qi, kj] = Q[qi, :] @ K[kj, :]^T * scale
7645             float smax = -INFINITY;
7646             float *S = (float *)malloc((size_t)seqlen * sizeof(float));
7647             for (int kj = 0; kj < seqlen; kj++) {
7648                 float s = 0.0f;
7649                 for (int d = 0; d < head_dim; d++)
7650                     s += ref_q[bi * seqlen * head_dim + qi * head_dim + d] *
7651                         ref_k[bi * seqlen * head_dim + kj * head_dim + d];
7652                 S[kj] = s * scale;
7653                 if (S[kj] > smax) smax = S[kj];
7654             }
7655             // softmax
7656             double sum_exp = 0.0;
7657             for (int kj = 0; kj < seqlen; kj++) sum_exp += (double)expf(S[kj] - smax);
7658             float inv_sum = 1.0f / (float)sum_exp;
7659             // O[qi, :] = sum_kj P[qi, kj] * V[kj, :]
7660             for (int d = 0; d < head_dim; d++) {
7661                 double o_acc = 0.0;
7662                 for (int kj = 0; kj < seqlen; kj++)
7663                     o_acc += (double)(expf(S[kj] - smax) * inv_sum) *
7664                         ref_v[bi * seqlen * head_dim + kj * head_dim + d];
7665                 ref_o[bi * seqlen * head_dim + qi * head_dim + d] = (float)o_acc;
7666             }

```

```

7667     free(S);
7668 }
7669 }
7670
7671 half *d_q, *d_k, *d_v, *d_o;
7672 check(cudaMalloc(&d_q, sz), "fa alloc Q");
7673 check(cudaMalloc(&d_k, sz), "fa alloc K");
7674 check(cudaMalloc(&d_v, sz), "fa alloc V");
7675 check(cudaMalloc(&d_o, sz), "fa alloc O");
7676 check(cudaMemcpy(d_q, h_q, sz, cudaMemcpyHostToDevice), "fa H2D Q");
7677 check(cudaMemcpy(d_k, h_k, sz, cudaMemcpyHostToDevice), "fa H2D K");
7678 check(cudaMemcpy(d_v, h_v, sz, cudaMemcpyHostToDevice), "fa H2D V");
7679
7680 // Template params for kHeadDim=64, kStagesK=2
7681 constexpr int kHeadDim = 64;
7682 constexpr int kStagesK = 2;
7683 constexpr int kPadQ = 8;
7684 constexpr int kPadK = 8;
7685 constexpr int kPadV = 8;
7686 constexpr int kMmaAtomM = 16;
7687 constexpr int kMmaAtomN = 8;
7688 constexpr int kMmaAtomK = 16;
7689 constexpr int kMmaTileSeqLenQ = 8;
7690 constexpr int kMmaTileSeqLenK = 1;
7691 constexpr int kMmaTileSeqLenP = 8;
7692 constexpr int kMmaTileHeadDimV = 1;
7693 constexpr int kValTileSeqLenQ = 1;
7694 constexpr int kValTileSeqLenK = 8;
7695 constexpr int kValTileSeqLenP = 1;
7696 constexpr int kValTileHeadDimV = kHeadDim / (8 * kMmaTileHeadDimV);
7697
7698 constexpr int Br = kMmaAtomM * kMmaTileSeqLenQ * kValTileSeqLenQ;
7699 constexpr int Bc = kMmaAtomN * kMmaTileSeqLenK * kValTileSeqLenK;
7700 if (seqlen < Br) return; // kernel requires seqlen >= tile size
7701 size_t smem_bytes =
7702     (Br * (kHeadDim + kPadQ) +
7703      kStagesK * Bc * (kHeadDim + kPadK) +
7704      Bc * (kHeadDim + kPadV)) * sizeof(half);
7705
7706 dim3 block(256);
7707 dim3 grid((seqlen + Br - 1) / Br, B * H);
7708
7709 // Test both accumulator variants: f16 (kMmaAccF32=0) and f32 (kMmaAccF32=1)
7710 for (int acc = 0; acc <= 1; ++acc) {
7711     const int kMmaAcc = acc;
7712     half *h_o = (half *)malloc(sz);
7713
7714     if (kMmaAcc == 0) {
7715         using FAKernel = void (*)(half *, half *, half *, half *, int, int);
7716         FAKernel fa_k = flash_attn_mma_stages_split_q<kHeadDim, kMmaAtomM, kMmaAtomN,
7717             kMmaAtomK, 0, kMmaTileSeqLenQ, kMmaTileSeqLenK, kMmaTileSeqLenP,
7718             kMmaTileHeadDimV, kValTileSeqLenQ, kValTileSeqLenK, kValTileSeqLenP,
7719             kValTileHeadDimV, kStagesK, kPadQ, kPadK, kPadV>;
7720         cudaFuncSetAttribute(fa_k, cudaFuncAttributeMaxDynamicSharedMemorySize, smem_bytes);
7721         fa_k<<<grid, block, smem_bytes>>>(d_q, d_k, d_v, d_o, seqlen, H);
7722     } else {
7723         using FAKernel = void (*)(half *, half *, half *, half *, int, int);
7724         FAKernel fa_k = flash_attn_mma_stages_split_q<kHeadDim, kMmaAtomM, kMmaAtomN,
7725             kMmaAtomK, 1, kMmaTileSeqLenQ, kMmaTileSeqLenK, kMmaTileSeqLenP,
7726             kMmaTileHeadDimV, kValTileSeqLenQ, kValTileSeqLenK, kValTileSeqLenP,
7727             kValTileHeadDimV, kStagesK, kPadQ, kPadK, kPadV>;
7728         cudaFuncSetAttribute(fa_k, cudaFuncAttributeMaxDynamicSharedMemorySize, smem_bytes);
7729         fa_k<<<grid, block, smem_bytes>>>(d_q, d_k, d_v, d_o, seqlen, H);

```

```

7730 }
7731 check(cudaGetLastError(), "fa launch");
7732 check(cudaDeviceSynchronize(), "fa sync");
7733
7734 check(cudaMemcpy(h_o, d_o, sz, cudaMemcpyDeviceToHost), "fa D2H");
7735
7736 float max_err = 0.0f;
7737 for (int i = 0; i < count; i++) {
7738     float err = fabsf(__half2float(h_o[i]) - ref_o[i]);
7739     if (err > max_err) max_err = err;
7740 }
7741 const char *acc_label = kMmaAcc ? "F32Acc" : "F16Acc";
7742 char label[64];
7743 snprintf(label, sizeof(label), "FA2 (kStagesK=2, Pad, %s)", acc_label);
7744 printf("| %-41s | %.6e | %-4s | %-19s |\n", label,
7745     max_err, max_err < 5e-1f ? "PASS" : "FAIL", "None");
7746 free(h_o);
7747 }
7748
7749 free(h_q); free(h_k); free(h_v);
7750 free(ref_q); free(ref_k); free(ref_v); free(ref_o);
7751 cudaFree(d_q); cudaFree(d_k); cudaFree(d_v); cudaFree(d_o);
7752 }
7753
7754
7755 #if defined(NOTES_V2_ENABLE_TMA_MMA_WS)
7756 // Test for flash_attn_tma_mma_ws_stages_split_q (SM120, D=64/128)
7757 // Reference: cuDNN SDPA (half) if available, else CPU FP32 (float) fallback.
7758 template <int kHeadDim>
7759 static void test_flash_attn_tma_mma_ws_impl(int seqlen, int head_dim) {
7760     int B = 1, H = 8;
7761     constexpr int kStagesK = 2;
7762     constexpr int kStagesV = 1;
7763     constexpr int kMmaAtomM = 16, kMmaAtomN = 8, kMmaAtomK = 16;
7764     constexpr int kMmaTileSeqLenQ = 8, kMmaTileSeqLenK = 1;
7765     constexpr int kMmaTileSeqLenP = 8, kMmaTileHeadDimV = 1;
7766     constexpr int kValTileSeqLenQ = 1, kValTileSeqLenK = 8;
7767     constexpr int kValTileSeqLenP = 1;
7768     constexpr int kValTileHeadDimV = kHeadDim / (8 * kMmaTileHeadDimV);
7769     constexpr int Br = kMmaAtomM * kMmaTileSeqLenQ * kValTileSeqLenQ; // 128
7770     constexpr int Bc = kMmaAtomN * kMmaTileSeqLenK * kValTileSeqLenK; // 64
7771     constexpr int kNumThreads = 384;
7772
7773     if (seqlen % Br != 0 || seqlen % Bc != 0 || seqlen < Br) {
7774         printf("| %-41s | %-12s | %-4s | %-19s |\n",
7775             "FA2 TMA MMA WS (unaligned)", "SKIP", "SKIP", "None");
7776         return;
7777     }
7778
7779     size_t sz = (size_t)B * H * seqlen * head_dim * sizeof(half);
7780     srand(42);
7781     half *h_q = (half *)malloc(sz);
7782     half *h_k = (half *)malloc(sz);
7783     half *h_v = (half *)malloc(sz);
7784     for (int i = 0; i < B * H * seqlen * head_dim; ++i) {
7785         h_q[i] = __float2half(((float)rand() / RAND_MAX) * 2.0f - 1.0f);
7786         h_k[i] = __float2half(((float)rand() / RAND_MAX) * 2.0f - 1.0f);
7787         h_v[i] = __float2half(((float)rand() / RAND_MAX) * 2.0f - 1.0f);
7788     }
7789
7790     half *d_q, *d_k, *d_v, *d_o;
7791     check(cudaMalloc(&d_q, sz), "fa_tma_ws alloc Q");
7792     check(cudaMalloc(&d_k, sz), "fa_tma_ws alloc K");

```

```

7793 check(cudaMalloc(&d_v, sz), "fa_tma_ws alloc V");
7794 check(cudaMalloc(&d_o, sz), "fa_tma_ws alloc O");
7795 check(cudaMemcpy(d_q, h_q, sz, cudaMemcpyHostToDevice), "fa_tma_ws H2D Q");
7796 check(cudaMemcpy(d_k, h_k, sz, cudaMemcpyHostToDevice), "fa_tma_ws H2D K");
7797 check(cudaMemcpy(d_v, h_v, sz, cudaMemcpyHostToDevice), "fa_tma_ws H2D V");
7798
7799 // Reference output: either cuDNN (half) or CPU (float)
7800 half *h_o_ref = nullptr;
7801 float *ref_o = nullptr;
7802 int count = B * H * seqlen * head_dim;
7803
7804 #if defined(NOTES_V2_ENABLE_CUDNN)
7805 {
7806     // Try cuDNN SDPA first; fall back to CPU if unsupported on this SM
7807     bool cudnn_ok = false;
7808     half *d_o_ref;
7809     check(cudaMalloc(&d_o_ref, sz), "fa_tma_ws alloc O_ref");
7810     {
7811         cudnnHandle_t cudnn_handle;
7812         cudnnCreate(&cudnn_handle);
7813         auto graph = std::make_shared<fe::graph::Graph>();
7814         graph->set_io_data_type(fe::DataType_t::HALF)
7815             .set_intermediate_data_type(fe::DataType_t::FLOAT)
7816             .set_compute_data_type(fe::DataType_t::FLOAT);
7817
7818         auto Q = graph->tensor(fe::graph::Tensor_attributes()
7819             .set_uid(1).set_dim({B, H, seqlen, head_dim})
7820             .set_stride({H * seqlen * head_dim, seqlen * head_dim, head_dim, 1}));
7821         auto K = graph->tensor(fe::graph::Tensor_attributes()
7822             .set_uid(2).set_dim({B, H, seqlen, head_dim})
7823             .set_stride({H * seqlen * head_dim, seqlen * head_dim, head_dim, 1}));
7824         auto V = graph->tensor(fe::graph::Tensor_attributes()
7825             .set_uid(3).set_dim({B, H, seqlen, head_dim})
7826             .set_stride({H * seqlen * head_dim, seqlen * head_dim, head_dim, 1}));
7827
7828         auto [O_sdpa, Stats] = graph->sdpa(Q, K, V,
7829             fe::graph::SDPA_attributes()
7830                 .set_name("sdpa_ref")
7831                 .set_attn_scale(1.0f / sqrtf((float)head_dim)));
7832
7833         O_sdpa->set_output(true).set_uid(4)
7834             .set_dim({B, H, seqlen, head_dim})
7835             .set_stride({H * seqlen * head_dim, seqlen * head_dim, head_dim, 1});
7836
7837         auto build_status = graph->build(cudnn_handle, {fe::HeurMode_t::A, fe::HeurMode_t::FALLBACK});
7838         if (build_status.is_good()) {
7839             std::unordered_map<fe::graph::Tensor_attributes::uid_t, void*> vp = {
7840                 {1, d_q}, {2, d_k}, {3, d_v}, {4, d_o_ref}};
7841             int64_t ws_size = 0;
7842             if (graph->get_workspace_size(ws_size).is_good()) {
7843                 int8_t *d_ws = nullptr;
7844                 if (ws_size > 0) check(cudaMalloc(&d_ws, ws_size), "fa_tma_ws workspace");
7845                 if (graph->execute(cudnn_handle, vp, d_ws).is_good()) {
7846                     check(cudaDeviceSynchronize(), "fa_tma_ws cudnn sync");
7847                     cudnn_ok = true;
7848                 }
7849                 if (d_ws) cudaFree(d_ws);
7850             }
7851         }
7852         cudnnDestroy(cudnn_handle);
7853     }
7854
7855     if (cudnn_ok) {

```

```

7856     h_o_ref = (half *)malloc(sz);
7857     check(cudaMemcpy(h_o_ref, d_o_ref, sz, cudaMemcpyDeviceToHost), "fa_tma_ws D2H ref");
7858 } else {
7859     fprintf(stderr, "cudnn SDPA unsupported on this SM, using CPU ref\n");
7860 }
7861 cudaFree(d_o_ref);
7862 }
7863 #endif
7864
7865 // CPU reference fallback
7866 if (!h_o_ref) {
7867     float *ref_q = (float *)malloc(sz * 4 / sizeof(half));
7868     float *ref_k = (float *)malloc(sz * 4 / sizeof(half));
7869     float *ref_v = (float *)malloc(sz * 4 / sizeof(half));
7870     ref_o = (float *)malloc(sz * 4 / sizeof(half));
7871     for (int i = 0; i < count; ++i) {
7872         ref_q[i] = __half2float(h_q[i]);
7873         ref_k[i] = __half2float(h_k[i]);
7874         ref_v[i] = __half2float(h_v[i]);
7875     }
7876     float scale = 1.0f / sqrtf((float)head_dim);
7877     for (int bi = 0; bi < B * H; ++bi) {
7878         for (int qi = 0; qi < seqlen; ++qi) {
7879             float smax = -INFINITY;
7880             float *S = (float *)malloc((size_t)seqlen * sizeof(float));
7881             for (int kj = 0; kj < seqlen; ++kj) {
7882                 float s = 0.0f;
7883                 for (int d = 0; d < head_dim; ++d)
7884                     s += ref_q[bi * seqlen * head_dim + qi * head_dim + d] *
7885                         ref_k[bi * seqlen * head_dim + kj * head_dim + d];
7886                 S[kj] = s * scale;
7887                 if (S[kj] > smax) smax = S[kj];
7888             }
7889             double sum_exp = 0.0;
7890             for (int kj = 0; kj < seqlen; ++kj)
7891                 sum_exp += (double)expf(S[kj] - smax);
7892             float inv_sum = 1.0f / (float)sum_exp;
7893             for (int d = 0; d < head_dim; ++d) {
7894                 double o_acc = 0.0;
7895                 for (int kj = 0; kj < seqlen; ++kj)
7896                     o_acc += (double)(expf(S[kj] - smax) * inv_sum) *
7897                         ref_v[bi * seqlen * head_dim + kj * head_dim + d];
7898                 ref_o[bi * seqlen * head_dim + qi * head_dim + d] = (float)o_acc;
7899             }
7900             free(S);
7901         }
7902     }
7903     free(ref_q);
7904     free(ref_k);
7905     free(ref_v);
7906 }
7907
7908 // TMA descriptors: box innermost 固定 64 half (128B), 满足
7909 // CU_TENSOR_MAP_SWIZZLE_128B 对 box innermost ≤ 128B 的硬约束。
7910 // D=64: blocks_width=1, 单次 TMA 覆盖整行。
7911 // D=128: blocks_width=2, kernel producer 沿 head_dim 连续发 2 次 TMA,
7912 //         minor_coord = 0/64, 写入 chunk-major smem 布局 [2, Br, 64]。
7913 // Q/K/V gmem 是 [B, H, seqlen, head_dim] row-major, 作为 2D
7914 // [B*H*seqlen, head_dim] 矩阵描述。blocks_height = B*H*seqlen/tile_major。
7915 // kernel 用 (Nb_id*H+Nh_id)*N 偏移 major_coord。
7916 constexpr int kTmaBoxMinor = 64; // box innermost = 64 half = 128B
7917 constexpr int kTmaChunksQ = kHeadDim / kTmaBoxMinor;
7918 constexpr int kTmaChunksKV = kHeadDim / kTmaBoxMinor;

```

```

7919 CUTensorMap *tma_q = allocate_and_create_tensor_map<Br, kTmaBoxMinor>(
7920     d_q, B * H * seqlen / Br, kTmaChunksQ);
7921 CUTensorMap *tma_k = allocate_and_create_tensor_map<Bc, kTmaBoxMinor>(
7922     d_k, B * H * seqlen / Bc, kTmaChunksKV);
7923 CUTensorMap *tma_v = allocate_and_create_tensor_map<Bc, kTmaBoxMinor>(
7924     d_v, B * H * seqlen / Bc, kTmaChunksKV);
7925
7926 using FAKernel = void (*)(half *, half *, half *, half *, int, int,
7927     const CUTensorMap *, const CUTensorMap *,
7928     const CUTensorMap *);
7929 FAKernel fa_k = flash_attn_tma_mma_ws_stages_split_q<
7930     kHeadDim, kMmaAtomM, kMmaAtomN, kMmaAtomK, 0, kMmaTileSeqLenQ,
7931     kMmaTileSeqLenK, kMmaTileSeqLenP, kMmaTileHeadDimV, kValTileSeqLenQ,
7932     kValTileSeqLenK, kValTileSeqLenP, kValTileHeadDimV, kStagesK, kStagesV,
7933     kNumThreads>;
7934
7935 // smem = Q[Br*d] + K[kStagesK*Bc*d] + V[kStagesV*Bc*d]
7936 size_t smem_bytes = (Br * kHeadDim + kStagesK * Bc * kHeadDim +
7937     kStagesV * Bc * kHeadDim) * sizeof(half);
7938 int device = 0, max_smem = 0;
7939 cudaFuncAttributes attributes{};
7940 cudaGetDevice(&device);
7941 cudaDeviceGetAttribute(&max_smem, cudaDevAttrMaxSharedMemoryPerBlockOptin,
7942     device);
7943 cudaFuncGetAttributes(&attributes, fa_k);
7944 bool smem_ok =
7945     (smem_bytes + attributes.sharedSizeBytes <= (size_t)max_smem);
7946 if (!smem_ok) {
7947     if (g_verbose)
7948         printf("| %-41s | %-12s | %-4s | %-19s |\n",
7949             "FA2 TMA MMA WS (SMEM SKIP)", "SMEM SKIP", "SKIP", "None");
7950 } else {
7951     dim3 block(kNumThreads);
7952     dim3 grid((seqlen + Br - 1) / Br, B * H);
7953     // Test both accumulator variants
7954     for (int acc = 0; acc <= 1; ++acc) {
7955         const int kMmaAcc = acc;
7956         half *h_o = (half *)malloc(sz);
7957
7958         if (kMmaAcc == 0) {
7959             using FAK = void (*)(half *, half *, half *, half *, int, int,
7960                 const CUTensorMap *, const CUTensorMap *,
7961                 const CUTensorMap *);
7962             FAK fk = flash_attn_tma_mma_ws_stages_split_q<
7963                 kHeadDim, kMmaAtomM, kMmaAtomN, kMmaAtomK, 0, kMmaTileSeqLenQ,
7964                 kMmaTileSeqLenK, kMmaTileSeqLenP, kMmaTileHeadDimV, kValTileSeqLenQ,
7965                 kValTileSeqLenK, kValTileSeqLenP, kValTileHeadDimV, kStagesK,
7966                 kStagesV, kNumThreads>;
7967             cudaFuncSetAttribute(fk, cudaFuncAttributeMaxDynamicSharedMemorySize,
7968                 smem_bytes);
7969             fk<<<grid, block, smem_bytes>>>(d_q, d_k, d_v, d_o, seqlen, H, tma_q,
7970                 tma_k, tma_v);
7971         } else {
7972             using FAK = void (*)(half *, half *, half *, half *, int, int,
7973                 const CUTensorMap *, const CUTensorMap *,
7974                 const CUTensorMap *);
7975             FAK fk = flash_attn_tma_mma_ws_stages_split_q<
7976                 kHeadDim, kMmaAtomM, kMmaAtomN, kMmaAtomK, 1, kMmaTileSeqLenQ,
7977                 kMmaTileSeqLenK, kMmaTileSeqLenP, kMmaTileHeadDimV, kValTileSeqLenQ,
7978                 kValTileSeqLenK, kValTileSeqLenP, kValTileHeadDimV, kStagesK,
7979                 kStagesV, kNumThreads>;
7980             cudaFuncSetAttribute(fk, cudaFuncAttributeMaxDynamicSharedMemorySize,
7981                 smem_bytes);

```

```

7982     fk<<<grid, block, smem_bytes>>>(d_q, d_k, d_v, d_o, seqlen, H, tma_q,
7983                                     tma_k, tma_v);
7984 }
7985 check(cudaGetLastError(), "fa_tma_ws launch");
7986 check(cudaDeviceSynchronize(), "fa_tma_ws sync");
7987
7988 check(cudaMemcpy(h_o, d_o, sz, cudaMemcpyDeviceToHost), "fa_tma_ws D2H");
7989 float max_err = 0.0f;
7990 bool checked = h_o_ref || ref_o;
7991 if (checked) {
7992     for (int i = 0; i < count; ++i) {
7993         float ref_val = h_o_ref ? __half2float(h_o_ref[i]) : ref_o[i];
7994         float err = fabsf(__half2float(h_o[i]) - ref_val);
7995         if (err > max_err) max_err = err;
7996     }
7997 }
7998 const char *acc_label = kMmaAcc ? "F32Acc" : "F16Acc";
7999 char label[64];
8000 snprintf(label, sizeof(label),
8001          "FA2 TMA MMA WS (%s)",
8002          acc_label);
8003 printf("| %-41s | %.6e | %-4s | %-19s |\n",
8004        label, max_err,
8005        (checked && max_err < 5e-1f) ? "PASS" : (checked ? "FAIL" : "SKIP"),
8006        "None");
8007 free(h_o);
8008 }
8009 }
8010
8011 free(h_q); free(h_k); free(h_v);
8012 free(h_o_ref); free(ref_o);
8013 cudaFree(d_q); cudaFree(d_k); cudaFree(d_v); cudaFree(d_o);
8014 cudaFree(tma_q); cudaFree(tma_k); cudaFree(tma_v);
8015 }
8016
8017 // D=64/128 dispatch wrapper
8018 static void test_flash_attn_tma_mma_ws(int seqlen, int head_dim) {
8019     if (head_dim == 64) {
8020         test_flash_attn_tma_mma_ws_impl<64>(seqlen, head_dim);
8021     } else if (head_dim == 128) {
8022         test_flash_attn_tma_mma_ws_impl<128>(seqlen, head_dim);
8023     } else {
8024         printf("| %-41s | %-12s | %-4s | %-19s |\n",
8025              "FA2 TMA MMA WS (D!=64/128)", "SKIP", "SKIP", "None");
8026     }
8027 }
8028
8029 // FA3-style dual-consumer correctness test (Br=64, Sk=Sv=2)
8030 // Tests both F16Acc and F32Acc against cuDNN SDPA reference.
8031 #if defined(NOTES_V2_ENABLE_CUDNN)
8032 namespace fe = cudnn_frontend;
8033 static float bench_cudnn_sdpa_tflops(half *d_q, half *d_k, half *d_v,
8034                                     half *d_o_ref, int B, int H, int seqlen,
8035                                     int head_dim, fe::DataType_t compute_type);
8036 #endif
8037 template <int kHeadDim, int kStagesK>
8038 static void test_flash_attn_3_tma_ws_impl(int seqlen, int head_dim) {
8039     int B = 1, H = 8;
8040     constexpr int kMmaAtomM = 16, kMmaAtomN = 8, kMmaAtomK = 16;
8041     constexpr int kMmaTileSeqLenQ = 4, kMmaTileSeqLenK = 1;
8042     constexpr int kMmaTileSeqLenP = 4, kMmaTileHeadDimV = 1;
8043     constexpr int kValTileSeqLenQ = 1, kValTileSeqLenK = 8;
8044     constexpr int kValTileSeqLenP = 1;

```

```

8045 constexpr int kValTileHeadDimV = kHeadDim / (8 * kMmaTileHeadDimV);
8046 constexpr int kStagesV = 1; // per-WG V: always 1 buffer
8047 constexpr int kNumConsumerWGs = 2;
8048 constexpr int Br = kMmaAtomM * kMmaTileSeqLenQ * kValTileSeqLenQ; // 64
8049 constexpr int Bc = kMmaAtomN * kMmaTileSeqLenK * kValTileSeqLenK; // 64
8050 constexpr int kNumThreads = 384;
8051
8052 if (seqlen % Br != 0 || seqlen % Bc != 0 || seqlen < Br) {
8053     printf("| %-41s | %-12s | %-4s | %-19s |\n",
8054         "FA3-style TMA MMA WS (unaligned)", "SKIP", "SKIP", "None");
8055     return;
8056 }
8057
8058 size_t sz = (size_t)B * H * seqlen * head_dim * sizeof(half);
8059 srand(42);
8060 half *h_q = (half *)malloc(sz);
8061 half *h_k = (half *)malloc(sz);
8062 half *h_v = (half *)malloc(sz);
8063 for (int i = 0; i < B * H * seqlen * head_dim; ++i) {
8064     h_q[i] = __float2half(((float)rand() / RAND_MAX) * 2.0f - 1.0f);
8065     h_k[i] = __float2half(((float)rand() / RAND_MAX) * 2.0f - 1.0f);
8066     h_v[i] = __float2half(((float)rand() / RAND_MAX) * 2.0f - 1.0f);
8067 }
8068
8069 half *d_q, *d_k, *d_v, *d_o;
8070 check(cudaMalloc(&d_q, sz), "fa3 alloc Q");
8071 check(cudaMalloc(&d_k, sz), "fa3 alloc K");
8072 check(cudaMalloc(&d_v, sz), "fa3 alloc V");
8073 check(cudaMalloc(&d_o, sz), "fa3 alloc O");
8074 check(cudaMemcpy(d_q, h_q, sz, cudaMemcpyHostToDevice), "fa3 H2D Q");
8075 check(cudaMemcpy(d_k, h_k, sz, cudaMemcpyHostToDevice), "fa3 H2D K");
8076 check(cudaMemcpy(d_v, h_v, sz, cudaMemcpyHostToDevice), "fa3 H2D V");
8077
8078 // Reference: cuDNN SDPA (half output)
8079 half *h_o_ref = nullptr;
8080 int count = B * H * seqlen * head_dim;
8081 #if defined(NOTES_V2_ENABLE_CUDNN)
8082 {
8083     half *d_o_ref;
8084     check(cudaMalloc(&d_o_ref, sz), "fa3 alloc O_ref");
8085     bench_cudnn_sdpa_tflops(d_q, d_k, d_v, d_o_ref, B, H, seqlen, head_dim,
8086         fe::DataType_t::FLOAT);
8087     h_o_ref = (half *)malloc(sz);
8088     check(cudaMemcpy(h_o_ref, d_o_ref, sz, cudaMemcpyDeviceToHost), "fa3 ref D2H");
8089     cudaFree(d_o_ref);
8090 }
8091 #endif
8092
8093 // TMA descriptors (Br=64 for Q, Bc=64 for K/V)
8094 constexpr int kTmaBoxMinor = 64;
8095 constexpr int kTmaChunks = kHeadDim / kTmaBoxMinor;
8096 CUTensorMap *tma_q = allocate_and_create_tensor_map<Br, kTmaBoxMinor>(
8097     d_q, B * H * seqlen / Br, kTmaChunks);
8098 CUTensorMap *tma_k = allocate_and_create_tensor_map<Bc, kTmaBoxMinor>(
8099     d_k, B * H * seqlen / Bc, kTmaChunks);
8100 CUTensorMap *tma_v = allocate_and_create_tensor_map<Bc, kTmaBoxMinor>(
8101     d_v, B * H * seqlen / Bc, kTmaChunks);
8102
8103 using FAKernel = void (*)(half *, half *, half *, half *, int, int,
8104     const CUTensorMap *, const CUTensorMap *,
8105     const CUTensorMap *);
8106
8107 // smem = Q + K[2WGs * Sk * Bc*D] + V[2WGs * 1 * Bc*D]

```

```

8108 size_t smem_bytes = (Br * kHeadDim +
8109                     kNumConsumerWGs * kStagesK * Bc * kHeadDim +
8110                     kNumConsumerWGs * kStagesV * Bc * kHeadDim) * sizeof(half);
8111 dim3 block(kNumThreads);
8112 dim3 grid((seqlen + Br - 1) / Br, B * H);
8113
8114 for (int acc = 0; acc <= 1; ++acc) {
8115     half *h_o = (half *)malloc(sz);
8116     FAKernel fk;
8117     if (acc == 0) {
8118         fk = flash_attn_3_tma_ws_stages_split_q<
8119             kHeadDim, kMmaAtomM, kMmaAtomN, kMmaAtomK, 0, kMmaTileSeqLenQ,
8120             kMmaTileSeqLenK, kMmaTileSeqLenP, kMmaTileHeadDimV, kValTileSeqLenQ,
8121             kValTileSeqLenK, kValTileSeqLenP, kValTileHeadDimV, kStagesK, kStagesV,
8122             kNumThreads>;
8123     } else {
8124         fk = flash_attn_3_tma_ws_stages_split_q<
8125             kHeadDim, kMmaAtomM, kMmaAtomN, kMmaAtomK, 1, kMmaTileSeqLenQ,
8126             kMmaTileSeqLenK, kMmaTileSeqLenP, kMmaTileHeadDimV, kValTileSeqLenQ,
8127             kValTileSeqLenK, kValTileSeqLenP, kValTileHeadDimV, kStagesK, kStagesV,
8128             kNumThreads>;
8129     }
8130     bool smem_ok = check_smem_feasible((const void *)fk, smem_bytes);
8131     if (!smem_ok) {
8132         const char *acc_label = acc ? "F32Acc" : "F16Acc";
8133         char label[64];
8134         snprintf(label, sizeof(label), "FA3-style TMA MMA WS (%s)",
8135                 acc_label);
8136         if (g_verbose)
8137             printf("| %-41s | %-12s | %-4s | %-19s |\n", label, "SMEM too large", "SKIP",
8138                    "None");
8139         free(h_o);
8140         continue;
8141     }
8142     cudaFuncSetAttribute(fk, cudaFuncAttributeMaxDynamicSharedMemorySize,
8143                         smem_bytes);
8144     fk<<<grid, block, smem_bytes>>>(d_q, d_k, d_v, d_o, seqlen, H, tma_q,
8145                                     tma_k, tma_v);
8146     check(cudaGetLastError(), "fa3 launch");
8147     check(cudaDeviceSynchronize(), "fa3 sync");
8148
8149     check(cudaMemcpy(h_o, d_o, sz, cudaMemcpyDeviceToHost), "fa3 D2H");
8150     float max_err = 0.0f;
8151     bool checked = h_o_ref != nullptr;
8152     if (checked) {
8153         for (int i = 0; i < count; ++i) {
8154             float err = fabsf(__half2float(h_o[i]) - __half2float(h_o_ref[i]));
8155             if (err > max_err) max_err = err;
8156         }
8157     }
8158     const char *acc_label = acc ? "F32Acc" : "F16Acc";
8159     char label[64];
8160     snprintf(label, sizeof(label), "FA3-style TMA MMA WS (%s)",
8161             acc_label);
8162     printf("| %-41s | %.6e | %-4s | %-19s |\n", label, max_err,
8163           (checked && max_err < 5e-1f) ? "PASS" : (checked ? "FAIL" : "SKIP"),
8164           "None");
8165     free(h_o);
8166 }
8167
8168 free(h_q); free(h_k); free(h_v); free(h_o_ref);
8169 cudaFree(d_q); cudaFree(d_k); cudaFree(d_v); cudaFree(d_o);
8170 cudaFree(tma_q); cudaFree(tma_k); cudaFree(tma_v);

```

```

8171 }
8172
8173 static void test_flash_attn_3_tma_ws(int seqlen, int head_dim) {
8174     // Try stages up to 4; test_flash_attn_3_tma_ws_impl internally checks
8175     // cudaDevAttrMaxSharedMemoryPerBlockOptin and skips oversized configs.
8176     if (head_dim == 64) {
8177         test_flash_attn_3_tma_ws_impl<64, 1>(seqlen, head_dim);
8178         test_flash_attn_3_tma_ws_impl<64, 2>(seqlen, head_dim);
8179         test_flash_attn_3_tma_ws_impl<64, 3>(seqlen, head_dim);
8180         test_flash_attn_3_tma_ws_impl<64, 4>(seqlen, head_dim);
8181     } else if (head_dim == 128) {
8182         test_flash_attn_3_tma_ws_impl<128, 1>(seqlen, head_dim);
8183         test_flash_attn_3_tma_ws_impl<128, 2>(seqlen, head_dim);
8184         test_flash_attn_3_tma_ws_impl<128, 3>(seqlen, head_dim);
8185         test_flash_attn_3_tma_ws_impl<128, 4>(seqlen, head_dim);
8186     } else {
8187         printf("| %-41s | %-12s | %-4s | %-19s |\n",
8188             "FA3-style TMA MMA WS (D!=64/128)", "SKIP", "SKIP", "None");
8189     }
8190 }
8191 #endif /* NOTES_V2_ENABLE_TMA_MMA_WS */
8192
8193 static float bench_hgemm_tflops(int M, int N, int K, float time_ms) {
8194     double flops = 2.0 * M * N * K;
8195     return (float)(flops / (double)time_ms / 1e9);
8196 }
8197
8198 static float bench_fa_tflops(int B, int H, int N, int D, float time_ms) {
8199     double flops = 4.0 * B * H * N * N * D;
8200     return (float)(flops / (double)time_ms / 1e9);
8201 }
8202
8203 static float bench_cublas_hgemm_tflops(cublasHandle_t handle, int M, int N, int K,
8204     half *d_a, half *d_b, half *d_c) {
8205     half alpha = __float2half(1.0f), beta = __float2half(0.0f);
8206     cudaStream_t stream;
8207     cudaStreamCreate(&stream);
8208     cublasSetStream(handle, stream);
8209     // warmup
8210     for (int w = 0; w < g_warmup; ++w)
8211         cublasGemmEx(handle, CUBLAS_OP_N, CUBLAS_OP_N, N, M, K, &alpha,
8212             d_b, CUDA_R_16F, N, d_a, CUDA_R_16F, K, &beta,
8213             d_c, CUDA_R_16F, N, CUBLAS_COMPUTE_16F, CUBLAS_GEMM_DEFAULT);
8214     cudaStreamSynchronize(stream);
8215     // timed repeat
8216     cudaEvent_t start, stop;
8217     cudaEventCreate(&start);
8218     cudaEventCreate(&stop);
8219     cudaEventRecord(start, stream);
8220     for (int r = 0; r < g_repeat; ++r)
8221         cublasGemmEx(handle, CUBLAS_OP_N, CUBLAS_OP_N, N, M, K, &alpha,
8222             d_b, CUDA_R_16F, N, d_a, CUDA_R_16F, K, &beta,
8223             d_c, CUDA_R_16F, N, CUBLAS_COMPUTE_16F, CUBLAS_GEMM_DEFAULT);
8224     cudaEventRecord(stop, stream);
8225     cudaEventSynchronize(stop);
8226     float time_ms = 0;
8227     cudaEventElapsedTime(&time_ms, start, stop);
8228     time_ms /= g_repeat;
8229     cudaEventDestroy(start);
8230     cudaEventDestroy(stop);
8231     cublasSetStream(handle, nullptr);
8232     cudaStreamDestroy(stream);
8233     return bench_hgemm_tflops(M, N, K, time_ms);

```

```

8234 }
8235
8236 // =====
8237 // Bench: HGEMM MMA (basic m16n8k16 + multistage pipeline)
8238 // =====
8239 template <int kStages, int kBlockSwizzle>
8240 static bool launch_timed_hgemm_mma(
8241     half *d_a, half *d_b_t, half *d_c, half *h_c,
8242     half *h_c_ref, int M, int N, int K, size_t size_c,
8243     cudaEvent_t start, cudaEvent_t stop, float &max_err,
8244     float &time_ms
8245 ) {
8246     constexpr int BM = 128, BN = 128, BK = 16;
8247     using Kernel = void (*)(half *, half *, half *, int, int, int);
8248     Kernel k = hgemm_mma_stages_tn<16, 8, 16, 2, 4, 4, 4, kStages, kBlockSwizzle>;
8249     size_t smem = kStages * (BM * BK + BN * BK) * sizeof(half);
8250     if (!check_smem_feasible((const void *)k, smem)) return false;
8251     dim3 block(256);
8252     const int tiles_n = (N + BN - 1) / BN;
8253     constexpr int kSwizzleN = 16;
8254     int gx = kBlockSwizzle ? div_ceil(tiles_n, kSwizzleN) : tiles_n;
8255     int gz = kBlockSwizzle ? kSwizzleN : 1;
8256     dim3 grid(gx, (M + BM - 1) / BM, gz);
8257     for (int w = 0; w < g_warmup; w++)
8258         k<<<grid, block, smem>>>(d_a, d_b_t, d_c, M, N, K);
8259     cudaDeviceSynchronize();
8260     cudaEventRecord(start);
8261     for (int r = 0; r < g_repeat; r++)
8262         k<<<grid, block, smem>>>(d_a, d_b_t, d_c, M, N, K);
8263     cudaEventRecord(stop);
8264     cudaEventSynchronize(stop);
8265     cudaEventElapsedTime(&time_ms, start, stop);
8266     time_ms /= g_repeat;
8267     check(cudaMemcpy(h_c, d_c, size_c, cudaMemcpyDeviceToHost), "bench mma D2H");
8268     max_err = 0.0f;
8269     for (int i = 0; i < M * N; i++) {
8270         float err = fabsf(__half2float(h_c[i]) - __half2float(h_c_ref[i]));
8271         if (err > max_err) max_err = err;
8272     }
8273     return true;
8274 }
8275
8276 static void bench_hgemm_mma(int M, int N, int K) {
8277     size_t size_a = (size_t)M * K * sizeof(half);
8278     size_t size_b = (size_t)K * N * sizeof(half);
8279     size_t size_c = (size_t)M * N * sizeof(half);
8280     half *h_a = (half *)malloc(size_a);
8281     half *h_b = (half *)malloc(size_b);
8282     half *h_c_ref = (half *)malloc(size_c);
8283     srand(42);
8284     for (int i = 0; i < M * K; i++)
8285         h_a[i] = __float2half(((float)rand() / RAND_MAX) * 2.0f - 1.0f);
8286     for (int i = 0; i < K * N; i++)
8287         h_b[i] = __float2half(((float)rand() / RAND_MAX) * 2.0f - 1.0f);
8288     size_t size_b_t = (size_t)N * K * sizeof(half);
8289     half *h_b_t = (half *)malloc(size_b_t);
8290     for (int n = 0; n < N; n++)
8291         for (int k = 0; k < K; k++)
8292             h_b_t[n * K + k] = h_b[k * N + n];
8293     half *d_a, *d_b, *d_b_t, *d_c;
8294     check(cudaMalloc(&d_a, size_a), "bench mma alloc A");
8295     check(cudaMalloc(&d_b, size_b), "bench mma alloc B");
8296     check(cudaMalloc(&d_b_t, size_b_t), "bench mma alloc B_t");

```

```

8297 check(cudaMalloc(&d_c, size_c), "bench mma alloc C");
8298 check(cudaMemcpy(d_a, h_a, size_a, cudaMemcpyHostToDevice), "bench mma H2D A");
8299 check(cudaMemcpy(d_b, h_b, size_b, cudaMemcpyHostToDevice), "bench mma H2D B");
8300 check(cudaMemcpy(d_b_t, h_b_t, size_b_t, cudaMemcpyHostToDevice), "bench mma H2D B_t");
8301 cublasHandle_t handle;
8302 cublasCreate(&handle);
8303 half alpha_h = __float2half(1.0f), beta_h = __float2half(0.0f);
8304 cublasGemmEx(handle, CUBLAS_OP_N, CUBLAS_OP_N, N, M, K, &alpha_h, d_b, CUDA_R_16F, N,
8305             d_a, CUDA_R_16F, K, &beta_h, d_c, CUDA_R_16F, N, CUBLAS_COMPUTE_16F,
8306             CUBLAS_GEMM_DEFAULT);
8307 check(cudaMemcpy(h_c_ref, d_c, size_c, cudaMemcpyDeviceToHost), "bench mma D2H ref");
8308 half *h_c = (half *)malloc(size_c);
8309 float cublas_tflops = bench_cublas_hgemm_tflops(handle, M, N, K, d_a, d_b, d_c);
8310 cudaEvent_t start, stop;
8311 cudaEventCreate(&start);
8312 cudaEventCreate(&stop);
8313 for (int stages : {2, 3}) {
8314     for (int swizzle : {0, 1}) {
8315         float max_err = 0, time_ms = 0;
8316         bool ok = false;
8317         if (stages == 2)
8318             ok = swizzle ? launch_timed_hgemm_mma<2, 1>(d_a, d_b_t, d_c, h_c, h_c_ref, M, N,
8319                 K, size_c, start, stop, max_err, time_ms)
8320             : launch_timed_hgemm_mma<2, 0>(d_a, d_b_t, d_c, h_c, h_c_ref, M, N,
8321                 K, size_c, start, stop, max_err, time_ms);
8322         else
8323             ok = swizzle ? launch_timed_hgemm_mma<3, 1>(d_a, d_b_t, d_c, h_c, h_c_ref, M, N,
8324                 K, size_c, start, stop, max_err, time_ms)
8325             : launch_timed_hgemm_mma<3, 0>(d_a, d_b_t, d_c, h_c, h_c_ref, M, N,
8326                 K, size_c, start, stop, max_err, time_ms);
8327         char label[64];
8328         snprintf(label, sizeof(label), "HGEMM MMA (S=%d, SW=%d)", stages, swizzle);
8329         if (!ok) {
8330             if (g_verbose)
8331                 printf("| %-41s | %-12s | %-4s | %-19s |\n", label, "SMEM SKIP", "SKIP", "None");
8332         } else {
8333             float tflops = bench_hgemm_tflops(M, N, K, time_ms);
8334             char tflops_str[32];
8335             snprintf(tflops_str, sizeof(tflops_str), "%.1f/%.1f (%.2fx)", tflops, cublas_tflops, tflops /
8336                 cublas_tflops);
8337             printf("| %-41s | %.6e | %-4s | %-19s |\n", label, max_err,
8338                 max_err < 1.0f ? "PASS" : "FAIL", tflops_str);
8339         }
8340     }
8341     cudaEventDestroy(start);
8342     cudaEventDestroy(stop);
8343     free(h_a);
8344     free(h_b);
8345     free(h_b_t);
8346     free(h_c);
8347     free(h_c_ref);
8348     cudaFree(d_a);
8349     cudaFree(d_b);
8350     cudaFree(d_b_t);
8351     cudaFree(d_c);
8352     cublasDestroy(handle);
8353 }
8354
8355 // =====
8356 // Bench: HGEMM MMA Swizzle + Register Double Buffering (kValTileK=4, BK=64)
8357 // =====
8358 template <int kStages, int kBlockSwizzle>

```

```

8359 static bool launch_timed_hgemm_swizzle(
8360     half *d_a, half *d_b_t, half *d_c, half *h_c,
8361     half *h_c_ref, int M, int N, int K, size_t size_c,
8362     cudaEvent_t start, cudaEvent_t stop, float &max_err,
8363     float &time_ms
8364 ) {
8365     constexpr int BM = 128, BN = 128, BK = 64;
8366     using Kernel = void (*)(half *, half *, half *, int, int, int);
8367     Kernel k = hgemm_mma_stages_tn_swizzle<16, 8, 16, 2, 4, 4, 4, 4, kStages, kBlockSwizzle>;
8368     size_t smem = kStages * (BM * BK + BN * BK) * sizeof(half);
8369     if (!check_smem_feasible((const void *)k, smem)) return false;
8370     cudaFuncSetAttribute((const void *)k, cudaFuncAttributeMaxDynamicSharedMemorySize, smem);
8371     dim3 block(256);
8372     const int tiles_n = (N + BN - 1) / BN;
8373     constexpr int kSwizzleN = 16;
8374     int gx = kBlockSwizzle ? div_ceil(tiles_n, kSwizzleN) : tiles_n;
8375     int gz = kBlockSwizzle ? kSwizzleN : 1;
8376     dim3 grid(gx, (M + BM - 1) / BM, gz);
8377     for (int w = 0; w < g_warmup; w++)
8378         k<<<grid, block, smem>>>(d_a, d_b_t, d_c, M, N, K);
8379     cudaDeviceSynchronize();
8380     cudaEventRecord(start);
8381     for (int r = 0; r < g_repeat; r++)
8382         k<<<grid, block, smem>>>(d_a, d_b_t, d_c, M, N, K);
8383     cudaEventRecord(stop);
8384     cudaEventSynchronize(stop);
8385     cudaEventElapsedTime(&time_ms, start, stop);
8386     time_ms /= g_repeat;
8387     check(cudaMemcpy(h_c, d_c, size_c, cudaMemcpyDeviceToHost), "bench swizzle D2H");
8388     max_err = 0.0f;
8389     for (int i = 0; i < M * N; i++) {
8390         float err = fabsf(__half2float(h_c[i]) - __half2float(h_c_ref[i]));
8391         if (err > max_err) max_err = err;
8392     }
8393     return true;
8394 }
8395
8396 static void bench_hgemm_swizzle(int M, int N, int K) {
8397     size_t size_a = (size_t)M * K * sizeof(half);
8398     size_t size_b = (size_t)K * N * sizeof(half);
8399     size_t size_c = (size_t)M * N * sizeof(half);
8400     half *h_a = (half *)malloc(size_a);
8401     half *h_b = (half *)malloc(size_b);
8402     half *h_c_ref = (half *)malloc(size_c);
8403     srand(42);
8404     for (int i = 0; i < M * K; i++)
8405         h_a[i] = __float2half(((float)rand() / RAND_MAX) * 2.0f - 1.0f);
8406     for (int i = 0; i < K * N; i++)
8407         h_b[i] = __float2half(((float)rand() / RAND_MAX) * 2.0f - 1.0f);
8408     size_t size_b_t = (size_t)N * K * sizeof(half);
8409     half *h_b_t = (half *)malloc(size_b_t);
8410     for (int n = 0; n < N; n++)
8411         for (int k = 0; k < K; k++)
8412             h_b_t[n * K + k] = h_b[k * N + n];
8413     half *d_a, *d_b, *d_b_t, *d_c;
8414     check(cudaMalloc(&d_a, size_a), "bench swizzle alloc A");
8415     check(cudaMalloc(&d_b, size_b), "bench swizzle alloc B");
8416     check(cudaMalloc(&d_b_t, size_b_t), "bench swizzle alloc B_t");
8417     check(cudaMalloc(&d_c, size_c), "bench swizzle alloc C");
8418     check(cudaMemcpy(d_a, h_a, size_a, cudaMemcpyHostToDevice), "bench swizzle H2D A");
8419     check(cudaMemcpy(d_b, h_b, size_b, cudaMemcpyHostToDevice), "bench swizzle H2D B");
8420     check(cudaMemcpy(d_b_t, h_b_t, size_b_t, cudaMemcpyHostToDevice), "bench swizzle H2D B_t");
8421     cublasHandle_t handle;

```

```

8422 cublasCreate(&handle);
8423 half alpha_h = __float2half(1.0f), beta_h = __float2half(0.0f);
8424 cublasGemmEx(handle, CUBLAS_OP_N, CUBLAS_OP_N, N, M, K, &alpha_h, d_b, CUDA_R_16F, N,
8425             d_a, CUDA_R_16F, K, &beta_h, d_c, CUDA_R_16F, N, CUBLAS_COMPUTE_16F,
8426             CUBLAS_GEMM_DEFAULT);
8427 check(cudaMemcpy(h_c_ref, d_c, size_c, cudaMemcpyDeviceToHost), "bench swizzle D2H ref");
8428 half *h_c = (half *)malloc(size_c);
8429 float cublas_tflops = bench_cublas_hgemm_tflops(handle, M, N, K, d_a, d_b, d_c);
8430 cudaEvent_t start, stop;
8431 cudaEventCreate(&start);
8432 cudaEventCreate(&stop);
8433 for (int stages : {2, 3}) {
8434     for (int swizzle : {0, 1}) {
8435         float max_err = 0, time_ms = 0;
8436         bool ok = false;
8437         if (stages == 2)
8438             ok = swizzle ? launch_timed_hgemm_swizzle<2, 1>(d_a, d_b_t, d_c, h_c, h_c_ref, M,
8439                 N, K, size_c, start, stop, max_err, time_ms)
8440             : launch_timed_hgemm_swizzle<2, 0>(d_a, d_b_t, d_c, h_c, h_c_ref, M,
8441                 N, K, size_c, start, stop, max_err, time_ms);
8442         else
8443             ok = swizzle ? launch_timed_hgemm_swizzle<3, 1>(d_a, d_b_t, d_c, h_c, h_c_ref, M,
8444                 N, K, size_c, start, stop, max_err, time_ms)
8445             : launch_timed_hgemm_swizzle<3, 0>(d_a, d_b_t, d_c, h_c, h_c_ref, M,
8446                 N, K, size_c, start, stop, max_err, time_ms);
8447         char label[64];
8448         snprintf(label, sizeof(label), "HGEMM Swizzle+Reg2x (S=%d, SW=%d)", stages, swizzle);
8449         if (!ok) {
8450             if (g_verbose)
8451                 printf("| %-41s | %-12s | %-4s | %-19s |\n", label, "SMEM SKIP", "SKIP", "None");
8452         } else {
8453             float tflops = bench_hgemm_tflops(M, N, K, time_ms);
8454             char tflops_str[32];
8455             snprintf(tflops_str, sizeof(tflops_str), "%.1f/%.1f (%.2fx)", tflops, cublas_tflops, tflops /
8456                 cublas_tflops);
8457             printf("| %-41s | %.6e | %-4s | %-19s |\n", label, max_err,
8458                 max_err < 1.0f ? "PASS" : "FAIL", tflops_str);
8459         }
8460     }
8461 }
8462 cudaEventDestroy(start);
8463 cudaEventDestroy(stop);
8464 free(h_a);
8465 free(h_b);
8466 free(h_b_t);
8467 free(h_c);
8468 free(h_c_ref);
8469 cudaFree(d_a);
8470 cudaFree(d_b);
8471 cudaFree(d_b_t);
8472 cudaFree(d_c);
8473 cublasDestroy(handle);
8474 }
8475 #if defined(NOTES_V2_ENABLE_CUTE)
8476 // =====
8477 // Bench: HGEMM CuTe
8478 // =====
8479 static void bench_hgemm_cute(int M, int N, int K) {
8480     size_t size_a = (size_t)M * K * sizeof(half);
8481     size_t size_b = (size_t)K * N * sizeof(half);
8482     size_t size_c = (size_t)M * N * sizeof(half);
8483     half *h_a = (half *)malloc(size_a);

```

```

8484 half *h_b = (half *)malloc(size_b);
8485 half *h_c_ref = (half *)malloc(size_c);
8486 srand(42);
8487 for (int i = 0; i < M * K; i++)
8488     h_a[i] = __float2half(((float)rand() / RAND_MAX) * 2.0f - 1.0f);
8489 for (int i = 0; i < K * N; i++)
8490     h_b[i] = __float2half(((float)rand() / RAND_MAX) * 2.0f - 1.0f);
8491 size_t size_b_t = (size_t)N * K * sizeof(half);
8492 half *h_b_t = (half *)malloc(size_b_t);
8493 for (int n = 0; n < N; n++)
8494     for (int k = 0; k < K; k++)
8495         h_b_t[n * K + k] = h_b[k * N + n];
8496 half *d_a, *d_b, *d_b_t, *d_c;
8497 check(cudaMalloc(&d_a, size_a), "bench cute alloc A");
8498 check(cudaMalloc(&d_b, size_b), "bench cute alloc B");
8499 check(cudaMalloc(&d_b_t, size_b_t), "bench cute alloc B_t");
8500 check(cudaMalloc(&d_c, size_c), "bench cute alloc C");
8501 check(cudaMemcpy(d_a, h_a, size_a, cudaMemcpyHostToDevice), "bench cute H2D A");
8502 check(cudaMemcpy(d_b, h_b, size_b, cudaMemcpyHostToDevice), "bench cute H2D B");
8503 check(cudaMemcpy(d_b_t, h_b_t, size_b_t, cudaMemcpyHostToDevice), "bench cute H2D B_t");
8504 cublasHandle_t handle;
8505 cublasCreate(&handle);
8506 half alpha_h = __float2half(1.0f), beta_h = __float2half(0.0f);
8507 cublasGemmEx(handle, CUBLAS_OP_N, CUBLAS_OP_N, N, M, K, &alpha_h, d_b, CUDA_R_16F, N,
8508     d_a, CUDA_R_16F, K, &beta_h, d_c, CUDA_R_16F, N, CUBLAS_COMPUTE_16F,
8509     CUBLAS_GEMM_DEFAULT);
8510 check(cudaMemcpy(h_c_ref, d_c, size_c, cudaMemcpyDeviceToHost), "bench cute D2H ref");
8511 half *h_c = (half *)malloc(size_c);
8512 float cublas_tflops = bench_cublas_hgemm_tflops(handle, M, N, K, d_a, d_b, d_c);
8513 cudaEvent_t start, stop;
8514 cudaEventCreate(&start);
8515 cudaEventCreate(&stop);
8516 for (int stages : {2, 3}) {
8517     for (int swizzle : {0, 1}) {
8518         char label[64];
8519         snprintf(label, sizeof(label), "HGEMM CuTe Swizzle (S=%d, SW=%d)", stages, swizzle);
8520         bool ok = false;
8521         float time_ms = 0, max_err = 0;
8522         if (stages == 2) {
8523             auto k = swizzle ? launch_hgemm_mma_stages_tn_cute<half, 2, 1>
8524                 : launch_hgemm_mma_stages_tn_cute<half, 2, 0>;
8525             for (int w = 0; w < g_warmup; w++) k(d_a, d_b_t, d_c, M, N, K);
8526             cudaDeviceSynchronize();
8527             cudaEventRecord(start);
8528             for (int r = 0; r < g_repeat; r++) k(d_a, d_b_t, d_c, M, N, K);
8529             cudaEventRecord(stop);
8530             cudaEventSynchronize(stop);
8531             cudaEventElapsedTime(&time_ms, start, stop);
8532             time_ms /= g_repeat;
8533             ok = true;
8534         } else {
8535             auto k = swizzle ? launch_hgemm_mma_stages_tn_cute<half, 3, 1>
8536                 : launch_hgemm_mma_stages_tn_cute<half, 3, 0>;
8537             for (int w = 0; w < g_warmup; w++) k(d_a, d_b_t, d_c, M, N, K);
8538             cudaDeviceSynchronize();
8539             cudaEventRecord(start);
8540             for (int r = 0; r < g_repeat; r++) k(d_a, d_b_t, d_c, M, N, K);
8541             cudaEventRecord(stop);
8542             cudaEventSynchronize(stop);
8543             cudaEventElapsedTime(&time_ms, start, stop);
8544             time_ms /= g_repeat;
8545             ok = true;
8546         }

```

```

8547     if (ok) {
8548         check(cudaMemcpy(h_c, d_c, size_c, cudaMemcpyDeviceToHost), "bench cute D2H");
8549         max_err = 0.0f;
8550         for (int i = 0; i < M * N; i++) {
8551             float err = fabsf(__half2float(h_c[i]) - __half2float(h_c_ref[i]));
8552             if (err > max_err) max_err = err;
8553         }
8554         float tflops = bench_hgemm_tflops(M, N, K, time_ms);
8555         char tflops_str[32];
8556         snprintf(tflops_str, sizeof(tflops_str), "%.1f/%.1f (%.2fx)", tflops, cublas_tflops, tflops /
cublas_tflops);
8557         printf("| %-41s | %.6e | %-4s | %-19s |\n", label, max_err,
8558             max_err < 1.0f ? "PASS" : "FAIL", tflops_str);
8559     } else {
8560         printf("| %-41s | %-12s | %-4s | %-19s |\n", label, "LAUNCH ERR", "FAIL", "None");
8561     }
8562 }
8563 }
8564 cudaEventDestroy(start);
8565 cudaEventDestroy(stop);
8566 free(h_a);
8567 free(h_b);
8568 free(h_b_t);
8569 free(h_c);
8570 free(h_c_ref);
8571 cudaFree(d_a);
8572 cudaFree(d_b);
8573 cudaFree(d_b_t);
8574 cudaFree(d_c);
8575 cublasDestroy(handle);
8576 }
8577 #endif
8578
8579 #if defined(NOTES_V2_ENABLE_WGMMMA)
8580 // =====
8581 // Bench: HGEMM WGMMMA (m64n128k16 + TMA + Warp Specialization, SM90+)
8582 // =====
8583 template <int kStages, int kBlockSwizzle>
8584 static bool launch_timed_hgemm_wgmma(half *d_c, half *h_c, half *h_c_ref,
8585     CUTensorMap *tma_a, CUTensorMap *tma_b,
8586     int M, int N, int K, size_t size_c,
8587     cudaEvent_t start, cudaEvent_t stop,
8588     float &max_err, float &time_ms) {
8589     constexpr int BM = 128, BN = 128, BK = 64, kNumThreads = 256;
8590     using Kernel = void (*)(int, int, int, half *, const CUTensorMap *,
8591         const CUTensorMap *);
8592     Kernel k = hgemm_wgmma_stages_tn<64, 128, 16, BM, BN, BK, kNumThreads, kStages,
8593         kBlockSwizzle>;
8594     size_t smem = kStages * (BM * BK + BN * BK) * sizeof(half);
8595     if (!check_smem_feasible((const void *)k, smem)) return false;
8596     cudaFuncSetAttribute(k, cudaFuncAttributeMaxDynamicSharedMemorySize, smem);
8597     dim3 block(kNumThreads);
8598     const int tiles_n = (N + BN - 1) / BN;
8599     constexpr int kSwizzleN = 16;
8600     int gx = kBlockSwizzle ? div_ceil(tiles_n, kSwizzleN) : tiles_n;
8601     int gz = kBlockSwizzle ? kSwizzleN : 1;
8602     dim3 grid(gx, (M + BM - 1) / BM, gz);
8603     for (int w = 0; w < g_warmup; w++)
8604         k<<<grid, block, smem>>>(M, N, K, d_c, tma_a, tma_b);
8605     cudaDeviceSynchronize();
8606     cudaEventRecord(start);
8607     for (int r = 0; r < g_repeat; r++)
8608         k<<<grid, block, smem>>>(M, N, K, d_c, tma_a, tma_b);

```

```

8609     cudaEventRecord(stop);
8610     cudaEventSynchronize(stop);
8611     cudaEventElapsedTime(&time_ms, start, stop);
8612     time_ms /= g_repeat;
8613     check(cudaMemcpy(h_c, d_c, size_c, cudaMemcpyDeviceToHost), "bench wgmma D2H");
8614     max_err = 0.0f;
8615     for (int i = 0; i < M * N; i++) {
8616         float err = fabsf(__half2float(h_c[i]) - __half2float(h_c_ref[i]));
8617         if (err > max_err) max_err = err;
8618     }
8619     return true;
8620 }
8621
8622 static void bench_hgemm_wgmma(int M, int N, int K) {
8623     constexpr int BM = 128, BN = 128, BK = 64;
8624     if (M % BM != 0 || N % BN != 0 || K % BK != 0) {
8625         printf("| %-41s | %-12s | %-4s | %-19s |\n", "HGEMM WGMMA (unaligned)", "SKIP", "SKIP",
8626             "None");
8627         return;
8628     }
8629     size_t size_a = (size_t)M * K * sizeof(half);
8630     size_t size_b = (size_t)K * N * sizeof(half);
8631     size_t size_c = (size_t)M * N * sizeof(half);
8632     half *h_a = (half *)malloc(size_a);
8633     half *h_b = (half *)malloc(size_b);
8634     half *h_c_ref = (half *)malloc(size_c);
8635     srand(42);
8636     for (int i = 0; i < M * K; i++)
8637         h_a[i] = __float2half(((float)rand() / RAND_MAX) * 2.0f - 1.0f);
8638     for (int i = 0; i < K * N; i++)
8639         h_b[i] = __float2half(((float)rand() / RAND_MAX) * 2.0f - 1.0f);
8640     size_t size_b_t = (size_t)N * K * sizeof(half);
8641     half *h_b_t = (half *)malloc(size_b_t);
8642     for (int n = 0; n < N; n++)
8643         for (int k = 0; k < K; k++)
8644             h_b_t[n * K + k] = h_b[k * N + n];
8645     half *d_a, *d_b, *d_b_t, *d_c;
8646     check(cudaMalloc(&d_a, size_a), "bench wgmma alloc A");
8647     check(cudaMalloc(&d_b, size_b), "bench wgmma alloc B");
8648     check(cudaMalloc(&d_b_t, size_b_t), "bench wgmma alloc B_t");
8649     check(cudaMalloc(&d_c, size_c), "bench wgmma alloc C");
8650     check(cudaMemcpy(d_a, h_a, size_a, cudaMemcpyHostToDevice), "bench wgmma H2D A");
8651     check(cudaMemcpy(d_b, h_b, size_b, cudaMemcpyHostToDevice), "bench wgmma H2D B");
8652     check(cudaMemcpy(d_b_t, h_b_t, size_b_t, cudaMemcpyHostToDevice), "bench wgmma H2D B_t");
8653     cublasHandle_t handle;
8654     cublasCreate(&handle);
8655     half alpha_h = __float2half(1.0f), beta_h = __float2half(0.0f);
8656     cublasGemmEx(handle, CUBLAS_OP_N, CUBLAS_OP_N, N, M, K, &alpha_h, d_b, CUDA_R_16F, N,
8657         d_a, CUDA_R_16F, K, &beta_h, d_c, CUDA_R_16F, N, CUBLAS_COMPUTE_16F,
8658         CUBLAS_GEMM_DEFAULT);
8659     check(cudaMemcpy(h_c_ref, d_c, size_c, cudaMemcpyDeviceToHost), "bench wgmma D2H ref");
8660     UTensorMap *tma_a = allocate_and_create_tensor_map(d_a, M / BM, K / BK);
8661     UTensorMap *tma_b = allocate_and_create_tensor_map(d_b_t, N / BN, K / BK);
8662     half *h_c = (half *)malloc(size_c);
8663     float cublas_tflops = bench_cublas_hgemm_tflops(handle, M, N, K, d_a, d_b, d_c);
8664     cudaEvent_t start, stop;
8665     cudaEventCreate(&start);
8666     cudaEventCreate(&stop);
8667     for (int stages : {1, 2, 3}) {
8668         for (int swizzle : {0, 1}) {
8669             float max_err = 0, time_ms = 0;
8670             bool ok = false;
8671             if (stages == 2)

```

```

8672     ok = swizzle
8673     ? launch_timed_hgemm_wgmma<2, 1>(d_c, h_c, h_c_ref, tma_a, tma_b, M, N,
8674       K, size_c, start, stop, max_err, time_ms)
8675     : launch_timed_hgemm_wgmma<2, 0>(d_c, h_c, h_c_ref, tma_a, tma_b, M, N, K,
8676       size_c, start, stop, max_err, time_ms);
8677 else
8678     ok = swizzle
8679     ? launch_timed_hgemm_wgmma<3, 1>(d_c, h_c, h_c_ref, tma_a, tma_b, M, N,
8680       K, size_c, start, stop, max_err, time_ms)
8681     : launch_timed_hgemm_wgmma<3, 0>(d_c, h_c, h_c_ref, tma_a, tma_b, M, N, K,
8682       size_c, start, stop, max_err, time_ms);
8683 char label[64];
8684 snprintf(label, sizeof(label), "HGEMM TMA WGMMMA WS (S=%d, SW=%d)", stages, swizzle);
8685 if (!ok) {
8686     if (g_verbose)
8687         printf("| %-41s | %-12s | %-4s | %-19s |\n", label, "SMEM SKIP", "SKIP", "None");
8688 } else {
8689     float tflops = bench_hgemm_tflops(M, N, K, time_ms);
8690     char tflops_str[32];
8691     snprintf(tflops_str, sizeof(tflops_str), "%.1f/%.1f (%.2fx)", tflops, cublas_tflops, tflops /
cublas_tflops);
8692     printf("| %-41s | %.6e | %-4s | %-19s |\n", label, max_err,
8693       max_err < 1.0f ? "PASS" : "FAIL", tflops_str);
8694 }
8695 }
8696 }
8697 cudaEventDestroy(start);
8698 cudaEventDestroy(stop);
8699 free(h_a);
8700 free(h_b);
8701 free(h_b_t);
8702 free(h_c);
8703 free(h_c_ref);
8704 cudaFree(d_a);
8705 cudaFree(d_b);
8706 cudaFree(d_b_t);
8707 cudaFree(d_c);
8708 cudaFree(tma_a);
8709 cudaFree(tma_b);
8710 cublasDestroy(handle);
8711 }
8712 #endif
8713
8714 #if defined(NOTES_V2_ENABLE_TMA_MMA_WS)
8715 // =====
8716 // Bench: HGEMM TMA MMA WS (mma.sync + TMA + Warp Specialization, SM120)
8717 // =====
8718 template <int kStages, int kBlockSwizzle>
8719 static bool bench_launch_tma_mma_ws(
8720     int M, int N, int K, half *d_a, half *d_b_t,
8721     half *d_c, half *h_c, half *h_c_ref, size_t size_c,
8722     CUTensorMap *tma_a, CUTensorMap *tma_b,
8723     cudaEvent_t start, cudaEvent_t stop,
8724     float &max_err, float &time_ms) {
8725     constexpr int BM = 128, BN = 128, BK = 64, kNumThreads = 256;
8726     constexpr size_t payload_bytes = kStages * (BM * BK + BN * BK) * sizeof(half);
8727     using Kernel = void (*)(int, int, int, half *, const CUTensorMap *,
8728       const CUTensorMap *);
8729     Kernel kernel = hgemm_tma_mma_ws_tn<16, 8, 16, 2, 2, 4, 8, 4, kStages,
8730       kNumThreads, kBlockSwizzle>;
8731     if (!check_smem_feasible((const void *)kernel, payload_bytes)) return false;
8732     cudaFuncSetAttribute(kernel, cudaFuncAttributeMaxDynamicSharedMemorySize,
8733       payload_bytes);

```

```

8734 dim3 block(kNumThreads);
8735 constexpr int kSwizzleN = 16;
8736 const int n_tiles = N / BN;
8737 const int grid_x = kBlockSwizzle ? div_ceil(n_tiles, kSwizzleN) : n_tiles;
8738 const int grid_z = kBlockSwizzle ? kSwizzleN : 1;
8739 dim3 grid(grid_x, M / BM, grid_z);
8740 for (int w = 0; w < g_warmup; w++)
8741     kernel<<<grid, block, payload_bytes>>>(M, N, K, d_c, tma_a, tma_b);
8742 cudaDeviceSynchronize();
8743 cudaEventRecord(start);
8744 for (int r = 0; r < g_repeat; r++)
8745     kernel<<<grid, block, payload_bytes>>>(M, N, K, d_c, tma_a, tma_b);
8746 cudaEventRecord(stop);
8747 cudaEventSynchronize(stop);
8748 cudaEventElapsedTime(&time_ms, start, stop);
8749 time_ms /= g_repeat;
8750 check(cudaMemcpy(h_c, d_c, size_c, cudaMemcpyDeviceToHost), "bench tma mma ws D2H");
8751 max_err = 0.0f;
8752 for (int i = 0; i < M * N; ++i)
8753     max_err = fmaxf(max_err, fabsf(__half2float(h_c[i]) - __half2float(h_c_ref[i])));
8754 return true;
8755 }
8756
8757 static void bench_hgemm_tma_mma_ws(int M, int N, int K) {
8758     constexpr int BM = 128, BN = 128, BK = 64;
8759     if (M % BM != 0 || N % BN != 0 || K % BK != 0) {
8760         printf("| %-41s | %-12s | %-4s | %-19s |\n", "HGEMM TMA MMA WS (unaligned)", "SKIP",
8761             "SKIP", "None");
8762         return;
8763     }
8764     const size_t size_a = size_t(M) * K * sizeof(half);
8765     const size_t size_b = size_t(K) * N * sizeof(half);
8766     const size_t size_b_t = size_t(N) * K * sizeof(half);
8767     const size_t size_c = size_t(M) * N * sizeof(half);
8768     half *h_a = static_cast<half *>(malloc(size_a));
8769     half *h_b = static_cast<half *>(malloc(size_b));
8770     half *h_b_t = static_cast<half *>(malloc(size_b_t));
8771     half *h_c = static_cast<half *>(malloc(size_c));
8772     half *h_c_ref = static_cast<half *>(malloc(size_c));
8773     srand(42);
8774     for (int i = 0; i < M * K; ++i)
8775         h_a[i] = __float2half((float(rand()) / RAND_MAX) * 2.0f - 1.0f);
8776     for (int i = 0; i < K * N; ++i)
8777         h_b[i] = __float2half((float(rand()) / RAND_MAX) * 2.0f - 1.0f);
8778     for (int n = 0; n < N; ++n)
8779         for (int k = 0; k < K; ++k)
8780             h_b_t[n * K + k] = h_b[k * N + n];
8781     half *d_a, *d_b, *d_b_t, *d_c;
8782     check(cudaMalloc(&d_a, size_a), "bench tma mma ws alloc A");
8783     check(cudaMalloc(&d_b, size_b), "bench tma mma ws alloc B");
8784     check(cudaMalloc(&d_b_t, size_b_t), "bench tma mma ws alloc B_t");
8785     check(cudaMalloc(&d_c, size_c), "bench tma mma ws alloc C");
8786     check(cudaMemcpy(d_a, h_a, size_a, cudaMemcpyHostToDevice), "bench tma mma ws H2D A");
8787     check(cudaMemcpy(d_b, h_b, size_b, cudaMemcpyHostToDevice), "bench tma mma ws H2D B");
8788     check(cudaMemcpy(d_b_t, h_b_t, size_b_t, cudaMemcpyHostToDevice),
8789         "bench tma mma ws H2D B_t");
8790     cublasHandle_t handle;
8791     cublasCreate(&handle);
8792     half alpha = __float2half(1.0f), beta = __float2half(0.0f);
8793     cublasGemmEx(handle, CUBLAS_OP_N, CUBLAS_OP_N, N, M, K, &alpha, d_b, CUDA_R_16F, N,
8794         d_a, CUDA_R_16F, K, &beta, d_c, CUDA_R_16F, N, CUBLAS_COMPUTE_16F,
8795         CUBLAS_GEMM_DEFAULT);
8796     check(cudaMemcpy(h_c_ref, d_c, size_c, cudaMemcpyDeviceToHost),

```

```

8797     "bench_tma_mma_ws_D2H_reference");
8798     CUTensorMap *tma_a = allocate_and_create_tensor_map(d_a, M / BM, K / BK);
8799     CUTensorMap *tma_b = allocate_and_create_tensor_map(d_b_t, N / BN, K / BK);
8800     float cublas_tflops = bench_cublas_hgemm_tflops(handle, M, N, K, d_a, d_b, d_c);
8801     cudaEvent_t start, stop;
8802     cudaEventCreate(&start);
8803     cudaEventCreate(&stop);
8804     for (int stages : {1, 2, 3}) {
8805         for (int swizzle : {0, 1}) {
8806             float max_err = 0, time_ms = 0;
8807             bool ok = false;
8808             if (stages == 2)
8809                 ok = swizzle ? bench_launch_tma_mma_ws<2, 1>(M, N, K, d_a, d_b_t, d_c, h_c,
8810                     h_c_ref, size_c, tma_a, tma_b,
8811                     start, stop, max_err, time_ms)
8812                     : bench_launch_tma_mma_ws<2, 0>(M, N, K, d_a, d_b_t, d_c, h_c,
8813                     h_c_ref, size_c, tma_a, tma_b,
8814                     start, stop, max_err, time_ms);
8815             else
8816                 ok = swizzle ? bench_launch_tma_mma_ws<3, 1>(M, N, K, d_a, d_b_t, d_c, h_c,
8817                     h_c_ref, size_c, tma_a, tma_b,
8818                     start, stop, max_err, time_ms)
8819                     : bench_launch_tma_mma_ws<3, 0>(M, N, K, d_a, d_b_t, d_c, h_c,
8820                     h_c_ref, size_c, tma_a, tma_b,
8821                     start, stop, max_err, time_ms);
8822             char label[64];
8823             snprintf(label, sizeof(label), "HGEMM TMA MMA WS (S=%d, SW=%d)", stages, swizzle);
8824             if (!ok) {
8825                 if (g_verbose)
8826                     printf("| %-41s | %-12s | %-4s | %-19s |\n", label, "SMEM SKIP", "SKIP", "None");
8827             } else {
8828                 float tflops = bench_hgemm_tflops(M, N, K, time_ms);
8829                 char tflops_str[32];
8830                 snprintf(tflops_str, sizeof(tflops_str), "%.1f/%.1f (%.2fx)", tflops, cublas_tflops, tflops /
8831                     cublas_tflops);
8832                 printf("| %-41s | %.6e | %-4s | %-19s |\n", label, max_err,
8833                     max_err < 1.0f ? "PASS" : "FAIL", tflops_str);
8834             }
8835         }
8836     }
8837     cudaEventDestroy(start);
8838     cudaEventDestroy(stop);
8839     free(h_a);
8840     free(h_b);
8841     free(h_b_t);
8842     free(h_c);
8843     free(h_c_ref);
8844     cudaFree(d_a);
8845     cudaFree(d_b);
8846     cudaFree(d_b_t);
8847     cudaFree(d_c);
8848     cudaFree(tma_a);
8849     cudaFree(tma_b);
8850     cublasDestroy(handle);
8851 }
8852 #endif
8853 // =====
8854 // Bench: FlashAttention-2 Split-Q (template on kHeadDim for dispatch)
8855 // =====
8856 template <int kHeadDim, int kStagesK = 2, int kPadQ = 8, int kPadK = 8,
8857         int kPadV = 8, int kMmaAccF32 = 0>
8858 static void bench_fa_launch(int B, int H, int seqlen, int head_dim,

```

```

8859     half *h_o_ref, float *ref_o, half *d_q, half *d_k, half *d_v, half *d_o,
8860     float cudnn_tflops_f16) {
8861     static_assert(kHeadDim == 64 || kHeadDim == 128, "Only D=64 and D=128 are supported");
8862     constexpr bool kSwizzleQ = kPadQ == 0;
8863     constexpr bool kSwizzleK = kPadK == 0;
8864     constexpr bool kSwizzleV = kPadV == 0;
8865     constexpr int kMmaAtomM = 16;
8866     constexpr int kMmaAtomN = 8;
8867     constexpr int kMmaAtomK = 16;
8868     constexpr int kMmaTileSeqLenQ = 8;
8869     constexpr int kMmaTileSeqLenK = 1;
8870     constexpr int kMmaTileSeqLenP = 8;
8871     constexpr int kMmaTileHeadDimV = 1;
8872     constexpr int kValTileSeqLenQ = 1;
8873     constexpr int kValTileSeqLenK = 8;
8874     constexpr int kValTileSeqLenP = 1;
8875     constexpr int kValTileHeadDimV = kHeadDim / (8 * kMmaTileHeadDimV);
8876     constexpr int Br = kMmaAtomM * kMmaTileSeqLenQ * kValTileSeqLenQ;
8877     constexpr int Bc = kMmaAtomN * kMmaTileSeqLenK * kValTileSeqLenK;
8878     constexpr const char *layout_name =
8879         kSwizzleQ && kSwizzleK && kSwizzleV ? "Swizzle" :
8880         kSwizzleQ && kSwizzleK ? "SwizzleQK" :
8881         kSwizzleQ && kSwizzleV ? "SwizzleQV" :
8882         kSwizzleK && kSwizzleV ? "SwizzleKV" :
8883         kSwizzleQ ? "SwizzleQ" :
8884         kSwizzleK ? "SwizzleK" :
8885         kSwizzleV ? "SwizzleV" : "Pad";
8886
8887     // Kernel requires seqlen >= Br (tile size); skip gracefully for short seqlen
8888     if (seqlen < Br) {
8889         char label[64];
8890         snprintf(label, sizeof(label), "FA2 (%s)",
8891             layout_name);
8892         printf("| %-41s | %-12s | %-4s | %-19s |\n", label,
8893             "seqlen<Br", "SKIP", "None");
8894         return;
8895     }
8896     size_t smem_bytes =
8897         (Br * (kHeadDim + kPadQ) +
8898         kStagesK * Bc * (kHeadDim + kPadK) +
8899         Bc * (kHeadDim + kPadV)) * sizeof(half);
8900
8901     dim3 block(256);
8902     dim3 grid((seqlen + Br - 1) / Br, B * H);
8903
8904     cudaStream_t timing_stream;
8905     cudaStreamCreate(&timing_stream);
8906     cudaEvent_t start, stop;
8907     cudaEventCreate(&start);
8908     cudaEventCreate(&stop);
8909
8910     using FAKernel = void (*)(half *, half *, half *, half *, int, int);
8911     FAKernel fa_k = flash_attn_mma_stages_split_q<
8912         kHeadDim, kMmaAtomM, kMmaAtomN, kMmaAtomK, kMmaAccF32,
8913         kMmaTileSeqLenQ, kMmaTileSeqLenK,
8914         kMmaTileSeqLenP, kMmaTileHeadDimV, kValTileSeqLenQ, kValTileSeqLenK,
8915         kValTileSeqLenP, kValTileHeadDimV, kStagesK, kPadQ, kPadK, kPadV>;
8916     bool smem_ok = check_smem_feasible((const void *)fa_k, smem_bytes);
8917     if (smem_ok) {
8918         cudaFuncSetAttribute(fa_k, cudaFuncAttributeMaxDynamicSharedMemorySize, smem_bytes);
8919     }
8920
8921     // Warmup

```

```

8922 if (smem_ok) {
8923     for (int w = 0; w < g_warmup; w++)
8924         fa_k<<<grid, block, smem_bytes, timing_stream>>>(d_q, d_k, d_v, d_o, seqlen, H);
8925     check(cudaStreamSynchronize(timing_stream), "fa warmup sync");
8926 }
8927
8928 // Timed repeat
8929 cudaEventRecord(start, timing_stream);
8930 if (smem_ok) {
8931     for (int r = 0; r < g_repeat; r++)
8932         fa_k<<<grid, block, smem_bytes, timing_stream>>>(d_q, d_k, d_v, d_o, seqlen, H);
8933 }
8934 cudaEventRecord(stop, timing_stream);
8935 cudaEventSynchronize(stop);
8936
8937 float time_ms = 0;
8938 cudaEventElapsedTime(&time_ms, start, stop);
8939 time_ms /= g_repeat;
8940
8941 size_t sz = (size_t)B * H * seqlen * head_dim * sizeof(half);
8942 half *h_o = (half *)malloc(sz);
8943 check(cudaMemcpy(h_o, d_o, sz, cudaMemcpyDeviceToHost), "bench fa D2H");
8944
8945 int count = B * H * seqlen * head_dim;
8946 char label[64];
8947 snprintf(label, sizeof(label), "FA2 (S=%d, %s, %s)", kStagesK,
8948         layout_name, kMmaAccF32 ? "F32Acc" : "F16Acc");
8949 float max_err = 0.0f;
8950 bool checked = h_o_ref || ref_o;
8951 if (smem_ok && checked) {
8952     for (int i = 0; i < count; i++) {
8953         float ref_val = h_o_ref ? __half2float(h_o_ref[i]) : ref_o[i];
8954         float err = fabsf(__half2float(h_o[i]) - ref_val);
8955         if (err > max_err) max_err = err;
8956     }
8957 }
8958 if (smem_ok && checked) {
8959     float tflops = bench_fa_tflops(B, H, seqlen, head_dim, time_ms);
8960     char tflops_str[32];
8961     if (cudnn_tflops_f16 > 0)
8962         snprintf(tflops_str, sizeof(tflops_str), "%.1f/%.1f (%.2fx)", tflops, cudnn_tflops_f16, tflops /
8963             cudnn_tflops_f16);
8964     else
8965         snprintf(tflops_str, sizeof(tflops_str), "%.1f", tflops);
8966     printf("| %-41s | %.6e | %-4s | %-19s |\n", label, max_err,
8967         max_err < 5e-1f ? "PASS" : "FAIL", tflops_str);
8968 } else if (smem_ok) {
8969     float tflops = bench_fa_tflops(B, H, seqlen, head_dim, time_ms);
8970     char tflops_str[32];
8971     snprintf(tflops_str, sizeof(tflops_str), "%.1f", tflops);
8972     printf("| %-41s | %-12s | %-4s | %-19s |\n", label, "unchecked", "SKIP", tflops_str);
8973 } else {
8974     if (g_verbose)
8975         printf("| %-41s | %-12s | %-4s | %-19s |\n", label, "SMEM too large", "SKIP", "None");
8976 }
8977
8978 cudaEventDestroy(start);
8979 cudaEventDestroy(stop);
8980 cudaStreamDestroy(timing_stream);
8981 free(h_o);
8982 }
8983 #if defined(NOTES_V2_ENABLE_TMA_MMA_WS)

```

```

8984 // Bench for flash_attn_tma_mma_ws_stages_split_q (SM120, D=64/128)
8985 template <int kHeadDim, int kStagesK, int kStagesV = 1, int kMmaAccF32 = 0>
8986 static void bench_fa_tma_mma_ws_launch(int B, int H, int seqlen, int head_dim,
8987                                     half *h_o_ref, float *ref_o,
8988                                     half *d_q, half *d_k, half *d_v,
8989                                     half *d_o, float cudnn_tflops_f16) {
8990     constexpr int kMmaAtomM = 16, kMmaAtomN = 8, kMmaAtomK = 16;
8991     constexpr int kMmaTileSeqLenQ = 8, kMmaTileSeqLenK = 1;
8992     constexpr int kMmaTileSeqLenP = 8, kMmaTileHeadDimV = 1;
8993     constexpr int kValTileSeqLenQ = 1, kValTileSeqLenK = 8;
8994     constexpr int kValTileSeqLenP = 1;
8995     constexpr int kValTileHeadDimV = kHeadDim / (8 * kMmaTileHeadDimV);
8996     constexpr int Br = kMmaAtomM * kMmaTileSeqLenQ * kValTileSeqLenQ;
8997     constexpr int Bc = kMmaAtomN * kMmaTileSeqLenK * kValTileSeqLenK;
8998     constexpr int kNumThreads = 384;
8999
9000     if (seqlen < Br || seqlen % Br != 0 || seqlen % Bc != 0) {
9001         char label[64];
9002         snprintf(label, sizeof(label),
9003                 "FA2 TMA MMA WS (Sk=%d, Sv=%d, unaligned)", kStagesK, kStagesV);
9004         printf("| %-41s | %-12s | %-4s | %-19s |\n", label, "SKIP", "SKIP", "None");
9005         return;
9006     }
9007
9008     // TMA descriptors: box innermost 固定 64 half (128B), 满足
9009     // CU_TENSOR_MAP_SWIZZLE_128B 对 box innermost ≤ 128B 的硬约束。
9010     // D=64: blocks_width=1, 单次 TMA 覆盖整行。
9011     // D=128: blocks_width=2, kernel producer 沿 head_dim 连续发 2 次 TMA,
9012     //         minor_coord = 0/64, 写入 chunk-major smem 布局 [2, Br, 64]。
9013     // Q/K/V gmem 是 [B, H, seqlen, head_dim] row-major, 作为 2D
9014     // [B*H*seqlen, head_dim] 矩阵描述。blocks_height = B*H*seqlen/tile_major。
9015     // kernel 用 (Nb_id*H+Nh_id)*N 偏移 major_coord。
9016     constexpr int kTmaBoxMinor = 64; // box innermost = 64 half = 128B
9017     constexpr int kTmaChunks = kHeadDim / kTmaBoxMinor;
9018     CUTensorMap *tma_q = allocate_and_create_tensor_map<Br, kTmaBoxMinor>(
9019         d_q, B * H * seqlen / Br, kTmaChunks);
9020     CUTensorMap *tma_k = allocate_and_create_tensor_map<Bc, kTmaBoxMinor>(
9021         d_k, B * H * seqlen / Bc, kTmaChunks);
9022     CUTensorMap *tma_v = allocate_and_create_tensor_map<Bc, kTmaBoxMinor>(
9023         d_v, B * H * seqlen / Bc, kTmaChunks);
9024
9025     using FAKernel = void (*)(half *, half *, half *, half *, int, int,
9026                             const CUTensorMap *, const CUTensorMap *,
9027                             const CUTensorMap *);
9028     FAKernel fa_k = flash_attn_tma_mma_ws_stages_split_q<
9029         kHeadDim, kMmaAtomM, kMmaAtomN, kMmaAtomK, kMmaAccF32, kMmaTileSeqLenQ,
9030         kMmaTileSeqLenK, kMmaTileSeqLenP, kMmaTileHeadDimV, kValTileSeqLenQ,
9031         kValTileSeqLenK, kValTileSeqLenP, kValTileHeadDimV, kStagesK, kStagesV,
9032         kNumThreads>;
9033
9034     size_t smem_bytes = (Br * kHeadDim + kStagesK * Bc * kHeadDim +
9035                         kStagesV * Bc * kHeadDim) * sizeof(half);
9036     bool smem_ok = check_smem_feasible((const void *)fa_k, smem_bytes);
9037     if (smem_ok) {
9038         cudaFuncSetAttribute(fa_k, cudaFuncAttributeMaxDynamicSharedMemorySize,
9039                             smem_bytes);
9040     }
9041
9042     dim3 block(kNumThreads);
9043     dim3 grid((seqlen + Br - 1) / Br, B * H);
9044
9045     cudaStream_t timing_stream;
9046     cudaStreamCreate(&timing_stream);

```

```

9047 cudaEvent_t start, stop;
9048 cudaEventCreate(&start);
9049 cudaEventCreate(&stop);
9050
9051 if (smem_ok) {
9052     for (int w = 0; w < g_warmup; ++w)
9053         fa_k<<<grid, block, smem_bytes, timing_stream>>>(d_q, d_k, d_v, d_o,
9054                                                         seqlen, H, tma_q, tma_k,
9055                                                         tma_v);
9056     check(cudaStreamSynchronize(timing_stream), "fa_tma_ws warmup sync");
9057 }
9058 cudaEventRecord(start, timing_stream);
9059 if (smem_ok) {
9060     for (int r = 0; r < g_repeat; ++r)
9061         fa_k<<<grid, block, smem_bytes, timing_stream>>>(d_q, d_k, d_v, d_o,
9062                                                         seqlen, H, tma_q, tma_k,
9063                                                         tma_v);
9064 }
9065 cudaEventRecord(stop, timing_stream);
9066 cudaEventSynchronize(stop);
9067
9068 float time_ms = 0;
9069 cudaEventElapsedTime(&time_ms, start, stop);
9070 time_ms /= g_repeat;
9071
9072 size_t sz = (size_t)B * H * seqlen * head_dim * sizeof(half);
9073 half *h_o = (half *)malloc(sz);
9074 check(cudaMemcpy(h_o, d_o, sz, cudaMemcpyDeviceToHost), "fa_tma_ws D2H");
9075 int count = B * H * seqlen * head_dim;
9076
9077 char label[64];
9078 snprintf(label, sizeof(label), "FA2 TMA MMA WS (Sk=%d, Sv=%d, %s)",
9079          kStagesK, kStagesV, kMmaAccF32 ? "F32Acc" : "F16Acc");
9080 float max_err = 0.0f;
9081 bool checked = h_o_ref || ref_o;
9082 if (smem_ok && checked) {
9083     for (int i = 0; i < count; ++i) {
9084         float ref_val = h_o_ref ? __half2float(h_o_ref[i]) : ref_o[i];
9085         float err = fabsf(__half2float(h_o[i]) - ref_val);
9086         if (err > max_err) max_err = err;
9087     }
9088 }
9089 if (smem_ok && checked) {
9090     float tflops = bench_fa_tflops(B, H, seqlen, head_dim, time_ms);
9091     char tflops_str[32];
9092     if (cudnn_tflops_f16 > 0)
9093         snprintf(tflops_str, sizeof(tflops_str), "%.1f/%.1f (%.2fx)", tflops, cudnn_tflops_f16, tflops /
9094                  cudnn_tflops_f16);
9095     else
9096         snprintf(tflops_str, sizeof(tflops_str), "%.1f", tflops);
9097     printf("| %-41s | %.6e | %-4s | %-19s |\n", label, max_err,
9098           max_err < 5e-1f ? "PASS" : "FAIL", tflops_str);
9099 } else if (smem_ok) {
9100     float tflops = bench_fa_tflops(B, H, seqlen, head_dim, time_ms);
9101     char tflops_str[32];
9102     snprintf(tflops_str, sizeof(tflops_str), "%.1f", tflops);
9103     printf("| %-41s | %-12s | %-4s | %-19s |\n", label, "unchecked", "SKIP",
9104           tflops_str);
9105 } else {
9106     if (g_verbose)
9107         printf("| %-41s | %-12s | %-4s | %-19s |\n", label, "SMEM too large", "SKIP",
9108               "None");
9109 }

```

```

9109     cudaEventDestroy(start);
9110     cudaEventDestroy(stop);
9111     cudaStreamDestroy(timing_stream);
9112     free(h_o);
9113     cudaFree(tma_q);
9114     cudaFree(tma_k);
9115     cudaFree(tma_v);
9116 }
9117
9118 // D=64/128 dispatch wrapper for bench
9119 template <int kStagesK, int kStagesV = 1, int kMmaAccF32 = 0>
9120 static void bench_fa_tma_mma_ws_dispatch(int B, int H, int seqlen, int head_dim,
9121     half *h_o_ref, float *ref_o,
9122     half *d_q, half *d_k, half *d_v,
9123     half *d_o, float cudnn_tflops_f16) {
9124     if (head_dim == 64) {
9125         bench_fa_tma_mma_ws_launch<64, kStagesK, kStagesV, kMmaAccF32>(B, H, seqlen, head_dim, h_o_ref,
9126             ref_o, d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9127     } else if (head_dim == 128) {
9128         bench_fa_tma_mma_ws_launch<128, kStagesK, kStagesV, kMmaAccF32>(B, H, seqlen, head_dim, h_o_ref,
9129             ref_o, d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9130     } else {
9131         char label[64];
9132         snprintf(label, sizeof(label),
9133             "FA2 TMA MMA WS (Sk=%d, Sv=%d, D!=64/128)", kStagesK, kStagesV);
9134         printf("| %-41s | %-12s | %-4s | %-19s |\n", label, "SKIP", "SKIP", "None");
9135     }
9136 }
9137
9138 // FA3-style dual-consumer bench (Br=64, per-WG independent K pipeline, Sv=1)
9139 template <int kHeadDim, int kStagesK, int kMmaAccF32 = 0>
9140 static void bench_fa_3_tma_ws_launch(int B, int H, int seqlen, int head_dim,
9141     half *h_o_ref, float *ref_o,
9142     half *d_q, half *d_k, half *d_v,
9143     half *d_o, float cudnn_tflops_f16) {
9144     constexpr int kMmaAtomM = 16, kMmaAtomN = 8, kMmaAtomK = 16;
9145     constexpr int kMmaTileSeqLenQ = 4, kMmaTileSeqLenK = 1;
9146     constexpr int kMmaTileSeqLenP = 4, kMmaTileHeadDimV = 1;
9147     constexpr int kValTileSeqLenQ = 1, kValTileSeqLenK = 8;
9148     constexpr int kValTileSeqLenP = 1;
9149     constexpr int kValTileHeadDimV = kHeadDim / (8 * kMmaTileHeadDimV);
9150     constexpr int kStagesV = 1; // per-WG V: always 1 buffer
9151     constexpr int kNumConsumerWGs = 2;
9152     constexpr int Br = kMmaAtomM * kMmaTileSeqLenQ * kValTileSeqLenQ; // 64
9153     constexpr int Bc = kMmaAtomN * kMmaTileSeqLenK * kValTileSeqLenK; // 64
9154     constexpr int kNumThreads = 384;
9155
9156     if (seqlen < Br || seqlen % Br != 0 || seqlen % Bc != 0) {
9157         char label[80];
9158         snprintf(label, sizeof(label),
9159             "FA3-style TMA MMA WS (Sk=%d, Sv=%d, unaligned)", kStagesK, kStagesV);
9160         printf("| %-41s | %-12s | %-4s | %-19s |\n", label, "SKIP", "SKIP", "None");
9161         return;
9162     }
9163
9164     constexpr int kTmaBoxMinor = 64;
9165     constexpr int kTmaChunks = kHeadDim / kTmaBoxMinor;
9166     CUTensorMap *tma_q = allocate_and_create_tensor_map<Br, kTmaBoxMinor>(
9167         d_q, B * H * seqlen / Br, kTmaChunks);
9168     CUTensorMap *tma_k = allocate_and_create_tensor_map<Bc, kTmaBoxMinor>(
9169         d_k, B * H * seqlen / Bc, kTmaChunks);
9170     CUTensorMap *tma_v = allocate_and_create_tensor_map<Bc, kTmaBoxMinor>(

```

```

9172     d_v, B * H * seqlen / Bc, kTmaChunks);
9173
9174     using FAKernel = void (*)(half *, half *, half *, half *, int, int,
9175                             const CUTensorMap *, const CUTensorMap *,
9176                             const CUTensorMap *);
9177     FAKernel fa_k = flash_attn_3_tma_ws_stages_split_q<
9178         kHeadDim, kMmaAtomM, kMmaAtomN, kMmaAtomK, kMmaAccF32, kMmaTileSeqLenQ,
9179         kMmaTileSeqLenK, kMmaTileSeqLenP, kMmaTileHeadDimV, kValTileSeqLenQ,
9180         kValTileSeqLenK, kValTileSeqLenP, kValTileHeadDimV, kStagesK, kStagesV,
9181         kNumThreads>;
9182
9183     // smem = Q + K[2WGs * Sk * Bc*D] + V[2WGs * 1 * Bc*D]
9184     size_t smem_bytes = (Br * kHeadDim +
9185                         kNumConsumerWGs * kStagesK * Bc * kHeadDim +
9186                         kNumConsumerWGs * kStagesV * Bc * kHeadDim) * sizeof(half);
9187     bool smem_ok = check_smem_feasible((const void *)fa_k, smem_bytes);
9188     if (smem_ok) {
9189         cudaFuncSetAttribute(fa_k, cudaFuncAttributeMaxDynamicSharedMemorySize,
9190                             smem_bytes);
9191     }
9192
9193     dim3 block(kNumThreads);
9194     dim3 grid((seqlen + Br - 1) / Br, B * H);
9195
9196     cudaStream_t timing_stream;
9197     cudaStreamCreate(&timing_stream);
9198     cudaEvent_t start, stop;
9199     cudaEventCreate(&start);
9200     cudaEventCreate(&stop);
9201
9202     if (smem_ok) {
9203         for (int w = 0; w < g_warmup; ++w)
9204             fa_k<<<grid, block, smem_bytes, timing_stream>>>(d_q, d_k, d_v, d_o,
9205                                                             seqlen, H, tma_q, tma_k,
9206                                                             tma_v);
9207         check(cudaStreamSynchronize(timing_stream), "fa3_tma_ws warmup sync");
9208     }
9209     cudaEventRecord(start, timing_stream);
9210     if (smem_ok) {
9211         for (int r = 0; r < g_repeat; ++r)
9212             fa_k<<<grid, block, smem_bytes, timing_stream>>>(d_q, d_k, d_v, d_o,
9213                                                             seqlen, H, tma_q, tma_k,
9214                                                             tma_v);
9215     }
9216     cudaEventRecord(stop, timing_stream);
9217     cudaEventSynchronize(stop);
9218
9219     float time_ms = 0;
9220     cudaEventElapsedTime(&time_ms, start, stop);
9221     time_ms /= g_repeat;
9222
9223     size_t sz = (size_t)B * H * seqlen * head_dim * sizeof(half);
9224     half *h_o = (half *)malloc(sz);
9225     check(cudaMemcpy(h_o, d_o, sz, cudaMemcpyDeviceToHost), "fa3_tma_ws D2H");
9226     int count = B * H * seqlen * head_dim;
9227
9228     char label[80];
9229     snprintf(label, sizeof(label), "FA3-style TMA MMA WS (Sk=%d, Sv=%d, %s)",
9230             kStagesK, kStagesV, kMmaAccF32 ? "F32Acc" : "F16Acc");
9231     float max_err = 0.0f;
9232     bool checked = h_o_ref || ref_o;
9233     if (smem_ok && checked) {
9234         for (int i = 0; i < count; ++i) {

```

```

9235     float ref_val = h_o_ref ? __half2float(h_o_ref[i]) : ref_o[i];
9236     float err = fabsf(__half2float(h_o[i]) - ref_val);
9237     if (err > max_err) max_err = err;
9238 }
9239 }
9240 if (smem_ok && checked) {
9241     float tflops = bench_fa_tflops(B, H, seqlen, head_dim, time_ms);
9242     char tflops_str[32];
9243     if (cudnn_tflops_f16 > 0)
9244         snprintf(tflops_str, sizeof(tflops_str), "%.1f/%.1f (%.2fx)", tflops, cudnn_tflops_f16, tflops /
cudnn_tflops_f16);
9245     else
9246         snprintf(tflops_str, sizeof(tflops_str), "%.1f", tflops);
9247     printf("| %-41s | %.6e | %-4s | %-19s |\n", label, max_err,
max_err < 5e-1f ? "PASS" : "FAIL", tflops_str);
9248 } else if (smem_ok) {
9249     float tflops = bench_fa_tflops(B, H, seqlen, head_dim, time_ms);
9250     char tflops_str[32];
9251     snprintf(tflops_str, sizeof(tflops_str), "%.1f", tflops);
9252     printf("| %-41s | %-12s | %-4s | %-19s |\n", label, "unchecked", "SKIP",
tflops_str);
9253 } else {
9254     if (g_verbose)
9255         printf("| %-41s | %-12s | %-4s | %-19s |\n", label, "SMEM too large", "SKIP",
"None");
9256 }
9257 }
9258
9259 cudaEventDestroy(start);
9260 cudaEventDestroy(stop);
9261 cudaStreamDestroy(timing_stream);
9262 free(h_o);
9263 cudaFree(tma_q);
9264 cudaFree(tma_k);
9265 cudaFree(tma_v);
9266 }
9267
9268 // D=64/128 dispatch wrapper for FA3-style bench
9269 template <int kMmaAccF32 = 0>
9270 static void bench_fa_3_tma_ws_dispatch(int B, int H, int seqlen, int head_dim,
half *h_o_ref, float *ref_o,
half *d_q, half *d_k, half *d_v,
half *d_o, float cudnn_tflops_f16) {
9271 // Try stages up to 4; bench_fa_3_tma_ws_launch internally checks
9272 // cudaDevAttrMaxSharedMemoryPerBlockOptin and skips configs that
9273 // exceed the actual device limit (e.g. Hopper 224KB vs Blackwell 101KB).
9274 if (head_dim == 64) {
9275     bench_fa_3_tma_ws_launch<64, 1, kMmaAccF32>(B, H, seqlen, head_dim, h_o_ref,
ref_o, d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9276     bench_fa_3_tma_ws_launch<64, 2, kMmaAccF32>(B, H, seqlen, head_dim, h_o_ref,
ref_o, d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9277     bench_fa_3_tma_ws_launch<64, 3, kMmaAccF32>(B, H, seqlen, head_dim, h_o_ref,
ref_o, d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9278     bench_fa_3_tma_ws_launch<64, 4, kMmaAccF32>(B, H, seqlen, head_dim, h_o_ref,
ref_o, d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9279 } else if (head_dim == 128) {
9280     bench_fa_3_tma_ws_launch<128, 1, kMmaAccF32>(B, H, seqlen, head_dim, h_o_ref,
ref_o, d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9281     bench_fa_3_tma_ws_launch<128, 2, kMmaAccF32>(B, H, seqlen, head_dim, h_o_ref,
ref_o, d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9282     bench_fa_3_tma_ws_launch<128, 3, kMmaAccF32>(B, H, seqlen, head_dim, h_o_ref,
ref_o, d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9283     bench_fa_3_tma_ws_launch<128, 4, kMmaAccF32>(B, H, seqlen, head_dim, h_o_ref,
ref_o, d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9284 }
9285 }
9286
9287 // D=64/128 dispatch wrapper for FA3-style bench
9288 template <int kMmaAccF32 = 0>
9289 static void bench_fa_3_tma_ws_dispatch(int B, int H, int seqlen, int head_dim,
half *h_o_ref, float *ref_o,
half *d_q, half *d_k, half *d_v,
half *d_o, float cudnn_tflops_f16) {
9290 // Try stages up to 4; bench_fa_3_tma_ws_launch internally checks
9291 // cudaDevAttrMaxSharedMemoryPerBlockOptin and skips configs that
9292 // exceed the actual device limit (e.g. Hopper 224KB vs Blackwell 101KB).
9293 if (head_dim == 64) {
9294     bench_fa_3_tma_ws_launch<64, 1, kMmaAccF32>(B, H, seqlen, head_dim, h_o_ref,
ref_o, d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9295     bench_fa_3_tma_ws_launch<64, 2, kMmaAccF32>(B, H, seqlen, head_dim, h_o_ref,
ref_o, d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9296     bench_fa_3_tma_ws_launch<64, 3, kMmaAccF32>(B, H, seqlen, head_dim, h_o_ref,
ref_o, d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9297     bench_fa_3_tma_ws_launch<64, 4, kMmaAccF32>(B, H, seqlen, head_dim, h_o_ref,
ref_o, d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9298 } else if (head_dim == 128) {
9299     bench_fa_3_tma_ws_launch<128, 1, kMmaAccF32>(B, H, seqlen, head_dim, h_o_ref,
ref_o, d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9300     bench_fa_3_tma_ws_launch<128, 2, kMmaAccF32>(B, H, seqlen, head_dim, h_o_ref,
ref_o, d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9301     bench_fa_3_tma_ws_launch<128, 3, kMmaAccF32>(B, H, seqlen, head_dim, h_o_ref,
ref_o, d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9302     bench_fa_3_tma_ws_launch<128, 4, kMmaAccF32>(B, H, seqlen, head_dim, h_o_ref,
ref_o, d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9303 }
9304 }

```

```

9297 } else {
9298     char label[64];
9299     snprintf(label, sizeof(label), "FA3-style TMA MMA WS (D!=64/128)");
9300     printf("| %-41s | %-12s | %-4s | %-19s |\n", label, "SKIP", "SKIP", "None");
9301 }
9302 }
9303 #endif /* NOTES_V2_ENABLE_TMA_MMA_WS */
9304
9305 #if defined(NOTES_V2_ENABLE_CUDNN)
9306 static float bench_cudnn_sdpa_tflops(half *d_q, half *d_k, half *d_v,
9307                                     half *d_o_ref, int B, int H, int seqlen,
9308                                     int head_dim, fe::DataType_t compute_type) {
9309     cudnnHandle_t handle;
9310     cudnnCreate(&handle);
9311     auto graph = std::make_shared<fe::graph::Graph>();
9312     graph->set_io_data_type(fe::DataType_t::HALF)
9313         .set_intermediate_data_type(fe::DataType_t::FLOAT)
9314         .set_compute_data_type(compute_type);
9315
9316     auto Q = graph->tensor(fe::graph::Tensor_attributes()
9317         .set_uid(1).set_dim({B, H, seqlen, head_dim})
9318         .set_stride({H * seqlen * head_dim, seqlen * head_dim, head_dim, 1}));
9319     auto K = graph->tensor(fe::graph::Tensor_attributes()
9320         .set_uid(2).set_dim({B, H, seqlen, head_dim})
9321         .set_stride({H * seqlen * head_dim, seqlen * head_dim, head_dim, 1}));
9322     auto V = graph->tensor(fe::graph::Tensor_attributes()
9323         .set_uid(3).set_dim({B, H, seqlen, head_dim})
9324         .set_stride({H * seqlen * head_dim, seqlen * head_dim, head_dim, 1}));
9325
9326     auto [O_sdpa, Stats] = graph->sdpa(Q, K, V,
9327         fe::graph::SDPA_attributes()
9328             .set_name("sdpa_ref")
9329             .set_attn_scale(1.0f / sqrtf((float)head_dim)));
9330
9331     O_sdpa->set_output(true).set_uid(4)
9332         .set_dim({B, H, seqlen, head_dim})
9333         .set_stride({H * seqlen * head_dim, seqlen * head_dim, head_dim, 1});
9334
9335     auto build_status = graph->build(handle, {fe::HeurMode_t::A, fe::HeurMode_t::FALLBACK});
9336     float tflops = -1.0f;
9337     if (build_status.is_good()) {
9338         std::unordered_map<fe::graph::Tensor_attributes::uid_t, void*> vp = {
9339             {1, d_q}, {2, d_k}, {3, d_v}, {4, d_o_ref}};
9340         int64_t ws_size = 0;
9341         if (graph->get_workspace_size(ws_size).is_good()) {
9342             int8_t *d_ws = nullptr;
9343             if (ws_size > 0) check(cudaMalloc(&d_ws, ws_size), "bench cudnn ws");
9344             if (graph->execute(handle, vp, d_ws).is_good()) {
9345                 check(cudaDeviceSynchronize(), "bench cudnn sync");
9346                 for (int w = 1; w < g_warmup; ++w)
9347                     (void)graph->execute(handle, vp, d_ws);
9348                 cudaDeviceSynchronize();
9349                 cudaEvent_t ev_s, ev_e;
9350                 cudaEventCreate(&ev_s); cudaEventCreate(&ev_e);
9351                 cudaEventRecord(ev_s);
9352                 for (int r = 0; r < g_repeat; ++r)
9353                     (void)graph->execute(handle, vp, d_ws);
9354                 cudaEventRecord(ev_e);
9355                 cudaEventSynchronize(ev_e);
9356                 float time_ms = 0;
9357                 cudaEventElapsedTime(&time_ms, ev_s, ev_e);
9358                 time_ms /= g_repeat;
9359                 tflops = bench_fa_tflops(B, H, seqlen, head_dim, time_ms);

```

```

9360     cudaEventDestroy(ev_s); cudaEventDestroy(ev_e);
9361 }
9362 if (d_ws) cudaFree(d_ws);
9363 }
9364 }
9365 cudnnDestroy(handle);
9366 return tflops;
9367 }
9368 #endif
9369
9370 static void bench_flash_attn(int B, int H, int N, int D) {
9371     int seqlen = N, head_dim = D;
9372
9373     if (head_dim != 64 && head_dim != 128) {
9374         printf("| %-41s | %-12s | %-4s | %-19s |\n", "FlashAttention-2", "unsupported D", "SKIP", "None");
9375         return;
9376     }
9377
9378     size_t sz = (size_t)B * H * seqlen * head_dim * sizeof(half);
9379
9380     srand(42);
9381     half *h_q = (half *)malloc(sz);
9382     half *h_k = (half *)malloc(sz);
9383     half *h_v = (half *)malloc(sz);
9384     for (int i = 0; i < B * H * seqlen * head_dim; i++) {
9385         h_q[i] = __float2half(((float)rand() / RAND_MAX) * 2.0f - 1.0f);
9386         h_k[i] = __float2half(((float)rand() / RAND_MAX) * 2.0f - 1.0f);
9387         h_v[i] = __float2half(((float)rand() / RAND_MAX) * 2.0f - 1.0f);
9388     }
9389
9390     // Device allocations — shared by kernel and reference
9391     half *d_q, *d_k, *d_v, *d_o;
9392     check(cudaMalloc(&d_q, sz), "bench fa alloc Q");
9393     check(cudaMalloc(&d_k, sz), "bench fa alloc K");
9394     check(cudaMalloc(&d_v, sz), "bench fa alloc V");
9395     check(cudaMalloc(&d_o, sz), "bench fa alloc O");
9396     check(cudaMemcpy(d_q, h_q, sz, cudaMemcpyHostToDevice), "bench fa H2D Q");
9397     check(cudaMemcpy(d_k, h_k, sz, cudaMemcpyHostToDevice), "bench fa H2D K");
9398     check(cudaMemcpy(d_v, h_v, sz, cudaMemcpyHostToDevice), "bench fa H2D V");
9399
9400     // Reference output: either cuDNN (half) or CPU (float)
9401     half *h_o_ref = nullptr;
9402     float *ref_o = nullptr;
9403     float cudnn_tflops_f16 = -1.0f, cudnn_tflops_f32 = -1.0f;
9404     int ref_count = B * H * seqlen * head_dim;
9405
9406     #if defined(NOTES_V2_ENABLE_CUDNN)
9407     if (!g_fa_skip_check) {
9408         half *d_o_ref;
9409         check(cudaMalloc(&d_o_ref, sz), "bench fa alloc O_ref");
9410         cudnn_tflops_f16 = bench_cudnn_sdpa_tflops(d_q, d_k, d_v, d_o_ref,
9411                                                     B, H, seqlen, head_dim,
9412                                                     fe::DataType_t::HALF);
9413         bool cudnn_ok = (cudnn_tflops_f16 > 0);
9414         if (cudnn_ok) {
9415             cudnn_tflops_f32 = bench_cudnn_sdpa_tflops(d_q, d_k, d_v, d_o_ref,
9416                                                         B, H, seqlen, head_dim,
9417                                                         fe::DataType_t::FLOAT);
9418             h_o_ref = (half *)malloc(sz);
9419             check(cudaMemcpy(h_o_ref, d_o_ref, sz, cudaMemcpyDeviceToHost), "bench fa D2H ref");
9420         } else {
9421             fprintf(stderr, "cudnn SDPA unsupported on this SM, using CPU ref\n");
9422         }
9423     }

```

```

9423     cudaFree(d_o_ref);
9424 }
9425 #endif
9426
9427 // CPU reference fallback
9428 if (!g_fa_skip_check && !h_o_ref) {
9429     float *ref_q = (float *)malloc(sz * 4 / sizeof(half));
9430     float *ref_k = (float *)malloc(sz * 4 / sizeof(half));
9431     float *ref_v = (float *)malloc(sz * 4 / sizeof(half));
9432     ref_o = (float *)malloc(sz * 4 / sizeof(half));
9433     for (int i = 0; i < ref_count; i++) {
9434         ref_q[i] = __half2float(h_q[i]);
9435         ref_k[i] = __half2float(h_k[i]);
9436         ref_v[i] = __half2float(h_v[i]);
9437     }
9438
9439     float scale = 1.0f / sqrtf((float)head_dim);
9440     for (int bi = 0; bi < B * H; bi++) {
9441         for (int qi = 0; qi < seqlen; qi++) {
9442             float smax = -INFINITY;
9443             float *S = (float *)malloc((size_t)seqlen * sizeof(float));
9444             for (int kj = 0; kj < seqlen; kj++) {
9445                 float s = 0.0f;
9446                 for (int d = 0; d < head_dim; d++)
9447                     s += ref_q[bi * seqlen * head_dim + qi * head_dim + d] *
9448                        ref_k[bi * seqlen * head_dim + kj * head_dim + d];
9449                 S[kj] = s * scale;
9450                 if (S[kj] > smax) smax = S[kj];
9451             }
9452             double sum_exp = 0.0;
9453             for (int kj = 0; kj < seqlen; kj++)
9454                 sum_exp += (double)expf(S[kj] - smax);
9455             float inv_sum = 1.0f / (float)sum_exp;
9456             for (int d = 0; d < head_dim; d++) {
9457                 double o_acc = 0.0;
9458                 for (int kj = 0; kj < seqlen; kj++)
9459                     o_acc += (double)(expf(S[kj] - smax) * inv_sum) *
9460                        ref_v[bi * seqlen * head_dim + kj * head_dim + d];
9461                 ref_o[bi * seqlen * head_dim + qi * head_dim + d] = (float)o_acc;
9462             }
9463             free(S);
9464         }
9465     }
9466     free(ref_q);
9467     free(ref_k);
9468     free(ref_v);
9469 }
9470
9471 if (head_dim == 64) {
9472     if (g_fa_layout == FALayout::All || g_fa_layout == FALayout::Pad) {
9473         cudaDeviceSynchronize();
9474         bench_fa_launch<64, 1, 8, 8, 8, 0>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9475                                             d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9476         cudaDeviceSynchronize();
9477         bench_fa_launch<64, 2, 8, 8, 8, 0>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9478                                             d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9479         cudaDeviceSynchronize();
9480         bench_fa_launch<64, 1, 8, 8, 8, 1>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9481                                             d_q, d_k, d_v, d_o, cudnn_tflops_f32);
9482         cudaDeviceSynchronize();
9483         bench_fa_launch<64, 2, 8, 8, 8, 1>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9484                                             d_q, d_k, d_v, d_o, cudnn_tflops_f32);
9485     }

```

```

9486 if (g_fa_layout == FALayout::All || g_fa_layout == FALayout::SwizzleQ) {
9487     cudaDeviceSynchronize();
9488     bench_fa_launch<64, 2, 0, 8, 8, 0>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9489         d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9490     cudaDeviceSynchronize();
9491     bench_fa_launch<64, 2, 0, 8, 8, 1>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9492         d_q, d_k, d_v, d_o, cudnn_tflops_f32);
9493 }
9494 if (g_fa_layout == FALayout::All || g_fa_layout == FALayout::SwizzleK) {
9495     cudaDeviceSynchronize();
9496     bench_fa_launch<64, 2, 8, 0, 8, 0>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9497         d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9498     cudaDeviceSynchronize();
9499     bench_fa_launch<64, 2, 8, 0, 8, 1>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9500         d_q, d_k, d_v, d_o, cudnn_tflops_f32);
9501 }
9502 if (g_fa_layout == FALayout::All || g_fa_layout == FALayout::SwizzleV) {
9503     cudaDeviceSynchronize();
9504     bench_fa_launch<64, 2, 8, 8, 0, 0>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9505         d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9506     cudaDeviceSynchronize();
9507     bench_fa_launch<64, 2, 8, 8, 0, 1>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9508         d_q, d_k, d_v, d_o, cudnn_tflops_f32);
9509 }
9510 if (g_fa_layout == FALayout::All || g_fa_layout == FALayout::SwizzleQK) {
9511     cudaDeviceSynchronize();
9512     bench_fa_launch<64, 2, 0, 0, 8, 0>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9513         d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9514     cudaDeviceSynchronize();
9515     bench_fa_launch<64, 2, 0, 0, 8, 1>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9516         d_q, d_k, d_v, d_o, cudnn_tflops_f32);
9517 }
9518 if (g_fa_layout == FALayout::All || g_fa_layout == FALayout::SwizzleQV) {
9519     cudaDeviceSynchronize();
9520     bench_fa_launch<64, 2, 0, 8, 0, 0>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9521         d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9522     cudaDeviceSynchronize();
9523     bench_fa_launch<64, 2, 0, 8, 0, 1>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9524         d_q, d_k, d_v, d_o, cudnn_tflops_f32);
9525 }
9526 if (g_fa_layout == FALayout::All || g_fa_layout == FALayout::SwizzleKV) {
9527     cudaDeviceSynchronize();
9528     bench_fa_launch<64, 2, 8, 0, 0, 0>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9529         d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9530     cudaDeviceSynchronize();
9531     bench_fa_launch<64, 2, 8, 0, 0, 1>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9532         d_q, d_k, d_v, d_o, cudnn_tflops_f32);
9533 }
9534 if (g_fa_layout == FALayout::All || g_fa_layout == FALayout::Swizzle) {
9535     cudaDeviceSynchronize();
9536     bench_fa_launch<64, 2, 0, 0, 0, 0>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9537         d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9538     cudaDeviceSynchronize();
9539     bench_fa_launch<64, 2, 0, 0, 0, 1>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9540         d_q, d_k, d_v, d_o, cudnn_tflops_f32);
9541 }
9542 #if defined(NOTES_V2_ENABLE_TMA_MMA_WS)
9543 {
9544     cudaDeviceSynchronize();
9545     bench_fa_tma_mma_ws_dispatch<1, 1, 0>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9546         d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9547     cudaDeviceSynchronize();
9548     bench_fa_tma_mma_ws_dispatch<2, 1, 0>(B, H, seqlen, head_dim, h_o_ref, ref_o,

```

```

9549         d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9550     cudaDeviceSynchronize();
9551     bench_fa_tma_mma_ws_dispatch<3, 1, 0>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9552         d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9553     cudaDeviceSynchronize();
9554     bench_fa_tma_mma_ws_dispatch<4, 1, 0>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9555         d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9556     cudaDeviceSynchronize();
9557     bench_fa_tma_mma_ws_dispatch<2, 2, 0>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9558         d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9559     // F32Acc variants
9560     cudaDeviceSynchronize();
9561     bench_fa_tma_mma_ws_dispatch<1, 1, 1>(B, H, seqlen, head_dim, h_o_ref,
9562         ref_o, d_q, d_k, d_v, d_o, cudnn_tflops_f32);
9563     cudaDeviceSynchronize();
9564     bench_fa_tma_mma_ws_dispatch<2, 1, 1>(B, H, seqlen, head_dim, h_o_ref,
9565         ref_o, d_q, d_k, d_v, d_o, cudnn_tflops_f32);
9566     cudaDeviceSynchronize();
9567     bench_fa_tma_mma_ws_dispatch<3, 1, 1>(B, H, seqlen, head_dim, h_o_ref,
9568         ref_o, d_q, d_k, d_v, d_o, cudnn_tflops_f32);
9569     cudaDeviceSynchronize();
9570     bench_fa_tma_mma_ws_dispatch<4, 1, 1>(B, H, seqlen, head_dim, h_o_ref,
9571         ref_o, d_q, d_k, d_v, d_o, cudnn_tflops_f32);
9572     cudaDeviceSynchronize();
9573     bench_fa_tma_mma_ws_dispatch<2, 2, 1>(B, H, seqlen, head_dim, h_o_ref,
9574         ref_o, d_q, d_k, d_v, d_o, cudnn_tflops_f32);
9575     // FA3-style dual-consumer
9576     cudaDeviceSynchronize();
9577     bench_fa_3_tma_ws_dispatch<0>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9578         d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9579     cudaDeviceSynchronize();
9580     bench_fa_3_tma_ws_dispatch<1>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9581         d_q, d_k, d_v, d_o, cudnn_tflops_f32);
9582 }
9583 #endif
9584 } else {
9585     if (g_fa_layout == FALayout::All || g_fa_layout == FALayout::Pad) {
9586         cudaDeviceSynchronize();
9587         bench_fa_launch<128, 1, 8, 8, 8, 0>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9588             d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9589         cudaDeviceSynchronize();
9590         bench_fa_launch<128, 2, 8, 8, 8, 0>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9591             d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9592         cudaDeviceSynchronize();
9593         bench_fa_launch<128, 1, 8, 8, 8, 1>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9594             d_q, d_k, d_v, d_o, cudnn_tflops_f32);
9595         cudaDeviceSynchronize();
9596         bench_fa_launch<128, 2, 8, 8, 8, 1>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9597             d_q, d_k, d_v, d_o, cudnn_tflops_f32);
9598     }
9599     if (g_fa_layout == FALayout::All || g_fa_layout == FALayout::SwizzleQ) {
9600         cudaDeviceSynchronize();
9601         bench_fa_launch<128, 2, 0, 8, 8, 0>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9602             d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9603         cudaDeviceSynchronize();
9604         bench_fa_launch<128, 2, 0, 8, 8, 1>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9605             d_q, d_k, d_v, d_o, cudnn_tflops_f32);
9606     }
9607     if (g_fa_layout == FALayout::All || g_fa_layout == FALayout::SwizzleK) {
9608         cudaDeviceSynchronize();
9609         bench_fa_launch<128, 2, 8, 0, 8, 0>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9610             d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9611         cudaDeviceSynchronize();

```

```

9612     bench_fa_launch<128, 2, 8, 0, 8, 1>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9613         d_q, d_k, d_v, d_o, cudnn_tflops_f32);
9614 }
9615 if (g_fa_layout == FALayout::All || g_fa_layout == FALayout::SwizzleV) {
9616     cudaDeviceSynchronize();
9617     bench_fa_launch<128, 2, 8, 8, 0, 0>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9618         d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9619     cudaDeviceSynchronize();
9620     bench_fa_launch<128, 2, 8, 8, 0, 1>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9621         d_q, d_k, d_v, d_o, cudnn_tflops_f32);
9622 }
9623 if (g_fa_layout == FALayout::All || g_fa_layout == FALayout::SwizzleQK) {
9624     cudaDeviceSynchronize();
9625     bench_fa_launch<128, 2, 0, 0, 8, 0>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9626         d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9627     cudaDeviceSynchronize();
9628     bench_fa_launch<128, 2, 0, 0, 8, 1>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9629         d_q, d_k, d_v, d_o, cudnn_tflops_f32);
9630 }
9631 if (g_fa_layout == FALayout::All || g_fa_layout == FALayout::SwizzleQV) {
9632     cudaDeviceSynchronize();
9633     bench_fa_launch<128, 2, 0, 8, 0, 0>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9634         d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9635     cudaDeviceSynchronize();
9636     bench_fa_launch<128, 2, 0, 8, 0, 1>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9637         d_q, d_k, d_v, d_o, cudnn_tflops_f32);
9638 }
9639 if (g_fa_layout == FALayout::All || g_fa_layout == FALayout::SwizzleKV) {
9640     cudaDeviceSynchronize();
9641     bench_fa_launch<128, 2, 8, 0, 0, 0>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9642         d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9643     cudaDeviceSynchronize();
9644     bench_fa_launch<128, 2, 8, 0, 0, 1>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9645         d_q, d_k, d_v, d_o, cudnn_tflops_f32);
9646 }
9647 if (g_fa_layout == FALayout::All || g_fa_layout == FALayout::Swizzle) {
9648     cudaDeviceSynchronize();
9649     bench_fa_launch<128, 2, 0, 0, 0, 0>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9650         d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9651     cudaDeviceSynchronize();
9652     bench_fa_launch<128, 2, 0, 0, 0, 1>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9653         d_q, d_k, d_v, d_o, cudnn_tflops_f32);
9654 }
9655 #if defined(NOTES_V2_ENABLE_TMA_MMA_WS)
9656 {
9657     cudaDeviceSynchronize();
9658     bench_fa_tma_mma_ws_dispatch<1, 1, 0>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9659         d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9660     cudaDeviceSynchronize();
9661     bench_fa_tma_mma_ws_dispatch<2, 1, 0>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9662         d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9663     cudaDeviceSynchronize();
9664     bench_fa_tma_mma_ws_dispatch<3, 1, 0>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9665         d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9666     cudaDeviceSynchronize();
9667     bench_fa_tma_mma_ws_dispatch<2, 2, 0>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9668         d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9669     // F32Acc variants
9670     cudaDeviceSynchronize();
9671     bench_fa_tma_mma_ws_dispatch<1, 1, 1>(B, H, seqlen, head_dim, h_o_ref,
9672         ref_o, d_q, d_k, d_v, d_o, cudnn_tflops_f32);
9673     cudaDeviceSynchronize();
9674     bench_fa_tma_mma_ws_dispatch<2, 1, 1>(B, H, seqlen, head_dim, h_o_ref,

```

```

9675         ref_o, d_q, d_k, d_v, d_o, cudnn_tflops_f32);
9676     cudaDeviceSynchronize();
9677     bench_fa_tma_mma_ws_dispatch<3, 1, 1>(B, H, seqlen, head_dim, h_o_ref,
9678         ref_o, d_q, d_k, d_v, d_o, cudnn_tflops_f32);
9679     cudaDeviceSynchronize();
9680     bench_fa_tma_mma_ws_dispatch<2, 2, 1>(B, H, seqlen, head_dim, h_o_ref,
9681         ref_o, d_q, d_k, d_v, d_o, cudnn_tflops_f32);
9682     // FA3-style dual-consumer
9683     cudaDeviceSynchronize();
9684     bench_fa_3_tma_ws_dispatch<0>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9685         d_q, d_k, d_v, d_o, cudnn_tflops_f16);
9686     cudaDeviceSynchronize();
9687     bench_fa_3_tma_ws_dispatch<1>(B, H, seqlen, head_dim, h_o_ref, ref_o,
9688         d_q, d_k, d_v, d_o, cudnn_tflops_f32);
9689 }
9690 #endif
9691 }
9692 cudaDeviceSynchronize();
9693
9694 free(h_q);
9695 free(h_k);
9696 free(h_v);
9697 free(h_o_ref);
9698 free(ref_o);
9699 cudaFree(d_q);
9700 cudaFree(d_k);
9701 cudaFree(d_v);
9702 cudaFree(d_o);
9703 }
9704
9705 // Host 端 v1/v2 等价性测试: 遍历 kColStride/i/j, 断言 swizzle_v1_impl == swizzle_v2_impl.
9706 // 用于在启用 NOTES_V2_ENABLE_SWIZZLE_V2 前后验证 v2 与 v1 bit-exact 等价.
9707 // kColStride=8 在 v2 内回退 v1, 故 8 也应恒等.
9708 template <int kColStride>
9709 static void swizzle_equiv_check_one(long &total, long &fail) {
9710     for (int i = 0; i < 256; ++i) {
9711         for (int j = 0; j < kColStride; ++j) {
9712             int v1 = swizzle_v1_impl<kColStride>(i, j);
9713             int v2 = swizzle_v2_impl<kColStride>(i, j);
9714             ++total;
9715             if (v1 != v2) {
9716                 ++fail;
9717                 if (fail <= 10)
9718                     printf(" MISMATCH cs=%d i=%d j=%d: v1=%d v2=%d\n", kColStride, i, j, v1, v2);
9719             }
9720         }
9721     }
9722 }
9723
9724 static void test_swizzle_equiv() {
9725     long total = 0, fail = 0;
9726     swizzle_equiv_check_one<8>(total, fail);
9727     swizzle_equiv_check_one<16>(total, fail);
9728     swizzle_equiv_check_one<32>(total, fail);
9729     swizzle_equiv_check_one<64>(total, fail);
9730     printf("| %-41s | %-12s | %-4s | %-19s |\n",
9731         "Swizzle v1/v2 equiv (host)", "N/A",
9732         fail == 0 ? "PASS" : "FAIL",
9733         fail == 0 ? "ALL PASS" : "FAIL");
9734     printf(" total=%ld fail=%ld\n", total, fail);
9735 }
9736
9737 int main(int argc, char *argv[]) {

```

```

9738 #if defined(NOTES_V2_ENABLE_WGMMMA) || defined(NOTES_V2_ENABLE_TMA_MMA_WS)
9739     cuInit(0);
9740 #endif
9741
9742 for (int i = 1; i < argc; i++) {
9743     if (strcmp(argv[i], "--bench-hgemm") == 0) {
9744         g_bench_hgemm = true;
9745     } else if (strcmp(argv[i], "--bench") == 0) {
9746         g_bench_hgemm = true;
9747         g_bench_fa = true;
9748     } else if (strcmp(argv[i], "--bench-fa") == 0) {
9749         g_bench_fa = true;
9750     } else if (strcmp(argv[i], "--bench-hgemm-all") == 0) {
9751         g_bench_hgemm_all = true;
9752     } else if (strcmp(argv[i], "--bench-fa-all") == 0) {
9753         g_bench_fa = true;
9754         g_fa_layout = FALayout::All;
9755     } else if (strcmp(argv[i], "--bench-all") == 0) {
9756         g_bench_all = true;
9757         g_fa_layout = FALayout::All;
9758     } else if (strcmp(argv[i], "--mnk") == 0 && i + 1 < argc) {
9759         sscanf(argv[++i], "%d,%d,%d", &g_bench_M, &g_bench_N, &g_bench_K);
9760     } else if (strcmp(argv[i], "--bhnd") == 0 && i + 1 < argc) {
9761         sscanf(argv[++i], "%d,%d,%d,%d", &g_bench_B, &g_bench_H, &g_bench_Nfa, &g_bench_D);
9762     } else if (strcmp(argv[i], "--fa-layout") == 0 && i + 1 < argc) {
9763         const char *layout = argv[++i];
9764         if (strcmp(layout, "all") == 0)
9765             g_fa_layout = FALayout::All;
9766         else if (strcmp(layout, "pad") == 0)
9767             g_fa_layout = FALayout::Pad;
9768         else if (strcmp(layout, "swizzle-q") == 0)
9769             g_fa_layout = FALayout::SwizzleQ;
9770         else if (strcmp(layout, "swizzle-k") == 0)
9771             g_fa_layout = FALayout::SwizzleK;
9772         else if (strcmp(layout, "swizzle-v") == 0)
9773             g_fa_layout = FALayout::SwizzleV;
9774         else if (strcmp(layout, "swizzle-qk") == 0)
9775             g_fa_layout = FALayout::SwizzleQK;
9776         else if (strcmp(layout, "swizzle-qv") == 0)
9777             g_fa_layout = FALayout::SwizzleQV;
9778         else if (strcmp(layout, "swizzle-kv") == 0)
9779             g_fa_layout = FALayout::SwizzleKV;
9780         else if (strcmp(layout, "swizzle") == 0)
9781             g_fa_layout = FALayout::Swizzle;
9782         else {
9783             fprintf(stderr, "Unsupported FA layout: %s\n", layout);
9784             return EXIT_FAILURE;
9785         }
9786     } else if (strcmp(argv[i], "--fa-skip-check") == 0) {
9787         g_fa_skip_check = true;
9788     } else if (strcmp(argv[i], "--swizzle-eq-check") == 0) {
9789         g_swizzle_eq_check = true;
9790     } else if (strcmp(argv[i], "--verbose") == 0) {
9791         g_verbose = true;
9792     } else if (strcmp(argv[i], "--warmup") == 0 && i + 1 < argc) {
9793         g_warmup = atoi(argv[++i]);
9794     } else if (strcmp(argv[i], "--repeat") == 0 && i + 1 < argc) {
9795         g_repeat = atoi(argv[++i]);
9796     }
9797 }
9798
9799 if (g_swizzle_eq_check) {
9800     printf("=== notes-v2.cu swizzle v1/v2 equivalence check ===\n");

```

```

9801     printf("| %-41s | %-12s | %-4s | %-19s |\n", "Kernel", "Max Err", "Pass", "TFLOPS/cu{BLAS,DNN}");
9802     printf("|-----|-----|-----|-----|\n");
9803     test_swizzle_equiv();
9804     printf("=== Done ===\n");
9805     return 0;
9806 }
9807
9808 if (g_bench_hgemm || g_bench_fa || g_bench_all) {
9809     printf("=== notes-v2.cu bench mode ===\n");
9810     printf("HGEMM: M=%d N=%d K=%d   FA: B=%d H=%d N=%d D=%d\n",
9811           g_bench_M, g_bench_N, g_bench_K,
9812           g_bench_B, g_bench_H, g_bench_Nfa, g_bench_D);
9813     printf("| %-41s | %-12s | %-4s | %-19s |\n", "Kernel", "Max Err", "Pass", "TFLOPS/cu{BLAS,DNN}");
9814     printf("|-----|-----|-----|-----|\n");
9815
9816     if (g_bench_hgemm || g_bench_hgemm_all || g_bench_all) {
9817 #if defined(NOTES_V2_ENABLE_CUTE)
9818         bench_hgemm_cute(g_bench_M, g_bench_N, g_bench_K);
9819 #endif
9820         if (g_bench_hgemm_all || g_bench_all) {
9821             bench_hgemm_mma(g_bench_M, g_bench_N, g_bench_K);
9822             bench_hgemm_swizzle(g_bench_M, g_bench_N, g_bench_K);
9823 #if defined(NOTES_V2_ENABLE_WGMMA)
9824             bench_hgemm_wgmma(g_bench_M, g_bench_N, g_bench_K);
9825 #endif
9826 #if defined(NOTES_V2_ENABLE_TMA_MMA_WS)
9827             bench_hgemm_tma_mma_ws(g_bench_M, g_bench_N, g_bench_K);
9828 #endif
9829         }
9830     }
9831     if (g_bench_fa || g_bench_all)
9832         bench_flash_attn(g_bench_B, g_bench_H, g_bench_Nfa, g_bench_D);
9833
9834     printf("=== Bench done ===\n");
9835     return 0;
9836 }
9837
9838 #if defined(NOTES_V2_ENABLE_TMA_MMA_WS)
9839 if (argc >= 2 && strcmp(argv[1], "--tma-mma-ws") == 0) {
9840     int M = 128, N = 128, K = 64;
9841     if (argc > 4) {
9842         M = atoi(argv[2]);
9843         N = atoi(argv[3]);
9844         K = atoi(argv[4]);
9845     }
9846     printf("=== SM120 TMA MMA WS validation ===\n");
9847     printf("| %-41s | %-12s | %-4s | %-19s |\n", "Kernel", "Max Err", "Pass", "TFLOPS/cu{BLAS,DNN}");
9848     printf("|-----|-----|-----|-----|\n");
9849     test_hgemm_tma_mma_ws(M, N, K);
9850     return 0;
9851 }
9852 #endif
9853 int M = 1024, N = 1024, K = 1024;
9854 if (argc > 3) { M = atoi(argv[1]); N = atoi(argv[2]); K = atoi(argv[3]); }
9855
9856 printf("=== notes-v2.cu verification harness ===\n");
9857 printf("| %-41s | %-12s | %-4s | %-19s |\n", "Kernel", "Max Err", "Pass", "TFLOPS/cu{BLAS,DNN}");
9858 printf("|-----|-----|-----|-----|\n");
9859
9860 test_block_reduce(N);
9861 test_dot(N);
9862 test_relu(1024);

```

```

9863 test_elementwise(1024);
9864 test_histogram(1024);
9865 test_merge_attn_states(512, 16, 128);
9866 test_softmax(256);
9867 test_rms_norm(8, 128);
9868 test_layer_norm(8, 128);
9869 test_rope(8, 128);
9870 test_mat_transpose(256, 256);
9871 test_mat_transpose_padded(256, 256);
9872 test_sgemv(256, 128);
9873 test_sgemm(M, N, K);
9874 test_hgemm_mma(M, N, K);
9875 test_hgemm_swizzle(M, N, K);
9876 #if (defined(NOTES_V2_ENABLE_CUTE))
9877     test_hgemm_cute(M, N, K);
9878 #endif
9879 #if (defined(NOTES_V2_ENABLE_WGMMMA))
9880     test_hgemm_wgmma(M, N, K);
9881 #endif
9882 #if defined(NOTES_V2_ENABLE_TMA_MMA_WS)
9883     test_hgemm_tma_mma_ws(M, N, K);
9884 #endif
9885     test_flash_attn(1024, 64);
9886 #if defined(NOTES_V2_ENABLE_TMA_MMA_WS)
9887     test_flash_attn_tma_mma_ws(1024, 64);
9888     test_flash_attn_tma_mma_ws(1024, 128);
9889     test_flash_attn_3_tma_ws(1024, 64);
9890     test_flash_attn_3_tma_ws(1024, 128);
9891 #endif
9892
9893     printf("=== All tests done ===\n");
9894     return 0;
9895 }
9896
9897 // =====
9898 // Quick build & run reference
9899 // =====
9900 // # sm_86 + CuTe (Ampere, RTX 30/40 系列, 无 WGMMMA) :
9901 // nvcc -std=c++20 -O2 -arch=sm_86 -DNOTES_V2_ENABLE_CUTE \
9902 // -I ../../third-party/cutlass/include \
9903 // -lcublas -lcuda notes-v2.cu -o notes_v2_cute_sm86.bin
9904 //
9905 // # sm_89 单独编译 + 运行:
9906 // nvcc -std=c++20 -O2 -arch=sm_89 -lcublas -lcuda notes-v2.cu -o notes_v2_sm89.bin
9907 //
9908 // # sm_89 + CuTe (需要 CUTLASS include 路径, 编译 CUDA_HOME 指向的 CUTLASS 或
9909 // 项目内置 third-party/cutlass) :
9910 // nvcc -std=c++20 -O2 -arch=sm_89 -DNOTES_V2_ENABLE_CUTE \
9911 // -I ../../third-party/cutlass/include \
9912 // -lcublas -lcuda notes-v2.cu -o notes_v2_cute_sm89.bin
9913 //
9914 // # sm_90a 单独编译 + 运行 (需要 Hopper GPU, H100/H200 均可) :
9915 // nvcc -std=c++20 -O2 -gencode arch=compute_90a,code=sm_90a \
9916 // -DNOTES_V2_ENABLE_WGMMMA -lcublas -lcuda notes-v2.cu -o notes_v2_sm90.bin
9917 //
9918 // # sm_90a + CuTe + WGMMMA:
9919 // nvcc -std=c++20 -O2 -gencode arch=compute_90a,code=sm_90a \
9920 // -DNOTES_V2_ENABLE_CUTE -DNOTES_V2_ENABLE_WGMMMA \
9921 // -I ../../third-party/cutlass/include \
9922 // -lcublas -lcuda notes-v2.cu -o notes_v2_cute_sm90.bin
9923 //
9924 // # sm_120a TMA + warp-specialized mma.sync (CUDA Toolkit >= 13.2;
9925 // RTX PRO 5000 / RTX 5090):

```

```

9926 // nvcc -std=c++20 -O2 -arch=sm_120a -DNOTES_V2_ENABLE_TMA_MMA_WS \
9927 // -lcublas -lcuda notes-v2.cu -o notes_v2_tma_mma_ws_sm120.bin
9928 // ./notes_v2_tma_mma_ws_sm120.bin --tma-mma-ws 1024 1024 1024
9929 //
9930 // # sm_120a + CUTE + TMA_MMA_WS + cuDNN SDPA (CUDA Toolkit >= 13.2;
9931 // requires cudnn-frontend submodule: git submodule update --init):
9932 // nvcc -std=c++20 -O2 -arch=sm_120a \
9933 // -DNOTES_V2_ENABLE_CUTE -DNOTES_V2_ENABLE_TMA_MMA_WS -DNOTES_V2_ENABLE_CUDNN \
9934 // -I ../../third-party/cutlass/include -I ../../third-party/cudnn-frontend/include \
9935 // -L/usr/local/cuda-13.2/targets/x86_64-linux/lib/stubs \
9936 // -lcublas -lcudnn -lnvrtc -lcuda notes-v2.cu -o notes_v2_cute_ws_sm120a.bin
9937 // # Default FA bench (kPadQ=kPadK=kPadV=8 only):
9938 // ./notes_v2_cute_ws_sm120a.bin --bench --bench-fa --bhnd 1,48,4096,64
9939 // # All FA layout variants:
9940 // ./notes_v2_cute_ws_sm120a.bin --bench --bench-fa-all --bhnd 1,48,4096,64
9941 // # FA TMA MMA WS bench (SM120, D=64 only, kStagesK=2/3):
9942 // ./notes_v2_cute_ws_sm120a.bin --bench-fa-tma-ws --bhnd 1,32,4096,64

```